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Aspects of relativistic MHD wind theory

Li, Zhi-Yun, Ph.D.

University of Colorado at Boulder, 1993

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ASPECTS OF
RELATIVISTIC MHD WIND THEORY

by

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B.S., University of Science and Technology of China, P.R. China, 1987

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A Thesis submitted to the

Faculty of the Graduate School of the

University of Colorado in partial fulfillment

of the requirements for the degree of

Doctor of Philosophy

Department of Physics

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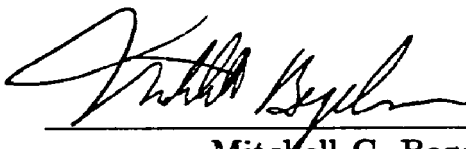
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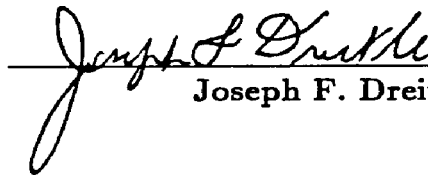
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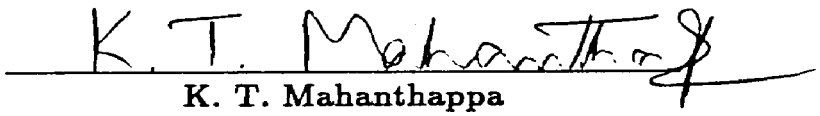
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To China, my motherland

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Li, Zhi-Yun (Ph.D., Physics)

Aspects of Relativistic MHD Wind Theory

Thesis directed by Professor Mitchell C. Begelman

We study the properties of the steady-state, relativistic outflows associated with active galactic nuclei (AGNs) and pulsars in the ideal magnetohydrodynamic (MHD) limit, assuming axisymmetry. We find that all relativistic MHD winds are collimated asymptotically towards the rotation axis, either to cylinders or to asymptotically current-free paraboloidal surfaces. The energy density of the electromagnetic field in the latter is small compared to the plasma kinetic energy density, and such a flow structure may exist in Crab-like free pulsar winds, where the flux of the plasma kinetic energy is inferred to dominate over the Poynting flux.

We obtain complete wind solutions first semi-analytically, by assuming that the flux surfaces in the relativistic flow are self-similar, and then numerically for a case in which the field lines are assumed to emerge vertically from an equatorial disk and are collimated parallel to the rotation axis far from the disk. We find that the electromagnetic energy is converted into plasma kinetic energy much more efficiently in a collimated flow than in a purely radial flow (which is the geometry generally assumed in earlier works), and indeed more than half of the total energy can end up in the kinetic form. We show that it is mainly the difference in the ambient pressure distribution that differentiates the relatively well-collimated jets in AGNs from the seemingly poorly-collimated Crab-like free pulsar winds.

We study the effects of external drag, due to either radiation or mass entrainment, on relativistic MHD winds. In a radial flow geometry, owing

to the location of the fast magnetosonic point at infinity, the radiation drag will not change the terminal speed of the flow, but will merely enhance the extraction of energy and angular momentum from the central rotating object and redistribute the locations of flow acceleration. If a flux tube diverges faster than radially, then it is possible for the drag to reduce the terminal flow speed greatly and in some cases even to stop the flow completely. It is conceivable that this reduction in flow speed may suppress the radio emission from most quasi-stellar objects (QSOs).

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CHAPTER I

INTRODUCTION

Outflows are observed in a wide variety of astrophysical systems, ranging from the solar wind in our neighborhood to powerful jets associated with distant Quasi-Stellar Objects (QSOs). These outflows can be produced by various mechanisms, including being driven by thermal pressure (e.g., solar wind, see Parker 1963), radiative force (e.g., O-star winds, see Abbott 1988 for a review), and electromagnetic fields. The last is a generic mechanism believed to operate whenever large amounts of open field lines are anchored onto a rapidly rotating central object. In essence, the plasma loaded on the open field lines is slung outward along the field lines by centrifugal force, and may be further accelerated by a magnetic pressure gradient to high speed. This mechanism has been discussed extensively in the context of magnetic braking of stars by stellar winds (e.g., Mestel 1968). The magnetic braking of the sun by the solar wind was first studied quantitatively by Weber & Davis (1967) using MHD theory. In a parallel work, Michel (1969) extended the analysis of Weber & Davis to the special relativistic regime and constructed the first MHD model for pulsar winds. When winds become collimated, they are called jets. The application of MHD wind theory to jets associated with extragalactic radio sources began with the unipolar inductor models of Lovelace (1976) and Blandford (1976), who were the first to demonstrate explicitly that these jets can be produced electromagnetically. A large number of theoretical works have since been devoted to the study of the properties of electromagnetically-driven outflows in this context (e.g., Lovelace, Wang, & Sulkanen 1987; Camenzind 1989; Li, Chiueh, & Begelman

1992 and references therein). After more than a decade of intensive research, however, many of the important aspects of these flows are only beginning to be understood. In this thesis, we shall summarize the current status of the MHD wind theory, and report the progress we made towards understanding a subset of MHD flows that have relativistic bulk motions. Relativistic flow velocities are inferred for some of the jets in the extragalactic radio sources and for pulsar winds such as that from the Crab pulsar, which is believed to power the Crab Nebula.

Before outlining the relativistic MHD wind theory we have developed in § 1.4, we shall first briefly review the two main classes of outflows to which the theory is intended to apply: relativistic jets from extragalactic radio sources (§ 1.1) and relativistic pulsar winds (§ 1.2). Since many of our conclusions on relativistic flows are equally applicable to nonrelativistic flows, we shall also describe some other types of nonrelativistic outflows observed in our own Galaxy, which are also likely driven electromagnetically (§ 1.3).

1.1. Jets in Extragalactic Radio Sources

1.1.1. An Overview of the Observed Properties

Jets are thin features that appear to link the nuclei of radio galaxies or quasars to extended regions of emission far outside the nucleus. The observational properties and theoretical interpretations of these features have been reviewed, for example, by Bridle & Perley (1984) and Begelman, Blandford, & Rees (1984). More information can be found in a recent book devoted entirely to this subject: "Beams and Jets in Astrophysics" edited by Hughes (1991), from which some of the material presented in this subsection is drawn.

The first known extragalactic jet, emanating from the giant elliptical galaxy M87, was discovered optically by Curtis (1918), although most of our knowledge about these objects and their associated sources comes from radio observations. On large scales (down to fraction of an arcsecond, or kpc scale for objects at redshift of $z \sim 0.1$), the observations are done mostly with the Very Large Array (VLA) in New Mexico, complemented by the Jodrell Bank MERLIN array for smaller sources where higher resolution is required and by the Westerbork array in Holland for very large extended sources. Smaller structures, on the milli-arcsecond scale (or pc scale for most sources), are observed with Very-Long-Baseline-Interferometry (VLBI). To illustrate the basic components that an extragalactic radio source might have on the large scale, we show in Figure 1.1 a 6-cm wavelength radiograph from the VLA of the famous "prototypical" classical double source Cygnus A. In this particular source, there is a radio "core" that centers on the host galaxy, and two thin "jets" that connect the core to the two "lobes" of extended emission that are far beyond and straddle the host galaxy. Bright "hot spots" are also visible in each lobe, which presumably mark the locations where the twin jets are terminated by impacting on the intergalactic medium (Blandford & Rees 1974). Not all of these components are observable in other radio sources.

It was noted a long time ago by Fanaroff & Riley (1974) that the morphologies of radio galaxies are correlated with radio luminosity. In particular, they introduced a simple classification depending on a critical luminosity which, at a frequency 178 MHz, is about $L_c = 5 \times 10^{32}$ erg s^{-1} Hz^{-1}. The so-called FR I objects with $L < L_c$ are usually edge-darkened, which means that the ratio of the separation of the peaks of the radio lobe emission to the total source size significantly is less than unity. Most of these low luminosity

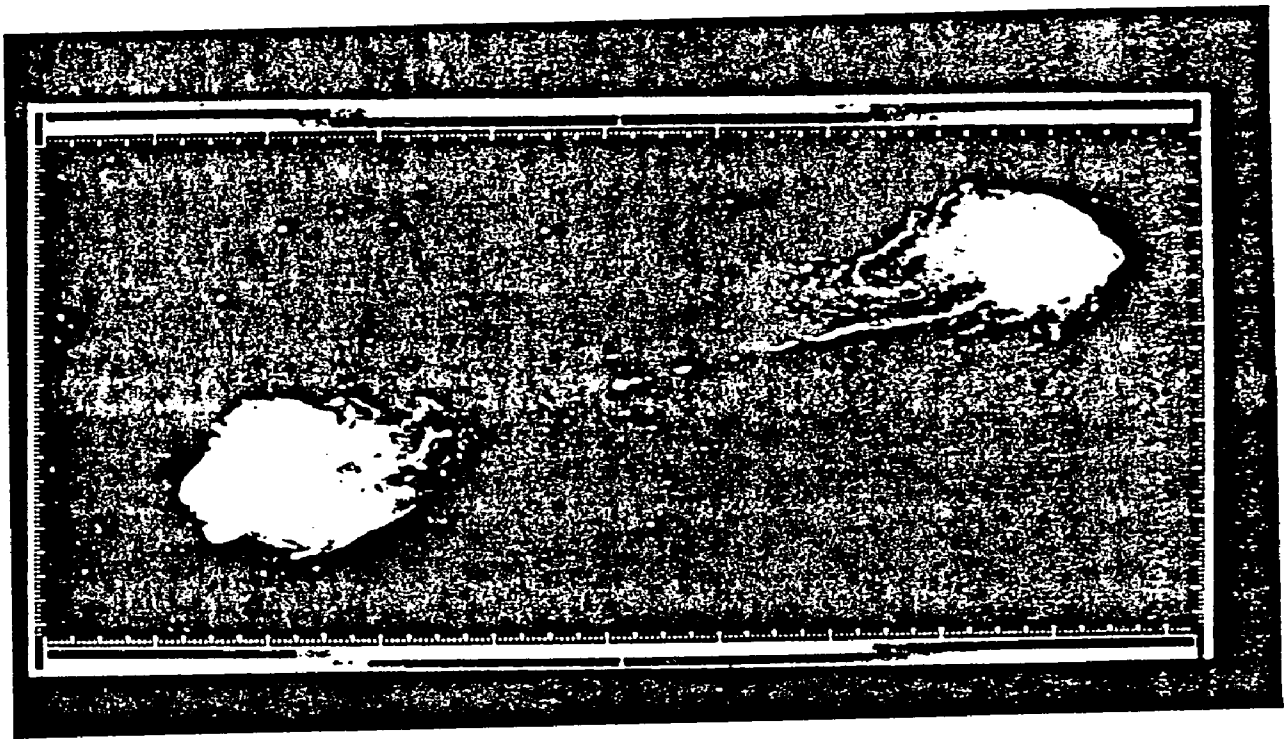


Fig. 1.1--Radiograph of Cygnus A, the "prototypical" classical double radio source, which consists of a radio "core", two thin "jets", and two extended "lobes". The "hot spots" mark the locations in the lobes where the jets are presumably terminated. We study the structure of the jets in their production region, on scales that are too small to be resolved in this picture.

objects have one-sided jets close to the core but the jets become two-sided after a few kpc. There is often a distinct gap between the radio core and the beginning of the jets. The jets are observed to be highly polarized, with the degree of polarization in the range of 10-40% at 5 GHz, which is strongly suggestive of a synchrotron origin for the emission. In some of the jets the magnetic field is inferred to be predominantly parallel to the jet initially and becomes perpendicular to the jet further out, whereas in others the field is perpendicular to the jet close to the axis of the jet but parallel to the jet near the edge. In contrast, the more powerful FR II sources have jets that are better collimated and are usually edge-brightened and one-sided. In addition, the magnetic field in these jets is inferred to be parallel to the jet throughout, except inside bright knots. Furthermore, the jet detection rate in FR II galaxies is less than 10%, much lower than that in the weaker FR I galaxies ($\sim$ 80%). The lobe-dominated quasars have luminosities larger than L_c in general, and their properties are similar to FR II radio galaxies.

There is also a class of sources whose emission is core-dominated. The very high surface brightness in such objects makes them ideal candidates for VLBI studies. The milli-arcsecond jets found in these objects are all on the same side of the core as the larger scale jets (if such jets exist) and the jets observed on different scales can be misaligned by large angles. Large misalignment is commonly interpreted as due to the amplification of a small misalignment when the jet is directed close to the line of sight.

A startling phenomenon, which is directly relevant to the research covered in this thesis, is the so-called "superluminal motion" found in many of the VLBI jets. Bright knots of emission in these jets are often seen to

move away from a stationary core at transverse speeds in the range of 2-10 times the speed of light c. It is commonly accepted that this apparent faster-than-light motion results from the knot moving relativistically along a trajectory pointing close to the line of sight. The apparent speed β_{app} (normalized by c) is related to the actual knot speed β by the well-known formula (cf. Blandford 1990)

$$\beta_{app} = \frac{\beta \sin\theta}{1 - \beta\cos\theta} \tag{1.1}$$

where θ is the angle between the knot velocity and the line of sight. The inferred Lorentz factor for the bright knots, which should also be indicative of the speed of the underlying flow, falls mostly into the range of 2-10. Relativistic jet speed on pc scales can also explain the one-sidedness of VLBI jets if it is due to Doppler boosting, and ease the problems associated with inverse-Compton radiation and high brightness temperatures implied by the rapid low-frequency variability observed in some sources. Jet speeds on larger (i.e., kpc) scale are notoriously difficult to obtain, and indirect arguments indicate that they have a wide range from about 300 km s^{-1} to close to the speed of light. In some cases, the jet is inferred to be relativistic on pc (VLBI) scales but nonrelativistic on kpc (VLA) scales. How this transition occurs remains unclear.

1.1.2. Jet-Producing Mechanisms

In a classic paper, Blandford & Rees (1974) proposed the "twin exhaust" model for the production of jets in extragalactic radio sources. In their model, two anti-parallel de Laval nozzles are supposed to form (Landau & Lifshitz 1959), along which the high-pressure gas in the nuclear region

of an extragalactic radio source escapes outward. Supersonic flow speed is obtained through converting the internal energy of an initially "hot" fluid to directed kinetic energy, and good collimation is always achieved as long as the pressure drop in the confining medium is not too steep. However, it seems clear now that powerful jets in extragalactic radio sources cannot be produced in this way in general (Blandford 1990). The main difficulty with this type of thermally-driven models is that the external confining medium would generate excessive X-ray emission. Given that the jets are already collimated on pc scales as indicated by VLBI observations, the X-ray luminosity in the confining ambient medium could exceed the observed upper limits by several orders of magnitude (Blandford 1990). Furthermore, it is not clear whether the nozzles can survive the strong Kelvin-Helmholtz instabilities expected near their walls.

A different approach is to assume that the jets are driven radiatively but are confined by some other means. Both an intense radiation field and excellent means for flow confinement are available in the narrow funnels formed by thick tori around massive black holes in active galactic nuclei (AGNs, Lynden-Bell 1978). However, from detailed analyses (e.g., Sikora & Wilson 1981) it was realized that the radiation field in the funnel is usually not anisotropic enough, so that it is very difficult to obtain flow Lorentz factor as high as ~ 10 needed to explain the VLBI observations. Furthermore, the thick tori which form the funnels are shown to be subject to a set of virulent instabilities (Papaloizou & Pringle 1984). A variant of the mechanism that avoids the instability problem is to assume that the radiation comes from near the black hole and/or the inner part of a thin accretion disk (Phinney 1987). The terminal flow Lorentz factor in this case could be somewhat

higher, although it is still difficult to reach 10, as needed. Furthermore, it is not clear how the flow can become collimated in this case.

Given that the above two best-studied mechanisms fail to work satisfactorily, the electromagnetic origin has emerged in recent years as the leading candidate for the production of jets in extragalactic radio sources (e.g., Blandford 1989). Depending on the conditions near their bases and in their environments, jets with a wide range of terminal speeds are possible. The need for an external confinement is obviated by the presence of self-induced toroidal magnetic fields.

1.2. Pulsar Winds

Radio pulsars are observed to spin down over some characteristic time scale which ranges from a few thousands of years for Crab-like young pulsars to tens of millions of years for old pulsars. The bulk of the spindown luminosity is believed to be carried away from the pulsar by a magnetized relativistic wind. Wind-excited nebulae have been observed around many pulsars, with the Crab Nebula being the most famous example. Other examples of pulsar-wind-driven-nebulae include the center-filled "plerions", an increasing number of which have been discovered in the Galaxy and the Large Magellanic Cloud (Kafatos & Henry 1985), and the spectacular bow-shock shaped nebula around the eclipsing milli-second pulsar PSR 1957+20 (Kulkarni & Hester 1988). The wind from the latter is also believed to be ablating away its companion (Phinney et al. 1988). More recently, Wang, Li, & Begelman (1993) have found an X-ray trail associated with the nearby old pulsar PSR 1929+10, which can be explained naturally as the shocked pulsar

wind material swept downstream by ram pressure as the pulsar moves supersonically through its ambient medium. To understand all of these pulsar wind-related phenomena, one has to understand the pulsar winds first.

Our knowledge about pulsar winds comes mostly from the study of the Crab Nebula (Figure 1.2). An one-dimensional, spherically symmetric MHD model for the pulsar wind-blown bubble was first proposed by Rees & Gunn (1974). This model was investigated in depth by Kennel & Coroniti (1984), who found amazingly that such a simple model can reproduce the spectral and spatial distribution of continuum emission (from infrared to X-ray) from the nebula reasonably well. In these models, the shock that terminates the free pulsar wind is identified with the optical "wisps" observed by Scargle (1969), which are located about 0.1 pc from the Crab pulsar (assuming a distance of 2 kpc to the Crab, from Trimble 1968). In order for the shocked pulsar wind material to slow down to an expansion speed of about 2000 km s^{-1} near the outer edge of the remnant, the free pulsar wind upstream of the shock has to be weakly magnetized. The inferred ratio of magnetic energy to kinetic energy, σ, must be on the order of 0.01 or less. From the image of diffuse X-ray emission, which is elongated along the inferred pulsar equator, it was inferred that the bulk of the Crab spindown energy is transported outward in the equatorial region of the pulsar (Aschenbach & Brinkmann 1975; Begelman & Li 1992).

The small ratio of magnetic energy to kinetic energy of the free pulsar wind inferred at the shock imposes a very restrictive constraint on all of the pulsar wind models. It definitely rules out Michel's (1969) cold radial wind model, which predicts that the flow is magnetically-dominated all

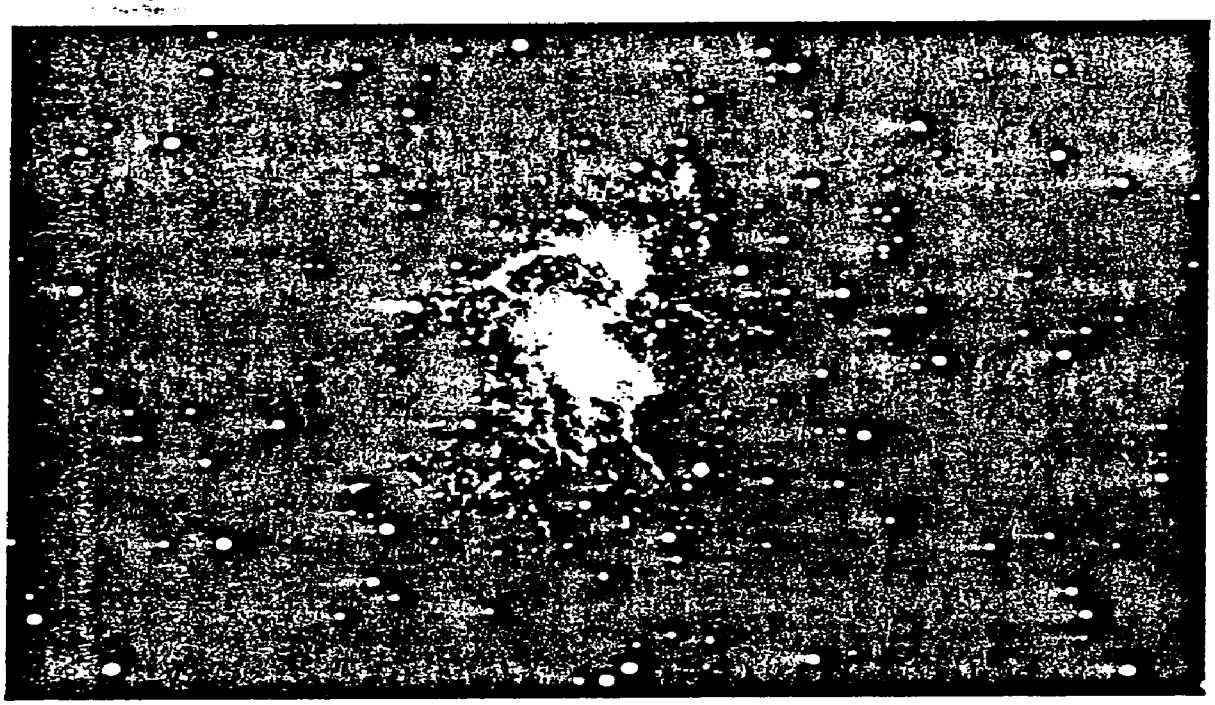


Fig. 1.2–The optical image of the Crab Nebula. The amorphous emission is believed to be synchrotron-radiation from shocked ultra-relativistic pulsar wind materials. It is the structure of the free pulsar wind (before the terminating shock) that we study in this thesis.

the way. To avoid this problem, Kennel, Fujimura, & Okamoto (1983) proposed that the wind can be mainly thermally-driven, in which case σ can be arbitrarily small. Since the Crab pulsar wind is ultrarelativistic, with an inferred Lorentz factor as large as 10^6 (Kennel & Coroniti 1984), the plasma would have to be unreasonably "hot" to start with. This requirement may be obviated somewhat if the "hot" plasma is accelerated continuously by reconnection of magnetic fields of opposite polarity in the equatorial region of the pulsar wind (Coroniti 1990), which is expected to occur when the magnetic axis is almost perpendicular to the spin axis of the pulsar. Alternatively, the energetics of the wind may be dominated by ions which are pulled out, and accelerated to high energy, by the electric force in the equatorial region of the pulsar (Arons 1991). We propose here that the small σ can result from purely geometric effects of the magnetic field in a cold MHD wind. Nearly complete conversion of magnetic energy to kinetic energy is possible when the wind is gradually collimated (Begelman & Li 1993; and chapter IV).

1.3. Other Related Outflows in Our Galaxy

Collimated outflows are observed in many Galactic systems, including young stellar objects (YSOs), binary star systems (e.g., SS433), supernova remnants (e.g., the Crab "jet"), and possibly the Galactic Center. Here we shall concentrate on the optical jets from YSOs which are most likely driven electromagnetically (see Ray & Mundt 1993 for a review). These jets have velocities of several hundred km s^{-1} and length-to-width ratios up to 50. Deep CCD images in Hα and low-excitation forbidden lines (e.g., [S II]) have revealed many detailed structures of the bright "knots" in the jets and the bow shock regions (where the jets are terminated). From these observations,

many properties of the underlying outflows are inferred. This situation is in contrast with that of the jets in extragalactic radio sources, whose densities, speeds, compositions, etc., are rarely known.

How these high-speed, well-collimated YSO jets are produced remains controversial. They could either be 1) radiatively-driven; 2) electromagnetically-driven; or 3) just "ordinary" stellar winds. Since flows obtained by either the first or the last mechanisms do not seem to have sufficient energy and/or momentum (e.g., Padman, Lasenby, & Green 1991), most of the recent work has been concentrated on the second mechanism (Königl 1989; Pelletier & Pudritz 1992; Lovelace, Contopoulos, & Romanova 1993; Safier 1993; Wardle & Königl 1993), which has two main advantages over the others. First, magnetic fields in the flow not only accelerate but also tend to collimate the flow (Heyvaerts & Norman 1989). This obviates the need for other types of confinement. Second, a magnetized wind launched from an accretion disk could carry away enough angular momentum from the disk, and thus resolve the long-standing "angular momentum removal" problem. In this regard, outflows are indispensable parts of the accretion processes, and indeed there is now strong evidence that outflows and disks are correlated with each other (cf. Pelletier & Pudritz 1992).

Finally, we mention in passing that these ionized optical jets are presumably related to the more slowly-moving molecular bipolar outflows (with a speed of few tens of km s^{-1}, see Lada 1985 for a review) that are also observed in some YSOs. One interesting possibility is that YSOs produce only the optical jets to begin with, but that as these objects age, the momentum in these jets is transferred gradually to entrained ambient molecular gas (Stahler 1993), presumably through some kind of turbulent mixing near

their boundaries. Similar processes have been invoked by De Young (1993) to unify the FR I and FR II radio galaxies.

1.4. Theory of MHD Jets and Winds

1.4.1. Basic Concepts

The tool for studying electromagnetically-driven outflows is the so-called magnetohydrodynamics (MHD) theory, in which the motion of the system can be described in terms of a single conducting fluid with the usual hydrodynamic variables such as density, velocity, and pressure. The validity of the MHD approximation will be shown in the next subsection for pulsar winds and AGN jets. The basic MHD equations can be found in textbooks such as Jackson (1975) for the nonrelativistic case and Anile (1989) for the relativistic case. When the conductivity of the fluid is infinite (the so-called "ideal MHD limit"), one can show that the magnetic field lines are frozen into the fluid and are carried along with it. In the axisymmetric, steady-state flows that will be considered throughout this thesis, this "flux-freezing" condition implies that the fluid plasma can flow only along surfaces spanned by poloidal magnetic field lines which revolve with fixed angular velocity around the symmetry axis. These surfaces, called "flux surfaces", are nested around the rotation axis (Figure 1.3). The area between two neighboring flux surfaces is termed a "flux tube", which is equivalent to the term "flux shell" in the some pulsar wind literatures (Michel 1982). A flux tube with fixed shape is analogous to a nozzle with rigid wall, inside which the flow plasma is accelerated along a flux tube by a combination of thermal, centrifugal, and electromagnetic forces. How fast the flow in a tube is moving will depend on

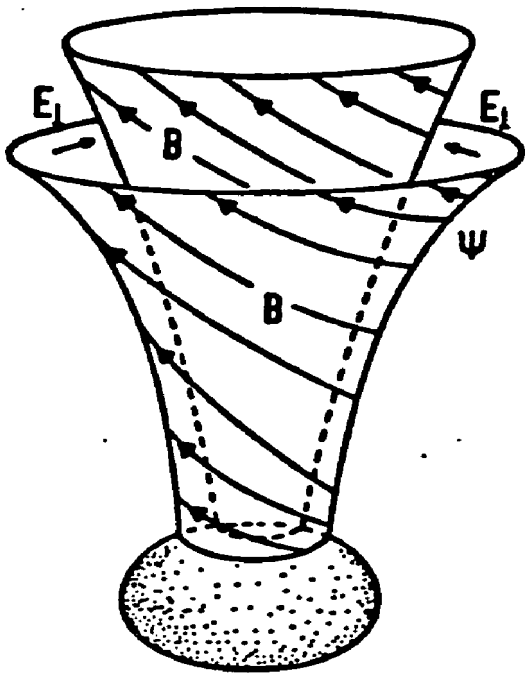


Fig. 1.3–The structure of relativistic MHD winds (adopted from Camenzind 1990). The wind ejected by a rotation central object consists of a nested family of magnetic flux surfaces. Due to rotation, toroidal fields are also produced. This rotation of magnetic field lines provides an electric field perpendicular to the flux surfaces. The wind plasma is flowing along these flux surfaces (but does not follow the field lines).

the shape of the tube (just like inside a nozzle), which is yet to be determined by demanding the balance of the flux surfaces in the cross-field direction. This force balance in turn depends on how fast the flow plasma is moving along each flux tube. It is this complex inter-dependence of flow acceleration along the flux surfaces and force balance across the flux surfaces that renders the structure of MHD winds so difficult to solve self-consistently, even when axisymmetry and steady-state are imposed.

1.4.2. Validity of MHD approximation in pulsar winds and AGN jets

In order for the MHD approximation to hold, charges of opposite sign in a flow must not separate from each other. This condition requires that the charge density of either sign should be much larger than the net charge density that is needed to "short out" the electric field along the magnetic field in the fluid frame — the so-called Goldreich-Julian charge density ρ_{GJ}. Numerically, $n_{GJ} \equiv \rho_{GJ}/e \sim 7\times10^{-2}B/P$ cm^{-3} well inside the light-cylinder radius (Goldreich & Julian 1969), where the magnetic field strength B is in units of Gauss and the rotation period P of the magnetic field is in units of seconds. To estimate the number density of the charged particles in a relativistic flow, we assume that the flow carries a mechanical luminosity of L. It then follows that the number flux is $\dot{N} = L/(\gamma mc^2)$, where γ and m are the Lorentz factor and mass of a charged particle. This yields a number proper density of $n = \dot{N}/(Av) \gtrsim \dot{N}/(Ac)$ where A and v are the cross-sectional area and velocity of the flow, respectively. The ratio of the two charge number densities then becomes

$$\epsilon_c \equiv \frac{n}{n_{GJ}} \gtrsim 5 \times 10^{-31} \frac{LP}{\gamma BmA}. \tag{1.2}$$

We now show that ϵ_c is likely to be much larger than unity for both the Crab pulsar wind and AGN jets. If most of the Crab pulsar spindown luminosity is carried out by an ultrarelativistic electron-positron wind, then we have

$$\epsilon_c \gtrsim 7 \times 10^3 \gamma_5^{-1} B_{12}^{-1} r_6, \tag{1.3}$$

where γ_5 is the Lorentz factor in units of 10^5 and B_{12} is the field strength on the neutron star surface in units of 10^{12} G. We have assumed a quasi-spherical flow with $A \sim 4\pi r^2$ and r_6 is the radius in units of 10^6 cm. In the case of AGN jets, the ratio in equation (1.2) becomes

$$\epsilon_c \gtrsim 10^9 M_8^{1/2} (L/L_E)(P/P_d)\gamma_j^{-1}(B/B_E)^{-1}(A/A_d)^{-1}, \tag{1.4}$$

assuming an ordinary electron-proton plasma for the jet and a Lorentz factor of $10\gamma_j$. The reference values in equation (1.4) are the fiducial values around a massive black hole of $10^8 M_8$ solar mass: Eddington luminosity $L_E = 10^{46} M_8$ erg s^{-1} and the corresponding equipartition magnetic field strength $B_E = 2 \times 10^4 M_8^{-1/2}$ G (Begelman, Blandford, & Rees 1984), the rotation period of the innermost stable orbit $P_d \sim 5 \times 10^4 M_8$ s, and the area of the accretion disk out to ~ 100 gravitational radii $A_d = 7 \times 10^{30} M_8^2$ cm^2.

1.4.3. Overview of Previous Works on MHD Wind Theory and Outline of the Thesis

The MHD wind theory has been developed, over the last three decades or so, in the context of several different types of astrophysical outflows, which include 1) nonrelativistic "ordinary" stellar winds; 2) relativistic pulsar winds; 3) nonrelativistic collimated outflows above accretion disks; and

4) relativistic jets associated with extragalactic radio sources. In this subsection, we shall quickly go over some of the most important insights gained from previous investigations of the MHD outflows in all these categories and put the present work in perspective.

a) Stellar Winds

It was realized early in the development of MHD stellar wind theory (Weber & Davis 1967) that there exist three critical points in an MHD flow, where the flow speed matches the propagation speed of the three types of disturbances: slow-magnetosonic, intermediate (or Alfvénic), and fast-magnetosonic. The behavior of an MHD flow is strongly affected by the conditions at these critical points. To obtain "critical" trans-magnetosonic solutions, the flow has to pass all three critical points *smoothly*. This imposes constraints on the mass flux, the energy and angular momentum fluxes, etc., in the flow. In the pioneering study by Weber & Davis, only the equatorial region of the solar wind, where the flux surfaces are assumed to be perfectly radial, was considered. It was shown that the magnetic field in the flow, while contributing little to the acceleration of the solar wind, nevertheless greatly enhances the rate of angular momentum extraction from the sun. However, Suess & Nerney (1973) argued that the flux surfaces near the equator could not remain radial, but would be deflected gradually toward the rotation axis as a result of the build-up of toroidal magnetic field in the flow. The poleward deflection of the flux surfaces was demonstrated explicitly by Sakurai (1985), who generalized numerically the one dimensional Weber-Davis model to two-dimensional (i.e., axisymmetric) stellar winds. This self-collimation

has been shown to be a general tendency for nonrelativistic MHD winds by Heyvaerts & Norman (1989).

The mathematical properties of the MHD equations governing axisymmetric stellar winds was given by Heinemann & Olbert (1978), who found a "cusp" critical point inside the slow magnetosonic point in addition to the other three critical points mentioned above. It was shown in this paper that the MHD equations can be cast into a single transfield (or cross-field) equation for a poloidal magnetic flux function. The equation is a second order quasi-linear partial differential equation, which is elliptic in the regions inside the cusp point and between the slow and fast points, and hyperbolic in all other regions. It is the complex nature of this equation of mixed type that hampers efforts to seek self-consistent solutions. Although serious attempts in this direction have been undertaken by Sakurai (1985, 1887) with restrictive initial field geometry (split-monopole), it should be stressed that the theory of MHD stellar wind remains far from complete. Self-consistent solutions under more general conditions have to be investigated before we can predict stellar wind properties, as has been done for the radiatively-driven O-Star winds (Kudritzki et al. 1992).

b) Nonrelativistic Jets

The study of astrophysical jets using the nonrelativistic MHD theory was first carried out by Chan & Henriksen (1980), who considered weakly-magnetized flows whose flow variables are assumed to be self-similar with respect to the *cylindrical radius*. Similar assumptions, which have been adopted by Koupelis & Van Horn (1989) and Lovelace, Contoupolos, & Berk (1992) in their studies of narrow jets, reduce the governing equations to a

set of ordinary differential equations. In essence, these approaches all deal with the change of cross-sectionally-averaged properties along the jets. Such an approach may not yield reliable results, however, since the flow properties could vary rapidly from the axis to the outmost flux surface (Li 1993; chapter VI). An alternative type of self-similar MHD flows, in which the flow variables are self-similar with respect to the *spherical radius*, was found by Blandford & Payne (1982). They were the first to demonstrate explicitly, and in a self-consistent way, that flow collimation and acceleration are achievable simultaneously for MHD outflows from Keplerian accretion disks. They also stressed that the flows can extract angular momentum from the accretion disk, and thus enable matter to be accreted independent of the presence of viscosity. Their results have been used in many later studies of YSOs (Pudritz & Norman 1985; Königl 1989) and AGNs (Emmering, Blandford, & Shlosman 1992), although the self-similar assumption in the model requires a very restrictive distribution of the poloidal magnetic field above the disk: its strength has to decrease rapidly with radius as $\propto R^{-5/4}$. It is not obvious that such a magnetic field distribution can be set up naturally (Begelman 1993). Recently, Pelletier & Pudritz (1992) have constructed similar models in which an MHD wind carries away all of the angular momentum released in an accretion disk.

c) Pulsar Winds

It is mainly in the force-free (i.e., massless) limit that MHD equations have been used to study the pulsar magnetospheres and their associated outflows (cf. Michel 1982). Although such an approximation may be valid close to the pulsar, it is not clear whether the magnetic field keeps dominating

the flow plasma up to large distances from the pulsar. In fact, at about 0.1 pc ($\sim 10^{12}$ times the pulsar radius), the magnetic energy flux in the Crab pulsar wind seems to be less than 1% of the kinetic energy flux of the plasma (Rees & Gunn 1974; Kennel & Coroniti 1984). Thus, it is more appropriate to include the plasma inertia in studying the global properties of Crab-like pulsar winds. A first step in such a direction was made by Michel (1969), who extended the conventional MHD wind theory to the special relativistic regime. In a treatment parallel to that of Weber & Davis (1967), he found a family of "minimum torque" solutions for radial pulsar winds in the absence of both thermal pressure (i.e., cold) and gravity. It was realized subsequently (Goldreich & Julian 1970) that these solutions are in fact the zero temperature limit of the critical solutions which pass through the fast-magnetosonic point. They are characterized by a single dimensionless parameter — the so-called "Michel's magnetization parameter", $\sigma \equiv \Phi^2\Omega^2/(4\pi f c^3)$, where Φ, Ω and f are the magnetic flux, rotation rate and mass flux of the radial flow, respectively — which is essentially the maximum Lorentz factor that the plasma in a flow can obtain when all of the energy in the flow is in the kinetic form. To obtain an ultrarelativistic pulsar wind, it is thus required that $\sigma \gg 1$. However, for Michel's "minimum torque" solutions, the final flow Lorentz factor, denoted by γ_e, is only $\sigma^{1/3}$, which is far less than σ. Such an inefficient flow acceleration is an artifact of the perfectly-radial wind assumption, as we shall show in chapter IV. In the more general case where flow collimation is allowed, we find that it is more likely that $\gamma_e \sim \sigma$. In particular, all of the flow energy could end up in kinetic form, provided the flux tubes in the flow diverge faster than radially.

The most extensive analysis of relativistic MHD winds of radial geometry was carried out by Kennel, Fujimura, & Okamoto (1983), who examined the nature of the various critical points in detail. The generalization of flows along radial flux surfaces assumed in this and earlier works (Michel 1969; Goldreich & Julian 1970) to flux surfaces of arbitrarily prescribed shape can be found in Okamoto (1978), who also noted that the singularity at the light cylinder radius, $R_L = c/\Omega$ (where Ω is the angular velocity of the pulsar), in the force-free limit is removed when the inertia of the flow is taken into account. In this sense, force-free and non-force-free flows are distinctively different, a point that has already been made by Camenzind (1987). In a recent study of pulsar magnetospheres, Sulkanen & Lovelace (1992) found a family of force-free jet solutions that are self-confined entirely within the light cylinder radius (see also Appl & Camenzind 1993). Presumably, these solutions would cross the light cylinder radius and join smoothly onto the asymptotic solutions discussed by Begelman & Li (1993; chapter IV) when the plasma inertia is included, although how this is accomplished exactly has yet to be worked out in a self-consistent way.

d) Relativistic Jets

Although the idea of producing relativistic jets electromagnetically has been around for a long time (Lovelace 1976; Blandford 1976), it is only in recent years that attempts have been made to study this scenario in detail. We are only beginning to understand the main properties of relativistic MHD jets, including their acceleration and collimation.

The basic relativistic MHD equations and their associated constants of motion have been given in various forms by many authors (Michel 1969;

Goldreich & Julian 1970; Okamoto 1978; Phinney 1983; Kennel, Fujimura, & Okamoto 1983; Bekenstein & Eichler 1985; Lovelace et al. 1986; Camenzind 1986; Li, Chiueh, & Begelman 1992). In view of the great difficulties in obtaining self-consistent solutions (§ 1.4.1), it is not surprising that most of the early works are done either for prescribed flux tube geometries, disregarding the cross-field force balance (cf. Phinney 1983), or in the force-free limit (Blandford 1976; Bekenstein & Eichler 1985; Lovelace, Wang, & Sulkanen 1987, Appl & Camenzind 1993) where the flow inertia is neglected and thus the problem of flow acceleration along the flux tubes cannot be addressed at all. Only in a few rather restrictive cases have both the acceleration of the flow plasma along the flux surfaces and the balance of forces across the flux surfaces been determined simultaneously (Camenzind 1987; Li, Chiueh, & Begelman 1992; Li 1993). Examples of self-consistent numerical solutions were first given by Camenzind (1987), who assumed that the rapidly rotating magnetic fields are only sparsely populated with plasma (but not force-free). Although this assumption is unlikely to be true in general, it does render the treatment of the critical surfaces in the flow much easier. One artifact of this assumption is that all field lines have to cross the light cylinder surface horizontally, which is typical of force-free flows (cf. Michel 1982). As a result of this limitation and the fact that it is very computer-time consuming to obtain numerical solutions, the parameter space is not explored. Nor is it easy to interpret the exemplary solution presented by Camenzind.

A number of analytic or semi-analytic approaches have also been taken, which turn out to be more fruitful in illustrating the basic properties of relativistic MHD winds, as demonstrated in this thesis and by Appl & Camenzind (1993). In an analysis parallel to that of Heyvaerts & Norman

(1989) for nonrelativistic MHD winds, we show in chapter III (from Chiueh, Li, & Begelman 1991) that all flux surfaces in steady-state, axisymmetric, relativistic MHD winds are collimated asymptotically, either to cylinders of finite radius when the electric current enclosed by the flow is finite, or to "paraboloidal" surfaces that extend transversely to infinity when the current enclosed is vanishingly small. The structures of current-carrying cylindrical jets are examined in detail by Appl & Camenzind (1993) for prescribed distributions of electric current density. Flows that are dominated by kinetic energy, such as those with asymptotically-paraboloidal geometry, are studied in chapter IV (from Begelman & Li 1993), with application to Crab-like pulsar winds. We conclude that the kinetic energy-dominated solutions are likely to be the natural asymptotes for flows that are either confined by external pressures which decrease rapidly enough with distance, or not confined at all. In order to show whether the asymptotic solutions are able to connect to the base of the flow, one has to obtain complete solutions from the base of the flow to large distances. This is first done in chapter V (from Li, Chiueh, & Begelman 1992), where we extend the nonrelativistic self-similar solutions of Blandford & Payne (1982) to special relativistic regime. We find well-collimated solutions that obtain near equipartition between electromagnetic and plasma kinetic energy fluxes. However, the requirement of self-similarity is very restrictive, and the construction of solutions under more flexible conditions is desired. This is done in chapter VI (from Li 1993), where the dependences of the flow acceleration and collimation on the distributions of the rotation rate of the flux surfaces, the mass loading rate, etc., are explored. Chapters VII (from Li, Begelman, & Chiueh 1992) and VIII (from Li & Begelman 1993, in preparation) are devoted to studying of

interactions of relativistic MHD flows with the ambient radiation fields that they are expected to traverse in the region where they are produced. This study is motivated by the suggestion that Compton "drag" may slow down relativistic jets to such a degree that most of the radio emission associated with the jets is suppressed (Blandford 1992). This might be responsible for the low radio emission in the majority of QSOs (which are often referred to as "radio-quiet"). We find that drag due to radiation alone may not be sufficient to quench the jets in most cases, and other forms of drag force, such as that due to entrainment of external material, are needed. Alternatively, the central engines for radio-loud and radio-quiet QSOs may be intrinsically different. We conclude in chapter IX and discuss further work that needs to be done in the field of relativistic MHD winds and jets in the future.

CHAPTER II

BASIC RELATIVISTIC MHD WIND EQUATIONS

As mentioned in § 1.4.2, the basic relativistic MHD wind equations have been given and explored by many authors. In this chapter, we shall cast these well-known equations into a new form that is convenient for our treatment of the problem. Specifically, we shall introduce the curvature of poloidal magnetic field lines into the cross-field equation (see eq. [2.10]), which renders it easier for us to obtain self-consistent wind/jet solutions.

The structure of the chapter is as follows. In § 2.1, we rearrange the relativistic ideal MHD equations (including both the thermal pressure and the gravity from a central object) into an energy equation that governs the flow motion along each flux surface and a modified Grad-Shafranov equation that controls the balance of various forces in the cross-field direction. We then include in the momentum equation a drag force term resulted from Compton scattering of the relativistic outflows on their ambient radiation field (§ 2.2), if such field exists. These modified equations form the base of our discussion of radiation-dragged relativistic MHD winds.

2.1. Derivation of the Energy and the Modified Grad-Shafranov Equations

Throughout the discussion in this chapter, we adopt the standard cylindrical coordinate system (R, ϕ, Z), with R measuring the cylindrical radius and Z measuring the distance along the cylindrical axis. Assume the

flow plasma to be a perfect conductor, for which the equation of flux-freezing condition $\nabla \times (\mathbf{V} \times \mathbf{B}/\gamma) = 0$ holds. This implies that

$$\frac{V_\phi - \gamma R\Omega(R,Z)}{V_p} = \frac{B_\phi}{B_p}$$

or, equivalently,

$$\mathbf{V} = \gamma R\Omega(R,Z)\hat{\mathbf{e}}_\phi + \frac{k(R,Z)}{\rho}\mathbf{B}, \tag{2.1}$$

where $\mathbf{V}$ is the spatial part of the 4-velocity, γ is the Lorentz factor, $\hat{\mathbf{e}}_\phi$ is the azimuthal unit vector, ρ is the proper plasma density, and the subscripts 'p' and 'ϕ' stand for poloidal and toroidal, respectively. The quantity $\Omega(R,Z)$ is just the angular velocity of a field line (Lovelace et al. 1986) which can be shown to be constant on a field line. A moving field line spans a flux surface, as shown in Figure 1.3. The quantity $k(R,Z)$ is simply the ratio of mass flux to magnetic flux, and is also a constant along a flux surface since both the mass flux and the magnetic flux are conserved. If we label different flux surfaces by a poloidal flux function $\Psi(R,Z)$ defined from

$$\mathbf{B}_p = \frac{1}{R}\nabla\Psi \times \hat{\mathbf{e}}_\phi, \tag{2.2}$$

then we have $\Omega(R,Z) = \Omega(\Psi)$ and $k(R,Z) = k(\Psi)$, i.e., both the field line angular velocity and the mass flux per unit magnetic flux are functions of magnetic poloidal flux function only. Two additional conserved quantities along a given flux surface are the total specific energy e and total specific angular momentum l. They are obtained from the special relativistic momentum equation for a stationary flow (Goldreich & Julian 1970; Kennel, Fujimura, Okamoto 1983)

$$\rho\mathbf{V}\cdot\nabla(\xi\mathbf{V}) = \rho_e\mathbf{E} + \frac{\mathbf{j}\times\mathbf{B}}{c} - \nabla P + \gamma\rho\nabla\left(\xi\gamma\frac{GM_*}{(R^2+Z^2)^{1/2}}\right) \tag{2.3}$$

where $\xi \equiv 1 + \int dP/\rho c^2$ is the specific enthalpy, ρ and P are the proper density and proper pressure, respectively, M_* is the mass of the central star or black hole, ρ_e is the charge density in the laboratory frame, whereas $\mathbf{E}$ is the electric field. The presence of an electric force of comparable magnitude with the magnetic field in the MHD description is a result of special relativistic effects. The quantities $\mathbf{E}$ and ρ_e are determined from

$$\mathbf{E} = -\frac{\mathbf{V} \times \mathbf{B}}{c\gamma} = -\frac{R\Omega \hat{\mathbf{e}}_\phi \times \mathbf{B}_p}{c} = -\frac{\Omega}{c}\nabla\Psi,$$

and

$$\rho_e = \frac{\nabla \cdot \mathbf{E}}{4\pi} = -\frac{1}{4\pi c}\nabla \cdot (\Omega \nabla \Psi),$$

which are combined to give the electric force

$$\mathbf{F}_E \equiv \rho_e \mathbf{E} = \frac{\Omega}{4\pi c^2}\nabla \cdot (\Omega \nabla \Psi)\nabla\Psi. \tag{2.4}$$

It is clear that the electric force comes from the rotation of the poloidal magnetic field and is in the cross-field direction. Therefore, it does not change either the energy or angular momentum along a flux surface. However, the electric force turns out to be crucial for maintaining the force balance between different flux surfaces.

The conserved total specific energy e is obtained by taking the dot product of the momentum equation (2.3) with $\xi\mathbf{V}/\rho$, which yields

$$e(\Psi) = \xi\gamma c_*^2 - \frac{R\Omega B_\phi}{4\pi k}, \tag{2.5}$$

where we have defined

$$c_*^2 \equiv c^2 - \frac{GM_*}{(R^2 + Z^2)^{1/2}}.$$

The conserved total specific angular momentum l is obtained by taking the azimuthal component of eq. (2.3), which yields

$$l(\Psi) = R\left[\xi V_\phi - \frac{B_\phi}{4\pi k}\right] = R\left[\left(\frac{\xi k}{\rho} - \frac{1}{4\pi k}\right) B_\phi + \xi\gamma R\Omega\right]. \qquad (2.6)$$

The two equations thus obtained (eqs. [2.5] and [2.6]) can be used to express the modified Lorentz factor of the flow $\xi\gamma$ and the toroidal component of the magnetic field strength B_ϕ in terms of e and l:

$$\xi\gamma = \frac{e}{c_*^2} - \frac{(1 - \Omega e R^2/lc_*^2)l\Omega/c_*^2}{1 - 4\pi k^2\xi/\rho - R^2\Omega^2/c_*^2} \qquad (2.7)$$

and

$$B_\phi = -\frac{4\pi lk}{R}\frac{(1 - \Omega e R^2/lc_*^2)}{1 - 4\pi k^2\xi/\rho - R^2\Omega^2/c_*^2}. \qquad (2.8)$$

With the help of the flux-freezing condition, we can rewrite the identity $\gamma^2 = 1 + V_p^2/c^2 + V_\phi^2/c^2$ using eqs. (2.7) and (2.8) as

$$\frac{1}{\xi^2}\left[\frac{e}{c_*^2} - \frac{l\Omega/c_*^2(1 - R^2\Omega e/lc_*^2)}{1 - 4\pi k^2\xi/\rho - R^2\Omega^2/c_*^2}\right]^2 = 1 + \left(\frac{kB_p}{\rho c}\right)^2$$

$$+ \left(\frac{l}{\xi Rc}\right)^2\left[1 - \frac{1 - R^2\Omega e/lc_*^2}{1 - 4\pi k^2\xi/\rho - R^2\Omega^2/c_*^2}\right]^2. \qquad (2.9)$$

Because ξ is a function of ρ, eq. (2.9) relates the flow density ρ to the poloidal component of the magnetic field strength B_p (which reflects the geometries of field lines) along a flux surface. Since $V_p = kB_p/\rho$, once the four flux constants (Ω, k, e, and l) on a flux surface are known, the acceleration of the flow (i.e., V_p) along that flux surface can be determined if the field geometry (that is, B_p) is given. We shall hereafter refer to equation (2.9) (and its various simplified forms to be discussed later) as *the energy equation.* In the absence of thermal pressure and gravity, and under the assumption of

a radial wind geometry, it is easy to show that the energy equation (2.9) is equivalent to the quadratic equation (20) of Michel (1969), and the Michel's dimensionless magnetization parameter σ equals $(B_p R^2 \Omega^2)/(4\pi k c^3)$.

To obtain the so-called *modified Grad-Shafranov (GS) equation*, we project the momentum equation (2.3) in the direction of the unit vector $\hat{\mathbf{e}}_c \equiv \mathbf{B}_p/B_p \times \hat{\mathbf{e}}_\phi$, which is in the cross-field direction pointing towards the rotation axis. After some lengthy manipulations, we arrive at the following form for the GS equation

$$\kappa B_p^2 (1 - \frac{4\pi k^2 \xi}{\rho} - \frac{\Omega^2 R^2}{c^2}) - \hat{\mathbf{e}}_c \cdot \left\{ (1 - \frac{\Omega^2 R^2}{c^2}) \nabla \frac{B_p^2}{2} + \nabla \frac{B_\phi^2}{2} \right.$$

$$\left. + 4\pi\gamma\rho \nabla[\xi\gamma(c_*^2 - c^2)] + 4\pi\rho c^2 \nabla\xi \right\}$$

$$= \frac{B_p R\Omega}{c^2} \nabla\Psi \cdot \nabla\Omega + \frac{2B_p\Omega^2}{c^2} \nabla\Psi \cdot \nabla R - (\frac{4\pi\rho}{R} V_\phi^2 - \frac{B_\phi^2}{R}) \hat{\mathbf{e}}_c \cdot \nabla R, \quad (2.10)$$

where κ is the local curvature of the poloidal magnetic field line, which is defined from $\kappa B_p^2 = \hat{\mathbf{e}}_c \cdot (\mathbf{B}_p \cdot \nabla)\mathbf{B}_p$.

Using equations (2.2) and (2.9) together with the definition of κ, one can write the GS equation (2.10) as a 2-D second-order partial differential equation for Ψ, in which all second-order terms appear on the left-hand side (Lovelace et al. 1986). It can be shown that the nonrelativistic flow equations (Blandford & Payne 1982; Sakurai 1985) are recovered in the limit that the speed of light $c \to \infty$. As in the nonrelativistic case (e.g., Weber & Davis 1967), this equation admits three critical points along each field line, corresponding to the slow magnetosonic, Alfvén, and fast magnetosonic wave modes (Lovelace et al. 1986), respectively. By demanding that the flow pass smoothly through these points, some arbitrary parameters at the base of the

flow can be fixed. However, these three critical surfaces in an 2-D flow are generally so complex that no attempt has yet been made to study flows with the most general boundary conditions, and all past studies involve major simplifying assumptions. In view of the great importance in understanding the basic properties of the electromagnetically-driven relativistic jets/winds that are believed to occur in astrophysical systems (see chapter I), we shall devote a major portion of this dissertation (from chapter III to chapter VI) to the study of various aspects of the modified GS equation and the energy equation.

2.2. Modified Wind Equations When Radiation "Drag" is Present

We consider only the "drag" effects of an ambient radiation field through which a relativistic MHD jet/wind has to transverse, and assume that any flow acceleration due to the anisotropy of radiation field is negligible compared with the acceleration by the hydromagnetic processes. In this case, the only modification to the flow momentum equation (2.3) would be the addition of an "effective" drag force per unit mass $\mathbf{F}_D$, so that

$$\rho\mathbf{V}\cdot\nabla(\xi\mathbf{V}) = \rho_e\mathbf{E}+\frac{\mathbf{j}\times\mathbf{B}}{c}-\nabla P+\gamma\rho\nabla\left(\xi\gamma\frac{GM_*}{(R^2+Z^2)^{1/2}}\right)+\gamma\rho\mathbf{F}_D. \quad (2.11)$$

For the Compton drag due to optically thin Thomson scattering, one has $\mathbf{F}_D \propto \gamma\mathbf{V}$ where it should be kept in mind that $\mathbf{V}$ is the spatial component of 4-velocity. One can then define a velocity independent, dimensionless drag coefficient ν so that

$$\mathbf{F}_D = -\gamma\mathbf{V}\Omega\nu. \quad (2.12)$$

Since the drag force is parallel to the flow velocity, it has no component in the cross-field direction, and thus will not enter the GS equation explicitly.

However, by changing flow variables along each flux surface, the drag force can affect the cross-field force balance indirectly. The quantities affected directly by the presence of drag force are the total specific energy e and the total specific angular momentum l, both of which are no longer constant on a given flux surface. They change with the length, S, along a flux surface according to (from momentum equation [2.11])

$$\frac{de}{dS} = -\gamma\Omega\frac{V_p^2 + V_\phi^2}{V_p}\nu(S) \tag{2.13}$$

and

$$\frac{dl}{dS} = -\gamma^2\Omega R\frac{V_\phi}{V_p}\nu(S) \tag{2.14}$$

where the drag coefficient ν as a function of position S has to be prescribed. Therefore, equations in the last section (§ 2.1) have exact the same forms even in the presence of radiation drag force of the form given by equation (2.12), provided that it is kept in mind that two (e and l) of the four previous flux constants (Ω, k, e, and l) are no longer constant, and have to be determined self-consistently from equations (2.13) and (2.14). We shall use these equations to study the structure of radiation-dragged relativistic MHD winds of radial (chapter VII) and arbitrary (chapter VIII) geometries.

Finally, we point out in passing that "drag" due to entrainment of external material can be formulated in a similar manner (§ 8.3.1), although the quantity k (which is the mass flux per unit magnetic flux) is no longer constant along a given flux tube.

CHAPTER III

GENERAL ASYMPTOTIC STRUCTURE OF RELATIVISTIC MHD WINDS

In the asymptotic flow region that lies far beyond the so called "light cylinder radius" $R_L = c/\Omega$ (i.e. the radius at which flow plasma can no longer corotate with the magnetic field, since otherwise the toroidal flow speed would exceed the speed of light), both the energy and the Grad-Shafranov equations are not only much easier to handle mathematically but also more transparent physically. Analysis of flow structure in this region would yield important insights as to what types of asymptotic magnetic field geometries are compatible with the requirements of energy and angular momentum conservations along each flux surface, and force balance in the cross-field direction. Our conclusion here that all flux surfaces in a magnetized relativistic wind tend to be collimated towards the rotation axis is complementary to similar results for nonrelativistic winds (Heyvaerts & Norman 1989). This property of self-collimation put the MHD wind models for a wide variety of astrophysical jets on a firm base. Although not pronounced in the case of nonrelativistic winds, flow collimation also drastically enhances the efficiency of energy conversion from Poynting flux stored in the electromagnetic field to the flow plasma kinetic energy flux, which is extremely low in uncollimated radial (or more exactly conical) winds (Michel 1969).

In the asymptotic region, it is reasonable to assume that the thermal pressure and the gravity are negligible, so that $\xi = 1$ (i.e., cold wind approximation) and $M_* = 0$. In this limit, one can rewrite the energy equation (2.9)

in a much simplified form far beyond the light cylinder $R \gg R_L = c/\Omega$,

$$\left[\frac{4\pi keV_p}{4\pi kc^2V_p + B_pR^2\Omega^2}\right]^2 = 1 + \frac{V_p^2}{c^2}, \tag{3.1}$$

where the left hand side of the equation is simply the flow Lorentz factor. We first want to know what types of field geometry are compatible with the energy equation cast in above form, if a flow of finite 4-speed V_p is to be obtained. To do this, we distinguish four asymptotic behaviors of the magnetic flux surfaces: Asymptotically horizontal (AH) flux surfaces have $R/Z \to \infty$. Asymptotically vertical (AV) flux surfaces, characterized by $Z/R \to \infty$, are subclassified into "paraboloidal" surfaces (if $R \to \infty$) and "cylindrical" surfaces (if $R \to$ *const.*). Intermediate between AV and AH flux surfaces are a class of surfaces characterized by both R and Z going to ∞, but $0 < R/Z < \infty$; these will be called "conical". Regions containing the various types of flux surfaces, if they coexist, must be nested about the rotation axis in the following order: cylindrical, paraboloidal, conical, AH.

The energy equation (3.1) can be used to rule out the existence of AH flux surfaces. Consider the flux contained between two such surfaces. Since the only asymptotic poloidal field component is B_R, we have, at some $R \to \infty$,

$$\Delta\Psi \equiv \Psi_2 - \Psi_1 = \int_{Z_1}^{Z_2} B_pRdz. \tag{3.2}$$

If B_pR^2 is bounded over some region (Ψ_1, Ψ_2), we denote its maximum value by Q, then

$$\Delta\Psi \le Q\int_{Z_1}^{Z_2}\frac{dz}{R} = Q\left(\frac{Z_2}{R} - \frac{Z_1}{R}\right) \to 0 \tag{3.2}$$

as $R \to \infty$, which contradicts the assumption of finite magnetic flux between the surfaces. Thus, $B_pR^2 \to \infty$ for AH flux surfaces. However, it is apparent that, for a non-singular distribution of flux constants k, Ω, e and a

finite terminal 4-speed V_p, equation (3.1) cannot be satisfied if $B_p R^2 \to \infty$. Therefore, AH flux surfaces are forbidden. Physically, a wind solution is impossible between AH surfaces because there is always a range of radii over which the inward magnetic tension force dominates over the outward pressure force, to the extent that it stops the flow completely. Equivalently, the coupling between B_p and B_ϕ implies that the Poynting flux soaks up all of the available energy beyond a certain radius.

Generally, conical flux surfaces have a finite and nonvanishing $B_p R^2$. This is readily understood since for a conical surface $B_p R^2 = B_r r^2 \sin^2\theta$, where r is the spherical radius, θ is the polar angle along the flux surface, and $B_r r^2$ is finite due to the conservation of magnetic flux. The condition for conical asymptotes is given by Heyvaerts & Norman (1989, proposition [d]), although their statement there is slightly misleading. A more appropriate statement would have been "If *all* flux surfaces bounded by the flux surface of interest and the equator (instead of "If some flux surface ...") have finite values of ρR^2 (or, equivalently, $B_p R^2$), then this flux surface must be conical."

For AV surfaces, three different types of fields exist, namely, cylindrical, current-carrying paraboloidal, and asymptotically current-free paraboloidal. The asymptotic value of $B_p R^2|_{Z\to\infty}$ is generally finite and nonvanishing for cylindrical and current-carrying (type I) paraboloidal flux surfaces, but is vanishing for asymptotically current-free (type II) paraboloidal surfaces. The geometrical difference between type I and type II paraboloidal fields is in the degree to which the poloidal field lines diverge. As we shall show, this difference determines the asymptotic ratio of the kinetic energy flux to the Poynting flux, which is of great observational significance in pulsar

winds. To illustrate both types of field, let us consider a general asymptotic expansion of the field line configuration (Heyvaerts & Norman 1989), the locus of which is

$$Z = f(\Psi)R^{\alpha(\Psi)} \tag{3.4}$$

with $\alpha(\Psi) > 1$. The poloidal field strength is approximately

$$B_p \cong \left|\frac{1}{R}\frac{\partial\Psi}{\partial R}\right| \cong \frac{\alpha}{R^2|\alpha' \ln R + (\ln f)'|}. \tag{3.5}$$

Type I field has $\alpha'(\Psi) = 0$, while type II field has $\alpha'(\Psi) \neq 0$, with B_pR^2 approaching zero as $(\ln R)^{-1}$ along the field line.

We next further constrain the asymptotic flux surfaces using the GS equation (2.10) which, in the case that $\xi = 1$ and $M_* = 0$, becomes

$$\kappa B_p^2(1 - \frac{4\pi k^2}{\rho} - \frac{\Omega^2R^2}{c^2}) - \hat{\mathbf{e}}_c \cdot \frac{1}{R^2}\nabla\left(\frac{B_\phi^2R^2}{2} - \frac{B_p^2R^4\Omega^2}{2c^2}\right)$$

$$= \hat{\mathbf{e}}_c \cdot \nabla\frac{B_p^2}{2} - \frac{4\pi\rho}{R}V_\phi^2\hat{\mathbf{e}}_c \cdot \nabla R. \tag{3.6}$$

We first consider the asymptotically cylindrical flux surfaces. We show that, far beyond the light cylinder, the terminal flow Lorentz factor on each flux surface is proportional to the total poloidal current enclosed within that surface. The asymptotic value of any quantity, denoted by subscript ∞, should be a function of Ψ only. Since the curvatures of the flux surfaces approach zero asymptotically and $\hat{\mathbf{e}}_c \to \nabla R$, the GS equation (3.6) is then reduced to

$$\frac{d}{d\Psi}\left(\frac{B_p^2R^4\Omega^2}{2c^2}\right)_\infty - \frac{d}{d\Psi}\left(\frac{B_\phi^2R^2}{2}\right)_\infty + \left(\frac{4\pi\rho V_\phi^2}{B_p}\right)_\infty - R_\infty^2\frac{d}{d\Psi}\left(\frac{B_p^2}{2}\right)_\infty = 0 \tag{3.7}$$

where the different terms on the left hand side of equation correspond to the electric force, the magnetic force due to the toroidal field, the centrifugal force, and the magnetic pressure due to the poloidal field, respectively. We can drop the subscript "∞" for clarity and rearrange the above equation into

$$\frac{d}{d\Psi}\left[(B_\phi R)^2 - \left(\frac{B_p R^2 \Omega}{c}\right)^2 + (B_p R)^2\right] = \frac{1}{R^2}\left[\frac{8\pi k (V_\phi R)^2}{V_p} + B_p R^2\right], \tag{3.8}$$

which, when integrated once, yields

$$A \equiv (B_\phi R)^2 - \left(\frac{B_p R^2 \Omega}{c}\right)^2 + (B_p R)^2 = A_0 - \Delta, \tag{3.9}$$

where

$$\Delta \equiv \int_\Psi^{\Psi_0} \frac{1}{R^2}\left[\frac{2k(V_\phi R)^2}{V_p} + (B_p R^2)\right] d\Psi$$

$$< \left(\frac{R_L}{R}\right)^2 \int_\Psi^{\Psi_0} \left[\frac{2kl^2\Omega^2}{c^2 V_p} + \frac{B_p R^2 \Omega^2}{c^2}\right] d\Psi \equiv A_1 \left(\frac{R_L}{R}\right)^2. \tag{3.10}$$

We have denoted the flux function on the last surface by Ψ_0, and A_0 is the value of A at $\Psi = \Psi_0$. We have also used the facts that R is a monotonically increasing function of Ψ and $|V_\phi R| < l$ (i.e., eq. [2.6]) in deriving the inequality. Since $B_p R^2$ is bounded for flux surfaces of all types except AH, the quantity A_1 should be finite if flux constants such as l and Ω are non-singular and k/V_p is finite on all flux surfaces. Therefore, for flux surfaces with $R/R_L \gg \max(1, \sqrt{A_1/A_0})$, the quantity $A = B_\phi^2 R^2 - B_p^2 R^4 c^2/\Omega^2 + O(R_L^2/R^2) = A_0 + O(R_L^2/R^2)$. On the other hand, the two components of the magnetic field strength B_p and B_ϕ are related by the energy equation (3.1) and equation (2.8), which in the limit $\xi = 1$, $M_* = 0$ and $R \gg R_L$ becomes

$$B_\phi R = -\frac{4\pi k e B_p R^2 \Omega}{4\pi k c^2 V_p + B_p R^2 \Omega^2}, \tag{3.11}$$

to the order of $(R_L/R)^2$. We therefore have

$$\left(\frac{B_\phi V_p}{B_p R\Omega}\right)^2 = 1 + \left(\frac{V_p}{c}\right)^2 + O\left(\frac{R_L^2}{R^2}\right) \cong \gamma^2. \tag{3.12}$$

Thus one obtains, from equations. (3.12) and (3.9), that

$$\frac{B_p R^2 \Omega}{V_p} = \sqrt{A_0} + O\left(\frac{R_L^2}{R^2}\right). \tag{3.13}$$

The above analysis holds as long as A_0 is non-zero. Equation (3.13) is just the relativistic generalization of the "solvability condition at infinity" derived by Heyvaerts & Norman (their eq. [26]). The asymptotic Lorentz factor is hence given by

$$\gamma(\Psi) \cong \frac{2}{c\sqrt{A_0}} I_\infty(\Psi), \tag{3.14}$$

where $I_\infty(\Psi) = c|B_\phi R|/2$ is the total poloidal current enclosed within the flux surface Ψ. We note that net current can flow on individual flux surfaces asymptotically, provided that the Lorentz factor varies from one surface to another.

We stress that our approximations hold only for those cylindrical surfaces that extend well beyond the light cylinder. For type I paraboloidal flux surfaces, equation (3.8) is still valid, and since $R \to \infty$ along all the flux surfaces, it follows that $A = A_0$ exactly and the equality in eq. (3.14) holds, as long as the quantity A_1 defined above does not diverge. It is straightforward to show that the same conclusions also hold for conical flux surfaces.

For nonvanishing A_0, we find that the current enclosed asymptotically by any type I paraboloidal or conical flux surface must be finite, according to eq. (3.14). Specifically, $I_\infty(0) \neq 0$, that is, the current density becomes singular on flux surfaces close to the rotation axis (i.e., as $\Psi \to 0$) for both the

conical and *type I paraboloidal* flux surfaces. However, for *cylindrical* fields, equations (3.13) and (3.14) do not apply to those innermost flux surfaces, which are located inside or not far from the light cylinder, and therefore the current density need not be singular. As a result, current-carrying cylindrical flux surfaces may be able to fill out space arbitrarily close to the rotational axis without singularity. Clearly, current-carrying conical and type I paraboloidal flux surfaces cannot fill out the polar region and hence cannot exist all by themselves. Can they avoid the polar region by enclosing cylindrical flux surfaces? The answer to this question depends on whether the cylindrical flux surfaces can extend transversely to infinity.

To examine this possibility, we begin by calculating the total magnetic flux Ψ_{tot} from the definition of the flux function for *cylindrical* surfaces,

$$\Psi_{tot} = \int_0^{R_0} B_p R^2 \frac{dR}{R}, \tag{3.15}$$

where R_0 is the radii of the outermost bounding cylinder. If the field fills the entire space (i.e., $R_0 \to \infty$) for a finite total magnetic flux Ψ_{tot}, we must demand that $B_p R^2 \to 0$ as $R \to R_0 \to \infty$. Otherwise the integration on the right hand side of equation (3.15) goes into infinity. Since $B_p R^2 = \rho R^2 V_p / k$ from equation (2.1), there are two ways for $B_p R^2$ to vanish if k is bounded, namely, either $V_p \to 0$, or $\rho R^2 \to 0$. The former has been shown to be possible for the current-carrying cylindrical asymptotes by Heyvaerts and Norman (their §IX) for nonrelativistic cases, and it can be shown that the same type of argument also holds for relativistic flows. The latter is equivalent to the requirement of a vanishing total poloidal current. To show this,

we can compute the total current I_∞ which, under the conditions $B_p R^2 \to 0$ and $V_p \neq 0$ as $R \to \infty$, takes the form

$$I_\infty^2 = \left(\frac{B_\phi R c}{2}\right)^2 = \frac{B_p^2 R^4 \Omega^2/4}{\left[1-(c^2/e)^2\right]}, \tag{3.16}$$

which goes to zero along with $B_p R^2$, that is, $A_0 = 0$. The cross-field equation has to be reconsidered as well, since the foregoing analysis about this equation assumed a nonvanishing A_0. It follows that, to the leading order, equation (3.7) becomes

$$\frac{d}{dR}\left[\left(\frac{c^2}{e}\right)^2 B_\phi^2 R^2\right] = 8\pi\rho R V_\phi^2, \tag{3.17}$$

in the limit of large R. Since $B_\phi^2 R^2$ is finite at a finite distance and decreases to zero toward the outermost flux surface, the left hand side of equation (3.17) is negative, but the right hand side is always positive. The equality can not be satisfied and thus the cross-field force can never be balanced within a current-free ($A_0 = 0$) cylindrical field that extends to an infinite transverse distance. In physical terms, the left hand side of equation (3.17) is the net outward force due to the imbalance between the outward magnetic pressure force and the inward electric force, and the right hand side is the outward centrifugal force. Rather than cancel, they add.

The difficulty associated with force imbalance at a large distance may be avoided if the asymptotically cylindrical flow has a finite transverse extent. First, it may be possible for a flow to be self-confined within a distance comparable to, or smaller than, the light cylinder radius. This is because at this distance the large-R expansion of the above analysis is no longer valid, eq. (19) does not hold, and the quantity A defined in eq. (3.9) can

be a negative quantity so as to satisfy force balance. This flow contains no net current, and exists in a vacuum (Sulkanen & Lovelace 1990). Second, a flow can be bounded by an external medium at some finite distance (not necessarily comparable to or smaller than the light cylinder radius); this has been demonstrated by Camenzind (1989). To counteract the external pressure at the boundary, B_ϕ immediately inside the boundary must be finite, so an externally confined jet carries a net current.

We may now address whether cylindrical flux surfaces can be enclosed by conical or type I paraboloidal surfaces (which are shown to be not able to fill out the polar region all by themselves. The asymptotic matching boundary must be located at $R \to \infty$. But, as we have just shown, V_p on the cylindrical flux surface at such a distance has to be zero. Hence the only way for the different types of flux surfaces to match together so as to coexist is to have the flow velocity vanish at the matching boundary, which is unacceptable in general. For one thing, the flow in the transition region cannot then be supermagnetosonic, since the magnetosonic speed,

$$V_A = \frac{1}{\gamma}\left(\frac{B_\phi^2}{4\pi\rho}\right)^{1/2} = \frac{|B_\phi R|}{\gamma\sqrt{4\pi\rho R^2}},$$

does not vanish at the same time $V_p = 0$. Therefore, we conclude that the existence of type I paraboloidal or conical flux surfaces is implausible.

Knowing that current-carrying (type I) paraboloidal flux surfaces are not allowed to occur, we finally want to examine the current-free (type II) paraboloidal surfaces. Since type II surfaces asymptotically enclose zero current, there are no difficulties associated with a singular current density on the rotational axis. To examine whether type II surfaces can fill the entire space, we must inspect the cross-field force balance equation at $R \to \infty$.

From the asymptotic expansion (eq. [3.4]), we have $B_pR^2 \cong g(\Psi)/\ln R$, where $g(\Psi) \equiv |\alpha/\alpha'|$. To the leading order of $O(R^{-3}\ln^{-2} R)$, the GS equation (3.6) becomes:

$$\frac{1}{B_pR^3}\nabla\Psi\cdot\nabla\left(\frac{B_pR^2\Omega}{V_{p\infty}}\right)^2 \cong 0, \tag{3.18}$$

where $V_{p\infty}$ is the final poloidal 4-velocity on a flux surface, therefore a function of Ψ only. The dependence of $V_{p\infty}$ on Ψ is determined by the conditions at the base of the flow and the critical surfaces. Upon substituting the approximate expression for B_pR^2 into equation (3.18), one has

$$g' + g\cdot\left(\frac{\Omega'}{\Omega} - \frac{V'_{p\infty}}{V_{p\infty}}\right) = 1,$$

or equivalently

$$g(\Psi) = \frac{V_{p\infty}}{\Omega}\left(\int\frac{\Omega}{V_{p\infty}}d\Psi + C\right),$$

which in turn determines the exponent $\alpha(\Psi)$, and thereby relates the asymptotic field structure to the conditions at the flow base and the critical surface. A similar expansion was obtained by Heyvaerts & Norman (1989, their eq. [74]).

Since $B_pR^2 \to 0$ asymptotically for the type II paraboloidal fields, from the simplified energy equation (3.1), we have the flow Lorentz factor $\gamma \to e/c^2$, that is, all the output energy eventually ends up in the form of flow plasma kinetic energy. This result differs drastically from the case of relativistic ***radial*** winds, in which most of the energy output is in the form of Poynting flux (Michel 1969). The fact that the kinetic energy flux is much greater than the Poynting flux has important implications for pulsar winds, as we discuss below.

In summary, we have performed a fully relativistic analysis of the asymptotic structure of stationary, axisymmetric hydromagnetic winds in this chapter. Our principal conclusions are:

1) If a flow fills the region containing the rotation axis, then the flux surfaces in the flow must do one of the following: (a) collimate to a set of current-carrying cylindrical surfaces extending to infinite transverse radius, provided that the velocity on the "last" flux surface (i.e., at infinite cylindrical radius) is zero; (b) collimate to a set of cylindrical surfaces extending to a finite radius, either self-confined or externally confined; (c) collimate to a current-free paraboloidal field configuration, which fills up the entire space.

2) If an asymptotically cylindrical flow carries a finite current at radii well beyond the light cylinder, then the Lorentz factor of the terminal flow speed on a given flux surface is proportional to the total current enclosed within this flux surface. The constant of proportionality is a global quantity which depends on the boundary conditions at the base of the flow and the criticality conditions.

3) If a flow is of type II paraboloidal (which contains no net current), then its asymptotic energy flux is carried entirely by the gas motion rather than the electromagnetic fields.

The arguments we have used to obtain these results do not depend on the integration of the flow through critical points, which has been used in the past to show that completely radial flow is impossible (Ardavan 1979). Although we considered only cold ($\xi = 1$) flows our results should not depend on this assumption, provided that the thermal pressure becomes insignificant compared to magnetic stresses asymptotically. However, we did not consider situations in which the jet radius oscillates in space (Chan & Henriksen 1980;

Achterberg, Blandford, & Goldreich 1983). Since this type of jet flow is likely to be subject to the formation of internal shocks (Achterberg, Blandford, & Goldreich 1983), an ideal MHD treatment is probably not realistic.

The existence of current-free flows can be of considerable relevance to pulsar winds. Plausible models for the dynamics and radiation from the Crab Nebula require that the wind be overwhelmingly dominated by kinetic energy by the time it reaches the standing shock at $r \sim 10^{17}$ cm (Rees & Gunn 1974; Kennel & Coroniti 1984). However, theories of radial winds (Michel 1969; Goldreich & Julian 1970; Kennel, Fujimura, & Okamoto 1983) predict that most of the energy should be in Poynting flux, unless the wind is dominated by thermal pressure. Recent models (e.g., Coroniti 1990) have suggested that magnetic reconnection or another type of dissipative process may provide the necessary heating and meanwhile reduce the dynamical role of the magnetic field. Since the current-free (type II) paraboloidal flow asymptotically carries an overwhelmingly dominant kinetic-energy flux, its existence offers an alternative model for solving this difficulty (Begelman & Li 1993). In view of the importance of this type of newly discovered field geometry, we shall devote the next chapter to the discussion of the properties of kinetically-dominated relativistic MHD winds in general, and the Type II paraboloidal flux surfaces in particular.

CHAPTER IV

ASYMPTOTIC DOMINATION OF COLD RELATIVISTIC MHD WINDS BY KINETIC ENERGY FLUX

4.1. Introduction

Shortly after the discovery of pulsars, it was proposed that a significant portion of the spindown energy could go into a relativistic wind propelled by the "magnetic slingshot" effect. In this mechanism, plasma injected onto open, rotating field lines is initially forced to corotate by magnetic tension, and is accelerated outward by centrifugal force (Weber & Davis 1967). Ultimately, the inertia of the matter bends back the field lines, resulting in a tightly wound spiral (or helical) field configuration at large distances. Asymptotically, the energy in the wind ($\equiv \mu c^2$ per unit mass) is shared between Poynting flux and particle kinetic energy. However, Michel (1969) showed that this sharing is far from equal. Considering radial, relativistic ($\mu \gg 1$) outflow in which the thermal pressure of the particles is neglected (a "cold" wind), he found that the Poynting flux dominates the kinetic energy flux in the ratio $\mu : \mu^{1/3}$.

This conclusion posed a problem for models of the Crab nebula in which the synchrotron emission was supposed to be powered by the energy dissipated at a strong standing shock in the wind (Rees & Gunn 1974). Observations of synchrotron-emitting "wisps" bounding an underluminous region surrounding the pulsar strongly suggest that such a shock is located approximately 0.1 pc from the pulsar. If the wind were dominated by Poynting flux, then it would be impossible for a standing shock to slow down the

wind to such a degree that it can match the subrelativistic expansion speed of the remnant (Kennel & Coroniti 1984). Instead, the backpressure of the accumulated magnetic flux would drive the shock inward towards the pulsar. Kennel & Coroniti concluded from their analysis that the wind therefore could *not* be driven primarily by the magnetic slingshot effect, but rather must be driven *thermally* by the pressure of relativistically hot plasma. Arbitrarily large ratios of kinetic energy to Poynting flux are possible in such winds (Kennel, Fujimura & Okamoto 1983).

In this chapter, we show that large ratios of kinetic to magnetic energy are, after all, possible in perfectly cold relativistic winds, and may be inevitable if the wind is allowed to propagate over a large enough distance. Hints of this trend have emerged only recently from self-consistent numerical models (Camenzind 1987, 1989; Li, Chiueh, Begelman 1992; Li 1993). Our approach here differs in that it is primarily analytic and attempts to show the generality of this phenomenon. The reason this result was not found by earlier analytic treatments (Michel 1969; Goldreich & Julian 1970) was that they considered only radial outflow. The case of radial outflow is a singular case, in which there is a nearly complete cancellation of outward magnetic pressure force by a component of the magnetic tension beyond the light cylinder radius, as is demonstrated in Figure 4.1 where the various forces contributing to flow acceleration are plotted. As a result of this cancellation, the fast magnetosonic point is reached only asymptotically in a cold radial wind (Goldreich & Julian 1970) and the acceleration of the flow by magnetic forces is severely limited. However, we show that when a flux tube diverges even slightly faster than radially, the fast magnetosonic point can occur close to the light cylinder. Furthermore, significant acceleration

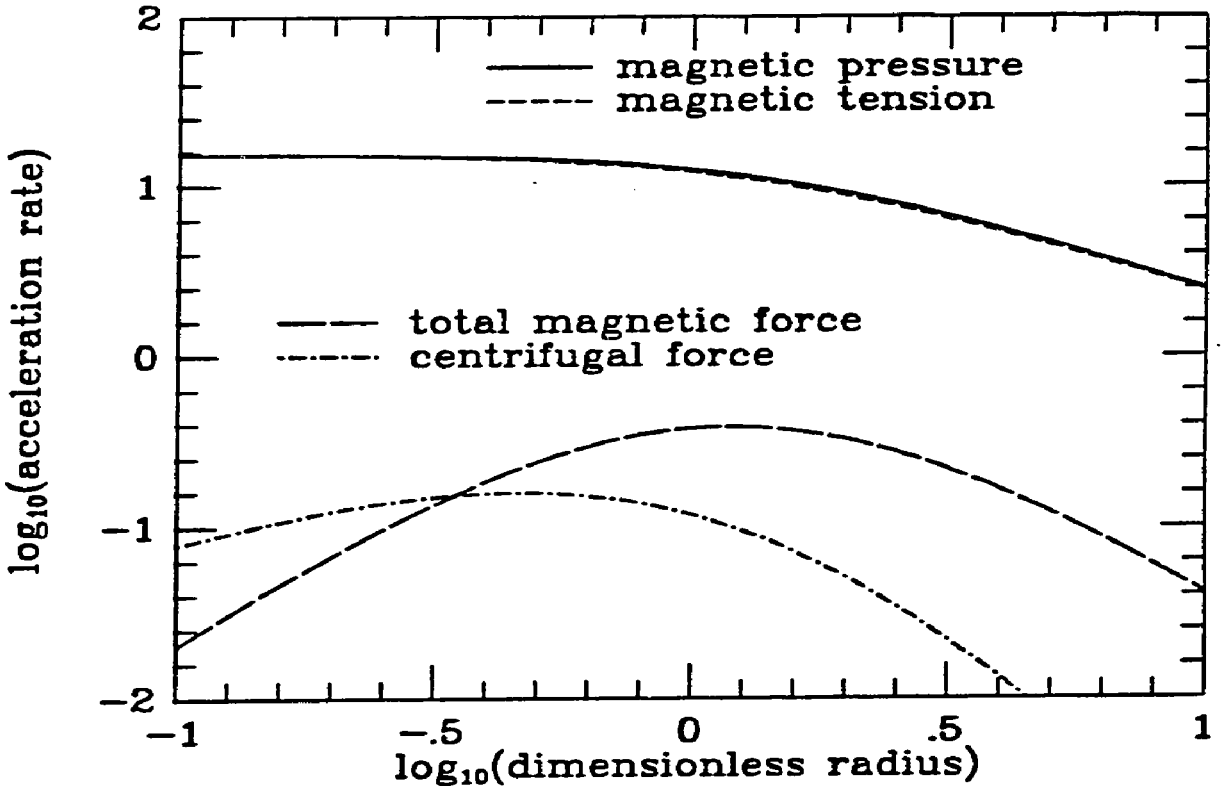


Fig. 4.1–Magnitudes of flow acceleration due to various forces (in units of $c\Omega$, where Ω is the rotation rate of the flux surface) are plotted against the dimensionless radius (in units of $R_L = c/\Omega$, the light cylinder radius) in a radial wind with the magnetization parameter of Michel (1969) $\sigma = 10$. The two largest forces are the magnetic pressure (solid line) and magnetic tension (which is negative, short dash line), which nearly cancel each other. The combined magnetic force (long dash line) and the centrifugal force (dot-dash line) are also plotted. Note the nearly perfect cancellation of magnetic tension and pressure.

of the flow is possible outside the magnetosonic point through a "magnetic nozzle" effect, provided that the flux tubes diverge sufficiently rapidly. The importance of the "magnetic nozzle" effect has already been pointed out by Camenzind (1989), and earlier by Holzer (1977) in the context of the nonrelativistic solar wind.

We show that, far beyond the light cylinder, a flow with finite total magnetic flux and nearly all energy in kinetic form can satisfy the generalized Grad-Shafranov equation which controls the cross-field force balance in a relativistic MHD wind. Our work here extends the asymptotic analysis of chapter III (Chiueh, Li, & Begelman 1991) to include the case where a confining medium exists whose pressure decreases to zero at infinity. We also study the region where the flow is approaching but has not yet reached its asymptote. We find that flows collimate towards their force-free asymptotes logarithmically with radius, and that as they do so their Poynting fluxes become vanishingly small. Logarithmic collimation has been noted by Sakurai (1985) in his numerical work on free nonrelativistic MHD winds and is apparent also in numerical solutions of relativistic winds (Li 1993). It has also been argued recently by Eichler (1993). We propose that the existence of this type of kinetic energy-dominated relativistic wind might provide at least a partial explanation for the low ratio of Poynting flux to plasma kinetic energy flux in the free wind of the Crab pulsar. Kinetic energy-dominated solutions would also have important implications for the study of relativistic jets from active galactic nuclei (AGNs) at large distances from the central source.

The plan of the chapter is as follows. In § 4.2 we show that the location of the fast magnetosonic point is extremely sensitive to the flux tube

divergence. Even very slight deviations from radial flow are sufficient to offset the nearly exact balance between the magnetic pressure gradient and the magnetic tension of a radial relativistic wind, and to move the magnetosonic point in to a radius close to the light cylinder. The general conditions for substantial acceleration to occur outside the fast magnetosonic point are then derived. Kinetic energy-dominated force-free asymptotic solutions are shown in § 4.3 to satisfy the relativistic Grad-Shafranov equation with reasonable boundary conditions, given *any* well-behaved distributions of flux constants. The latter point is crucial, since it suggests that virtually any flow with finite flux will asymptotically approach the kinetic energy-dominated state, provided that the confining external pressure vanishes asymptotically and that no toroidal field exists in the external medium. The latter condition requires that all of the current in the flow has to be screened by a current sheet near the outermost boundary of the flow. These flows are different from the self-collimating cylindrical flows (with finite width) considered recently by Appl & Camenzind (1993), in which the flow current is only partially screened at the boundary. The smooth merging of a non-force-free flow solution at large radius to its asymptotic solution is demonstrated explicitly for a special case. Finally, we discuss our results and their astrophysical implications in § 4.4.

4.2. Conditions for Relativistic MHD Flows to be Dominated by Kinetic Energy Flux

The efficiency with which the Poynting flux is converted into the kinetic energy flux is fundamentally different for relativistic and nonrelativistic MHD flows: rather efficient energy conversion can be achieved in a nonrelativistic flow even with radial (i.e., conical) geometries, whereas such conversion is not possible for a radial relativistic flow (Michel 1969; and see discussion in in § 4.1). This can be shown with a radial cold nonrelativistic wind without gravity. In this case, one can formulate the problem similar to that of Michel (1969) and obtain the total specific energy (excluding the rest mass of the flow)

$$e = \frac{V_\infty^2}{2} + \frac{\sigma c^3}{V_\infty}, \tag{4.1}$$

where V_∞ is the flow terminal poloidal speed and σ ($\ll 1$) is the same magnetization parameter as Michel's. Physically, the first term on the right hand side of equation (4.1) is the kinetic energy, whereas the second term is the electromagnetic energy. The "minimum torque" solution (with its fast magnetosonic point located at infinity) is obtained if we maximize the total energy e with respect to the speed V_∞, holding σ fixed. The result is that $V_\infty = \sigma^{1/3} c$, exactly the same as Michel's solution for the relativistic winds. At the Alfvén point ($R_A = \sqrt{6}/2\sigma^{1/3} R_L$), the flow speed V_A is already two-third of the terminal speed V_∞. The terminal kinetic energy is $\sigma^{2/3} c^2/2$, which is half of the electromagnetic energy. This final near equipartition in energy in nonrelativistic radial winds is in vast contrast with the relativistic radial winds where the terminal kinetic energy is only a tiny fraction ($= \sigma^{-2/3}$, and $\sigma \gg 1$) of the electromagnetic energy.

In order to find the conditions for relativistic MHD flows to be dominated by the kinetic energy flux, we now consider in some detail the energy equation (2.9) which governs flow properties along prescribed flux tubes. Besides yielding interesting results itself, this discussion also forms the foundation for the numerical works in the later chapters. Once again, we make the cold wind assumption and neglect gravity. To nondimensionalize quantities along a flux surface with angular speed Ω and k, we scale the magnetic field strength $\mathbf{B}$ by $4\pi kc$ so that $\mathbf{b} \equiv \mathbf{B}/(4\pi kc)$, and the spatial component of the 4-velocity $\mathbf{V}$ by the speed of light so that $\mathbf{u} \equiv \mathbf{V}/c$. Furthermore, we take the length scale to be the light cylinder radius $R_L = c/\Omega$ so that $x \equiv R/R_L$ and $y \equiv Z/R_L$. Finally, we replace the total specific energy and angular momentum with their dimensionless counterparts $\mu \equiv e/c^2$ and $\lambda \equiv l\Omega/c^2$. We can then cast the energy equation into the following dimensionless form,

$$(\mu - \lambda\eta)^2 = 1 + (a\tau)^2 + \left[\frac{\lambda}{x}(1-\eta)\right]^2, \tag{4.2}$$

where we have defined three two auxiliary dimensionless quantities

$$a \equiv b_p x^2, \ \tau \equiv \frac{4\pi k^2 R_L^2}{\rho R^2}, \ \eta \equiv \frac{1-(\mu x/\lambda)^2}{1-x^2(1+\tau)}. \tag{4.3a,b,c}$$

The quantity a characterizes the divergence of a flux tube which remains constant on radial tubes. It decreases along flux tubes that diverge faster than radial. The dimensionless quantity η is just the fraction of the total angular momentum carried out electromagnetically, whereas τ at large distance ($x \gg 1$) represents the ratio of the kinetic to electromagnetic energies in the flow. The two components of the dimensionless velocity and the dimensionless toroidal magnetic field strength are given, respectively, by

$$u_p = a\tau, \quad u_\phi = \lambda(1-\eta)/x, \quad b_\phi = -\lambda\eta/x. \tag{4.4a,b,c}$$

For given μ and λ, the flow solution is completely determined once the quantity a as a function of cylindrical radius x is specified. The total specific energy and angular momentum of the flow are not independent from each other, however. They are related through the initial condition at the base of the flow. If the flow is injected onto the flux tube with some finite speed $u_p = u_0$ at an initial radius $x = x_0$ at the base of the flow where $a = a_0$, then one has the following quadratic equation relating μ and λ,

$$1 + u_0^2 = [(1 - c_\mu)^2 - c_\mu^2/x_0^2]\mu^2 - 2[(1 - c_\mu)c_\lambda - (1 - c_\lambda)c_\mu/x_0^2]\mu\lambda + [c_\lambda^2 - (1 - c_\lambda)^2/x_0^2]\lambda^2 \quad (4.5)$$

where

$$c_\mu = -\frac{\mu x_0^2}{1 - (1 + \tau_0)x_0^2}, \quad c_\lambda = \frac{1}{1 - (1 + \tau_0)x_0^2}$$

and $\tau_0 = u_0/a_0$. In the case that $u_0 = 0$ and $x_0 = 0$, we have the simple relation that $\mu = \lambda + 1$, which simply reflects the fact that both the energy and the angular momentum are dominated by the magnetic field at the flow base. We should consider this simplest case only unless noted otherwise.

The energy equation (4.2) is a quartic algebraic equation of either τ or η. It seems that, for a prescribed distribution of $a(x)$, one can obtain a flow solution for any choice of μ (and thus λ). It turns out, however, that for only one special value of μ can a smooth trans-magnetosonic solution be achieved. Solutions with μ less than this critical energy μ_c crash at the fast magnetosonic point (to be defined shortly), whereas those with $\mu > \mu_c$ remain submagnetosonic. The critical energy μ_c is to be fixed by the regular condition at the so-called "critical point" of the energy equation (4.2). The

Alfvén point, which is reached when the relativistic Alfvén Mach number, defined as

$$M_A^2 \equiv \frac{V_p^2}{B_p^2(1-R^2/R_L^2)/(4\pi\rho)} = \frac{x^2\tau}{1-x^2}, \qquad (4.6)$$

equals unity, is not a critical point. This is because the condition $M_A = 1$ is satisfied automatically at the Alfvén radius $x_A = \sqrt{\lambda/\mu}$. In fact, when treated as a quartic of η, the energy equation (4.2) can pass the Alfvén point without any notice. The $(-R^2/R_L^2)$ term in the above (as well as below) definition of Mach number represents the pressure modification due to the electric field, which renders the Alfvén wave nonexistent beyond the light cylinder radius. On the other hand, the so-called "fast magnetosonic point", at which the fast magnetosonic Mach number M_f defined from

$$M_f^2 \equiv \frac{V_p^2}{[B_\phi^2 + B_p^2(1-R^2/R_L^2)]/(4\pi\rho)} = \frac{\tau b_p^2 x^2}{b_\phi^2 + b_p^2(1-x^2)} \qquad (4.7)$$

equals unity, is a real critical point to the energy equation (4.2). This is shown by differentiating equation (4.2) once with respect with radius, so that

$$\frac{d\tau}{dx} = \frac{[1-x^2(1+\tau)]\left[\frac{\lambda^2(1-\eta)^2}{x^3\tau} - a\tau\frac{da}{dx}\right] - 2\lambda^2\eta x[(1+\tau)\eta - \mu/\lambda]}{x^4\{b_\phi^2 + b_p^2[1-x^2(1+\tau)]\}}. \qquad (4.8)$$

It is evident that the denominator on the right hand side vanishes at the fast point $M_f = 1$. Topologically, two branches of roots of equation (4.2) cross at both the Alfvén point and the fast point. In the case of Alfvén point, both root branches are unphysical, whereas one branch of them is physical in the case of fast point. Therefore the Alfvén point can always be passed without imposing any additional constraint on the flow solution. To pass smoothly through the fast point, however, a regularity condition

has to be imposed so that the numerator on the right hand side of equation (4.8) vanishes simultaneously with the denominator. It is the presence of a nonvanishing da/dx term in the numerator in a nonradial flow that makes its fast point location qualitatively different from that in a radial wind.

As mentioned earlier, in the case of a radial cold wind (Michel 1969), the flux tube divergence quantity a remains constant along a flux tube and the fast point is never reached except asymptotically. To show that this is an artifact of the special geometry and results from nearly complete cancellation of the magnetic pressure force and the magnetic tension force, we write down the momentum equation (2.3) of the relativistic MHD winds in the poloidal direction, which governs the change of the poloidal 4-speed V_p with radius as

$$\frac{1}{2}\frac{dV_p^2}{dR} = \frac{V_\phi^2}{R} - \frac{1}{8\pi\rho R^2}\frac{d}{dR}\left(B_\phi^2 R^2\right), \tag{4.9}$$

or, in dimensionless form

$$\frac{1}{2}\frac{du_p^2}{dx} = \frac{u_\phi^2}{x} - \frac{\tau}{2}\frac{d}{dx}(b_\phi^2 x^2). \tag{4.10}$$

Apparently, the first term on the right hand side is the centrifugal force, whereas the second term is the combination of the magnetic pressure force and the magnetic tension force. The Figure 4.1 that we showed earlier is generated for a particular radial flux tube with constant $a(x) = 10$. The near cancellation of the magnetic forces in the figure is evident.

We now investigate how the location of the fast point depends on the properties of nonradial flows. For flows with geometry other than radial, da/dx no longer vanishes everywhere. We parameterize the deviation of the quantity a from constancy by $\delta(x) \equiv d\ln a/d\ln x$. At the fast point, given

the value of a_f (the subscript 'f' stands for "fast point"), the energy equation (4.2), together with the requirements that both the denominator and the numerator of equation (4.8) vanish simultaneously, yield three coupled algebraic equations for three unknowns: the critical energy μ_c, the auxiliary quantity τ_f, and the amount of deviation δ_f needed to move the fast point to a finite radius of $x_f = 1/t_f$. The equations are

$$\tau_f = \{[(t_f^2 - 1)\mu_c - t_f^2]/a_f\}^{2/3} + t_f^2 - 1 \tag{4.11}$$

$$2\tau_f(t_f^2 - \tau_f\mu_c)[(t_f^2 - 1)\mu_c - t_f^2] + t_f^2(t_f^2 - 1 - \tau_f) \times [1 - \tau_f(\mu_c - 1)]^2 + 2\tau_f^2 a_f^2(t_f^2 - 1 - \tau_f)^3(1 - \delta_f/2) = 0 \tag{4.12}$$

$$(t_f^2 - 1 - \tau_f)^2(1 + \tau_f^2 a_f^2) + t_f^2[1 - \tau_f(\mu_c - 1)]^2 - (t_f^2 - \tau_f\mu_c)^2 = 0. \tag{4.13}$$

The numerical results are presented in Figures 4.2*a* and *b*. It is apparent from Fig. 4.2*a* that only a negative δ_f can move the fast point to a finite radius, which means the divergence of the flux tube has to be faster, rather than slower, than that in a radial wind. Since the curve is almost horizontal near $t = 0$, even a flux tube divergence slightly faster than radial can have a large effect there. In fact, a small $\delta_f = -0.05$ can bring the fast point to about 5 light cylinder radii. Plotted in Fig. 4.2*b* is the critical total energy in units of a_f and the kinetic energy $\gamma_f = (t_f^2 - \mu_c\tau_f)/[t_f^2 - (1 + \tau_f)]$ in units of $\mu_c^{1/3}$. Clearly, the *total* critical energy obtained on the flux surface is close to the value of a at the fast point and is quite insensitive to where the fast point is located, while the *kinetic energy* at the critical point is close to the cubic root of the total energy. Therefore, even though the fast point is moved close to the light cylinder by the divergence of the flux tubes, the total energy flux there is still dominated by the Poynting flux. In order for the final

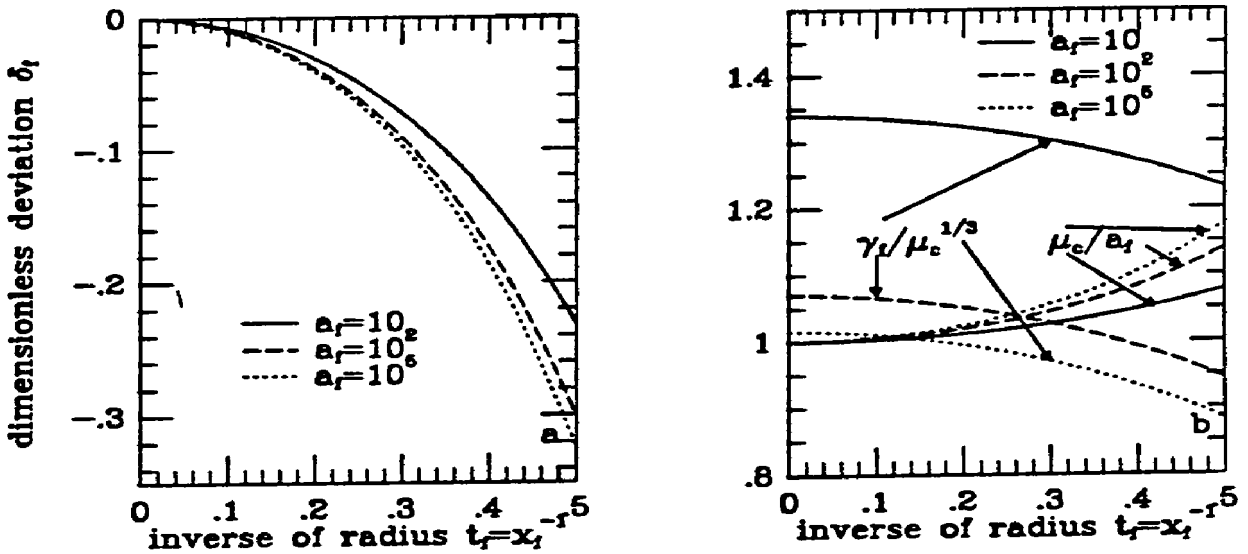


Fig. 4.2–(a) the amount of δ_f, which characterizes a flux tube's deviation from a perfect radial geometry, needed to bring the fast point to a finite radius $t_f = x_f^{-1}$, for three different values of $a_f = 10$, 10^2 and 10^5; (b) the corresponding critical total specific energy μ_c (in units of a_f) and the specific kinetic energy at the fast point γ_f (in units of $\mu_c^{1/3}$), as functions of the inverse of the fast point radius.

kinetic energy flux to be comparable to or even larger than the Poynting flux, significant acceleration must occur beyond the fast point. This is achieved through the so-called "magnetic nozzle" effect discussed below.

According to the equation (4.2), for given total energy μ and angular momentum l, the flow variables at any location along a flux tube depend only on the local value of a. Therefore in order to obtain the asymptotic ratio of the Poynting flux to the kinetic energy flux ζ_∞ where, from equation (2.5), the ratio is given by

$$\zeta \equiv -\frac{R\Omega B_\phi}{4\pi k c^2 \gamma}, \tag{4.14}$$

we have to know the asymptotic value of a, a_∞, and the asymptotic radius x_∞. Note that the latter could be infinite. We restrict our discussion to those flux tubes that extend well beyond the light cylinder with $x_\infty \gg 1$, so that the energy equation (4.2) is simplified to

$$\left(\frac{\tau_\infty}{1+\tau_\infty}\mu\right)^2 = 1 + (a_\infty \tau_\infty)^2 \tag{4.15}$$

and the ratio ζ and fast magnetosonic Mach number M_f^2 are also reduced to

$$\zeta_\infty = \frac{1}{\tau_\infty}, \qquad M_{f\infty}^2 = \tau_\infty^3 a_\infty^2. \tag{4.16}$$

Therefore, from equations (4.15) and (4.16), the derivative of ζ_∞ with respect to a_∞ is given by

$$\frac{d\zeta_\infty}{da_\infty} = \frac{a_\infty \tau_\infty (1+\tau_\infty)}{M_{f\infty}^2 - 1}. \tag{4.17}$$

Thus, if the flow in the tube is asymptotically supermagnetosonic ($M_f > 1$), a smaller final value of a implies that a larger fraction of total energy will be in kinetic form, while the opposite is true for a submagnetosonic flow.

Geometrically, the quantity a is related to the relative opening ΔS (the distance between two adjacent flux surfaces Ψ and $\Psi + \Delta\Psi$) of a flux tube with respect to its cylindrical radius R, because

$$a \propto B_p R^2 = R\frac{\Delta\Psi}{\Delta S}. \tag{4.18}$$

Since a smaller value of a corresponds to a larger opening of the streamlines, the behaviors of the flow inside a flux tube is analogous to hydrodynamic flow through a *de Laval* nozzle (Landau and Lifshitz 1959). For supermagnetosonic, ultra-relativistic ($\mu \gg 1$) flows with $a_\infty/\mu \ll 1$, we have

$$\zeta_\infty \approx \frac{a_\infty}{\mu}, \qquad M_{f\infty}^2 \approx \frac{\mu^3}{a_\infty}. \tag{4.19a,b}$$

These flows are therefore dominated by kinetic energy, i.e., $\zeta_\infty \ll 1$. The opposite is true for submagnetosonic flows in the same limits

$$\zeta_\infty \approx \mu, \qquad M_{f\infty}^2 \approx \frac{a_\infty^2}{\mu^3} \tag{4.20a,b}.$$

Since $\mu \cong a_f$ for a critical solution, we have $\zeta_\infty \approx a_\infty/a_f$ in the supermagnetosonic case, i.e., *nearly all energy could be converted eventually into kinetic form ($\zeta_\infty \ll 1$) provided that the field lines have such a geometry that the quantity "a" decreases significantly from its value at the fast point, a_f, to its asymptotic value a_∞ ($a_\infty/a_f \ll 1$) once the fast point is passed.* We thus obtain a general condition for a relativistic MHD wind to be dominated by kinetic energy flux. However, we have yet to show that such winds satisfy the cross-field force balance equation. We shall address this point in the next section.

4.3. Nearly-Radial, Kinetic Energy-Dominated Wind Solutions to the Grad-Shafranov Equation

A supermagnetosonic wind becomes strongly dominated by kinetic energy (with small ζ) only when the quantity a decreases by a large factor beyond the fast point (eq. [4.19]). Since a measures the divergence of a flux tube, it is conceivable that such smallness of a can be achieved on flux surfaces deflected gradually towards the rotation axis. Physically, this is because the flux tubes near the equator would have more space to fill once the inner flux surfaces close to the rotation axis become well collimated. Collimation toward the rotation axis has been shown to be a general tendency of magnetized winds (Heyvaerts & Norman 1989; and see chapter III), and this tendency has been demonstrated explicitly by several authors either with external confinement (Camenzind 1987, 1989; Li 1993) or without (Blandford & Payne 1982; Sakurai 1985; Li, Chiueh, & Begelman 1992). Our objective in this section is to determine the conditions under which such solutions of the relativistic Grad-Shafranov equation can develop vanishingly small ratios of Poynting flux to kinetic energy flux. We will comment on the differences between this analysis and previous work in § 4.4.

The asymptotic solutions we seek are related to the "current-free paraboloidal" asymptotes discussed in Heyvaerts & Norman (1989) and chapter III, which display the behavior $\zeta \rightarrow 0$ as $r \rightarrow \infty$. We refer to these asymptotes as "force-free" in the sense that the *cross-field* force balance is determined by electromagnetic stresses, with inertial forces playing a negligible role. However, we shall show that they are not "force-free" in the poloidal direction, since the flow continues to accelerate. Below we generalize these

solutions to include flows which have finite total flux and are confined by an external medium with pressure decreasing to zero at infinity. The structures of these asymptotic solutions are uniquely determined by the pressure boundary condition and the distribution of a particular flux function. We then concentrate on the region where the inertia of the flow is still important, i.e., on field lines which have not yet been collimated parallel to the rotation axis. An example is given to illustrate how solutions in this non-force-free (in the cross-field direction) region join smoothly onto the corresponding force-free asymptotes. In our analysis, we are not able to determine whether a given solutions satisfies the critical conditions, since we consider only the region well beyond the fast magnetosonic point. Therefore, it is not possible *a priori* to constrain the flow energy and angular momentum using the critical conditions. To make allowance for this, we shall leave unspecified the flux constants such as μ, λ and k along each flux surface for the duration of the general discussion. Remarkably, we will obtain a generic family of solutions capable of handling *any* well-behaved distributions of critical energy and other flux functions. This gives us confidence that our solution represents a natural asymptotic state for a wide range of flows.

Far beyond the light cylinder where $R \gg R_L$, the cross-field GS equation (eq. [3.6]) becomes

$$-\kappa \left(\frac{B_p^2 R^2 \Omega^2}{c^2} + \frac{k^2 B_p^2}{4\pi\rho} \right) + \frac{\nabla\Psi}{B_p R^3} \cdot \nabla \left(\frac{B_\phi^2 R^2}{2} - \frac{B_p^2 R^4 \Omega^2}{2c^2} \right) \simeq 0, \quad (4.21)$$

where κ is the curvature of the poloidal field lines and terms of order $(R_L/R)^2$ or higher are neglected. We can verify a *posteriori* that the solutions we seek have the following properties: 1) $B_\theta \ll B_r$ (nearly-radial poloidal field lines), where B_θ and B_r are the two components of the poloidal magnetic

field strength $\mathbf{B}_p$ in standard spherical polar coordinates (r, θ, ϕ); and 2) the Poynting flux $\propto B_p R^2 \rightarrow 0$ as $r \rightarrow \infty$ (kinetic energy-dominated). From condition (2) we conclude that the second term inside the first bracket of equation (4.21) is much larger than the first term, provided that $k(\Psi)V_\infty(\Psi)$ (where V_∞ is the value of V_p in the limit $r \rightarrow \infty$) is nonzero for all Ψ and that $\Omega(\Psi)$ is finite. For the relativistic winds dominated by kinetic energy that we are considering, we have $V_\infty \approx \mu c$. Using the energy equation in the region $R \gg R_L$, one can combine the two terms in the second bracket into one term, $(B_p R^2 \Omega / V_\infty)^2/2$, so that the cross-field balance equation (4.21) can be simplified to

$$\kappa V_\infty^2 \simeq \frac{\Omega}{k}\left(\frac{\nabla\Psi}{B_p R}\right)\cdot\nabla\left(\frac{B_p R^2 \Omega}{V_\infty}\right). \tag{4.22}$$

The left-hand side is simply the centripetal acceleration of the flow along the curved flux surfaces, whereas the right-hand side is the sum of the magnetic force due to the toroidal field and the electric force arising from special relativistic effects. This clearly shows that continuous collimation (and even reconvergence) of a flow with nonzero κ is ensured by the presence of an unbalanced electromagnetic force across the flux surfaces.

The curvature of the poloidal field line κ for nearly-radial winds in standard spherical polar coordinates (r, θ, ϕ) is given approximately by

$$\kappa \simeq -\frac{1}{B_r r^3}\frac{\partial(B_\theta r^3)}{\partial r}, \tag{4.23}$$

where higher order terms in B_θ are discarded. Making use of the relations

$$B_r = \frac{1}{r^2 \sin\theta}\frac{\partial\Psi}{\partial\theta}, \qquad B_\theta = -\frac{1}{r\sin\theta}\frac{\partial\Psi}{\partial r}, \tag{4.24}$$

and the fact that $B_p \simeq B_r$, we finally arrive at the equation that governs the flow behavior of nearly-radial, kinetic-energy-dominated MHD winds in the region $R \gg R_L$. To facilitate expression of the boundary conditions near the pole and on the outermost flux surface, we define an auxiliary flux function $\varphi \equiv (1 - \Psi/\Psi_0)$, where Ψ_0 is the total magnetic flux, which goes to 1 near the axis where $\Psi \to 0$ and vanishes on the last flux surface where $\Psi = \Psi_0$. We can then rewrite equation (4.22) as

$$\frac{1}{\xi(\varphi)} \frac{\partial \varphi}{\partial \theta} \frac{\partial}{\partial \theta} \left[\sin\theta \xi(\varphi) \frac{\partial \varphi}{\partial \theta} \right] \simeq -\nu(\varphi) \frac{\partial}{\partial r} \left(r^2 \frac{\partial \varphi}{\partial r} \right), \qquad (4.25)$$

to the lowest order, where we have defined $\xi(\varphi) = \xi(\Psi) \equiv \Omega(\Psi)/V_\infty(\Psi)$ and $\nu(\varphi) = \nu(\Psi) \equiv V_\infty^3(\Psi) k(\Psi)/[\Psi_0 \Omega^2(\Psi)]$. The boundary conditions are $\varphi = 0$ on the outermost flux surface and $\varphi \to 1$ as the cylindrical axis ($\theta = 0$) is approached. If the flow is confined to a region of space by some external pressure, then the location of the outermost flux surface must be determined self-consistently. We show how to do this in the force-free asymptotic limit in § 4.3.1. In the limiting case of no external confinement, the last flux surface is assumed to coincide with the equatorial plane ($\theta = \pi/2$). In § 4.3.2 we solve equation (4.25) for the full structure of the space-filling flow, for the special case of ξ, ν independent of φ. More generally, the initial conditions at the base of the flow and the critical conditions at the critical surfaces are absorbed into the distributions of these flux constants, which are free to be specified.

4.3.1. Asymptotic Force-Free Solutions

Equation (4.25) is a nonlinear second-order partial differential equation, which cannot be solved analytically in general. Nevertheless, it is

straightforward to extract the force-free asymptotic behavior of the solutions that interest us. The left-hand side of equation (4.25) represents the electromagnetic forces which, in the force-free limit, must balance to lowest order. In this limit,

$$\frac{\partial}{\partial\theta}\left[\sin\theta\xi(\varphi)\frac{\partial\varphi}{\partial\theta}\right]\approx 0. \tag{4.26}$$

We shall deal with the flow inertia term on the right-hand side of equation (4.25) later. Since all flux surfaces are nearly parallel to the rotation axis at large radius r, we can replace $\sin\theta$ in equation (4.26) by θ and integrate to obtain

$$\ln(\theta_0/\theta) = q_1(\varphi)g(r) \tag{4.27}$$

where θ_0 is a reference angle and the function $q_1(\varphi)$ is given by

$$q_1(\varphi) = \int_0^{\varphi}\xi(\varphi)d\varphi + c_1. \tag{4.28}$$

The free function $g(r)$ in equation (4.27) is to be determined using the boundary conditions. In principle, θ_0 can also depend on r, but since we are able to fit all flux functions and the pressure boundary condition described below using a single free function, we shall treat θ_0 as a constant. It is required that $g(r\to\infty)\to\infty$ in order that $\theta\to 0$ as $r\to\infty$ for a flux surface φ with nonzero $q_1(\varphi)$.

We demand that the last flux surface $\varphi = 0$ be contained by an external pressure distribution which is assumed to be a function of the spherical radius r only, i.e., $P_e(r)$. For relativistic flows in the region $R \gg R_L$ under consideration, it can be shown that the modified (by electric field) magnetic pressure is given by

$$P_m = \frac{B_\phi^2}{8\pi}+\frac{B_p^2}{8\pi}\left(1-\frac{R^2\Omega^2}{c^2}\right)\approx\frac{1}{8\pi}\left[\frac{\Psi_0}{g(r)r\theta}\right]^2\approx\frac{1}{8\pi}\left[\frac{\Psi_0 e^{q_1(\varphi)g(r)}}{g(r)r\theta_0}\right]^2 \tag{4.29}$$

where equation (4.27) has been used. The electric field reduces the total pressure to about $(B_\phi/\gamma)^2/(8\pi)$, where γ is the Lorentz factor of the flow, and to achieve pressure balance across the last flux surface most, if not all, of the flow current in a relativistic flow (i.e., $\gamma \gg 1$) has to be screened out by an induced current sheet near the flow boundary. Otherwise, the flow will be squeezed by an unbalanced external toroidal field, possibly all the way to within the light cylinder surface (Appl & Camenzind 1993). We consider only the case where all of the flow current is screened out, and thus the external toroidal field is zero. The pressure balance then requires

$$q_1(0)g(r) = \ln[g(r)] + \frac{1}{2}\ln[P_e(r)r^2] + \ln[\sqrt{8\pi}\theta_0/\Psi_0]. \qquad (4.30)$$

Since $g(r) \to \infty$ as $r \to \infty$, we have $g(r) \propto \ln[P_e(r)r^2]$ to the leading order.

To simplify the subsequent analysis, we shall assume a power-law pressure distribution, i.e., $P_e(r) = P_0 r^{-\alpha}$. In this case, we have $g(r) = c_2 \ln(r/r_0)$ (where r_0 is a reference radius and c_2 is a free constant) and, from equation (4.30), $q_1(0) \approx (1 - \alpha/2)/c_2$. Therefore, equation (4.27) becomes

$$\ln(\theta_0/\theta) = q(\varphi)\ln(r/r_0) \qquad (4.31)$$

for a power-law distribution of external pressure, where we have defined $q(\varphi) = c_2 q_1(\varphi)$. It is apparent that the exponent α has to be less than 2 for the above analysis to be self-consistent. Otherwise, $q(0) = c_2 q_1(0) \approx 1 - \alpha/2 < 0$, and the angle θ for some of the flux surfaces near $\varphi = 0$ will not approach zero as $r \to \infty$ according to equation (4.31).

The physical meaning of $q(\varphi)$ becomes transparent when we rewrite equation (4.31) as

$$\theta = \theta_0 (r/r_0)^{-q(\varphi)}, \qquad (4.32)$$

that is, $q(\varphi)$ is the rate at which the flux surface deflects towards to the rotation axis. The asymptotic cylindrical radii of the flux surfaces behave according to

$$R = r\sin\theta \approx r\theta = (r_0\theta_0)(r/r_0)^{1-q(\varphi)}. \tag{4.33}$$

The maximum value of $q(\varphi)$ is $q_0 \equiv q(\varphi = 1) = c_2(\int_0^1 \xi(\varphi)d\varphi + c_1)$. If $q_0 < 1$, then all flux surfaces would have $R \to \infty$ as $r \to \infty$. In particular, there would be a gap devoid of flow between the rotation axis and the first flux surface $\varphi = 1$. If, on the other hand, $q_0 > 1$, then there exists a special flux surface φ_c such that $q(\varphi_c) = 1$. Flux surfaces outside this surface will be "paraboloidal" in the sense that as $r \to \infty$, $\theta \to 0$, and the cylindrical radius $R \to \infty$ (see chapter III). Flux surfaces enclosed by φ_c would have $R \to 0$ as $r \to \infty$ if equation (4.33) continues to hold, i.e., they would reconverge onto the rotation axis. However, before these flux surfaces approach the light cylinder radius, the assumption of $R \gg R_L$ used in deriving the various equations in this section will break down, hence equation (4.33) can no longer be valid in this region. The question then is: could the inclusion of other terms neglected in the present discussion change the flux surfaces in the reconverging zone so much that they become cylindrical and coexist with the flux surfaces in the "paraboloidal" zone? The answer is no, as we showed in chapter III. The basic reason for this incompatibility is that cylindrical flux surfaces with finite radius necessarily enclose finite currents, whereas the "paraboloidal" surfaces discussed here have to be asymptotically current-free. The only other possible configuration is that all the flux surfaces inside φ_c crash onto the rotation axis even when other terms are included. It is hard to believe such singular solutions could describe realistic winds, and

we shall not discuss them any further. This leaves us with only the choice $q_0 = q(1) = 1$.

The physical meaning of the condition $q_0 = 1$ is that all the flux surfaces, except the one that lies exactly on the axis, will ultimately expand laterally and become "paraboloidal", although they do so at different rates. Those closer to the axis (i.e., with smaller Ψ) will travel a longer distance parallel to the axis before crossing a given cylindrical radius and expanding gradually to infinity. A flux surface infinitely close to the axis ($\Psi \to 0$) would travel an infinitely long distance along the axis before expanding laterally through a given cylinder. In this way, we obtain a set of well-behaved asymptotic solutions that can fill the entire space smoothly. It should be pointed out that our far-field solution does not apply to the region close to and within the light cylinder radius. Nevertheless, since the portion of the total flux that stays in this region becomes increasingly small with radius r, we conjecture that our solution is a good description of the overall flow structure at large distances.

The condition $q(1) = 1$, together with the condition on the last flux surface $q(0) \approx 1 - \alpha/2$, yields the following expression for $q(\varphi)$:

$$q(\varphi) \approx 1 - \frac{\alpha}{2}\left[1 - \frac{\int_0^\varphi \Omega(\varphi)/V_\infty(\varphi)d\varphi}{\int_0^1 \Omega(\varphi)/V_\infty(\varphi)d\varphi}\right]. \qquad (4.34)$$

It is apparent that a faster rate of decline in the pressure of the external medium (i.e., a larger α) implies a slower rate of collimation of the outermost flux surface, since $R \propto r^{\alpha/2}$. In the limit that α approaches 2 from below, we have $q(0) \to 0$. Therefore, flux surfaces arbitrarily close to the outermost flux surface should be correspondingly slow in becoming parallel to the axis. This implies that the flow is bounded by a cone of some arbitrary angle (or

by the equator in case of a free wind with no external confinement). Since it is impossible for flow to move parallel to the equatorial plane asymptotically as we showed in chapter III, the fastest magnetic pressure drop possible in a flow would be r^{-2} (to within a logarithmic factor). We suspect that if the ambient pressure decreases faster than r^{-2}, the magnetic pressure in the flow would always squeeze the medium all the way to the equatorial plane and maintain a pressure distribution of approximately r^{-2}. Therefore, we infer that, depending on the pressure distribution of the external confining medium, a relativistic MHD flow becomes either a jet that is collimated towards the rotation axis and is excluded from the equatorial region, or a wind that occupies all of space. For the specific case of a power-law distribution of external pressure at large distance, a jet forms whenever $\alpha \leq 2$; on the other hand, if $\alpha > 2$ or there is no confinement at all, a space-filling wind would form instead. Note, however, that even in a space-filling wind, any given flux surface (except the outermost one) will gradually collimate towards the axis.

Depending on the distribution of $\xi(\varphi) = \Omega(\varphi)/V_\infty(\varphi)$, one can arrange for relatively more flux surfaces to occupy the region closer to the axis of the flow or the region closer to the outer boundary. If $\xi(\varphi)$ close to the boundary is much smaller than near the axis, then there will be relatively more flux surfaces concentrated near the boundary than near the axis (that is, there will be more surfaces with $q(\varphi)$ close to $1 - \alpha/2$ than close to 1). We shall discuss the implication of this property for the Crab pulsar wind in § 4.4.

We now check whether the two conditions assumed before, 1) nearly-radial flow and 2) domination by kinetic energy, are valid for our asymptotic solutions. The two components of the poloidal magnetic field are given by

$$B_r \approx \frac{\Psi_0}{r^2 \ln(r/r_0)\theta^2 q'(\varphi)} \quad \text{and} \quad B_\theta \approx -\frac{\Psi_0 q(\varphi)}{r^2 \ln(r/r_0)\theta q'(\varphi)}. \tag{4.35}$$

It is clear that $|B_\theta| = |\theta q(\varphi) B_r| \ll |B_r|$ since $\theta \ll 1$ and $q(\varphi) \leq 1$. Therefore the first condition of nearly-radial flow is satisfied. To show that the second condition is also satisfied, we evaluate the Poynting flux, which is proportional to

$$a \propto B_p R^2 \approx B_r (r \sin\theta)^2 \approx \frac{\Psi_0}{q'(\varphi)\ln(r/r_0)}. \tag{4.36}$$

Therefore, the Poynting flux in these asymptotic solutions goes to zero logarithmically with radius and the flow is indeed kinetic energy-dominated. It should be noted that the poloidal current enclosed by a flux surface at cylindrical radius R also decreases with increasing r since $I_R \propto B_\phi R \propto B_p R^2 \propto [\ln(r/r_0)]^{-1}$. This implies that current density must cross the flux surfaces. This can be verified directly from the expression for the poloidal current density

$$\mathbf{j}_p \approx \frac{c}{4\pi c_2 \ln(r/r_0)} \left\{ \frac{d[V_\infty(\varphi)/c]}{d\varphi} \mathbf{B}_p - \frac{\Psi_0 V_\infty(\varphi)/c}{r^2 \theta \ln(r/r_0)} \hat{\mathbf{e}}_\theta \right\}. \tag{4.37}$$

It is the component of the current in the cross-field direction (namely $\hat{\mathbf{e}}_\theta$) that continues to accelerate the flow at the expense of the Poynting flux. We mention in passing that, by using equations (4.35) and (4.36), one can verify that the terms we discarded in deriving our approximate equations are of order $O(\{\ln[r/r_0]\}^{-1})$ or smaller compared to those terms retained.

4.3.2. Approach to the Asymptotes

We now study the behavior of the flow before it reaches the force-free asymptotic region. In this case, one has to solve equation (4.25) including the inertia term on the right-hand side. This cannot be done analytically for arbitrary distributions of $\xi(\varphi)$ and $\nu(\varphi)$. However, in the simplest case where both ξ and ν are independent of φ, equation (4.25) is separable in r and θ. We concentrate on free winds with no confining medium, for which the last flux surface is assumed to lie along the equator.

With $\xi(\varphi) = \xi_0$ and $\nu(\varphi) = \nu_0$, it is straightforward to verify by inspection that equation (4.25) admits *approximate* solutions of the following separable form

$$\varphi \approx \frac{f(\theta)}{\ln(r/r_0)} \tag{4.38}$$

in the region $\ln(r/r_0) \gg 1$. The ordinary differential equation to be satisfied by $f(\theta)$ is given by

$$\frac{df}{d\theta}\frac{d}{d\theta}\left(\sin\theta\frac{df}{d\theta}\right) = \nu_0 f, \tag{4.39}$$

and we have neglected terms of order $[\ln(r/r_0)]^{-1}$ and higher in the radial dependence. In the polar region where θ is small, equation (4.39) is satisfied by either $f(\theta) \propto \theta^2$ or $f(\theta) \propto \ln(1/\theta)$. It is apparent that the former has to be discarded since it gives $\varphi \propto f(\theta) \to 0$ as $\theta \to 0$, so we shall concentrate on the latter (which corresponds to the class of asymptotes we discussed earlier). In the latter case, one can replace the singular function $f(\theta)$ with a finite auxiliary function $h(\theta) \equiv -(\sin\theta/\nu_0)(df/d\theta)$, which satisfies

$$h'' + h'\left(\frac{h'}{h} - \frac{\cos\theta}{\sin\theta}\right) + 1 = 0 \tag{4.40}$$

where the prime denotes differentiation with respect to θ. In addition, we have

$$f = \nu_0 \frac{h'h}{\sin\theta}. \tag{4.41}$$

The second-order ordinary differential equation (4.40) for $h(\theta)$ can be integrated by a fourth-order Runge-Kutta method, which is initiated from near $\theta = 0$ using the expansion

$$h = h_0 + \frac{\theta^2}{2}\ln(\theta_0/\theta) + \cdots \tag{4.42}$$

The two dimensionless constants h_0 and θ_0 are to be fixed by the boundary conditions, one on the equator, the other near the pole. The condition on the equator is simply that $h'h|_{\theta=\pi/2} = 0$. The condition near the pole is that q_0 defined below equation (4.33) equals 1.

It is instructive to note that in the asymptotic region (where θ is small and the expansion [4.42] is valid) we recover the asymptotic expansion (§ 4.3.1). According to equation (4.41), the function $f(\theta)$ in this region becomes

$$f(\theta) \approx h_0\nu_0 \ln(\theta_0/\theta) \tag{4.43}$$

to the leading order. From equation (4.43), we also have

$$\varphi \approx \frac{h_0\nu_0 \ln(\theta_0/\theta)}{\ln(r/r_0)}, \tag{4.44}$$

which is equivalent to the following expression for the cylindrical radius R in the region near the axis:

$$R \approx r\theta \approx (r_0\theta_0)\left(\frac{r}{r_0}\right)^{1-\varphi/(h_0\nu_0)}. \tag{4.45}$$

Comparing equation (4.45) with equation (4.33), we see that $q(\varphi) = \varphi/(h_0\nu_0)$, and the condition $q(1) = 1$ near the axis fixes the parameter $h_0 = 1/\nu_0$. The other free parameter in expansion (4.42) is θ_0, which is to be determined using the condition $h'h|_{\theta=\pi/2} = 0$. For illustrative purposes, in Figure 4.3*a* we plot $h(\theta)$ corresponding to $\nu_0 = 0.1$, 1, and 3. The value of ξ_0 does not come into the computation, as can be seen from equation (4.25). Note that $f(\theta = \pi/2) = 0$ is ensured by $h'(\theta = \pi/2) = 0$ in each case. The smaller ν_0 is, the less $h(\theta)$ changes with θ. In the limit that $\nu_0 \to 0$, $h(\theta)$ remains constant, which is the force-free limit. In the corresponding plots of $f(\theta)$ (Figure 4.3*b*), it is apparent that $f(\theta)$ varies gradually with θ except near $\theta = 0$, where the logarithmic dependence on θ dominates. In Figure 4.4, we plot the shape of the poloidal field lines for the case $h_0 = \nu_0 = 1$, on two different scales: $r = 10^3 r_0$ (Figure 4.4*a*) and $r = 10^5 r_0$ (Figure 4.4*b*). Although the field lines appear to be radial on each scale, their collimation towards the cylindrical axis is evident when these two figures are compared. Eventually, all field lines except the one that lies exactly on the equator will be collimated towards the rotation axis. The closer to the equator a field line is, the more gradually it will curve.

It is straightforward to verify that solutions in the sub-asymptotic regime described here also satisfy the conditions of nearly-radial flux surfaces and large ratios of kinetic energy to Poynting flux. We have

$$B_r = -\frac{\Psi_0 f'(\theta)}{r^2 \ln(r/r_0)\sin\theta} \quad \text{and} \quad B_\theta = -\frac{\Psi_0 \varphi}{r^2 \ln(r/r_0)\sin\theta}\,, \tag{4.46}$$

so that $B_\theta = \varphi/f'(\theta)B_r = -h'(\theta)/\ln(r/r_0)B_r$. Since $h'(\theta)$ has a finite maximum and equals zero at both $\theta = 0$ and $\pi/2$, as can be seen from Figure 4.3*a*, we conclude that B_θ is much smaller than B_r as long as $\ln(r/r_0)$ is large

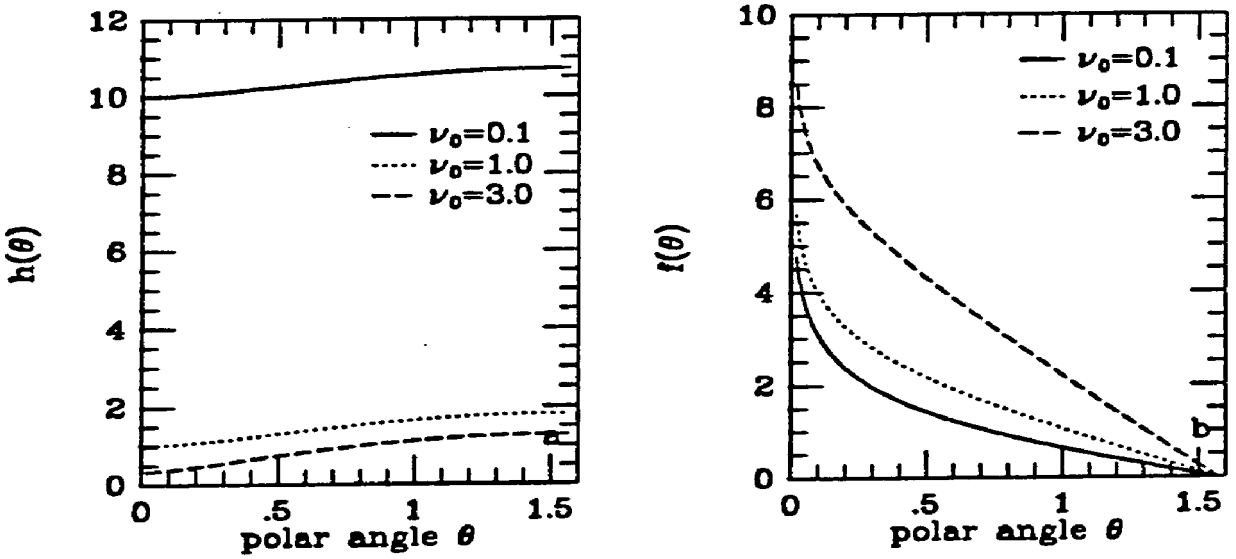


Fig. 4.3–Curves of the two functions of θ (equator at $\theta = \pi/2 = 1.57$) defined in the text, $h(\theta)$ (a) and $f(\theta)$ (b), are plotted for the case of constant ξ and ν. We have chosen $\nu_0 = 0.1$, 1.0, and 3.0.

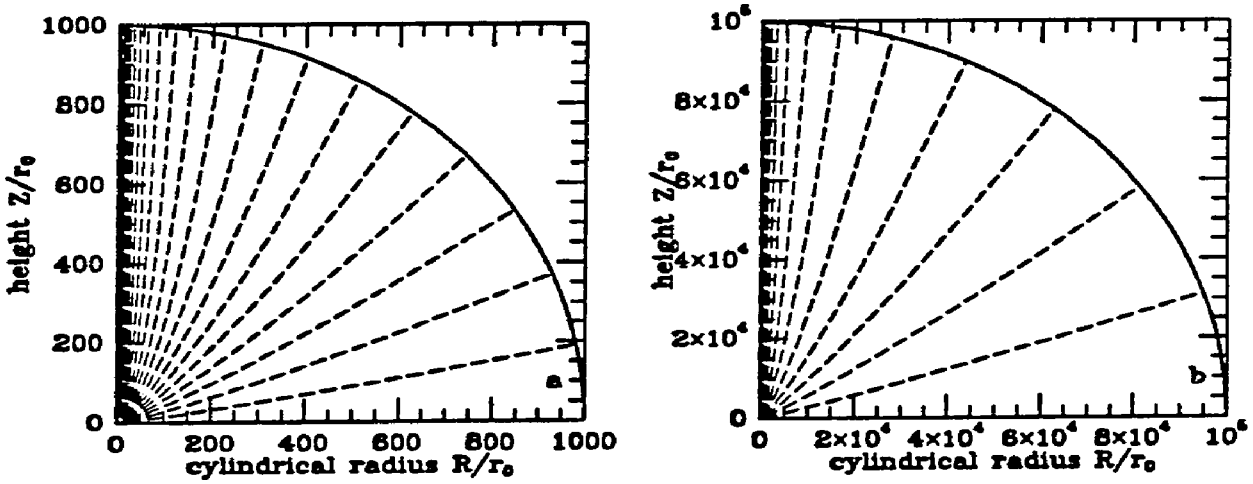


Fig. 4.4–The poloidal field lines for the case $\nu_0 = 1.0$ in Fig. 4.3 on two different scales: (a) on the scale of $10^3 r_0$, and (b) on the scale of $10^5 r_0$. In both panels, the value of the flux function is set to be 1 on the equator, and it decreases successively by steps of 0.05 counter-clockwise.

enough. Therefore the first condition is satisfied. The second condition is also satisfied because the Poynting flux, which is proportional to

$$a \propto B_p R^2 \approx B_r (r \sin\theta)^2 \approx -\frac{f'(\theta)\sin\theta}{\ln(r/r_0)} = \frac{\nu_0 h(\theta)}{\ln(r/r_0)} , \qquad (4.47)$$

goes to zero logarithmically with r since $h(\theta)$ is a finite, smooth function (see Figure 4.3*a*).

It should be stressed that the solutions and figures shown above are meant to be illustrative only, and that we do not expect constant ξ and ν in real relativistic MHD winds from astrophysical objects. These functions have to be determined self-consistently from conditions both near the base of the flow and at the critical surfaces. We do hope, however, that the solutions we obtain in this simplest case could be representative of the flow structure with more complex distributions of flux constants. These solutions may also provide analytical checks for numerical models of MHD winds well beyond the light cylinder that are now being developed (J. Hawley & C. Evans, private communication). Finally, we mention in passing that it is possible obtain solutions of separable form for power-law distributions of $\xi(\varphi)$ and $\nu(\varphi)$, although these solutions would not generally be able to fill the entire space.

4.4. Conclusions and Astrophysical Implications

4.4.1. Summary and Discussion

We have determined the conditions under which most of the Poynting flux in an initially magnetically-dominated wind can be converted into kinetic energy flux. In order for this to occur, the quantity $a = B_p R^2$ (which is

related to the rate of opening of a flux tube) has to decrease by a large factor in the region of the flow beyond the fast magnetosonic point. The fast point, which is at infinity in a cold radial wind, can be brought to a finite radius close to the light cylinder if the flux tube diverges even slightly faster than radially. We showed that the critical specific energy μ along a flux tube approximately equals the value of a at the fast point, while the bulk Lorentz factor of the plasma only attains a value of about $\mu^{1/3}$ within the same region. Thus, a relativistic ($\mu \gg 1$) MHD flow is still magnetically-dominated as it crosses the fast point, with a ratio of Poynting flux to kinetic energy flux of roughly $\mu : \mu^{1/3}$, not much different from that in a radial wind (Michel 1969). However, efficient conversion of Poynting flux to kinetic energy flux can occur outside the fast point through the so-called "magnetic nozzle" effect, in which the unbalanced magnetic pressure force yields further flow acceleration. This is in contrast to the radial wind case where the magnetic pressure and tension forces nearly cancel one another (see Fig. 4.1).

Having established that the efficient conversion of Poynting flux into kinetic energy is possible in principle, we studied whether it would occur in practice. The relativistic Grad-Shafranov equation, which describes cross-field force balance in magnetized winds, admits kinetic energy-dominated solutions with nearly-radial flux surfaces at large distances from the base of the flow. These solutions asymptotically become force-free in the cross-field direction, with all flux surfaces collimating gradually towards the rotation axis. We extended our earlier asymptotic analysis in chapter III and found force-free asymptotic solutions with or without external confinement for *any* well-behaved distribution of flux constants. In particular, we inferred that whether a kinetic energy-dominated flow becomes a jet (occupying a narrow

region of space) or a wind (filling the entire space) ultimately depends on the how fast the external pressure declines with distance, provided that no external toroidal magnetic field helps to confine the flow. Consideration of the initial value problem of wind formation suggests that, to the extent that ideal MHD theory is applicable, all of the flow current should be screened out by an induced current sheet near the flow boundary so that the external toroidal field is zero. However, a detailed analysis of the return current is beyond the scope of this thesis. If an external toroidal field exists, self-collimated flows are possible no matter how rapidly the external pressure declines with distance (Appl & Camenzind 1993).

We were able to show that there is a *sub-asymptotic* regime in which the centripetal force of the plasma flowing along a curved flux surface has to be balanced by the net electromagnetic force. The gradual merging of these non-force-free solutions onto their force-free asymptotes as r increases is demonstrated explicitly in the simplest case where the flux functions $\xi(\varphi)$ and $\nu(\varphi)$ are both held constant. An important piece of the analysis that remains to be done is the transition from a magnetically dominated flow to a kinetic energy-dominated flow, but we note that previous self-similar (Blandford & Payne 1982; Li, Chiueh, & Begelman 1992, see next chapter) and numerical (Camenzind 1987, 1989; Li 1993) models of magnetized winds have shown at least enough acceleration beyond the fast point to yield rough equipartition between Poynting flux and kinetic energy flux. Once these two fluxes become comparable, it seems probable that the flow would tend towards the kinetic energy-dominated asymptotes we have found. We therefore conjecture that *the kinetic energy-dominated asymptotes represent "natural"*

far-field structures of cold, relativistic MHD winds with finite total magnetic flux.

If our conjecture is correct, then why have the previous self-similar solutions yielded only a rough equipartition between kinetic energy flux and Poynting flux, i.e., $\zeta_\infty \sim O(1)$, rather than asymptotic conversion of all Poynting flux into kinetic energy? We suspect this result is an artifact of the assumption of self-similarity. The total magnetic flux in these solutions has to be infinite, whereas in our case the flux is finite. If an infinite supply of flux surfaces must fit into the available space, a is never able to decline by a very large factor. In contrast, numerical calculations of relativistic MHD winds confined by an external medium (Camenzind 1987, 1989; Li 1993) display a wide range of ζ, depending on the magnitude of the external pressure. The general trend seems to be that, for a flow with a given distribution of flux constants and no external toroidal field, the lower the external confining pressure is, the smaller the final ζ in the flow will become (Li 1993, see chapter VI). This is consistent with our kinetic energy-dominated solutions in which $\zeta \rightarrow 0$ where the external pressure goes to zero at infinity. Of course, in the existing numerical models the minimum value of ζ is limited by the size of the grid, since ζ declines only logarithmically with r. Nevertheless, in the case of a nonrelativistic free wind with no external confinement, Sakurai (1985) was able to show that all of the flow energy can end up in kinetic form. Although Sakurai's combined numerical and analytic analysis is nonrelativistic, it is reassuring to note that he considered the entire flow self-consistently, including the passage through the critical surfaces. Finally, we note that similar conclusions regarding the logarithmic collimation

of kinetically-dominated MHD flows have been reached by Eichler (1993) by considering "thin" force-free jets confined by an external medium.

It should be reiterated that although kinetic energy-dominated solutions with arbitrary distributions of flux constants can satisfy the Grad-Shafranov equation at large distances, the question of how flows become dominated by kinetic energy in the first place has not been addressed. Presumably, somewhere between the fast magnetosonic point and the sub-asymptotic regime considered in § 4.3, there must lie a magnetically-dominated region where flow acceleration occurs at a rate much faster than logarithmic. It is conceivable that ζ is quickly reduced to a value of order unity not too far beyond the light cylinder. A transition region where the kinetic and the electromagnetic energy fluxes are comparable follows, within which ζ is reduced even further. Then the kinetic energy-dominated region takes over, which continues to lower ζ logarithmically with distance.

4.4.2. Implications for AGN Jets

The existence of kinetic energy-dominated solutions has important implications for jets in AGNs. On the VLBI scale of the jet ($\gtrsim 1$ pc), ζ could be as low as $1/(\ln 10^5) \sim 0.1$. Since most of the flow energy is already in kinetic form, there is less need to invoke other dissipative mechanisms to tap the electromagnetic energy as in the case of, for example, a force-free jet (Lovelace, Wang, & Sulkanen 1987). Also, in our solutions, the current carried in the jet migrates gradually across the flux surfaces towards the outer boundary, as it accelerates the flow. By the time most of the magnetic energy is converted to kinetic energy, most of the current has already left the jet and returned to the base of the flow through either the outer part of the jet itself, or a

current sheet on the last flux surface. This is in contrast with a magnetically-dominated jet where the main current is closed only after the jet is loaded with external plasma (Benford 1984) or shocked (Goldreich & Julian 1969). More importantly, shocks formed in kinetically-dominated flows can be significantly different from shocks formed in either magnetically-dominated or even equipartition relativistic flows. Since the compression ratio behind the shock is much higher in the kinetically-dominated case, particle acceleration behind the shock could operate more efficiently and yield flatter synchrotron spectra, higher emissivities, etc. Furthermore, interactions between the jet and circumnuclear gas, e.g., entrainment of emission-line clouds, are likely also to be affected strongly by the relative magnitudes of electromagnetic and kinetic energy fluxes.

4.4.3. Implications for Crab Pulsar Wind and Related Objects

As mentioned in the introduction, the Crab pulsar wind must be dominated by kinetic energy flux before being shocked at $r_s \sim 0.1$ pc. Well beyond the light cylinder, it may be describable by our solutions with no external confinement. Since Ω is the same on all flux surfaces, equation (4.34) implies that a larger terminal 4-speed V_∞ near the equatorial plane than in the polar region should help to concentrate flux surfaces near the equator, where the bulk of the wind energy is believed to reside (e.g., Arons 1991). Just such a dependence of 4-speed on latitude is inferred by Begelman & Li (1992) from consideration of the wavelength-dependent morphology of the Crab synchrotron nebula. It is not clear, however, that a value of ζ (which is replaced by σ hereafter, as is conventional in the pulsar wind context) as low as the inferred value of 0.01 or less (Rees & Gunn 1974;

Kennel & Coroniti 1984) can be achieved in this way. From equation (4.19), the ratio of Poynting flux to kinetic energy flux upstream of the terminating shock (denoted by a subscript "s") is about $\sigma_s \sim (B_p R^2)_s/(B_p R^2)_f$. Since the portion of the wind powering the Crab does not appear to be highly collimated, we have $(B_p R^2)_s \sim \Psi_s/\pi$, where Ψ_s is the magnetic flux that carries most of the flow energy at the shock. We also have $(B_p R^2)_f \sim \Psi_c$, where Ψ_c is the total flux of open field lines. For the latter estimate we have assumed that the fast magnetosonic surface is not far from the light cylinder and that most of the open field lines cross the light cylinder within a height of $\sim R_L$ (Camenzind 1987). Therefore, we have a very rough estimate $\sigma_s \sim \Psi_s/\Psi_c$. Thus, to obtain $\sigma \ll 1$ in the Crab Nebula, plasma has to be injected preferentially onto only a small fraction of escaping field lines. For $\sigma = 10^{-2}$, this fraction would be about 1%, and it is not clear whether this is physically plausible. However, it does seem to be qualitatively consistent with the "outer gap" model for pair production in pulsar magnetospheres (Cheng, Ho, & Ruderman 1986), where the plasma is preferentially deposited and accelerated along flux surfaces close to the dividing surface between open and closed field lines. Alternatively, the loaded plasma could be an ion outflow concentrated near the equatorial plane of rotation (Arons 1991). This small fraction of the total flux surfaces would occupy most of the space by the time the free wind-terminating shock is reached, as indicated in Figure 4.4*b*.

In summary, we believe that a low-σ cold wind is possible in the Crab Nebula, as long as only a small fraction of the field lines in the equatorial zone are loaded with plasma. Once these escape to a large distance from the light cylinder they have a lot of space to inflate, allowing the "magnetic nozzle" mechanism to operate efficiently. Although the conditions necessary

to obtain $\sigma \lesssim 10^{-2}$ in the Crab pulsar wind might be extreme, a value of σ of order of 0.1 is certainly obtainable, in contrast to a cold radial wind.

CHAPTER V

COLD RELATIVISTIC MHD JETS: A FAMILY OF SELF-SIMILAR SOLUTIONS

So far, we have considered only flow structures in the far-field region well beyond the light cylinder radius. While these asymptotic (or semi-asymptotic) analyses are important in clarifying the ultimate flow collimation behaviors (chapter III) and in understanding some puzzling aspects of pulsar winds like that in the Crab Nebula (chapter IV), whether such asymptotic solutions be able to connect smoothly to the bases of the flows remains unclear. In this chapter, we shall construct self-consistent relativistic jet solutions all the way from their sources to the infinity, and demonstrate how MHD flows can achieve both highly relativistic speeds and a high degree of collimation. Due to the complex nature of the Grad-Shafranov equation which governs the flow structure (Lovelace et al. 1986), all previous investigations of relativistic MHD winds have involved some major simplifying assumptions, either by taking the force-free limit (Blandford 1976; Lovelace, Wang, & Sulkanen 1987; Sulkanen & Lovelace 1990), or by assuming that the magnetic fields are only sparsely populated with flow plasma (i.e., the Michel's magnetization parameter $\sigma \gg 1$, Camenzind 1987). Our approach here is no exception. Our main goals are to show that there indeed exist well-collimated flow solutions that go smoothly from the flow bases to infinity, with their flux surfaces asymptoting towards either cylinders or paraboloids, as predicted in chapter III, and that on these collimated flows flow acceleration is much more efficient than in radial winds. To make this complicated problem tractable, we again make the cold wind assumption and neglect

the gravity. It then turns out that the energy and the relativistic Grad-Shafranov equations admit a family of self-similar solutions. The meaning of self-similarity in terms of magnetic field structure will be evident shortly. The self-similar approach has been undertaken by Blandford & Payne (1982) for nonrelativistic flows. Being relativistic, the flow in our problem possesses a natural length scale, the light cylinder radius, which is absent in the non-relativistic theory. This turns out to impose a greater restriction on the class of self-similar flows than those found by Blandford & Payne. For one thing, the gravity of the central object cannot be included in our discussion, since it would introduce another length scale — the gravitational radius — into the problem. We should try to relax the self-similarity assumption under some rather restrictive circumstances in the next chapter.

This semi-analytic, self-similar approach enables us to follow flow acceleration of an electromagnetically-driven relativistic flow to much larger distances from the base of the flow than previously possible (Camenzind 1987). We shall demonstrate explicitly how the poleward deflection of the flux surfaces causes the fast magnetosonic point to move inward from infinity to a few light cylinder radii from the axis, in contrast to the radial wind case. While the fraction of the magnetic energy converted into plasma kinetic energy is still small at the fast point — not much different from that in the case of Michel's minimum torque solution — the "magnetic nozzle" effect discussed in the last chapter continues to convert magnetic energy into plasma kinetic energy beyond the fast point until the two are comparable. This eventual approximate equipartition in a relativistic magnetized wind was foreseen by Camenzind (1989) on the basis of his numerical models (Camenzind 1987).

5.1. Relativistic Self-Similar MHD Wind Equations

Let us now seek scale-free (or self-similar) solutions to the energy equation (2.9) and GS equation (2.10) under the assumptions that $M_* = 0$ and $\xi = 1$. Because of these simplifying assumptions, processes in the wind production region near the base of the flow, where gravity and thermal pressure may be important, cannot be adequately addressed.

In a self-similar flow, dimensional analysis demands that all dimensionless quantities should vary the same way along all flux surfaces. Consequently, the dimensionless quantities defined in the last chapter (§ 4.1), namely $\mu = e/c^2$, $\lambda = l\Omega/c^2$, $x = R/R_L$, $y = Z/R_L$, $\mathbf{u} = \mathbf{V}/c$, $\mathbf{b} = \mathbf{B}/(4\pi kc)$ and from equation (4.3), a, τ and η, are all independent of the poloidal flux function Ψ. In particular, the fact that the radius and the height of each flux surface change with the dimensionless quantities x and y according to

$$(R, Z) = \frac{c}{\Omega(\Psi)}(x, y)\,, \tag{5.1}$$

implies the following geometric meaning of this self-similarity. As shown in Figure 5.1, all poloidal field lines at the points intersected by any straight line emerging from the origin have the same slope. Self-similarity also demands that $\Omega(\Psi)$ scale as R^{-1}, so that the linear velocity is the same at the base of each flux tube. This condition, which can be very restrictive indeed, follows from the fully relativistic nature of our treatment (in which speed scales with the speed of light c and is independent of Ψ), and does not occur in the nonrelativistic model of Blandford & Payne (1982). Furthermore, the two

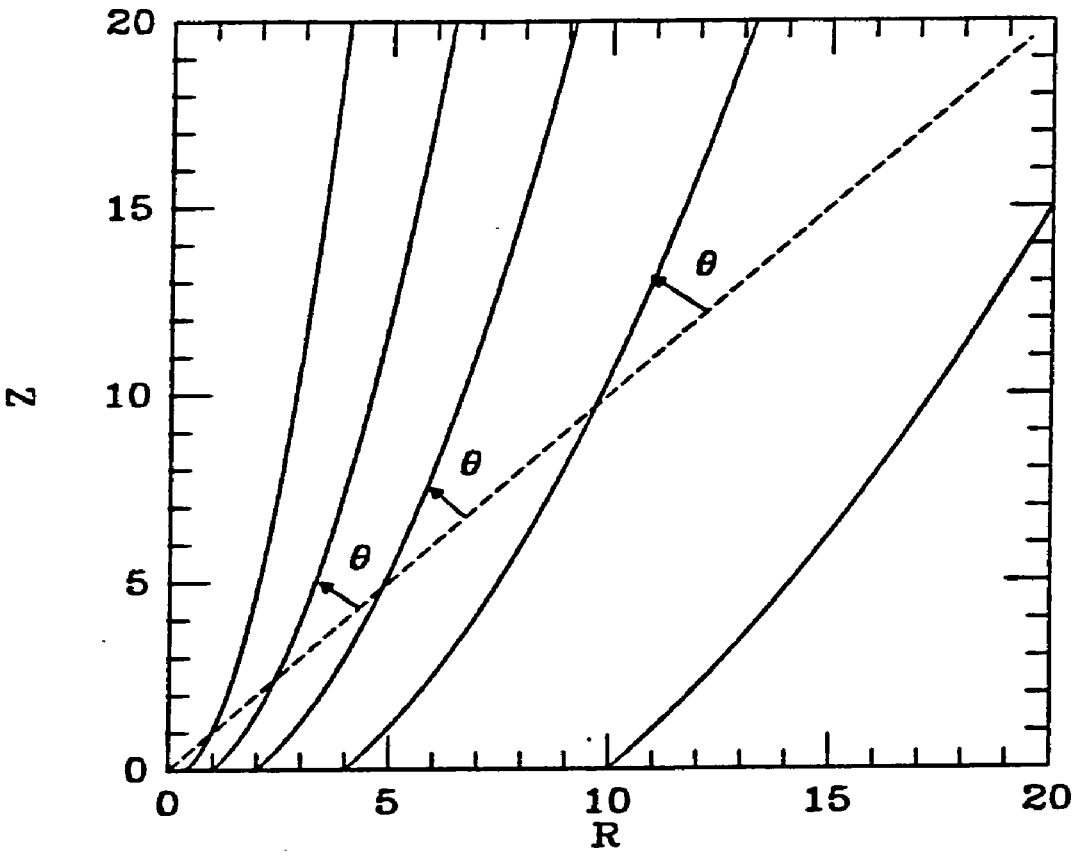


Fig. 5.1–A configuration of self-similar poloidal field lines (solid). The dashed line can be any straight line drawn from the origin, and the angles between the solid and dashed lines are all equal.

flux constants $\Omega(\Psi)$ and $k(\Psi)$ cannot be independent of each other. Their combined dimensionless quantity

$$\varphi \equiv \frac{[\Omega(\Psi)]^2 \Psi}{4\pi k(\Psi) c^3} \equiv \sigma \tag{5.2}$$

has to be independent of Ψ too. In the radial wind literature, φ is equivalent to σ, the so-called "magnetization parameter" whose magnitude characterizes how relativistic a flow can become (Michel 1969; Goldreich & Julian 1970).

The energy equation (4.2), in which all quantities are in dimensionless form, should have no flux function Ψ dependence and should be the same on each flux surface. This is true for any arbitrary distribution of the angular speed $\Omega(\Psi)$. However, to obtain the dimensionless GS equation, which has also to be independent of Ψ to be self-consistent, a power-law distribution of $\Omega(\Psi)$ is required. This requirement can be checked easily by inspection. Therefore, we let

$$\Omega = \Omega_* \chi^{-\beta}. \tag{5.3}$$

where Ω_* and β are constants and $\chi \equiv \Psi/\Psi_*$,. The other flux constants would scale with Ψ as

$$e(\Psi) = e_*, \quad l(\Psi) = l_* \chi^{\beta}, \quad k(\Psi) = k_* \chi^{1-2\beta}. \tag{5.4}$$

The quantities with a subscript "$*$" are evaluated along some arbitrary reference flux surface labelled by Ψ_*.

Before we cast the GS equation into a dimensionless form, we shall first express the curvature of the poloidal field line κ and the dimensionless poloidal magnetic field strength b_p in terms of the height y of the poloidal

field line and its first and second derivatives, $\dot{y}$ and $\ddot{y}$, with respect to the radius x. From pure geometric considerations, in a self-similar wind, the dimensionless combination

$$\kappa R = \frac{x\ddot{y}}{(1+\dot{y}^2)^{3/2}} \frac{\dot{y}}{|\dot{y}|}, \tag{5.5}$$

and the two components of $\mathbf{b}_p$, b_R and b_Z, are

$$b_R = -\frac{\sigma}{x(x\dot{y}-y)} \frac{d(\ln \Psi)}{d(\ln \Omega)}, \qquad b_R = -\frac{\sigma\dot{y}}{x(x\dot{y}-y)} \frac{d(\ln \Psi)}{d(\ln \Omega)} \tag{5.6a,b}$$

which are combined to yield

$$b_p = \left| \frac{\sigma}{\beta x^2} \frac{\sqrt{1+\dot{y}^{-2}}}{1-y/(x\dot{y})} \right|, \quad \text{or,} \quad a \equiv b_p x^2 = \left| \frac{\sigma}{\beta} \frac{\sqrt{1+\dot{y}^{-2}}}{1-y/(x\dot{y})} \right|. \tag{5.7a,b}$$

Note that if the flow becomes radial asymptotically, then one should have $\dot{y} \to y/x \to const.$ as $x \to \infty$, which leads to $b_p x^2 \to \infty$. In this case, as we have shown in chapter III, the energy equation cannot be satisfied, which means that radial flow (i.e., flow on cones) is incompatible with the assumption of self-similarity.

We can now nondimensionalize the GS equation (2.10) in the limits that $\xi = 1$ and $M_* = 0$. By factoring out the common Ψ dependence ($\propto \Psi^{2-5\beta}$) in each term, equation (2.10) can be reduced to

$$\frac{\kappa R}{x} b_p^2 [1-(1+\tau)x^2] + \frac{(1-\beta)p|b_p|}{\sigma x} - \frac{x+y\dot{y}}{2x^2\dot{y}\sqrt{1+\dot{y}^{-2}}(x\dot{y}-y)} \frac{dp}{dx}$$

$$= \frac{1}{x\sqrt{1+\dot{y}^{-2}}} \left(\frac{u_\phi^2}{\tau x^2} + b_p^2 \right) \tag{5.8}$$

where we have defined an effective magnetic pressure

$$p \equiv b_\phi^2 x^2 + b_p^2 x^2 (1-x^2)$$

which, according to results in the next chapter (eq. [6.1]), will be related to τ simply

$$p = \frac{1 - \delta/x^2}{\tau^2} \tag{5.9}$$

and δ is a constant defined as $1 - (\mu - \lambda)^2$. Therefore, the derivative of p (with respect to x) can be translated easily to the derivative of τ, which is given by equation (4.8). Since the quantity da/dx (remember $a = b_p x^2$) in equation (4.8) and κ in equation (5.8) are all dependent on $\ddot{y}$ (via eqs. [5.7a] and [5.5] respectively), one can rewrite eqs. (4.8) and (5.8) as two coupled first order equations for $d\tau/dx$ and $d\dot{y}/dx$ as follows

$$A_1 \frac{d\tau}{dx} + B_1 \frac{d\dot{y}}{dx} = C_1, \tag{5.10}$$

and

$$A_2 \frac{d\tau}{dx} + B_2 \frac{d\dot{y}}{dx} = C_2 \tag{5.11}$$

where

$$A_1 = x^4\{b_\phi^2 + b_p^2[1 - (1+\tau)x^2]\},\ B_1 = -\frac{\tau a^2(x + y\dot{y})[1 - (1+\tau)x^2]}{(1 - \dot{y}^2)(x\dot{y} - y)},$$

$$C_1 = [1 - (1+\tau)x^2]\left[\frac{\lambda^2(1-\eta)^2}{\tau x^3} - \frac{a^2\tau}{x}\right] - 2\lambda\eta x[\lambda\eta(1+\tau) - \mu],$$

and

$$A_2 = \frac{(1 - \delta/x^2)(x + y\dot{y})}{\tau^3 x^2 \dot{y}\sqrt{1 + \dot{y}^{-2}}(x\dot{y} - y)}, \qquad B_2 = \frac{[1 - (1+\tau)x^2]b_p^2}{\dot{y}^3(1 + \dot{y}^{-2})^{3/2}},$$

$$C_2 = \frac{u_\phi^2/(\tau x^2) + b_p^2}{x\sqrt{1 + \dot{y}^{-2}}} - \frac{(1-\beta)(1 - \delta/x^2)|b_p|}{\sigma x \tau^2} + \frac{\delta(x + y\dot{y})/(\tau^2 x^5 \dot{y})}{\sqrt{1 + \dot{y}^{-2}}(x\dot{y} - y)}.$$

Therefore, one has

$$\frac{d\tau}{dx} = \frac{C_1 B_2 - C_2 B_1}{A_1 B_2 - A_2 B_1} \tag{5.12}$$

and

$$\ddot{y} = \frac{A_1 C_2 - A_2 C_1}{A_1 B_2 - A_2 B_1}. \tag{5.13}$$

The critical point occurs where the denominator in eqs. (5.12) and (5.13) vanishes, i.e.,

$$D \equiv A_1 B_2 - A_2 B_1 = 0 \tag{5.14}$$

where

$$D = \frac{[1-(1+\tau)x^2]a^2}{\dot{y}^3(1+\dot{y}^{-2})^{3/2}} \left\{ b_\phi^2 + b_p^2[1-(1+\tau)x^2] + \left(\frac{x+y\dot{y}}{x\dot{y}-y}\right)[b_\phi^2 + b_p^2(1-x^2)] \right\}.$$

It is evident that the critical point occurs when either $1-(1+\tau)x^2$ or the expression inside the big curly bracket equals zero. The former condition simply states that the Alfvén point, where $1-(1+\tau)x^2 = 0$ is satisfied automatically, is a critical point, whereas the latter condition is more subtle to interpret. To see more clearly the meaning of the latter condition, we rearrange it into the following form

$$\begin{aligned} M_f'^{\,2} &\equiv \frac{\tau b_p^2 x^2}{[b_\phi^2 + b_p^2(1-x^2)]\{1+[(x+y\dot{y})/(x\dot{y}-y)]^2\}} \\ &= \frac{M_f^2}{1+[(x+y\dot{y})/(x\dot{y}-y)]^2} = 1, \end{aligned} \tag{5.15}$$

where M_f is the fast magnetosonic Mach number defined in equation (4.7) and M_f' is a modified fast magnetosonic Mach number. When we realize that $\sin\theta = [|x\dot{y}-y|/\sqrt{(x^2+y^2)(1+\dot{y}^2)}]$, where θ is the angle between the tangent of the field line at point (x,y) and the spherical radius passing through that point (see Fig. 5.1), then the physical interpretation for the second critical point becomes clear: it is the point where the modified fast magnetosonic Mach number $M_f' = M_f/\sin\theta$ equals unity. The peculiar angular modification of the critical point condition arises from the assumption

of self-similarity, which imposes the constraint that all flow properties along any radial cone should be similar. A wave propagating in the flow must have the same phase at all points on the cone, and therefore can only propagate in the polar direction. In this direction, the Alfvén wave has a wave speed $V_A \sin\theta$ and the fast magnetosonic wave has a speed V_f. Since a critical point is physically a location at which the flow speed matches the wave speed in the direction of propagation, the relevant component of the flow speed to compare is therefore $V_p \sin\theta$. It immediately follows that the critical points should occur at the the Alfvén point $M_A = 1$ and the modified fast point $M_f' = 1$. These arguments have been given by Blandford & Payne (1982), who first recognized the importance of the latter point.

There are two ways to tackle equations (5.10) and (5.11) mathematically. One can either eliminate the derivative $(d\tau/dx)$ in favor of $\ddot{y}$ and solve for a second order ordinary differential equation (eq. [5.13]) coupled with a quartic algebraic energy equation (4.2) for τ, as we have done in Li, Chiueh, & Begelman (1992), or eliminate $\ddot{y}$ in both equations and obtain two coupled first-order ordinary differential equations for τ and the shape of the poloidal field y with respect to x

$$\frac{d\tau}{dx} = f(\tau, \dot{y}, y, x) \tag{5.16}$$

$$\frac{dy}{dx} = \dot{y}(\tau, y, x) \tag{5.17}$$

where f is given by equation (5.12) and $\dot{y}$ is related to other quantities by following quadratic equation obtained from the expression for a (eq. [5.7b])

$$\left[\left(\frac{\beta a}{\sigma}\right)^2 - 1\right]\dot{y}^2 - 2\frac{y}{x}\left(\frac{\beta a}{\sigma}\right)^2 \dot{y} + \left[\frac{y^2}{x^2}\left(\frac{\beta a}{\sigma}\right)^2 - 1\right] = 0, \tag{5.18}$$

where a in turn is given by the energy equation (4.2) so that

$$a^2 = \frac{1}{\tau^2}\left[(\mu - \lambda\eta)^2 - \frac{\lambda^2(1-\eta)^2}{x^2} - 1\right]. \tag{5.19}$$

In this way, we avoid the consideration of the fast magnetosonic point which is always passed without notice in our solutions.

Let us look more closely at the Alfvén point which we claimed to be a critical point (from the presence of the factor $1-(1+\tau)x^2$ in the expression for the denominator D of equation [5.12]). However, since both B_1 and B_2 in the numerator of equation (5.12) also have the same factor $1-(1+\tau)x^2$, they cancel out each other. Therefore, the Alfvén point is not really a critical point to equation (5.12). But the Alfvén point does impose constraints on possible flow solutions. These constraints come from the requirement that at the Alfvén point $x = x_A = \sqrt{\mu/\lambda}$, τ has to take on some special value $\tau = \tau_A = 1/x_A^2 - 1 = (\mu-\lambda)/\lambda$. As we show in the numerical section § 5.4., this condition will ultimately fix the slope of the poloidal field lines as they leave the equatorial plane.

Before we integrate the coupled equations (5.16) and (5.17) numerically, we shall first analyze the behaviors of the flow solutions in both the near-field region close to the base of the flow (§ 5.2) and the far-field (i.e., asymptotic) region (§ 5.3).

5.2. Near-Field Expansion

Near the base of the flow, the magnetic field lines are rigid and nearly force-free. It can be shown that it is impossible for all field lines to emerge from the origin, to maintain cross-field balance and still to be self-similar. Therefore, we shall focus on situations where the field lines on a given flux

surface are anchored to a thin equatorial disk ($y = 0$) at some finite radius R_0 ($= x_0 c/\Omega[\Psi]$, where subscript "0" denotes quantities evaluated at base of the flow). If field lines leave the equatorial plane $y = y_0 = 0$ at $x = x_0$ with some finite slope $\dot{y}_0$, then the initial value for a would be $a_0 = \sigma\sqrt{1+\dot{y}_0^2}/(\beta\dot{y}_0)$. Since we have neglected gravity, there is no force to compete with the centrifugal force. Therefore, as long as the field lines have a component directing outward (i.e., $\dot{y}_0 > 0$), parcels of plasma can always be accelerated centrifugally, regardless of the angle at which the field line is anchored to the disk (in contrast to the case considered by Blandford & Payne [1982]).

With a_0 thus computed, one needs only to specify the initial injection speed u_0 to completely determine the flow behavior near the base of the flow (since $\tau_0 = u_0/a_0$). The total specify energy and angular momentum of the flow are related by equation (4.5). Once τ_0, $\dot{y}_0$, y_0 and x_0 are known, one can in principle integrate the equations (5.16) and (5.17) outwards for any given value of μ (and thus λ). This can be done easily numerically if the injection speed $u_0 \neq 0$. The case that needs some special attention is when $u_0 = 0$, where $(d\tau/dx) \propto (x - x_0)^{-1/2} \rightarrow \infty$ as $x \rightarrow x_0$. Physically, this scaling is a result of the plasma being accelerated from rest by a finite force. Mathematically, in the limit $x \rightarrow x_0$ and $y \rightarrow 0$, the expression (5.12) for $(d\tau/dx)$ yields

$$\frac{d\tau}{dx} = \frac{\tau}{2(x - x_0)}, \text{or, } \tau = c_0(x - x_0)^{1/2} \tag{5.20}$$

to the leading order and c_0 is a constant coefficient. We have used the relation $\mu = \lambda + \sqrt{1 - x_0^2}$ (and thus $\delta = x_0^2$) when $u_0 = 0$ in the deriving the above

equation. By expanding the energy equation (4.2) around $x = x_0$, one can show that the constant is

$$c_0 = \sqrt{\frac{2x_0}{\eta_0^2\lambda^2 x_0^2 + a_0^2(1-x_0^2)}} \tag{5.21}$$

where a_0 has been given above, and η_0 is

$$\eta_0 = \frac{1-\mu x_0^2/\lambda}{1-x_0^2}.$$

The expansions near $x = x_0$ (eqs. [20] and [21]) can be used to initiate the integration of equations (5.16) and (5.17). Numerically, we shoot simultaneously from the base of the flow $x = x_0$ outward and the Alfvén point $x = x_A$ inward, and match the two solutions at $x = x_A/2$. This process will fix the initial slope of poloidal field line $\dot{y}_0$ at $x = x_0$, as well as the distance of the Alfvén point away from the equatorial plane y_A. This way, for any given total specific energy μ (and thus λ), one can completely determine the flow solution up to the Alfvén point which satisfies the initial flow conditions and passes through the Alfvén point smoothly. How such solutions behave at large distances is the subject of the next section.

5.3. Far-Field Behaviors of Flow Solutions

According to chapter III, where we studied the asymptotic structures of relativistic MHD winds, flux surfaces in the flows are to be collimated to either "cylinders" (i.e., $R \to$ constant, as $Z \to \infty$), or "paraboloids" (i.e., $R/Z \to 0$ and $R \to \infty$, as $Z \to \infty$), or "cones" (i.e., $R/Z \to$ constant, as $Z \to \infty$). As discussed in § 5.1, conical flux surfaces are excluded in a self-similar wind. To see more clearly how solutions behave at large distances,

one can rewrite equations (5.16) and (5.17) as (replacing x by $\ln y$ as the independent variable)

$$\frac{d\tau}{d\ln y} = \frac{s_1\bar{b}_2\bar{c}_1 - \bar{b}_1\bar{c}_2}{\bar{d}}, \tag{5.22}$$

$$\frac{d\ln x}{d\ln y} = s_2, \tag{5.23}$$

where we have defined $s_1 \equiv x/(y\dot{y})$ and $s_2 \equiv y/(x\dot{y})$, and

$$\bar{b}_1 \equiv -\frac{\tau(1+s_1)(1+\dot{y}^{-2})^{1/2}}{1-s_2}, \qquad \bar{b}_2 \equiv 1,$$

$$\bar{c}_1 \equiv [(1+\tau) - x^{-2}]\left[a^2\tau - \frac{\lambda^2(1-\eta)^2}{\tau x^2}\right] - 2\lambda\eta[\lambda\eta(1+\tau) - \mu],$$

$$\bar{c}_2 \equiv \frac{1}{(1+\dot{y}^{-2})^{1/2}}\left[\frac{\lambda^2(1-\eta)^2}{\tau x^2} + \frac{a^2}{x^2}\right] - \frac{(1-\beta)(1-\delta/x^2)|a|}{\sigma\tau^2} + \frac{s_2(1+s_1)\delta}{\tau^2 x^2(1+\dot{y}^{-2})^{1/2}(1-s_2)},$$

$$\bar{d} \equiv \frac{p(1+x^2/y^2)(1+\dot{y}^{-2})}{(1-s_2)^2} - \frac{\tau a^2 x^2}{y^2}.$$

These barred quantities are all slowly-varying quantities approaching finite constants in the limits that $\dot{y} \to \infty$ as $y \to \infty$ (which is valid for all possible asymptotes), and that the radius $x \to x_\infty$ where x_∞ can be either a finite constant (cylinder) or infinity (paraboloid). In these limits, we always have $s_1 = x/(y\dot{y}) \to 0$.

Let us first consider the case where solutions approach cylinders. Two conditions must be satisfied, namely, $(d\tau/d\ln y) \to 0$ and $(d\ln x/d\ln y) \to 0$ as $y \to \infty$. The latter condition simply means that $s_1 \to 0$ according to equation (5.23), and the former condition implies, according to equation (2.22), that $\bar{c}_2$ has to vanish as $y \to \infty$, or equivalently

$$\left[\frac{\lambda^2(1-\eta_\infty)^2}{\tau_\infty} + a_\infty^2\right]\frac{1}{x_\infty^2} - \frac{(1-\beta)(1-\delta/x_\infty^2)|a_\infty|}{\sigma\tau_\infty^2} = 0, \tag{5.24}$$

where quantities with subscript "∞" are evaluated on the limiting radius x_∞. Clearly, the value of x_∞ depends critically on the exponent β, and for β close to unity, one has $x_\infty \gg 1$ and thus

$$x_\infty^2 \cong \frac{[\mu/(1+\tau_\infty) - \lambda]^2 \tau_\infty \beta + \sigma^2 \tau_\infty^2/\beta}{1-\beta}. \tag{5.35}$$

Neglecting terms of order x_∞^{-2} and higher, τ_∞ is given by the following quartic equation (from the energy equation [4.2])

$$\mu^2 \tau_\infty^2 = (1+\tau_\infty)^2 (1 + a_\infty^2 \tau_\infty^2), \tag{5.36}$$

where $a_\infty = \sigma/\beta$. If the terminal flow speed is highly relativistic so that $u_\infty = \tau_\infty a_\infty \gg 1$, then equation (5.26) is reduced to $\tau_\infty \cong \mu/a_\infty - 1$, or $u_\infty = \mu - a_\infty$. The exact value of x_∞ (and τ_∞) is obtained by solving for the equation $\bar{c}_2 = 0$ and the energy equation (4.2) numerically. The result is presented in Figure 5.2, where x_∞ as functions of β are plotted for three different values of σ ($\sigma = 5.0, 10.0$, and 50.0) with $x_0 = u_0 = 0.1$, under the assumption that $\mu = 2\sigma$. As we shall see later, this relation is approximately true for our "critical" cylindrical solutions provided $\sigma \geq 1$. To put the result in Figure 5.2 in physical perspective, we note that x_∞ increases monotonically with β. Presumably, this is because, in flows with smaller β, the magnetic field strength in the outer flow region is relatively stronger (compared to the case with larger β) according to the scaling $B \propto R^{1/\beta - 2}$, and thus would result in better flow collimation, i.e., smaller limiting radius.

One can expand equations (5.22) and (5.23) near x_∞ to obtain approximate far-field solutions. This is done in Appendix A, where the field line approaches the limiting radius x_∞ as a power-law, $y \propto (x_\infty - x)^{-\iota}$, where the exponent ι can be computed straightforwardly.

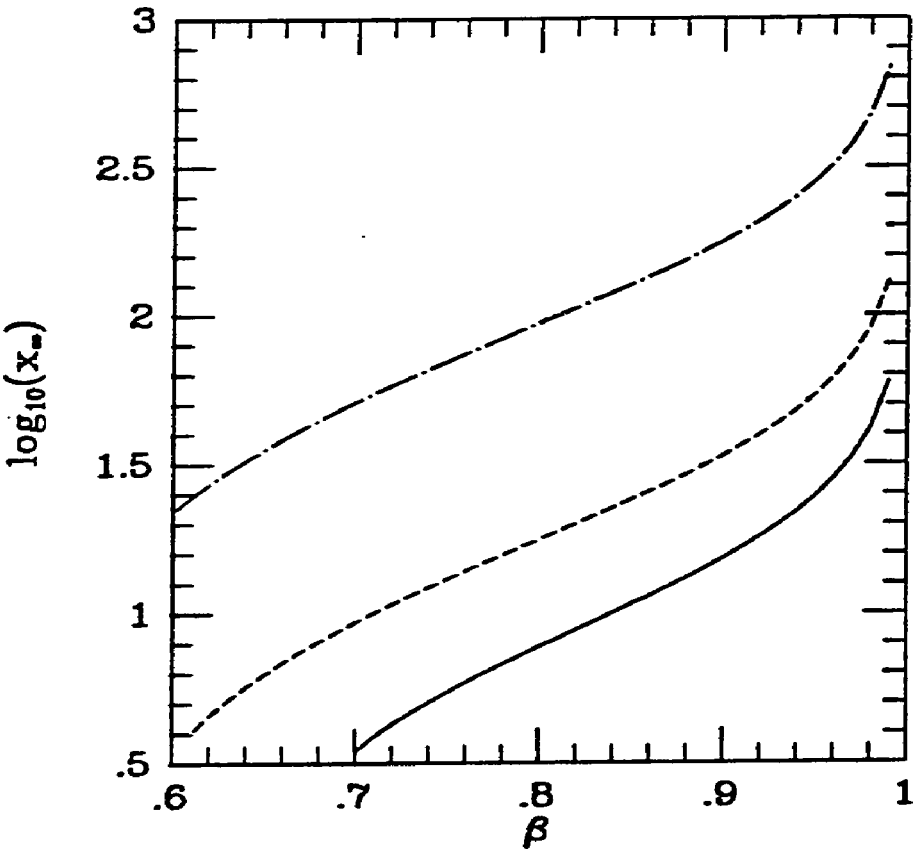


Fig. 5.2–The limiting cylinder radius x_∞ is plotted as a function of the parameter β for $\sigma = 5$ (solid), 10 (dashed), and 50 (chain-dotted). We assumed $\mu = 2\sigma$ and $x_0 = 0.1$.

As is apparent from both equation (5.24) and Figure 5.2, the value of x_∞ goes to infinity as $\beta \to 1$. Therefore, we conclude that self-similar paraboloidal solutions exist only when $\beta = 1$. As we demonstrate explicitly below, these paraboloidal solutions enclose a finite amount of electric current, and thus belong to the "Type I" paraboloids discussed in chapter III. Furthermore, one can show that all of the current I is concentrated asymptotically on the rotation axis. This is because the total current $I_\infty(\Psi)$ enclosed by an arbitrary flux surface Ψ asymptotically, $I_\infty \propto (B_\phi R)_\infty$, becomes independent of Ψ when $\beta = 1$. Asymptotically current-carrying paraboloidal flux surfaces were also found by Blandford & Payne (1982) in their far-field analysis of a nonrelativistic self-similar wind. The fast critical points for their solutions are always located at infinity. However, because of relativistic effects and the fact that our velocity scaling law ($\Omega[\Psi] \propto \Psi^{-1}$) differs from the one they used, the above statement is not necessarily true in our case. A numerical example to illustrate this point is given in § 5.4.2.

Finally, we mention in passing that we shall not consider flows with $\beta > 1$, which cannot extend asymptotically either to cylinders of finite radius, or paraboloids defined earlier. Furthermore, these solutions contain unphysical infinite current on their axis, since the current enclosed within a radius R asymptotically scales with R as $R^{(1-\beta)/\beta}$.

5.4. Numerical Examples of Self-Similar Solutions

5.4.1. Cylindrical Solutions When $\beta < 1$

Given σ, β (< 1) and a set of initial conditions at $x = x_0$, there is always an unique limiting radius x_∞ for each μ greater than unity, according to § 5.3. The solution started from the base of the flow with the

same μ, however, is not necessarily able to approach this limiting radius x_∞ asymptotically. In fact, it turns out that only one "critical" solution, with an unique specific total energy μ_c, can actually asymptote to the limiting cylinder. In this subsection, we shall determine numerically this "critical" energy μ_c and the associated "critical" cylindrical solution, and find out how solutions would behave if their specific total energy μ is larger or smaller than μ_c.

As discussed briefly in § 5.2, a flow solution with given σ, β, μ and initial conditions is completely determined by matching the two solutions from the base of the flow and from the Alfvén point. At the Alfvén point, $x = x_A = \sqrt{\lambda/\mu}$, and $\tau = \tau_A = (\mu - \lambda)/\lambda$ (see § 5.1), one can obtain easily the quantity a_A, which in turn yields a relation between $\dot{y}_A$ and y_A according to equation (5.7*b*). Therefore, the flow condition at the Alfvén point is completely determined once the height y_A is specified (§ 5.2). On the other hand, the flow condition at the base of the flow is determined once the slope of the poloidal field line $\dot{y}_0$ is specified. One can integrate the two governing equations (5.16) and (5.17) simultaneously from x_0 outwards and x_A inwards, and by demanding the two solutions have the same τ and y at some radius between x_0 and x_A (which is taken to be $x = x_A/2$), one can thus fix both y_A and $\dot{y}_0$.

Once smooth solution from the base of the flow to the Alfvén point is found by iteration, there is no difficulty in shooting outward from the Alfvén point, passing automatically through both the light cylinder and the conventional relativistic fast magnetosonic point $M_f = 1$. The fast point is located, in general, at only a few light cylinder radii from the rotation axis when σ (the relativistic parameter) is comparable to or larger than unity. In

practice, due to the rapid increase of the flow distance from the equatorial plane y with respect to its cylindrical radius x, we change the independent variable from x to $\ln y$, and switch to integrating equations (5.22) and (5.23) as soon as the Alfvén point is passed.

We find that the solutions at large distances from the base of the flow generally do not yield physically meaningful asymptotes. They fall into two categories, and solutions in both categories eventually bend back towards the rotation axis as the flow travels away from the equatorial plane, presumably because the outward-directed inertial force becomes too weak to counteract the unbalanced magnetic pinch force. One category of solutions has small values of μ, and the flows accelerate rapidly as they get pinched, until the modified fast magnetosonic point $M_f' = 1$ is reached (§ 5.1). The other category of solutions has larger μ's, and the flows decelerate as they become pinched; this continues until the flows stop completely. Nonetheless, there exists a "critical" value, μ_c, at the boundary of the two types of solutions, for which the outward inertial force just balances the electromagnetic force, in such a way that the flow expands outward until it approaches a limiting radius.

We illustrate the above general discussions with a "typical" example, where $\sigma = 10.0$, $\beta = 0.95$, and $x_0 = u_0 = 0.1$ are assumed. The numerical results are presented in Figures 5.3*a–d*. In the plots *a–c*, different curves correspond to different choices for the total specific energy: $\mu = 21.5$, 21.8, 21.84293077, 22.0, and 22.5, where the "critical" solution with $\mu_c = 21.84293077$ is plotted in solid line in each plot. In Figure 5.3*a*, we have plotted the shapes of the poloidal field line, where $\log_{10}(1+x)$ and $\log_{10}(1+x)$ are used as coordinates to show more clearly the flow behaviors

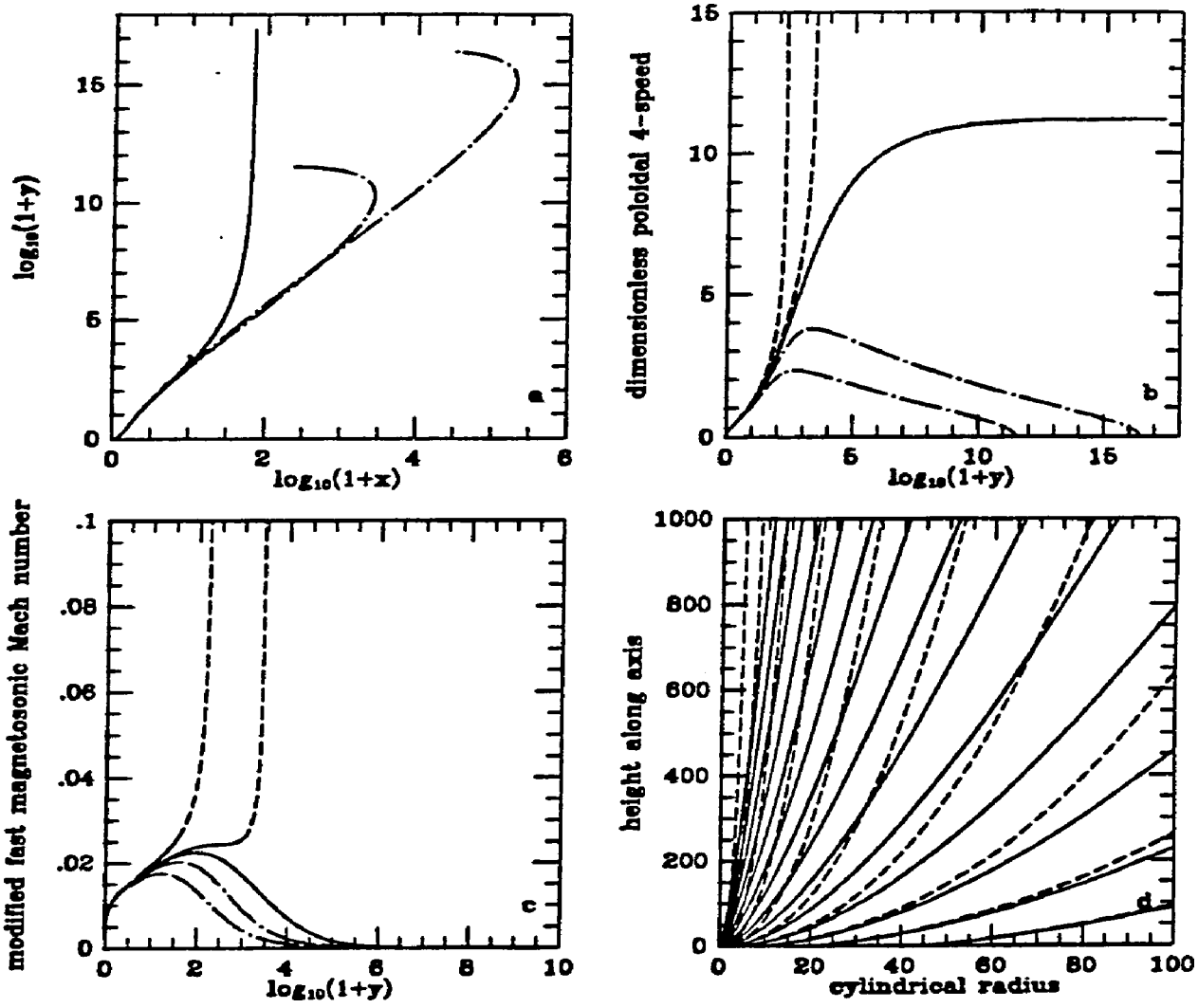


Fig. 5.3–For five different total energies: $\mu = 21.84293077$ (critical, solid), 21.5, 21.8 (dashed), and 22.0, 22.5 (chain-dotted), we plot the shapes of the poloidal field lines (a), the poloidal flow speeds (b) and the modified fast magnetosonic Mach numbers (c). In panel (d) we plot lines of the poloidal current density (dashed) together with the poloidal field lines for the critical solution. The crossing of the two sets of lines is evident.

both near the base and in the far-field region. Although not particularly evident from Fig. 5.3*a*, the field lines with $\mu < \mu_c$ do bend towards the rotation axis, just like those with $\mu > \mu_c$. In general, solutions with μ closer to μ_c start to bend towards the rotation axis at larger distances from the flow base than those with μ further away from μ_c. The fact that the two categories of solutions have distinctively different behaviors can be seen more clearly from the flow acceleration curves in Figure 5.3*b*, where the dimensionless poloidal 4-speed as functions of $\log_{10}(1+y)$ has been plotted for each μ. While the flow speed decreases gradually to zero as $\log_{10}(1+y)$ increases for solutions with $\mu > \mu_c$, those solutions with $\mu < \mu_c$ get accelerated rapidly before they are terminated. When we plot the modified fast magnetosonic Mach number M_f' (Figure 5.4*c*), we find that solutions with $\mu < \mu_c$ approach the modified fast point rapidly. Therefore, they are terminated because the modified fast point is encountered. On the other hand, solutions with $\mu > \mu_c$ are terminated presumably because their poloidal speeds vanish. In Figure 5.3*d*, we plot the lines of the poloidal current density for the critical solution (dashed) together with the poloidal field lines (solid). It is the crossing of the current density that yields the flow acceleration along the field lines.

An interesting feature of the "critical" cylindrical solution is that its acceleration is quite gradual (in fact, logarithmic) with its length y. In the above "typical" solution, flow acceleration continues up to a characteristic length of order 10^7 times the light cylinder radius (Fig. 5.3*b*). This is also the length where the radius x becomes very close to the limiting radius $x_\infty = 67.36$, and we find that field line indeed approaches x_∞ with a power-law $y \propto (x_\infty - x)^{-\iota}$ (see Appendix A), with $\iota = 5.39$. The fast magnetosonic point, which occurs at $x_f = 3.12$ and $y_f = 78.69$, is indeed located not far

from the light cylinder radius. The specific kinetic (i.e., the flow plasma Lorentz factor) at the fast point $\gamma_f = 2.838$ is very close to $\mu_c^{1/3} = 2.795$, which is also true in the case of a radial wind (Michel 1969). The flow terminal Lorentz factor, however, is much larger, with $\gamma_\infty = 11.24$, which is slightly more than half of the total specific energy of $\mu_c = 21.84$. Therefore, most of the flow acceleration happens beyond the fast magnetosonic point, presumably through the so-called "magnetic nozzle" effect (see discussion in chapter IV, the paragraph above eq. [4.19]) where the flux tube diverges more rapidly than radial. The eventual near equipartition of energy between the flow plasma and the electromagnetic field found in this particular example holds also for other cylindrical solutions, as long as the flow is relativistic ($\sigma > 1$) and the limiting radius is much larger than the light cylinder radius ($x_\infty \gg 1$, i.e., β is close to 1). We shall demonstrate this trend explicitly in § 5.4.3, where some general flow properties are considered.

To contrast the self-similar critical solution with the critical solution of a radial wind, we solve for τ from the energy equation (4.2), a quartic equation in τ. Plotted in Figure 5.4 are the real roots of the energy equation for the "critical" solution discussed above (solid lines), and for a radial wind (dashed lines, from Michel [1969]) with the same $\sigma = 10$. The physical roots in both cases are marked. Near the origin, each case has only two real roots, one of which has $\tau < 0$ and is therefore unphysical. Two additional roots appear at some point inside the Alfvén radius. One of the new roots intersects the physical root at the Alfvén point (labelled A). A second intersection between one of the new roots and the physical root occurs at the fast magnetosonic point, which is located at $x_f = 3.12$ (labelled F) for the

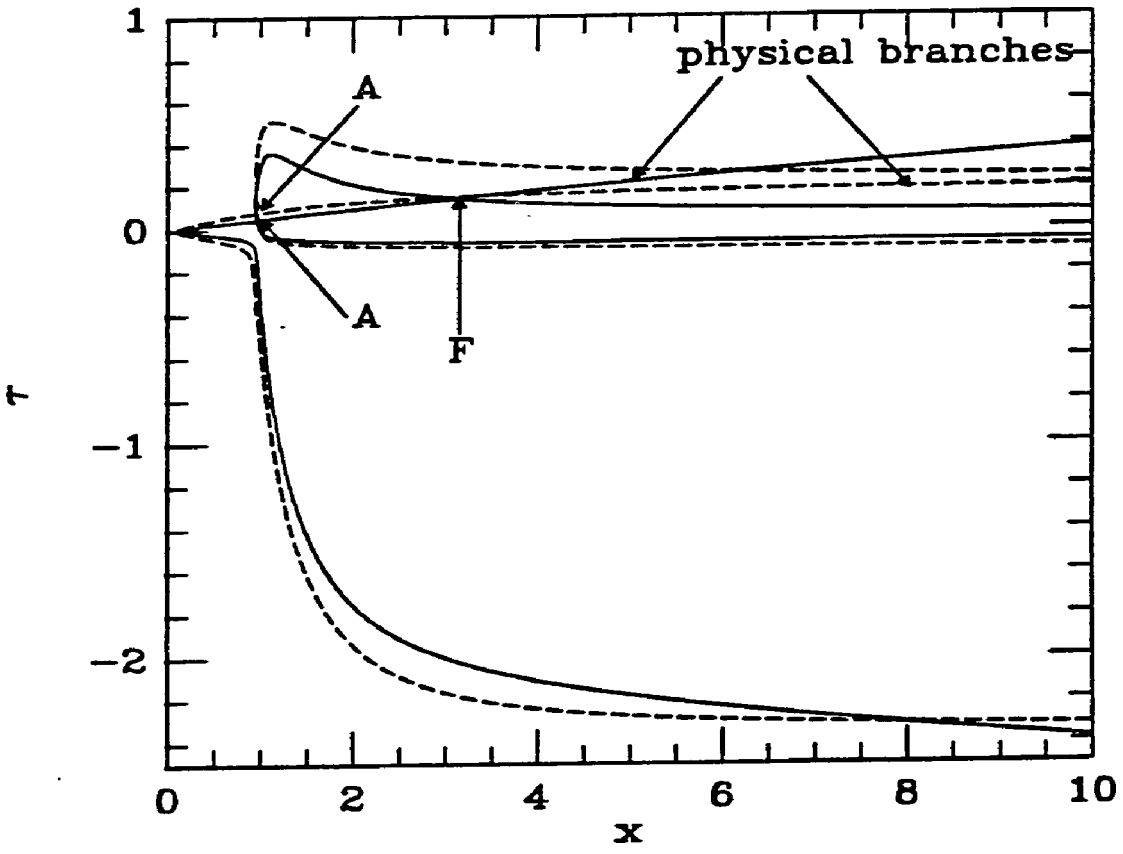


Fig. 5.4–Solution topologies of the energy equation (4.2) for the critical solution in Fig. 5.3 (solid) and a radial wind (dashed) with the same $\sigma = 10$. The Alfvén (A) and fast (F) points correspond to intersections between the physical branch of the solution of the equation and other solution branches.

self-similar case but which lies at infinity for the radial case (Goldreich & Julian 1970).

All solutions with $\mu \neq \mu_c$ have a turning point at some finite distance y_t and cylindrical radius x_t, where the transversely expanding poloidal field line starts to pinch back towards the rotation axis. This type of flow reconvergence has also be noted in the nonrelativistic self-similar flows (Blandford & Payne 1982), and maybe be a common feature of magnetized winds (Pelletier & Pudritz 1992). Flow reconvergence, when it occurs at very large distance from the base of the flow as in our case, may help to explain the formation of the bright radio knots observed in many extragalactic radio sources. The two types of completely different solution behaviors after the flow reconverges (as shown in Figs. 5.3*b* and *c*) may be understood in terms of the results present in chapter IV. According to equation (5.7*b*), the bending of the poloidal field line towards the rotation axis always reduces the quantity "a" and, from equations (4.19) and (4.20), supermagnetosonic (with $\mu < \mu_c$) and submagnetosonic (with $\mu > \mu_c$) solutions respond to the reduction in "a" quite differently: supermagnetosonic solutions increase their fast magnetosonic Mach number rapidly according to $M_f^2 \approx \mu^3/a_\infty$ until the modified fast point is encountered, whereas M_f (and thus u_p) of those submagnetosonic solutions in the latter case decreases with the quantity "a" and the kinetic energy in the flow is sucked back by the electromagnetic field until the flow stops. Finally, we note that we have not been able to find any solution with $\mu < \mu_c$ that passes smoothly through the modified fast magnetosonic point. Even if such smooth trans-modified fast magnetosonic solution exists, it is unlikely that such solution can approach any limiting cylinder of finite radius and thus may not have well-behaved asymptote.

5.4.2. Paraboloidal Solutions When $\beta = 1$

Paraboloidal flow solutions with $\beta = 1$ have basically similar solution topologies as the "typical" solutions discussed above. In particular, when σ and the initial condition $u_p = u_0$ at $x = x_0$ are specified, we find that there exists one "critical" value for μ, μ_c, below which solutions ($\mu < \mu_c$) accelerate rapidly as the flows reconverge. These solutions can not extend all the way to infinity, as before. The solutions with $\mu > \mu_c$, on the other hand, do extend to infinity, but always have terminal speed less than that of the critical solution with μ_c. In this sense, the critical solution of paraboloidal flows with $\beta = 1$ is analogous to the Michel's minimum-torque solution (which has the largest terminal speed of $\sigma^{1/3}$) but, of course, the final flow kinetic energy in the present case (which is close to σ) is much higher.

We plot in Figure 5.5 the properties of an illustrative example with $\sigma = 10$, $x_0 = u_0 = 0.1$, and of course, $\beta = 1$. Shown in Fig. 5.5*a* are the poloidal field lines with different values of μ: 21.9100, 21.9150, 21.9154, 21.9159, 22.0, and 23.0. Again, $\log_{10}(1+x)$ and $\log_{10}(1+y)$ are plotted. The approximate (due to finite maximum distance $y = 10^{21}$ used numerically) critical energy is $\mu_c = 21.9159$, and solutions with energy less than this bend back towards the rotation axis, whereas those with larger energy can extend to infinity. The corresponding dimensionless 4-speeds are plotted in Fig. 5.5*b*. It is clear that the speed of the critical solution is the highest among solutions that can extend to infinity.

Finally, we confirm our earlier statement that the fast magnetosonic point of the paraboloidal solution is located at a finite distance, $x_f = 3.14$, which is distinctively different from the family of nonrelativistic paraboloidal solutions found by Blandford & Payne (1982).

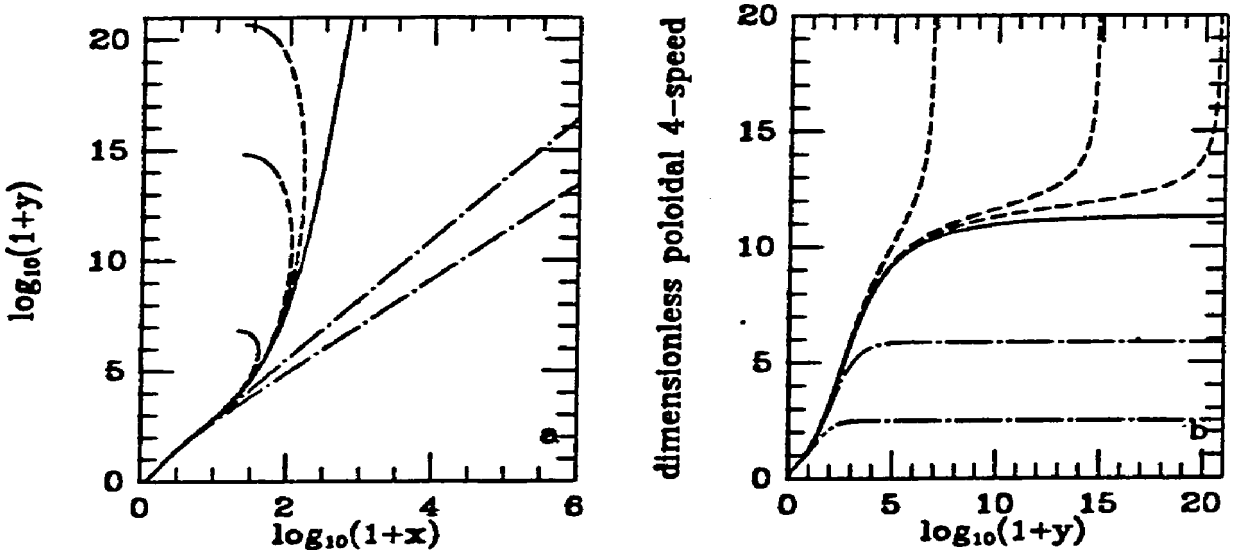


Fig. 5.5–In the case $\beta = 1$, for six different total energies: $\mu =$ 21.9159 (critical, solid), 21.91, 21.915, 21.9154 (dashed), and 22.5, 23.0 (chain-dotted), we plot the shapes of the poloidal field lines (a), and the modified fast magnetosonic Mach numbers (b). The solution topologies are quite similar to those shown in Fig. 5.3 where $\beta = 0.95$.

5.4.3. Energy Partition in the Flow

The value of the critical energy μ_c is determined by the value of β, the initial conditions x_0 and u_0, and more importantly, σ. It turns out that μ_c is insensitive to x_0 and u_0 as long as they are small compared to unity, i.e., provided the linear speed of the rotating magnetic field and the poloidal injection speed at the base of the flow be much less than the speed of light. Table 5.1 shows how μ_c, the terminal flow Lorentz factor γ_∞ (which is also the final specific kinetic energy of the flow), and the limiting cylinder radius x_∞ depend on σ and β, for $x_0 = u_0 = 0.1$. Several features are clear from Table 5.1. First, we notice that μ_c is quite insensitive to the value of β too; therefore the approximate relation $u_\infty \cong \mu_c - \sigma/\beta$ (which is valid when $x_\infty >> 1$, i.e., β close to 1, see § 5.3) implies that the larger β is, the greater acceleration there will be. Second, most of the μ_c's in Table 5.1 are about twice as large as their corresponding σ's, except for those of small σ (the column $\sigma = 1.0$ in Table 5.1). Thus, we deduce from the foregoing approximation for u_∞ that, if β is close to unity, the kinetic energy approaches about half of the total energy, or equivalently, that the kinetic energy flux will come nearly into equipartition with the Poynting flux, and both of which are about equal to σ.

5.5. Discussion

We have explored a class of relativistic, hydromagnetic jet flows which exhibit self-similar scaling in radius. We find, in agreement with Blandford & Payne's earlier work on self-similar nonrelativistic winds, that all such flows collimate along the rotation axis. It turns out that the asymptotic

Table 5.1

The dependences of the critical energy μ_c and the terminal flow Lorentz factor γ_∞ on parameters σ and β for $x_0 = 0.1$

flow variables	β	σ			
		1	5	20	50
μ_c	0.95	3.58	11.80	41.93	102.01
	0.90	3.55	11.75	41.85	101.93
	0.80	3.51	11.68	41.74	101.76
	0.70	3.50	11.63	41.68	101.69
γ_∞	0.95	2.42	6.47	20.85	49.37
	0.90	2.32	6.12	19.60	46.36
	0.80	2.10	5.32	16.70	39.24
	0.70	1.86	4.33	13.04	30.23

behavior of the flow solutions is controlled by the scaling of the rotation law, $\Omega \propto \Psi^{-\beta}$, or equivalently, the distribution of the magnetic field with radius, $\Psi \propto R^{1/\beta}$. The flux surfaces approach cylinders asymptotically when $\beta < 1$, and become paraboloidal when $\beta = 1$. We are not able to find solutions with $\beta > 1$ that can extend smoothly to infinity. Our flows can attain highly relativistic speeds, with most of the acceleration occurring beyond the fast magnetosonic point through the "magnetic nozzle" effect.

The direct applicability of our solutions to realistic astrophysical systems is limited by the restrictions necessary to obtain a self-similar solution. The assumption of self-similarity demands that no length scale be present in the problem. Since a special relativistic treatment imposes a velocity scale, the speed of light c, we require that all components of the velocity be the same at corresponding points of all flux surfaces intersected by a straight line from the origin. This implies that the angular velocity at the base of the flow must scale according to $\Omega \propto R^{-1}$. Thus, we are unable to treat a relativistic wind emanating from a Keplerian disk (as was treated by Blandford & Payne in the nonrelativistic limit), for example. Such a wind is certainly possible in principle, as we show in the next chapter where the self-similarity assumption is removed. We are also unable to include the effects of gravity and pressure within the self-similar framework.

Nevertheless, our solutions capture several important aspects of relativistic MHD winds which are likely to apply to non-self-similar flows as well. In particular, they exhibit the tendency towards asymptotic collimation and much more efficient flow acceleration than in a radial wind. Approximate equipartition between the kinetic energy flux and the Poynting flux of the flow is obtained in our solutions. Flow collimation was also found,

for example, by Sakurai (1985, 1987), who constructed numerical models of nonrelativistic winds, assuming a split monopole field geometry at the flow base. The most thorough investigations so far on relativistic winds driven by rotating magnetic fields were carried out by Camenzind (1986, 1987). In his model calculations, the magnetosphere is assumed to be sparsely populated with plasma. Not only is the Alfvén radius located at the light cylinder, but also the plasma current was negligibly small so that all field lines cross the light cylinders horizontally; the flow begins to collimate only beyond the light cylinders. The finite plasma current is included in our treatment, as is evident from the fact that field lines cross the light cylinder with a finite slope. Therefore, collimation may occur even well inside the light cylinder. A related difference between Camenzind's calculations and ours is that he assumed that the fast point lies far beyond the light cylinder in order to allow for a simple outer boundary conditions, whereas the fast point in our self-similar flows is located typically only a few light-cylinder radii from the rotational axis. We feel that it is more realistic for the fast point to be located not too far from the light cylinder, since it takes only slight deviation from perfect radial field geometry to bring the fast point close to the light cylinder radius (chapter IV). Common to both calculations is the fact that, at the fast point, the flow attains a Lorentz factor of order $\mu^{1/3}$ and most of the energy flux is stored in the form of Poynting flux. It is only through the so-called "magnetic nozzle effect" beyond the fast point that the Poynting flux continues to be converted into kinetic energy flux. This property is much more pronounced in relativistic flows than in nonrelativistic flows, as also discussed in chapter IV.

It is worth noting that neglecting the inertia of the wind plasma yields a family of force-free solutions which are confined within the light cylinder (Sulkanen & Lovelace 1990). These solutions are not simply the highly magnetized wind solutions in the high σ (i.e., low mass-loading rate) limit. The inclusion of the plasma inertia, however small it is, would remove the singularities at the light cylinder that prevent the force-free solutions from crossing. In fact, the important processes of flow acceleration, especially those occurring beyond the light cylinder, are not captured at all in force-free solutions.

To contrast our relativistic self-similar solutions with those found by Blandford & Payne (1982), we first note that the scalings of flow variables at the base are different in the two cases. It is therefore not surprising that we do not necessarily recover all of their results, even in the limit of low σ (nonrelativistic). In particular, the family of asymptotically paraboloidal solutions they found always have $M_f = 1$ at infinity, whereas a similar family of solutions (with $\beta = 1$) in our case do not have such a property in general. Furthermore, an additional electric field (due to special-relativistic effects) contributes to the cross-field force balance in our case. This new piece of physics permits us to have asymptotically cylindrical solutions, which are absent from their paper.

Finally, we estimate the energy output per unit time, $L(\Psi)$, inside some flux surface Ψ, making use of the fact that the total specific energy μ scales approximately with σ. This rough scaling is also true for a limited family of non-self-similar solutions to be discussed later in chapter VI. We then have the following relation

$$L(\Psi) = \int_0^\Psi k(\Psi)\mu c^2 d\Psi \approx 2\int_0^\Psi k(\Psi)\sigma c^2 d\Psi \qquad (5.37)$$

where $k(\Psi)$ is the mass flux per unit magnetic flux, which has the scaling $\Psi^{1-2\beta}$. Since μ (and σ) is independent of Ψ and $\beta < 1$, the above integral is dominated by the outer flux surfaces, except in the limit of $\beta \to 1$ where all flux surfaces contribute equally to the power output. Making use of the definition of σ (cf. eq. [5.2]), we have $L(\Psi) \approx \Psi^2\Omega^2/[4\pi(1-\beta)c]$, that is, the total power output enclosed within a flux surface is proportional to the square of the product of the angular velocity on that surface and the magnetic flux it encloses.

To summarize, our study of self-similar, relativistic winds has yielded the following main results:

(i) We demonstrate the existence of complete current-carrying, magnetized jet solutions. These results support of the conclusions of our previous work on the collimation of relativistic MHD winds (chapter III) which dealt only with asymptotically far-field solutions.

(ii) For these collimated relativistic jets, the conversion of electromagnetic energy into plasma kinetic energy is much more efficient than for radial winds. We find that the final kinetic energy flux is comparable to the Poynting flux in general.

CHAPTER VI

COLD RELATIVISTIC MHD JETS: A FAMILY OF NON-SELF-SIMILAR SOLUTIONS

We have extended Blandford & Payne's self-similar solutions to relativistic flows in the last chapter, where we show that it is possible to collimate a cold relativistic MHD wind while yielding substantially more flow acceleration than the case of a pure radial wind (Michel 1969). In fact, we find that the kinetic energy flux is comparable to the Poynting flux in these flows. However, the underlying assumption of self-similarity and the associated requirements on the distributions of the magnetic field strength and the mass loading rate, etc., in both models severely limit their applicabilities to realistic jets. It is thus the goal of the present paper to relax the assumption of self-similarity and to construct non-self-similar relativistic solutions to the Grad-Shafranov equation under somewhat more flexible conditions. To achieve this objective, we start from a given flux surface with prescribed flow properties close to the rotation axis and construct, from the GS equation, one flux surface after another so that they satisfy the cross-field force balance with this starting surface, and with each other. The flow properties, such as the flow speed and the magnetic field strength, are evaluated simultaneously. We assume the last flux surface to be a contact discontinuity between the jet flow and its ambient medium, where the pressure distribution on both sides should be the same. The importance of an ambient medium in aiding the confinement of astrophysical jets has been argued recently by Eichler (1993). As in chapter IV, we assume that all of the electric current in the flow is screened by a thin current sheet at the outmost periphery of the flow, so

that the toroidal magnetic field in the ambient medium is zero. Although it is likely that the jet/ambient medium interface might be subject to Kelvin-Helmholtz instabilities, we ignore this possibility here. In this way, we can investigate how the pressure distribution of the external medium is related to the acceleration and collimation of a magnetized jet. Since the environments of extragalactic radio jets in their production regions are not well determined at the present time, we feel that this approach is appropriate.

We organize the content of the chapter in the following order: in § 6.1, we show, from the basic MHD equations, how the flow properties along any given flux surface are controlled by the relativistically modified magnetic pressure along that flux surface, and how the variation of this modified pressure from one flux surface to another is governed by the Grad-Shafranov equation. We then lay out, in § 6.2, a flux-surface construction scheme which can, in principle, integrate the cross-field equation outward from an assumed initial flux surface near the rotation axis. In § 6.3, we first show examples of "typical" solutions, and then explore the dependence of the jet properties on various contributing factors, including the conditions on the starting flux surface, the distributions of the flux surface rotation rate and mass loading rate, and the external pressure. And finally, we summarize our main results and discuss their limitations in § 6.4.

6.1. Flow Dependence on the Modified Magnetic Pressure

To avoid excessive complications, we shall neglect both gravity and the internal pressure of the flow (the so-called "cold wind" assumption) in our discussion. These assumptions are partially justified by the consideration that, for highly relativistic jets from AGNs, the flow Alfvén speed should be

very close to the speed of light. For jets emanating from a region not too close (say, a few Schwarzschild radii) to the central massive black hole that is presumably present, both the escape speed and flow sound speed should be no more than a small fraction of the speed of light. We therefore simplify the conditions at the base of the flow by injecting the flow out of the equatorial plane with a speed given by some small fraction of c. In the case of jets associated with young stellar objects (Lada 1985), since the jet speeds are usually comparable to the escape speeds of the central objects, gravity is expected to play a greater role and thus has to be included.

The Grad-Shafranov (GS) equation, eq. (3.6), which describes the balance of forces in the cross-field direction, can be rearranged into

$$\frac{\partial P}{\partial \Psi} = \frac{1}{R^2}\left(\frac{2P}{B_p} - \frac{B_p}{4\pi} - \frac{\rho u_\phi^2}{B_p}\right)\hat{\mathbf{c}}\cdot\hat{\mathbf{e}}_R - \frac{(\kappa R)B_p}{4\pi R^2}\left(1 - \frac{4\pi k^2}{\rho} - \frac{R^2\Omega^2}{c^2}\right), \quad (6.1)$$

where κ is the curvature of the poloidal field line, $\hat{\mathbf{c}} = \mathbf{B}_p/B_p \times \hat{\mathbf{e}}_\phi$ is the cross-field unit vector pointing toward the rotation axis, while $\hat{\mathbf{e}}_R$ is the unit vector in the cylindrical radius R direction. The GS equation, written in the form of equation (6.1), expresses the variation of the "modified magnetic pressure", P, defined as

$$P \equiv \frac{B_p^2}{8\pi}\left(1 - \frac{R^2\Omega^2}{c^2}\right) + \frac{B_\phi^2}{8\pi}, \quad (6.2)$$

across the flux surfaces Ψ. The term $R^2\Omega^2/c^2$, which represents the deviation from the conventional definition of the magnetic pressure, arises from the special relativistic effect (i.e., the electric field), and vanishes in the non-relativistic limit. Before concentrating on solving the GS equation, we shall first examine how flow properties along a given flux surface are determined, if the distribution of P on that surface is known.

Along a flux surface with given shape $y = y(x)$ and given flux constants Ω and k, we can non-dimensionalize the "modified magnetic pressure" as

$$p \equiv \frac{P}{2\pi k^2 c^2} = [b_p^2(1-x^2) + b_\phi^2]. \tag{6.3}$$

With a prescribed form of $p(s)$ (where s measures the length of the poloidal field line from the flow base) for the pressure distribution, we can combine equation (6.3) with the energy equation (4.2) to yield an expression for the quantity η at each location (x, y) along that flux surface. This is done by eliminating b_p from both equations. After some lengthy manipulations, we arrive at a quadratic equation for η, which can be arranged into

$$\mathrm{A}\eta^2 + \mathrm{B}\eta + \mathrm{C} = 0, \tag{6.4}$$

with

$$\mathrm{A} = p(s)(1-x^2) + \frac{x^2-\chi}{1-x^2},\ \mathrm{B} = -2p(s)\left(1 - \frac{\mu x^2}{\lambda}\right),\ \mathrm{C} = \frac{p(s)(1-\mu x^2/\lambda)^2}{1-x^2}$$

where $\chi \equiv 1 - \delta^2$ is defined with $\delta \equiv \mu - \lambda$. Only one root of the above quadratic equation is physical, and that is

$$\eta = \frac{1 - \mu x^2/\lambda}{(1-x^2) - \sqrt{(x^2-\chi)/p(s)}} \tag{6.5a}$$

which yields immediately (from the definition of η) that

$$\tau = \sqrt{(1-\chi/x^2)/[p(s)x^2]}. \tag{6.5b}$$

The other root of equation (6.4), which is unphysical, is discarded.

It is apparent that the expression for η (eq. [6.5a]) has a critical point at the Alfvén point $x = x_A \equiv \sqrt{\lambda/\mu}$, where the numerator vanishes. At this

point, the denominator has to vanish as well if the toroidal magnetic field strength B_ϕ (and thus η) does not vanish here. This condition is equivalent to $M_A^2 = \tau x^2/(1-x^2) = 1$. Given the value for the modified pressure p at the Alfvén point (x_A, y_A), this requirement translates into a relation between the so-called critical energy μ_c and critical angular momentum λ_c, in order to ensure that the flow pass the Alfvén point smoothly. To determine μ_c and λ_c separately, another relation between them is needed. This is supplied by the conditions at the base of the flow. If we assume that the flux surface under consideration starts from the equatorial plane ($Z = 0$) at a distance $R_0 (\equiv x_0 R_L)$ away from the origin with an injection poloidal speed u_0, then, according to the energy equation (4.2), μ_c and λ_c have also to satisfy the following equation

$$(\mu_c - \lambda_c \eta_0)^2 = 1 + u_0^2 + \lambda_c^2 (1-\eta_0)^2/x_0^2, \tag{6.6}$$

where, from the definition of η, $\eta_0 = (1 - \mu x_0^2/\lambda)/[1-(1+\tau_0)x_0^2]$, and τ_0 is related to the modified pressure at the base of the flow p_0 by equation (6.3*b*)

$$\tau_0^2 = \frac{1-(\mu-\lambda)^2}{p_0 x_0^2}. \tag{6.7}$$

In particular, if the flow starts from rest ($\tau_0 = u_0 = 0$), then one has the simple relation $\mu_c = \lambda_c + \sqrt{1-x_0^2}$.

With μ_c and λ_c thus determined, the flow properties along a given flux surface with a prescribed pressure distribution $p(s)$ are then completely fixed. To illustrate how the pressure distribution along a given flux surface controls its flow properties, we now consider the case with $u_0 = x_0 = 0$, and prescribe a simple form of $p(x)x^2 = (x_b^2 + fx^2)/(x_b^2 + x^2)$ for the "modified magnetic pressure", with one free parameter f denoting the ratio of $p(x)x^2$

at $x = \infty$ to $p(x)x^2$ at $x = 0$ and the other free parameter x_b characterizing the radius around which most of the change in $p(x)x^2$ occurs. With these simplifications, we arrive at a cubic equation for the critical angular momentum λ_c, which reads

$$(1-f)\lambda_c^3 + 2f(1+\lambda_c)\lambda_c^2 + (2x_b^2 - f)(1+\lambda_c)^2\lambda_c - x_b^2(1+\lambda_c)^3 = 0. \quad (6.8)$$

The critical energy is given by $\mu_c = \lambda_c + 1$. Plotted in Figure 6.1 are the flow acceleration curves for the cases with $f = 0.5$, 0.1 and $x_b = 1.0$, 5.0. The Alfvén points and the fast magnetosonic points (where the fast Mach number $M_f^2 = \tau b_p^2 x^2/[b_\phi^2 + (1-x^2)b_p^2] = 1$) are labelled by the letters "A" and "F" on the plots, respectively. It is clear from the figure that x_b, which characterizes the the location around which $p(x)x^2$ decreases most, also determines where most of the flow acceleration occurs. We note in particular that the fast magnetosonic point is not a critical point when the distribution of magnetic pressure is prescribed and no regularity condition is required there in order to fix the critical energy μ_c and angular momentum λ_c (and thus the critical solutions). This is in contrast to the case where the poloidal magnetic field strength is prescribed. In that case, it is the fast magnetosonic point which controls the flow behavior instead of the Alfvén point (Begelman & Li 1993, see chapter IV).

Also apparent from Fig. 6.1 is the trend that the lower f is, the more the flow gets accelerated. In fact, the two solutions with $f = 0.5$ remain submagnetosonic throughout. This trend is not surprising, however, since $p(x)x^2$ is just proportional to the magnetic energy at large distances $x \gg 1$. Therefore, to yield large flow acceleration, it is necessary for the "modified

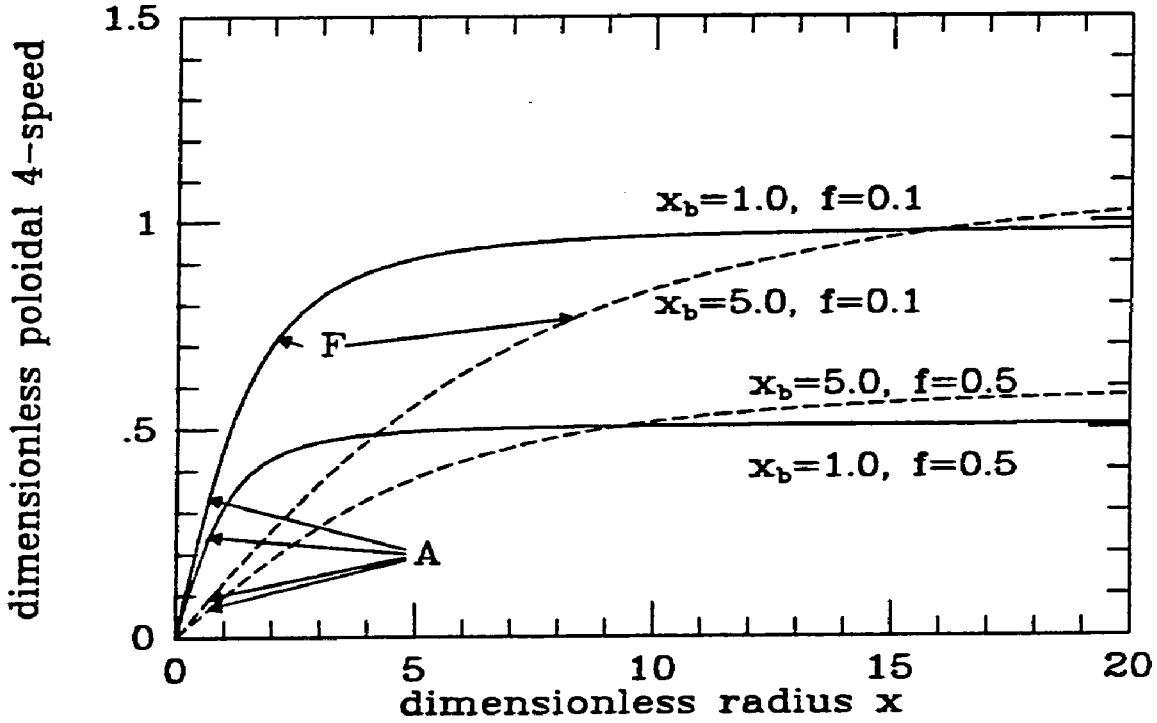


Fig. 6.1–Flow acceleration curves for different distributions of the modified magnetic pressure. The parameters x_b and f are defined in the text. The Alfvén and fast magnetosonic points are labeled by letters A and F respectively.

magnetic pressure" to drop to a low value. In a medium with pressure decreasing outwards, as might be expected in an AGN environment, an initially magnetically dominated jet would try to expand laterally, in order to lower its internal magnetic pressure and thus come into pressure balance with its surroundings. In principle, a magnetized cold jet can convert most of its Poynting flux to kinetic energy flux in this way. To demonstrate that this could indeed be the case, we have to solve the GS equation explicitly, to determine the self-consistent distribution of the "modified magnetic pressure". This is done in § 6.2 and § 6.3.

6.2. Construction of Self-Consistent Flux Surfaces

For the axisymmetric stationary flows under consideration, we need only to consider the flow structure in a meridional plane, where the magnetic flux surfaces are the same as the poloidal field lines. A 2-dimensional curvilinear coordinate system (Ψ, ξ) will be adopted, where poloidal field lines are represented by lines of constant Ψ, while lines orthogonal to the field lines are labelled by ξ, as depicted in Figure 6.2. As mentioned earlier, the change of the "modified magnetic pressure" along each line of constant ξ (i.e., in the cross-field direction) is governed by the GS equation (eq. [6.1]). If, for one reason or another, the pressure distribution $P(\Psi_i, \xi_j)$ on a set of discrete points ξ_j $(j = 1, 2, \cdots, N)$ along one given field line Ψ_i can be specified, then a flow structure that includes this field line and satisfies both the GS equation and the energy equation can be constructed numerically, for any distributions of flux constants $\Omega(\Psi)$ and $k(\Psi)$, and of the injection speed $u_0(\Psi)$. The basic scheme proceeds as follows: at each discrete point (Ψ_i, ξ_j) we first evaluate $(Q_i)_j$, the expression on the right hand side of the

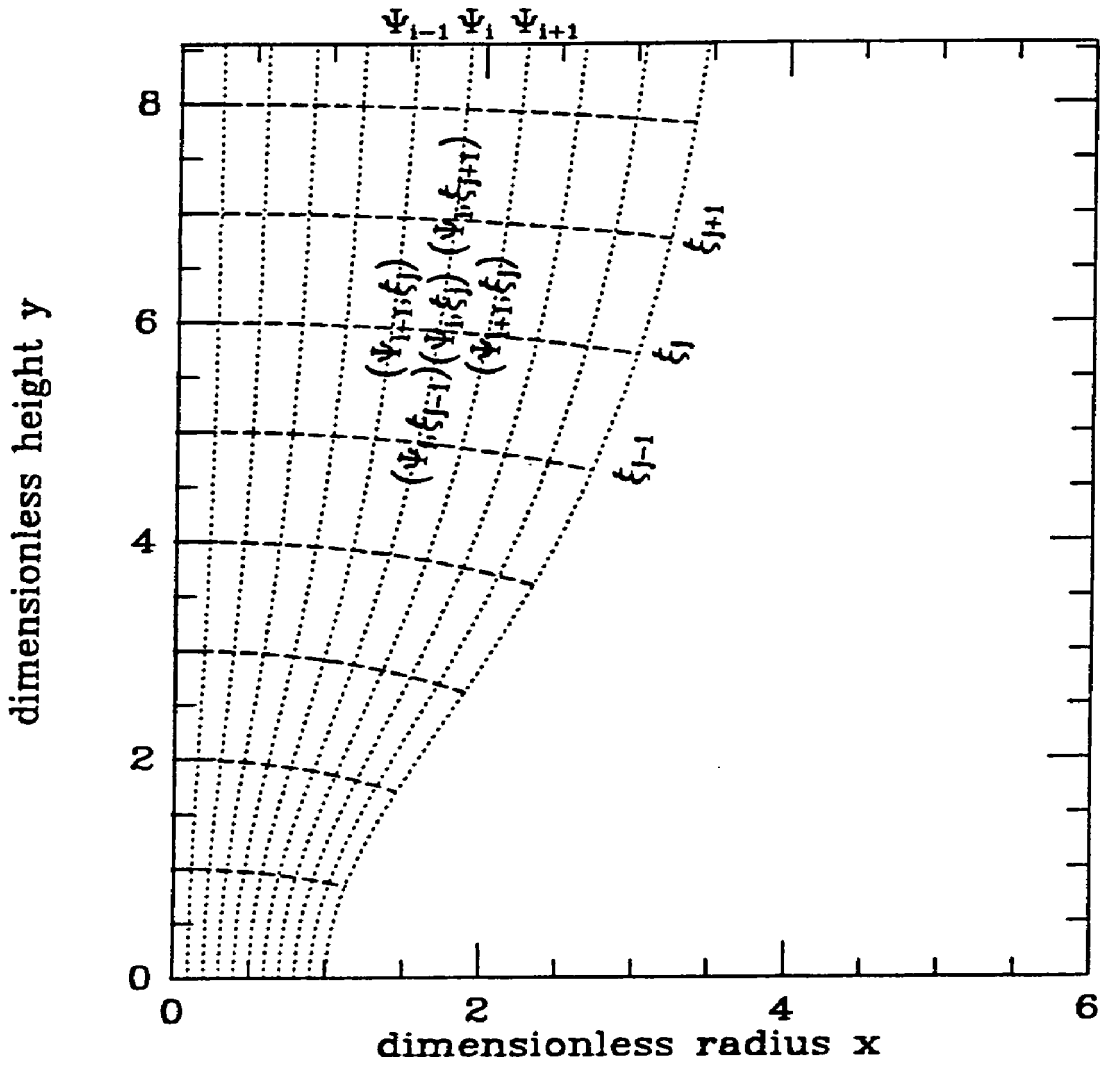


Fig. 6.2–Lines of constant Ψ (dotted, poloidal magnetic field lines) and ξ (dashed), projected on a meridional plane. Flux surface construction proceeds from one flux surface (Ψ_i) to another along lines of constant ξ.

GS equation (2.3), where the quantities dependent on field line shape, such as $\hat{\mathbf{c}} \cdot \hat{\mathbf{e}}_R$ and κ, are obtained by the "cubic spline" interpolation method (Press et al. 1989), whereas other flow quantities such as the flow speed and the two components of the field strength (B_p and B_ϕ) are determined according to the formulae presented in the last section. Conditions required at the two endpoints of the field line in order to carry out the interpolation are discussed below. To construct a neighboring field line Ψ_{i+1} ($=\Psi_i + \Delta\Psi$, where $\Delta\Psi$ is a small flux increment), we use the conservation of the magnetic flux inside the tube between the two field lines. If the two field lines are close enough together, one can evaluate the length of the line segment of constant ξ_j between the two field lines $\Delta S_j = \Delta\Psi/\langle B_p R\rangle_j$, where $\langle B_p R\rangle_j$ is an average value of the poloidal field strength times the cylindrical radius inside the tube. If the segment ΔS_j makes an angle θ_j with the cylindrical axis, then the coordinates of the points $[(R_{i+1})_j, (Z_{i+1})_j]$ on the new field line are related to the old ones $[(R_i)_j, (Z_i)_j]$ by

$$(R_{i+1})_j = (R_i)_j + \Delta S_j \sin\theta_j, \quad (Z_{i+1})_j = (Z_i)_j + \Delta S_j \cos\theta_j.$$

As a first step, we approximate $\langle B_p R\rangle_j$ by its known counterpart $[(B_p R)_i]_j$ on the field line Ψ_i, and θ_j by the angle $(\Theta_i)_j$ between the normal to the field line Ψ_i at ξ_j and the cylindrical axis. We have thus located the corresponding discrete points (Ψ_{i+1}, ξ_j) on the neighboring field line Ψ_{i+1}. The pressure $(P_{i+1})_j$ at these newly obtained points (Ψ_{i+1}, ξ_j) is simply $(P_i)_j + \langle Q\rangle_j \Delta\Psi$ according to the GS equation, where the average $\langle Q\rangle_j$ inside the tube can be approximated by its known counterpart $(Q_i)_j$ on the field line Ψ_i.

With the location of the discrete points (Ψ_{i+1}, ξ_j) on the new field line Ψ_{i+1} and their corresponding pressure thus obtained, we can then evaluate the flow quantities along this field line, and in particular, $[(B_pR)_{i+1}]_j$, $(Q_{i+1})_j$ and $(\Theta_{i+1})_j$. We then repeat the above process of constructing the Ψ_{i+1} field line, but using $\langle B_pR\rangle_j = \{[(B_pR)_i]_j + [(B_pR)_{i+1}]_j\}/2$, $\langle Q\rangle_j = [(Q_i)_j + (Q_{i+1})_j]/2$ and $\theta_j = [(\Theta_i)_j + (\Theta_{i+1})_j]/2$ instead. In principle, one can iterate until some required accuracy is obtained. For the slowly varying flow solutions to be discussed below (§ 6.3), we find that one iteration is sufficient.

We can repeat the whole process to construct the next field line Ψ_{i+2} from the properties on the field line Ψ_{i+1}, etc., until some final field line with a prescribed total flux Ψ_e is reached. Provided that the adjacent discrete points on each field line be close enough together and that the flux increment $\Delta\Psi$ at each step be small enough, the flow region thus obtained, which satisfies the energy equation identically, should also satisfy the GS equation closely. The choice of the starting flux surface will be discussed in § 6.3, where we give specific examples of flow construction according to this scheme.

Numerically, to obtain the geometric quantities on a field line (say Ψ_k) defined by a set of discrete points (Ψ_k, ξ_j) of known locations, special treatment has to be applied to the two endpoints of the field line. This is because both the slope and the curvature at any given discrete point (Ψ_k, ξ_j) depend on the locations of its two neighboring points $(\Psi_k, \xi_{j\pm1})$, but the endpoints have only one neighboring point each. Equivalently, we need to specify two more conditions at the endpoints in order to obtain a unique "cubic spline" interpolation for the shape of the field line from a set of discrete points of known positions. Since we are seeking collimated jet solutions, it

is natural for us to demand that the field line become vertical very far away from the base of the flow. We thus impose the condition $dR/dZ = 0$ at the "far" end of the field line, which we shall take to be at a height many orders of magnitude larger than the light cylinder radius. This condition of asymptotic cylindrical flow should be satisfied if the pressure in the ambient medium does not drop to zero entirely. In this case, it can be shown (Heyvaerts & Norman 1989; Chiueh, Li, & Begelman 1991) that the only asymptotic configuration that a magnetized wind might have is one with all of flux surfaces nested around the rotation axis cylindrically. Possible singular solutions where all flux surfaces collapse to the rotation axis (Blandford & Payne 1982) are not considered here. The condition at the other end of the field lines, i.e., at the base of the flow, can be much more complex and is not always easy to handle numerically. The cases that are most straightforward to interpret and are easiest to implement numerically are those with the flow emanating vertically from the equatorial plane. Such flows therefore have $dR/dZ = 0$ at the base, as well. Note that the requirement that the field line has to make an angle larger than 30° with the axis (Blandford & Payne 1982) does not apply here, since we inject the plasma from the base of the flow at a finite speed in the absence of gravity of the central object. Although realistic flows cannot satisfy this rather restrictive condition in general, and indeed modifications have to be made in order to model jets from disks with poloidal field strength decreasing outward (see discussion in § 6.4), in this preliminary investigation we shall nevertheless use this family of solutions to demonstrate the general dependence of the flow properties on the distributions of angular speed, mass loading rate, and ambient pressure, etc., and to gain some insights into the nature of the GS equation that controls the flow structure.

Finally, we mention in passing that the interesting case of a free wind (that is, one for which the external confinement is negligible) that fills out all of space, such as that of the Crab pulsar (where the flows from both hemispheres presumably balance each other on the equatorial plane), is beyond the scope of our discussion here. The reason is that the flux surfaces in these free winds might be asymptotically paraboloidal (see chapter III) instead of cylindrical, as we have assumed here. Also, in view of the fact that the immediate environments of jets in their production regions are not known at present, the so-called "free-boundary problem" (see Pizzo [1990] for a discussion in the context of modeling sunspots), where one has to determine the location of the last surface Ψ_e iteratively under a given pressure distribution of the ambient medium, is not attempted here. Instead, we shall address only the easier question of what kind of ambient pressure distribution is required to contain the flows we construct.

6.3. Jet Solutions to the Relativistic Grad-Shafranov Equation

6.3.1. A "Typical" Solution

In this section, we demonstrate how the general numerical scheme laid out above is carried out in practice for relativistic flows. First of all, we shall non-dimensionalize the GS equation (6.1) with respect to a "reference" flux surface Ψ_* (on which $\Omega_* = \Omega[\Psi_*]$ and $k_* = k[\Psi_*]$) so that

$$\frac{\partial \bar{p}}{\partial \bar{\varphi}} = \frac{2}{\bar{x}^2}\left\{\left[\frac{\bar{b}_\phi^2 - \alpha^2 \bar{b}_p^2 \bar{x}^2}{\bar{b}_p} - \frac{\bar{u}_\phi^2}{\bar{u}_p}\right]\hat{\mathbf{c}}\cdot\hat{\mathbf{e}}_R - (\kappa R)\bar{b}_p\left[1 - \alpha^2\bar{x}^2\left(1 + \frac{\beta^2\bar{\tau}}{\alpha^2}\right)\right]\right\} \tag{6.9}$$

where $\alpha(\Psi) \equiv \Omega(\Psi)/\Omega_*$ and $\beta(\Psi) \equiv k(\Psi)/k_*$, and quantities non-dimensionalized using combinations of Ω_* and k_* are denoted with an overbar. They are related to the dimensionless quantities (without an overbar) defined earlier on any flux surface Ψ (with α and β) by

$$\bar{p} \equiv \frac{P}{2\pi k_*^2 c^2} = \beta^2 p, \quad \bar{\varphi} \equiv \frac{\Psi \Omega_*^2}{4\pi k_* c^3} = \frac{\beta}{\alpha^2}\varphi, \quad \bar{\mathbf{b}} \equiv \frac{\mathbf{B}}{4\pi k_* c} = \beta \mathbf{b},$$

and

$$(\bar{x}, \bar{y}) \equiv \frac{\Omega_*}{c}(R, Z) = \frac{1}{\alpha}(x, y), \quad (\bar{u}_p, \bar{u}_\phi) = (u_p, u_\phi).$$

As is evident from the above definitions, since the barred dimensionless quantities are directly proportional to their dimensional counterparts, they will be used in presenting our results, unless noted otherwise. In particular, we can specify the distributions of the flux constants α and β in terms of $\bar{\varphi}$, instead of the flux function Ψ itself. Since the curvature κ has the dimension of $(\text{length})^{-1}$, the combined dimensionless quantity (κR) in the above equation will not be affected by any change in the length scale.

There are, in principle, five free functions that have to be specified before a solution can be constructed, namely, the distributions of the injection speed u_0 and the two flux constants (α and β), together with the shape of the starting flux surface and the pressure distribution on this surface. We note that the distribution of the poloidal magnetic field strength b_p at the base of the flow ($Z = 0$) is not a free function once the shape of the starting surface and its pressure distribution are prescribed. The critical energy and angular momentum distributions are determined self-consistently from the conditions at the Alfvén surface and the base of the flow. Given the large number of the free functions involved, it is impractical to survey all kinds

of functional forms possible. Instead, we shall try to parameterize them in an effort to capture their main effects on the solutions. Fortunately, after some numerical experimentation we find that the injection speed does not affect the flow structure much, as long as it is small compared to unity. This may not be the case when gravity is included. We hereafter adopt a working number of $u_0 = 0.05$, and thus remove one free function.

There are therefore four free functions remaining. We shall first discuss the two free functions associated with the starting flux surface. For definitiveness, we assume that the starting surface can be parametrized by

$$X = X_0\{1 + [(Y/Y_b)(1 - X/X_\infty)]^4\}^{N_X/16}, \tag{6.10}$$

where X and Y are the dimensionless radius and height of the starting surface. Equation (6.10) describes a line that emerges from the equatorial plane ($Y = 0$) vertically at X_0 and approaches the limiting radius X_∞ asymptotically ($Y \to \infty$). The exponent N_X characterizes how fast the flux surface is bent towards the limiting radius, whereas the parameter Y_b marks where this bending begins. We have adopted a value of 4 for the exponent of (Y/Y_b) to ensure not only that the flux surface threads the equatorial plane vertically, but also that the curvature of the flux surface near the base of the flow be small. Small curvature is desirable mainly for numerical reasons, since artificial numerical oscillations are found to be associated with rapid field line bending in this region. We shall elaborate on this point towards the end of this subsection.

Prescribing the pressure distribution on the starting surface is equivalent to specifying the distribution of the poloidal field strength B_p. A convenient way of doing this is to assume

$$B_p = B_0(X_0/X)^{N_B}, \tag{6.11}$$

where the exponent N_B characterizes how fast the poloidal field strength decreases along the starting flux surface relative to the base at X_0. The corresponding pressure required to support the starting flux surface from within can be calculated easily from the energy equation (4.2). This pressure can be supplied, for example, by a different flow driven entirely by the central rotating black hole (Phinney 1983) or, in case of a young stellar object, by a stellar wind (Shu et al. 1988). The presence of a supersonic flow inside the wind also helps to stabilize the magnetized wind under consideration, as pointed out by Eichler (1993). The difficult yet important problem of matching the two flows is beyond the scope of the present paper.

We shall label the starting flux surface with $\Psi = 0$, and choose it as the "reference" surface at the same time. Therefore, on this surface, dimensionless quantities with an overbar and those without are the same. Before we construct field lines according to the scheme presented in the last section, the other two remaining free functions, $\alpha(\bar{\varphi})$ and $\beta(\bar{\varphi})$, have to be specified. We find it more transparent to replace the distribution of the mass loading rate per magnetic flux $\beta(\bar{\varphi})$ by the dimensionless flux function $\varphi(\bar{\varphi}) = \bar{\varphi}[\alpha(\bar{\varphi})]^2/\beta(\bar{\varphi})$ (from eq. [5.2]) in our discussions. For simplicity, they are parametrized using the functional forms

$$\alpha(\bar{\varphi}) = \left[1 + \left(\frac{\bar{\varphi}}{\bar{\varphi}_\alpha}\right)^2\right]^{-N_\alpha/2}, \quad \varphi(\bar{\varphi}) = \bar{\varphi}\left[1 + \left(\frac{\bar{\varphi}}{\bar{\varphi}_\varphi}\right)^2\right]^{-N_\varphi/2}, \quad (6.12a, b)$$

both of which decrease outward as power-laws outside some characteristic flux surfaces ($\bar{\varphi}_\alpha$ and $\bar{\varphi}_\varphi$, respectively), but remain more or less constant within them.

We now concentrate on one "typical" solution with $N_\alpha = 1$, $N_\varphi = 0.5$, and $\bar{\varphi}_\alpha = \bar{\varphi}_\varphi = 0.5$, and with the following set of parameters to describe the starting surface: $X_0 = 0.20$, $X_\infty = 0.95$, $Y_b = 5.0$, $N_X = 2.0$, $B_0 = 5.0$, and $N_B = 3.0$. This particular case will serve as the "standard" solution with which all solutions with different parameters that we shall describe below (§ 6.3.2) are compared. Numerically, we keep doubling the number of the grid points on the starting flux surfaces so as to reduce the distance between adjacent points and keep decreasing the flux increment $\Delta\varphi$ at each step until the solution converges. The height of the far endpoint of the starting field line is taken to be far enough from the base of the flow ($Y_{max} = 10^5$ in this case) that its exact location on the limiting radius does not affect the solution structure. The main properties of the solution are summarized in Figure 6.3. In Fig. 6.3*a*, we have plotted the starting flux surface $\bar{\varphi} = 0$ and subsequent flux surfaces separated by $\Delta\bar{\varphi} = 0.5$ (i.e., $\bar{\varphi} = 0.5$, 1.0, $\cdots$, 5.0), together with the light cylinder, Alfvén and fast magnetosonic surfaces. Coordinates $\log_{10}(1 + \bar{x})$ and $\log_{10}(1 + \bar{y})$ are used in the figure in order to show more clearly where most of the field line bending occurs. The acceleration of the flow along these surfaces is displayed in Fig. 6.3*b*, where gradual (in fact, logarithmic) increase in speed away from the base of the flow is evident. The logarithmically slow collimation and acceleration of the flow in the kinetic energy–dominant region are in accord with general expectations from analytic calculations (Begelman & Li 1993; Eichler 1993). The fact that more final energy is in kinetic form (including the plasma rest

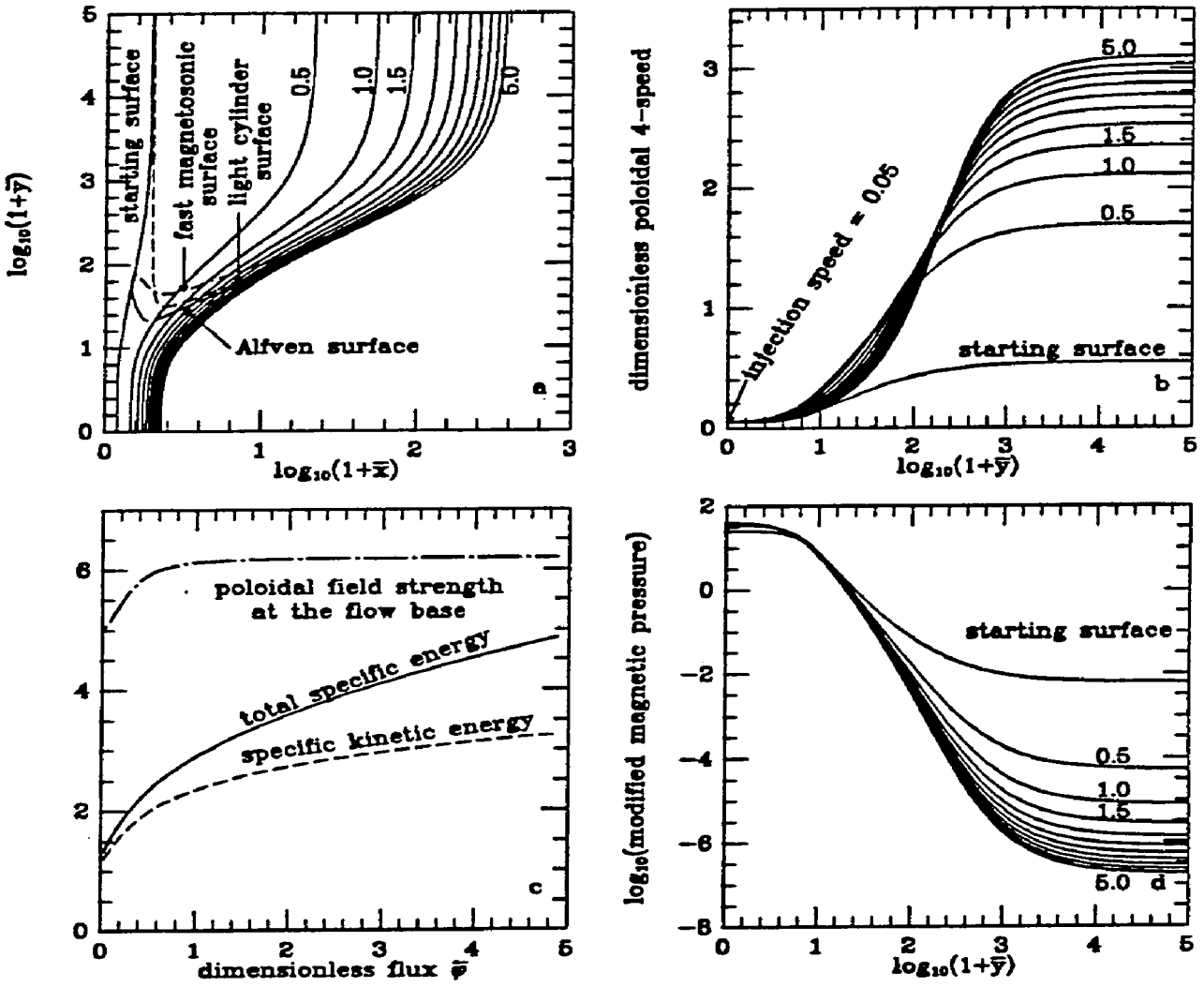


Fig. 6.3–Properties of the standard solution: (a) shapes of the flux surfaces with the dimensionless flux $\bar{\varphi}$ spaced by 0.5; (b) flow acceleration curves on these surfaces; (c) total specific energy, final specific kinetic energy (including the plasma rest mass), and poloidal magnetic field strength at the base of the flow plotted as functions of the dimensionless flux $\bar{\varphi}$, and (d) distributions of the dimensionless modified magnetic pressure $\log_{10}(\bar{p})$.

mass) than in electromagnetic form in this particular example is apparent from Fig. 6.3*c*, where the critical total specific energy μ_c is plotted side by side with the final specific kinetic energy of the flow (i.e., the terminal flow Lorentz factor γ_f), as functions of the dimensionless flux function $\bar{\varphi}$. Also plotted in Fig. 6.3*c* is the distribution of the poloidal field strength $\bar{b}_p$ at the base of the flow, which increases slowly outwards at first and then remains more or less constant. Finally, in Fig. 6.3*d* we show the distributions of the dimensionless "modified magnetic pressure" along various flux surfaces. The dimensionless pressure in the ambient medium must equal the pressure on the last flux surface $\bar{\varphi}_e = 5.0$, if the last surface is indeed a contact discontinuity.

Although the magnetic pressure near the base of the flow varies only slowly across flux surfaces (Fig. 6.3*c*), its value on the innermost flux surface can be many orders of magnitude larger than that near the edge, at large distances from the base (Fig. 6.3*d*). This feature, which we suspect to be common to many solutions that satisfy the GS equation, is a sure sign of the magnetic confinement of the flow. After analyzing the various terms on the right hand side of the GS equation (6.9) that contribute to the pressure change, we find that the term with $\hat{\mathbf{c}} \cdot \hat{\mathbf{e}}_R$ (which is proportional to pressure) always dominates the curvature term except inside the Alfvén surface, and its value at large distance is comparable to or even greater than the modified pressure $\bar{p}$. We believe this is why the pressure at large distance can drop so rapidly from one flux surface to another. Also evident from equation (6.9) is the potential numerical difficulty in dealing with the curvature term. Numerical errors in determining the curvature would affect the magnetic pressure, and thus B_p, which in turn feeds back to the determination of the curvature through magnetic flux conservation. Small irregularities on the

field line shape will thus be amplified, causing artificial oscillations. This problem is most severe inside the Alfvén surface, where the contribution to the pressure change from the curvature term could dominate the other term (eq. [6.9]). This is the basic reason that we choose the exponent of (Y/Y_b) in equation (6.10) to be 4 instead of, say, 2.

This simple example is suggestive of the following possible scenario for the production of relativistic jets in AGNs: near the base of the flow, energy and angular momentum are extracted from the central rotating object (a disk or black hole) by, and stored in, the corotating open magnetic fields. The stored energy and angular momentum are released gradually into the plasma in the magnetically dominated jet when it propagates into an ambient medium with rapidly decreasing pressure. In order to lower the internal magnetic pressure inside the jet to match the decreasing pressure outside, the poloidal current, which initially flows parallel to the field lines in the nearly force-free region at the base of the flow, has to flow across the field lines, and thus yields a $\mathbf{j} \times \mathbf{B}$ force that accelerates the plasma loaded on the magnetic field. Therefore, the pressure distribution of the ambient medium clearly plays an active role in determining how much and how fast the magnetic energy is released to the flow acceleration.

Although the flow obtained in above particular example is well-collimated all the way (with maximum half-opening angle less then about 5° in most parts of the flow, Fig. 6.3*a*), and the terminal flow speed (Fig. 6.3*b*) increases monotonically from the inner flux surface toward the edge of the flow, these trends depend sensitively on the particular free functions used, as we show below.

6.3.2. Some General Trends

In this subsection, we shall vary the four free functions (that is, $\alpha[\bar{\varphi}]$, $\varphi[\bar{\varphi}]$, the shape of the starting surface and its poloidal field strength distribution), one by one, to show their effects on the properties of the flow solutions. We reiterate that it is not our intention to cover the entire parameter space. Rather, only the main trends are illustrated with a few heuristic examples. Our strategy is to vary one free function at a time, while holding the other three fixed.

We first illustrate the effect of the distribution of the flux surface rotation rate by considering three exponents for $\alpha(\bar{\varphi})$ (eq. [6.12*a*]): $N_\alpha = 0.5$, 1.0, and 1.5, with $\bar{\varphi}_\alpha = 0.5$. The same distribution of $\varphi(\bar{\varphi})$ and starting surface are chosen as in Figure 6.3. The resulting distributions of the total critical energy μ_c and the final ratio of the kinetic energy flux to the Poynting flux are shown in Figure 6.4. It is apparent that the dependence of μ_c on the distribution of $\alpha(\bar{\varphi})$ is rather weak. The ratio of the kinetic energy flux to the Poynting flux (Fig. 6.4), as well as the final width of the flow and the minimum magnetic pressure on the last flux surface (Table 6.1), on the other hand, do differ significantly in each case. The trend seems to be that, the faster the outer flux surfaces rotate relative to the innermost ones, the more efficient is the conversion of electromagnetic energy into flow kinetic energy. This, in turn, leaves a lower magnetic pressure at the edge of the flow needing to be balanced by that in the ambient medium. The smaller ambient pressure required for the flows that rotate more rapidly is also consistent with the larger flow radius (Table 6.1) in these cases. For the case of $N_\alpha = 0.5$, we find that the flow does not become well-collimated

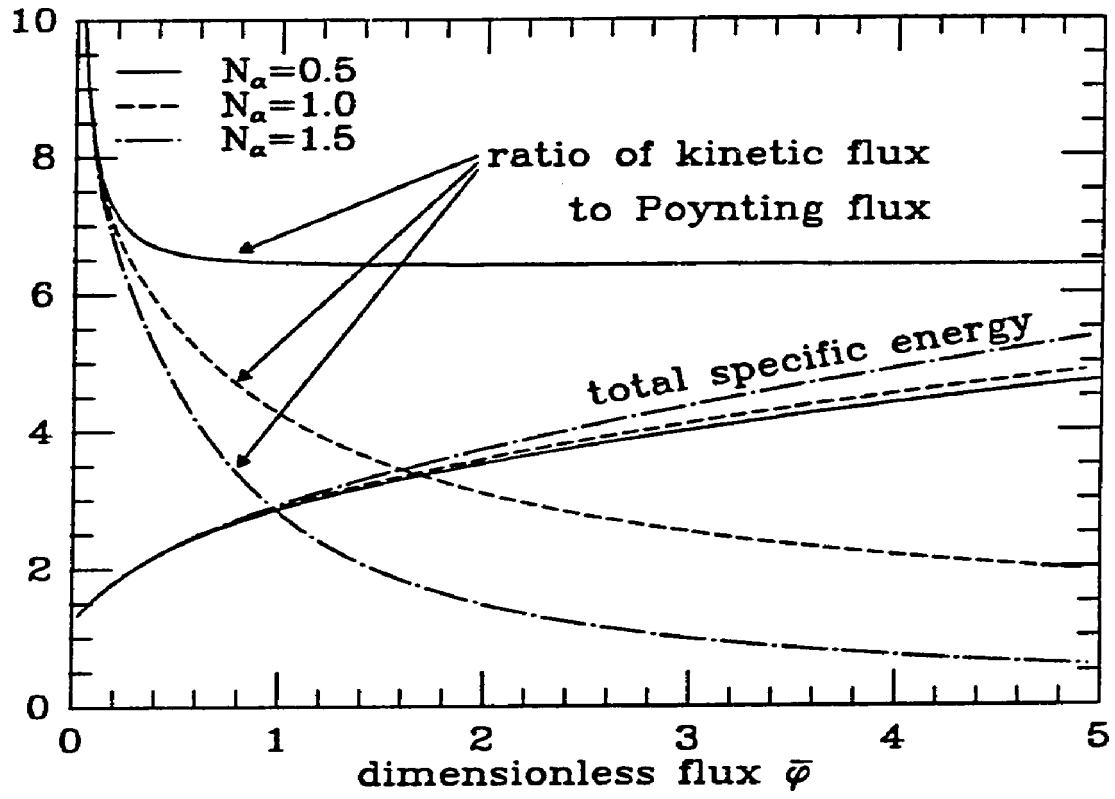


Fig. 6.4–Distributions of the total specific energy and the ratio of kinetic energy flux to Poynting flux for three different distributions of the angular speed of the flux surfaces (paramerterized according to eq. [6.12a]) with $N_\alpha = 0.5$ (solid), 1.0 (dashed), and 1.5 (chain-dot).

TABLE 6.1

Dimensionless flow width and minimum magnetic pressure on the last flux surface for different exponents N_α [eq. (6.12a)]

exponent N_α	flow width	minimum pressure
0.5	2.78×10^3	3.15×10^{-9}
1.0	3.56×10^2	2.09×10^{-7}
1.5	1.26×10^2	2.00×10^{-6}

TABLE 6.2

Dimensionless flow width and minimum magnetic pressure on the last flux surface for different exponents N_B [eq. (6.11)]

exponent N_B	flow width	minimum pressure
2.5	31.5	2.22×10^{-4}
3.0	3.56×10^2	2.09×10^{-7}
3.3	4.03×10^3	6.17×10^{-10}

until it reaches a length of order several thousand times the light cylinder radius (of the starting surface), in contrast to the "standard" solution which is already well-collimated not far from the base of the flow.

The shape of the starting flux surface does not affect the critical energy distribution μ_c much, either. If one varies only the exponent N_X in equation (6.10), which characterizes how fast the starting flux surface is bent towards the limiting radius, then the final state of the flow, including its final distributions of speed, magnetic pressure, kinetic energy flux, and Poynting flux, remains almost identical. The rapidity with which the solution converges onto the final state, however, is clearly controlled by the exponent N_X, as is exemplified by Figure 6.5. In Fig. 6.5, the pressure distributions at the first surface ($\bar{\varphi} = 0.0$), the last surface ($\bar{\varphi} = 5.0$), and one surface in between with $\bar{\varphi} = 1.0$ are plotted for $N_X = 1.5$ and $N_X = 3.0$, respectively, again keeping other parameters the same as in Figure 6.3. It seems that a slower rate of decrease in the ambient pressure would force the flow to change more gradually, and to travel further away from its base before reaching its final state. The effect of changing the limiting radius X_∞ of the starting surface is similar to that of changing the exponent N_B in equation (6.11), both of which affect the lowest pressure on the starting surface. We thus need only consider the variation of N_B. We have computed two solutions with $N_B = 2.5$ and 3.3, and compare them with the "standard" solution with $N_B = 3.0$. As direct consequences of lower pressure at large distance, solutions with higher N_B have larger flow radius and need much lower external pressure for confinement (Table 6.2, page 134). The distribution of the total critical energy is not much affected, as can be seen from Figure 6.6. The

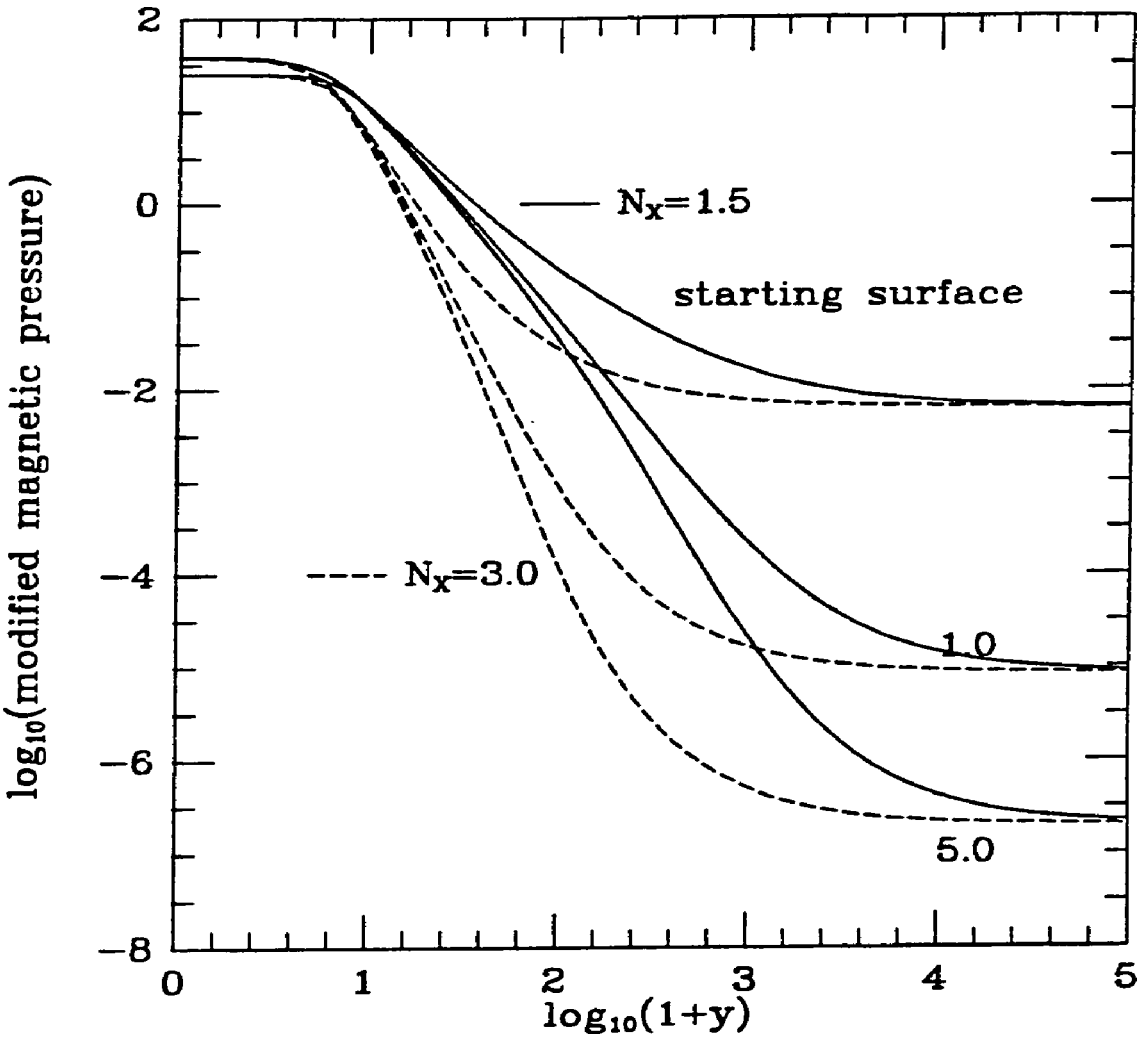


Fig. 6.5–Distributions of the modified magnetic pressure on three flux surfaces with $\bar{\varphi} = 0.0$ (the starting surface), 1.0, and 5.0 (the final surface) for two different shapes of the starting surface given by expression (6.10) with $N_X = 1.5$ (solid) and $N_X = 3.0$ (dashed).

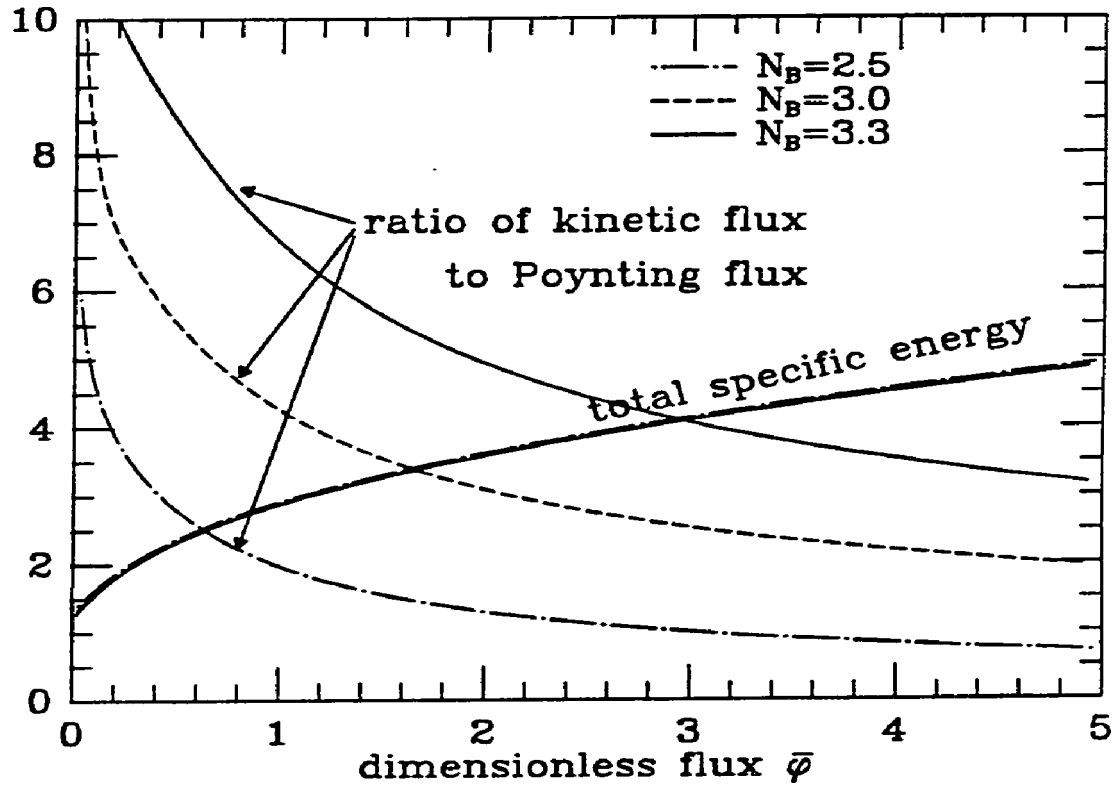


Fig. 6.6–Distributions of the total specific energy and the ratio of kinetic energy flux to Poynting flux for three different distributions of the poloidal magnetic field strength on the starting surface given by expression (6.11) with $N_B = 2.5$ (chain-dotted), 3.0 (dashed), and 3.3 (solid).

final ratio of the kinetic energy flux to the Poynting flux, on the other hand, differs considerably in each case, as expected.

Finally, we hold fixed the distribution $\alpha(\bar{\varphi})$ and the shape of the starting surface and its pressure distribution (same as in Figure 6.3), and consider three representive distributions of $\varphi(\bar{\varphi})$ as shown in Figure 6.7*a*: in the solid curve, φ increases linearly with $\bar{\varphi}$, whereas in the other two curves, φ increases with $\bar{\varphi}$ at first, but either approaches constant (dashed) or declines with $\bar{\varphi}$ (chain-dotted). It is clear (Fig. 6.7*b*) that the distribution of the total critical energy μ_c is controlled mainly by the distribution of $\varphi(\bar{\varphi})$, and follows the variation of $\varphi(\bar{\varphi})$ closely. Physically, the dimensionless quantity φ characterizes the ability of magnetic field lines to accelerate the flow plasma to high speed, and it reduces to Michel's (1969) magnetization parameter σ in a radial wind. Since $\varphi \propto \Psi\Omega^2 k^{-1}$, the flow can be potentially accelerated to an higher speed along a flux surface of given Ψ if the flux surface rotates faster and/or the mass loading rate per unit magnetic flux of the flux surface is lower. Depending on the distribution of $\varphi(\bar{\varphi})$, the flow terminal speed (shown in Fig. 6.7*c*) can either increase or decline towards the outer edge of the flow.

6.4. Conclusions and Discussion

We have obtained self-consistent relativistic jet solutions numerically by constructing flux surfaces that satisfy the relativistic GS equation. To avoid numerical difficulties, we have restricted ourselves to a family of solutions that emanate vertically from the equatorial plane, and have small curvature near the base of the flow. For these flows, we have shown that the distribution of the total energy output depends mainly on the distribution

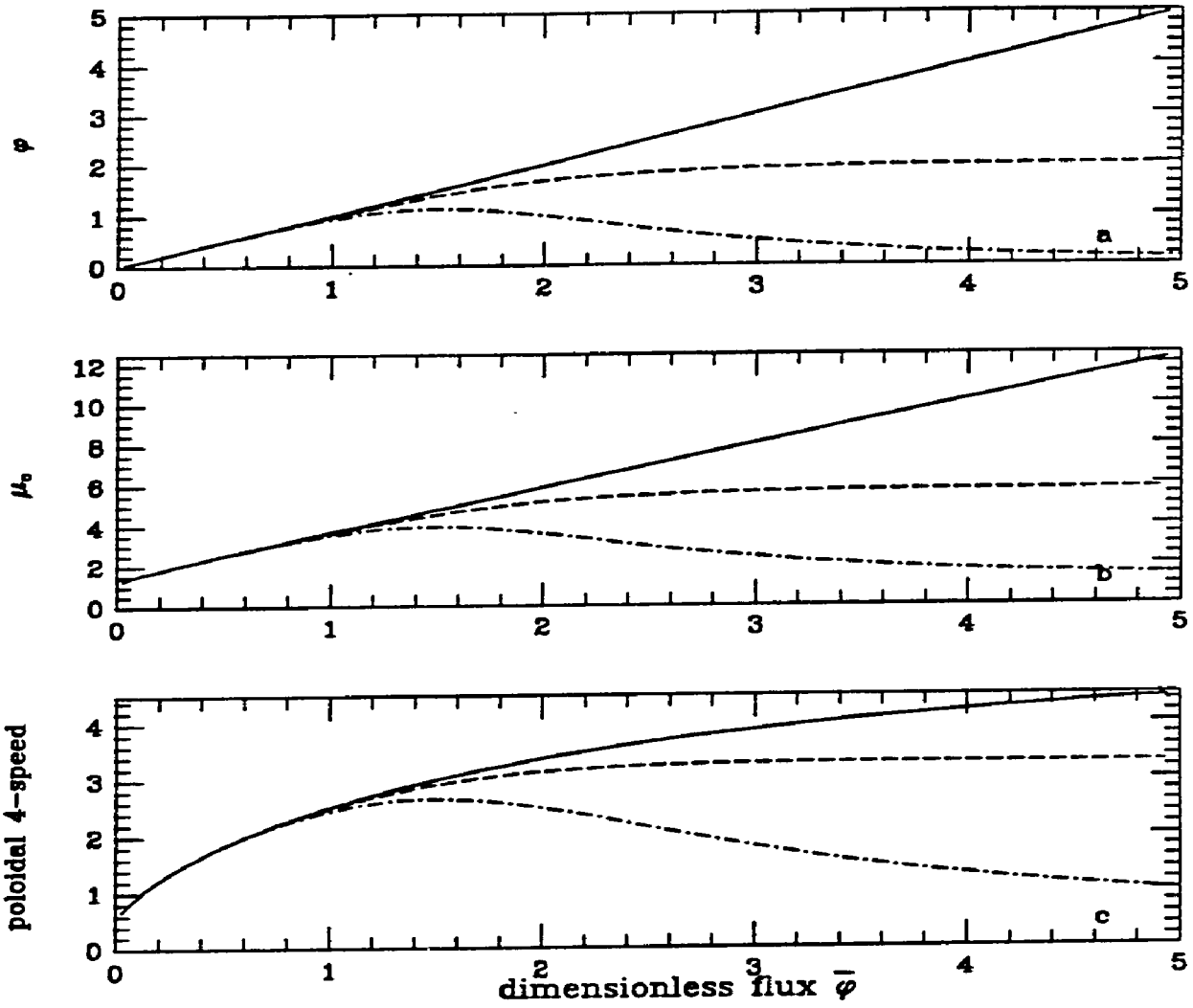


Fig. 6.7–(a) The prescribed distributions of the dimensionless flux function φ, and the resultant corresponding distributions of (b) the total specific energy μ_c and (c) the dimensionless poloidal 4-speed. The similarity between curves in (a) and (b) is evident. The total specific energy is controlled by φ, as discussed in the text.

of the so-called "dimensionless flux function" φ, a combination of the flux constants given by $\Psi\Omega^2/(4\pi kc^3)$ (eq. [5.2]). It is interesting to note that the same sort of dependence can also be detected in the numerical examples of Camenzind (1987), and we suspect this feature is generic to the solutions that satisfy the relativistic GS equation. The final width of the flow and the fraction of the total energy in kinetic form, on the other hand, depend sensitively on the initial conditions of the starting flux surface and the distribution of the angular speed of the flux surfaces. For systems with magnetic pressure on the starting surface declining more rapidly away from the flow base and/or the outer flux surfaces rotating faster relative to the starting surface, wider flows would be expected and more electromagnetic energy would be converted into plasma kinetic energy, thus lowering the magnetic pressure inside the flow which needs to be balanced by the external pressure. This external pressure could be supplied, in principle, by both the "ordinary" pressure in the ambient medium and the magnetic pressure of the toroidal field associated with the net current of the flow when it is not screened completely (Appl & Camenzind 1993, chapter IV). In case the external toroidal magnetic field is zero, a large fraction of the total energy will be in kinetic form when the pressure in the ambient medium is low enough, and the conversion of magnetic energy to the directed flow kinetic energy will be very efficient. This trend is in accord with our conjecture that flows with no external confinement should approach a current-free paraboloidal asymptote, in which the terminal electromagnetic energy flux can, in principle, decrease to an arbitrarily small value, although this decline is likely to be logarithmically slow (Begelman & Li 1993; Eichler 1993). The shape of the starting surface does not affect the final state of the flow much, although

it does control how fast the final flow speed is reached, and where most of the flow collimation occurs.

The main disadvantage of the present approach for solving the GS equation is that, after prescribing the properties of the starting surface, the flow structure will be totally determined by the conservation laws along each flux surface and the GS equation, and thus we cannot handle the flow boundary conditions properly. Specifically, the resultant pressure distribution on the last flux surface may not match that in the ambient medium, if the latter is known. The distribution of the poloidal field strength $(B_p)_0$ at the base of the flow thus obtained may not be compatible with the conditions that exist on the rotating object on which the flow is anchored. In particular, the nearly constant (or slightly outward increasing) $(B_p)_0$ that is common in our solutions (as a result of the strict assumption about the field line geometry near the base of the flow) may not be a very realistic approximation to the conditions above actual disks around massive black holes. For nonrelativistic flows, outward field strength declining as fast as $(B_p)_0 \propto R_0^{-5/4}$ is required above a self-similar Keplerian disk (Blandford & Payne 1982; Königl 1989), although whether such a steep outward decline can be established naturally remains an open question (Begelman 1993). However, we believe our conclusions would remain at least qualitatively true as long as $(B_p)_0$ does not decrease outward too rapidly, since these results reflect mostly the flow properties at large distances from the base of the flow. Further uncertainties arise from the choice of the starting surface. In principle, the position and the pressure distribution of the starting surface are to be determined self-consistently by balancing the disk wind under construction and the flow that it presumably encloses. Finally, we have not attempted a stability analysis

of the jet flows obtained, although supermagnetosonic flows that are accelerated gradually as we have described here may be prone to kink instabilities (Eichler 1993).

As idealized as it is, the present model is unique in several ways. First, this work is the only other true attempt to solve the generalized 2-D relativistic GS equation besides Camenzind (1987), without using either the force-free assumption (Blandford 1976) or the self-similar assumption (see chapter V). Our approach is complementary to Camenzind's who, admittedly, was trying to solve the much harder boundary value problem. Compared with Camenzind (1987), our method of flow construction is faster and less costly numerically. There is no need for us to assume that the magnetic fields are only sparsely populated by plasma, as in the calculation of Camenzind (1987). Besides, we carry our solution much further beyond the fast magnetosonic surface than Camenzind, and are thus able to show the dependence of the terminal flow properties on various flow conditions. We do find in our solutions the trend that was already apparent in Camenzind's work, namely, that the the total specific energy in the flow escaping along each flux surface scales roughly with the dimensionless quantity φ. If this scaling holds approximately for all relativistic winds, then one can estimate the total energy flux escaping in a relativistic wind of total flux Ψ_e as

$$\dot{E} \approx \int_0^{\Psi_e} \frac{\Psi[\Omega(\Psi)]^2}{4\pi c} d\Psi. \tag{6.13}$$

This expression is in agreement with the scaling $\dot{E} \sim \Omega^2\Psi^2/c$ obtained by Begelman (1992) using an independent argument.

Second, we can treat an arbitrary distribution of the flux constants $\Omega(\Psi)$ and $k(\Psi)$, in contrast to the case of self-similar flow where specific

power-law distributions are required for self-consistency (Blandford & Payne 1982; see chapter V). Third, our solutions should be capable of comparison with those solutions obtained from the quasi-one-dimensional methods (Chan & Henriksen 1980; Koupelis & Van Horn 1989; Lovelace et al. 1991), once their methods are applied to relativistic flows. The logarithmic flow acceleration which is common in our solutions is also found in the quasi-one-dimensional nonrelativistic cases (Lovelace et al. 1991), although we find that the field strength (and thus the flow magnetic pressure) varies greatly across the cross-section of the flow. At large distances from the base of the flow, the magnetic pressure is much higher near the center of the flow than near the edge. It is not clear how this large pressure gradient would affect the averaging processes used in the quasi-one-dimensional methods.

CHAPTER VII

RADIATION DRAG ON RADIAL, RELATIVISTIC MHD WINDS

7.1. Introduction

Relativistic winds or jets emanating from the regions close to accreting compact objects are expected to pass through an intense radiation field (Begelman, Blandford & Rees 1984). Losing energy and angular momentum to the photons through inverse Compton scattering, the flows may be decelerated or even quenched (Phinney 1987; Melia & Königl 1989). The Comptonized photons, which are boosted into a narrow cone along the relativistic flow direction, may have enough luminosity to be observable, providing an explanation for the continuum radiation from blazars (Begelman & Sikora 1987; Melia & Königl 1989).

The effects of scattering on the observable radiation and the effects of radiation drag on the dynamics of jets and winds are interconnected problems. In order to understand either, one must solve self-consistently for the dynamical effects of passage through the radiation field, taking into account all forces acting on the flow. Begelman & Sikora (1987) calculated the spectrum and polarization properties of radiation scattered by a relativistic wind with a specified geometry and velocity field, but did not address the dynamics self-consistently. Phinney (1987) considered the dynamics of a jet accelerated by radiation forces in a specified photon field, and illustrated how Compton drag will generally prevent the flow from attaining extremely high Lorentz factors ($\gtrsim 10$). However, Phinney's analysis applies only to flows accelerated

by radiation pressure, whereas it is suspected that hydromagnetic forces are the primary agent responsible for accelerating jets from accreting compact objects (Blandford 1989). Melia & Königl (1989) calculated the deceleration of an existing relativistic flow by radiation drag, noting that the flow could have been accelerated by non-radiative forces at some smaller radius. Unfortunately, since the radiation is presumably most intense in the regions where acceleration is occurring, it is misleading to treat the acceleration and deceleration regions as spatially separate. Instead, one should consider a flow subject simultaneously to both accelerating forces (e.g., hydromagnetic stresses) and the effects of radiation drag.

In this chapter, we study the effects of an ambient radiation field on a radial, relativistic MHD wind from a rotating central object. The basic theory of relativistic MHD winds has been developed in the previous chapters. In view of the difficulties in obtaining self-consistent solutions even for axisymmetric flows, we follow Michel (1969) and Goldreich & Julian (1970; hereafter GJ) in assuming a radial flow geometry, neglecting the force balance across flux surfaces. Flows with more general geometries will be considered in the next chapter. We assume that the flow is optically thin and treat the dynamical effect of the radiation field as an effective drag force acting against the flow. For specified mass fluxes and drag coefficients, we determine critical solutions which pass smoothly from the origin to infinity, and calculate their physical properties such as energy flux, angular momentum flux, and velocity field.

The results of our study are somewhat surprising. Instead of reducing the terminal velocity and energy flux of the wind, the action of radiation drag increases the torque on the central object, thus extracting more energy

and angular momentum at the source. The reason for this is that much of the energy flux in a radial wind is carried by the Poynting flux, with only a fraction being released as kinetic energy. By retarding the flow, radiation drag can cause the field lines to "wind up", thereby increasing the amount of energy stored in magnetic form. When the drag decreases (e.g., at large radii where the radiation field becomes dilute), the magnetic energy is released, accelerating the flow. Under certain conditions, radiation drag can actually cause the flow to accelerate by catalyzing the conversion of Poynting flux to kinetic energy. We analyze these effects to reveal their physical origins. Although the special properties of radial wind solutions probably enhance these effects, we suggest that they will be present at some level in any hydromagnetic wind model.

The plan of the chapter is as follows. In § 7.2 we incorporate the radiation drag force into the radial wind equations, and examine the effects of this drag on the critical points of the flow. The equations are solved numerically in § 7.3, and the results interpreted physically with the aid of simple analytic models. We also show that the amount of energy dissipated through radiation drag can be quite substantial, and may lead to observable effects. Finally, we summarize and discuss our conclusions in § 7.4.

7.2. Radial Winds With External Drag Force

7.2.1. Basic Equations

As mentioned above, we shall follow the earlier analyses of Michel (1969) and GJ in considering radial winds except that, instead of being confined to the equatorial plane, the plasma in our discussion can flow along cones of arbitrary polar angle. The interplay between the shape of the flux surfaces and the radiation drag force cannot be addressed self-consistently in this model. This last important point is touched upon only in the next chapter when an idealized drag distribution is applied to the self-similar solutions discussed in chapter V.

In standard spherical polar coordinates (r, θ, ϕ), the conservation of magnetic flux requires that

$$B_r r^2 = \Phi_0 = \text{constant}. \tag{7.1}$$

If the flux surface under consideration has an angle $\theta_0 \neq 0$, then $B_p = B_r = \Phi_0 \sin^2\theta_0 / R^2$ where R is the cylindrical radius. As we mentioned in the paragraph below equation (2.9), once the distribution of the poloidal field strength B_p as a function of R is given, the flow speed along the flux surface is completely determined by the "energy" equation (2.9). To nondimensionalize this equation and others, we shall again scale lengths by the light cylinder radius $R_L \equiv c/\Omega$, the field strength by $4\pi kc$, and the 4-speed by the speed of light, as we did in § 4.2. That is, $x \equiv R/R_L$, $\mathbf{b} \equiv \mathbf{B}/(4\pi kc)$, and $(u, v) \equiv (V_p/c, V_\phi)/c$. Recall that k is a flux constant denoting the mass flux per

unit magnetic flux. Written in dimensionless form, the total specific energy (eq. [2.5]) and angular momentum (eq. [2.6]) then become

$$\mu \equiv \frac{e}{c^2} = \xi\gamma g_* - b_\phi x, \tag{7.2}$$

where $g_* = c_*^2/c^2 = 1 - GM_*/(rc^2) = 1 - (GM_*/R_L c^2)/(x/\sin\theta_0)$ is the modification due to gravity, and

$$\lambda \equiv \frac{I\Omega}{c^2} = (\xi v - b_\phi)x. \tag{7.3}$$

The Lorentz factor γ, toroidal speed v and field strength b_ϕ that appeared in the above equations are given by (from equations [2.6]-[2.8])

$$\gamma = \frac{1}{\xi g_*}\left[\mu - \frac{\lambda(g_* - \mu x^2/\lambda)}{g_* - \xi g_* u/b_p - x^2}\right], \tag{7.4}$$

$$v = \frac{\lambda}{\xi x}\left[1 - \frac{g_* - \mu x^2/\lambda}{g_* - \xi g_* u/b_p - x^2}\right], \tag{7.5}$$

$$b_\phi = -\frac{\lambda}{x}\left[\frac{g_* - \mu x^2/\lambda}{g_* - \xi g_* u/b_p - x^2}\right]. \tag{7.6}$$

Finally, if we define $\sigma \equiv (\Phi_0 \sin^2\theta_0)/(4\pi k c R_L^2) = b_p x^2$, then we rewrite the "energy" equation (2.9) as

$$\frac{1}{\xi^2 g_*^2}\left[\mu - \frac{\lambda(g_* - \mu x^2/\lambda)}{g_* - (\xi g_* u/\sigma + 1)x^2}\right]^2 = 1 + u^2 + \left\{\frac{\lambda}{\xi x}\left[1 - \frac{g_* - \mu x^2/\lambda}{g_* - (\xi g_* u/\sigma + 1)x^2}\right]\right\}^2 \tag{7.7}$$

The specific enthalpy ξ is related to the poloidal speed u by the equation of state and the conservation of mass flux. Therefore, the flow speed u will be completely determined if the energy μ and angular momentum λ can somehow be coupled to u. In the case of optically thin flow where the drag

force is given by equation (2.12), this is accomplished through equations (2.13) and (2.14), which in dimensionless form, read

$$\frac{d\mu}{dx} = -\gamma \frac{u^2 + v^2}{u} \nu_R(x), \tag{7.8}$$

$$\frac{d\lambda}{dx} = -\gamma^2 \frac{vx}{u} \nu_R(x), \tag{7.9}$$

where $\nu_R(x) \equiv \nu(S)/\sin\theta_0$ and $\nu(S)$ is the dimensionless drag coefficient defined in § 2.2. For radiation drag due to optically thin Thomson scattering of a locally isotropic radiation background, ν_R is a function of r along any given flux surface. However, if the radiation field is anisotropic, then the radiation force terms are more complicated (Phinney 1982). Since we wish to isolate the effects of radiation drag on a flow which is accelerated by hydromagnetic stresses, we will neglect the effects of anisotropy in the radiation field, thus disregarding the contribution of radiation force to accelerating the flow. We comment on the validity of this assumption in § 7.3.3. We have thus obtained all equations that are needed to compute the flow acceleration along a radial flux surface subjected to an external drag force. In the absence of the drag force, i.e. $\nu_R(x) = 0$, it is straightforward to recapture Michel's eqs. [20] and [A4]–[A6] for ideal radial winds. In the presence of any prescribed non-zero drag distribution, these equations have to be solved iteratively for the structure of the wind. Before doing this, we shall first discuss the effects of the drag force on the critical conditions at the various critical points.

7.2.2. Critical Points

To facilitate the expression of the critical conditions, we shall write the energy and angular momentum as $\mu = \mu_0 - \Delta\mu$ and $\lambda = \lambda_0 - \Delta\lambda$, where

μ_0 and λ_0 are initial energy and angular momentum at $x = x_0$ whereas $\Delta\mu$ and $\Delta\lambda$ are the energy and angular momentum dissipated between x_0 and x by the drag force. Since $\mu_0 = \mu + \Delta\mu$ is constant on each flux surface, it is possible to use a differential equation for the flow of the form

$$\frac{du}{dx} = \left.\frac{\partial u}{\partial x}\right|_{\mu_0} = -\frac{(\partial\mu_0/\partial x)_u}{(\partial\mu_0/\partial u)_x} \tag{7.10}$$

to study the behavior of the critical points (Kennel et al. 1983). Critical points occur where $(\partial\mu_0/\partial x)_u$ and $(\partial\mu_0/\partial u)_x$ vanish simultaneously. Formally there are three critical points, which occur where the radial flow speed matches the radial phase velocity of the three magnetosonic wave modes: slow, intermediate and fast. The significance of these modes and the associated critical points is discussed in depth in Kennel et al. (1983) and references therein; we will confine ourselves to discussing the effects of drag on the critical points.

Since the critical points reflect the local propagation of signals in the flow, they should depend on the local behavior of μ_0. For a wind with drag, this argument carries the important implication that

$$\Delta\mu = \int_{x_0}^{x} \gamma \frac{u^2 + v^2}{u} \nu_R(x) dx, \tag{7.11}$$

$$\Delta\lambda = \int_{x_0}^{x} \gamma^2 \frac{vx}{u} \nu_R(x) dx, \tag{7.12}$$

should be treated as explicit functions of x and not as functionals of $u(x)$. Treating μ as an explicit function of λ, x, and u, we may express the critical conditions in the form:

$$\left.\frac{\partial\mu}{\partial u}\right|_{\lambda,x} = 0 \tag{7.13a}$$

$$\left.\frac{\partial\mu}{\partial x}\right|_{\lambda,u} = \left.\frac{\partial\mu}{\partial\lambda}\right|_{x,u} \frac{d\Delta\lambda}{dx} - \frac{d\Delta\mu}{dx}, \tag{7.13b}$$

where $d\Delta\mu/dx$ and $d\Delta\lambda/dx$ are to be obtained easily from equations (7.8) and (7.9). The above derivative conditions for a critical point are equivalent to the condition that the energy equation (7.7) has double roots (Parker 1963; Michel 1969), and we will use the latter criterion where it is more convenient.

a) Slow Magnetosonic Point

In the discussion below, we shall assume that the temperature is always subrelativistic, $\int dp/\rho c^2 \ll 1$, and that $1 - g_* \ll 1$. The speed at the slow magnetosonic point must therefore be subrelativistic. Let us consider the case in which the slow point (x_s) and the intermediate point (x_i) are well-separated in radius, with $x_s \ll x_i$. Assuming that u increases and temperature declines with radius, u_i (the speed at the intermediate point) and x_i should be weakly affected by the finite temperature, and therefore should be close to the values given by Michel (1969) for a cold wind, i.e. $x_i \sim \sqrt{\lambda/\mu}$ and $u_i \sim \sigma(\mu - \lambda)/\lambda$. We therefore assume that the conditions at the slow point satisfy $x_s^2 \ll \lambda/\mu$ and $u_s \ll \sigma(\mu - \lambda)/\lambda$. Under these approximations, the expression for the specific energy reduces to

$$\mu_0 \approx \lambda_0 + g_* + \frac{u^2}{2} + \int^p \frac{dp}{\rho c^2} - \frac{x^2}{2} + \int_{x_*}^{x} \nu_R u dx, \qquad (7.14)$$

where x_* is the inner radius of the flow.

The thermal behavior of the gas will probably be affected by the radiation field. Since the slow magnetosonic point is expected to occur near the base of the flow where the radiation field is most intense, the gas could undergo strong radiative heating (corresponding to an effective polytropic index

< 1) or could equilibrate at the inverse Compton temperature (corresponding to an isothermal equation of state). For a discussion of nonmagnetized winds which are driven through the critical point by strong radiative heating, the reader is referred to Begelman, McKee & Shields (1983). In this section, we wish only to illustrate the effects of finite temperature on a wind which is driven primarily by centrifugal force, and we therefore adopt a polytropic equation of state, $p = c^2\kappa\rho^\Gamma$, with $\Gamma > 1$. The enthalpy term $\int dp/\rho c^2$ then becomes $\Gamma\kappa\rho^{\Gamma-1}/(\Gamma-1) \equiv s^2/(\Gamma-1)$, where s is the dimensionless sound speed. One of the critical conditions, $\partial\mu/\partial u|_{\lambda,x} = 0$ (eq. [7.13a]), yields

$$u_s^2 = s_s^2 = s_*^2\left(\frac{u_* x_*^2}{u_s x_s^2}\right)^{\Gamma-1}, \tag{7.15}$$

where the subscript "$*$" denotes boundary conditions at some (inner) radius and we have used $\rho \propto (x^2 u)^{-1}$. Writing the point mass potential as $g_* = 1 - x_g/x$ where $x_g \equiv GM\sin\theta_0/R_L c^2$, we obtain

$$\frac{x_g}{x_s^2} - \frac{2s_s^2}{x_s} - x_s + \nu_s u_s = 0 \tag{7.16}$$

for the auxiliary critical condition, where ν_s is the drag coefficient evaluated at x_s.

Equation (7.16) shows that the presence of drag allows a wind solution to pass through a slow magnetosonic point *even if gravity is negligible.* In the drag-free case the position of the critical point is fixed by the balance between gravity and a combination of centrifugal and pressure forces (GJ), and hence gravity is essential in order for the critical point to exist. The drag force acts on a wind in a way similar to the gravitational force, in the sense that it retards radial acceleration. For the weak-drag case ($\nu_s u_s \ll x_g/x_s^2$), the

additional drag force increases the radius of the critical point. For the strong-drag case ($\nu_s u_s \gg x_g/x_s^2$), we can derive expressions for x_s and u_s in the two limits described by GJ: pressure-dominated and centrifugally dominated. To express these relations most economically we introduce a parameter q, where $q = \nu_s/2$ in the pressure-dominated limit ($x_s^2 \ll s_s^2$) and $q = 1/\nu_s$ in the centrifugally-dominated limit ($x_s^2 \gg s_s^2$). We then have

$$x_s \approx q^{-\frac{\Gamma+1}{3\Gamma-1}} s_*^{\frac{2}{3\Gamma-1}} (u_* x_*^2)^{\frac{\Gamma-1}{3\Gamma-1}} \tag{7.17a}$$

$$u_s \approx q x_s. \tag{7.17b}$$

Note that ν_s may depend on x_s, in which case equation (7.17a) is implicit.

As in the drag-free case (GJ), the energy integral (7.14), together with the critical conditions (7.17a,b), impose two constraints on the flow parameters μ_0, λ_0, u_*, and s_*. An additional constraint is imposed by the condition that the flow pass through the fast magnetosonic point (see below). Let us examine how the constraints associated with the slow magnetosonic point are affected by the presence of drag. For simplicity, we assume that $x_s \gg x_g$ and $s_* \gg s_s = u_s \gg u_*$. By comparing the expression for μ_0 at the inner boundary and at the slow magnetosonic point, we obtain

$$\int_{x_*}^{x_s} \nu_R u dx \approx \frac{s_*^2}{\Gamma - 1} - \frac{x_g}{x_*} + \frac{x_s^2}{2} \ . \tag{7.18}$$

We may parameterize the integrated dissipation by $\int \nu_R u dx = \alpha \nu_s u_s x_s$, where $\alpha \sim O(1)$ depends on the detailed behavior of the drag coefficient and the flow field. For self-consistency we assume $s_*^2 \gg x_g/x_*$, since otherwise the critical point will be located too near x_*. Using the critical condition (7.17b), we obtain

$$x_s \approx \frac{s_*}{\beta}, \tag{7.19}$$

where $\beta^2 \equiv (\Gamma - 1)(1 - \alpha\nu_s^2)/2$ in the pressure-dominated limit and $\beta^2 \equiv (\Gamma - 1)(1 - 2\alpha)/2$ in the centrifugally-dominated limit. Equation (7.19) implies that a critical point will not occur if $\beta^2 < 0$, *i.e.*, if the dissipation is too strong. Equation (7.17*a*) can be used to eliminate x_s, with the result that

$$\frac{s_*^3}{u_* x_*^2} \approx \beta^{\frac{3\Gamma-1}{\Gamma-1}} q^{-\frac{\Gamma+1}{\Gamma-1}} . \tag{7.20}$$

b) Intermediate Critical Point

Let us assume that the intermediate critical point is far enough from the origin that the effect of gravity is negligible. After some manipulations we can rearrange the energy equation (7.7) into

$$\frac{\xi^2(1+u^2)}{\mu}\left[P(\sigma+\xi u)+Q\right]^2 - \left[\xi u P + Q\right]^2 + x^2 Q^2 = 0, \tag{7.21}$$

where $P \equiv \mu x^2 - \lambda$, and $Q \equiv \lambda\xi u - \sigma\Theta$ (where $\Theta \equiv \mu - \lambda$). This choice of variables makes it easy to apply the criterion (stated earlier) that a critical point occurs where equation (7.7) has double roots. Thus, when $P = 0$ ($x^2 = \lambda/\mu$), equation (7.21) becomes

$$\frac{Q^2}{\mu}\left[\xi^2(1+u^2) - \sigma\Theta\right]_{x=\sqrt{\lambda/\mu}} = 0, \tag{7.22a}$$

which contains a double root for u at $Q = 0$ ($\xi u = \Theta/\lambda$). When we set $Q = 0$, we have

$$\frac{P^2}{\mu}\left[\xi^2(1+u^2)(\sigma+\xi u)^2 - \mu\xi^2 u^2\right]_{u=\Theta/(\xi\lambda)} = 0, \tag{7.22b}$$

which contains a double root for x at $P = 0$. The simultaneous double roots in x and u occur automatically, *without the adjustment of any flow parameters*. This implies that the passage of the flow through the intermediate

critical point (often called the Alfvén point) does not impose any constraints on the flow. This is in contrast to the conditions at the slow and fast magnetosonic points, where it is necessary to constrain the flow parameters in order to obtain simultaneous double roots in x and u. Note that the denominator in equations (7.4)-(7.6) vanishes at the intermediate critical point, but the fact that $Q = P = 0$ implies that the numerators vanish as well. Thus, all three quantities γ, v and b_ϕ are finite at the critical point, as they should be. Finally, we point out that the drag affects the location of the intermediate critical point only indirectly, through the variation of λ and μ with radius.

c) Fast Magnetosonic Point

If the internal sound speed of the plasma is small compared to its terminal velocity, the fast magnetosonic point will be located well outside the intermediate critical point, $x_f \gg x_i$. Based on our results for the intermediate critical point, we adopt the approximations $\mu x^2 \gg \lambda$, $x^2(\xi u/\sigma + 1) \gg 1$, and again neglect gravity. Keeping terms to order $\xi - 1$ and $(u/\sigma+1)^{-1}x^{-2}$, we have

$$\mu \approx \frac{(1+u^2)^{1/2}(\sigma+u)}{u} + (1+u^2)^{1/2}\int \frac{dp}{\rho c^2} + \frac{1}{2u^3x^2}\left[\lambda u - \sigma(1+u^2)^{1/2}\right]\left[\frac{\lambda u(\sigma+u)}{(1+u^2)^{1/2}} - \sigma(\sigma+3u)\right]. \quad (7.23)$$

For computational simplicity, we assume an isothermal equation of state with constant sound speed a (as in GJ), so that

$$\int_{x_*}^{x} \frac{dp}{\rho c^2} = -a^2 \ln\left(\frac{ux^2}{u_* x_*^2}\right). \quad (7.24)$$

Recall that the critical condition $\partial\mu/\partial u|_{\lambda,x} = 0$ does not depend on the drag force in the vicinity of the critical point. Therefore, to lowest order we obtain the standard result

$$u_f = \sigma^{1/3} + O[a^2, (\sigma^{-2/3} + 1)^{-1} x_f^{-2}] \tag{7.25}$$

(Michel 1969; GJ). To calculate the first-order terms, we consider the subrelativistic ($u, \sigma \ll 1$) and ultrarelativistic ($u, \sigma \gg 1$) limits separately. Expanding the velocity-dependent terms in (7.23) and keeping only terms to first order with $\sigma \sim O(u^3)$, we obtain

$$\mu \approx 1 + \frac{\sigma}{u} + \frac{u^2}{2} - a^2 \ln\left(\frac{ux^2}{u_* x_*^2}\right) + \frac{1}{2u^2x^2}(\lambda u - \sigma)(\lambda u - 3\sigma), \tag{7.26}$$

in the subrelativistic limit, where we have assumed that $\lambda u_f/\sigma \sim O(1)$ (GJ). In the ultrarelativistic limit, we have

$$\mu \approx \sigma + u + \frac{\sigma}{2u^2} - ua^2 \ln\left(\frac{ux^2}{u_* x_*^2}\right) + \frac{1}{2u^3x^2}\left[(\lambda - \sigma)u - \frac{\sigma}{2u}\right]\left[(\lambda - \sigma)\sigma + (\lambda - 3\sigma)u - \frac{\lambda\sigma}{2u^2}\right], \tag{7.27}$$

where we have assumed $\lambda/\sigma - 1 \sim O(u^{-2})$ (GJ).

Although it is straightforward to solve the critical condition $\partial\mu/\partial u|_{\lambda,x} = 0$ to first order, it is sufficient for our purposes to adopt the zeroth-order expression for u_f (eq. [7.25]). The derivatives $\partial\mu/\partial x$ are also easily computed, but instead of vanishing at the critical point they must be set equal to the terms on the right-hand-side of equation (7.13*b*). To lowest order, the drag derivatives are given by $d\Delta\mu/dx \approx \nu_R u$, $d\Delta\lambda/dx \approx \nu_R(\lambda u - \sigma)/u^2$ in the subrelativistic limit; and $d\Delta\mu/dx \approx \nu_R u^2$, $d\Delta\lambda/dx \approx \nu_R u(\lambda - \sigma)$ in the ultrarelativistic limit. One can show that only

the $d\Delta\mu/dx$ term contributes to the right-hand-side of equation (7.13*b*) to lowest order. The critical condition can be cast as a cubic equation for x_f,

$$\nu_f x_f^3 - \frac{2a^2 x_f^2}{\sigma^{1/3}} + Y = 0, \tag{7.28}$$

where $Y \equiv \sigma^{-1/3}(\lambda - \sigma^{2/3})(3\sigma^{2/3} - \lambda)$ in the subrelativistic limit and $Y \equiv \frac{1}{4}\left[2(\lambda - \sigma) - \sigma^{1/3}\right]\left[6 - \lambda/\sigma - 2(\lambda - \sigma)\sigma^{-1/3}\right]$ in the ultrarelativistic limit. The quantity ν_f is the local value of the drag coefficient at the fast magnetosonic point. In the limit $\nu_f \to 0$ we recover simple quadratic equations for x_f which agree with the results of GJ in the subrelativistic limit (GJ eq. [32]). Formally, our results appear to be different from those of GJ in the ultrarelativistic limit because, in their case, both μ and λ are conserved quantities, and can be related to each other uniquely through $\mu = 1 + \lambda$ if the flow starts from rest at the origin. However, in the presence of drag, this relation can hold only for μ_0 and λ_0, not for μ and λ. In the limit that drag is negligible *everywhere*, we do recover GJ eq. [25], and our result for x_f agrees with GJ eq. [23].

Before deriving approximate solutions to equation (7.28), we must make an important distinction between the *local* and the *global* effects of drag on the outer critical conditions and on the constants of the flow. Clearly the presence of local drag at x_f has an important effect on the location of the critical point if $\nu_f \gtrsim a^2/(\sigma^{1/3}x_f)$. However, the cumulative effect of drag throughout the flow can have an important effect on the constraint imposed by energy conservation, even if $\nu_f = 0$. The energy conservation constraint leads to an equation analogous to equation (7.18), which applied at the slow magnetosonic point. Denote the energy and the integrated drag terms for energy and angular momentum evaluated at x_f by μ_f, $(\Delta\mu)_f$ and $(\Delta\lambda)_f$,

respectively, and let $\Delta_f \equiv (\Delta\mu)_f - (\Delta\lambda)_f$. To lowest order, equation (7.26) gives $\mu_f \approx 1 + \frac{3}{2}\sigma^{2/3}$ for the subrelativistic limit, while equation (7.27) gives $\mu_f \approx \sigma + \frac{3}{2}\sigma^{1/3}$ for the ultrarelativistic limit. Substituting μ_f and $\mu_0 = 1 + \lambda_0$ into the energy conservation equation (2.6) and recalling that $\lambda_f = \lambda_0 - (\Delta\lambda)_f$, we find

$$\lambda_f \approx \frac{3}{2}\sigma^{2/3} + \Delta_f \tag{7.29}$$

in the subrelativistic limit; and

$$\lambda_f \approx \sigma + \frac{3}{2}\sigma^{1/3} + \Delta_f \tag{7.30}$$

in the ultrarelativistic limit. Therefore, the quantity Y can be reexpressed in terms of Δ_f, and becomes $Y = \frac{1}{4}(1\sigma^{2/3} + 2\Delta_f)(3\sigma^{2/3} - 2\Delta_f)\sigma^{-1/3}$ in the subrelativistic limit, and $Y = \frac{1}{2}\sigma^{-1/3}(\sigma^{2/3} - \Delta_f^2)$ in the ultrarelativistic limit. The value of Δ_f must be computed self-consistently by solving the flow equation with a specific form for the drag.

Let us now solve for x_f in the presence of drag of a special form. As will be discussed in more detail in § 7.3.3, it is reasonable to assume that the drag coefficient declines at least as fast as $\nu_R(x) \sim \nu_0/x^2$ at a large distance. Furthermore, if we assume that most of the dissipation occurs well inside the fast point so that the location of the fast point does not affect Δ_f much, i.e., Y is more or less independent of x_f, then we can easily solve for x_f from equation (7.28):

$$x_f \cong \sigma^{1/3}\frac{\nu_0 \pm \sqrt{\nu_0^2 + 8a^2 Y/\sigma^{1/3}}}{4a^2}. \tag{7.31}$$

It is interesting to notice that, if $Y < 0$, two fast points are possible. In the limit of zero temperature, one fast point is ejected to infinity, while the

other remains at a finite distance. However, as will be demonstrated later in § 7.3.3, the quantity Δ_f is always small enough so that Y can never be negative, regardless of the strength of the drag. We therefore demand that the minus sign in equation (7.31) be rejected, hence there is only one fast magnetosonic point. In the limit of zero drag and finite temperature, equation (7.31) reduces to the ideal result of GJ. If the temperature goes to zero, the fast point will be ejected to infinity, as in Michel's minimum-torque solution, even in the presence of the drag. It should be pointed out that double fast points can exist for flux tubes that are nonradial, as shown explicitly in the next chapter.

In summary, we have shown, from analyses of the critical points, that: 1) the critical conditions at the slow magnetosonic point impose two conditions on the four parameters μ_0, λ_0, u_* and s_*; 2) the critical conditions at the Alfvén point are automatically satisfied; and 3) the critical conditions at the fast point will not only fix its location, but will also impose one more constraint on the above four parameters. In the limit of low temperature, the total specific energy as well as the velocity at the fast critical point are hardly influenced by the presence of the drag force, although the specific angular momentum output may be greatly enhanced.

In the following section, we shall solve self-consistently for the structure of the flow in the presence of drag of some prescribed form. For the sake of simplicity, only zero-temperature flows in the absence of gravity are considered.

7.3. Dynamics of Cold, Dragged Radial Winds

The presence of a large drag force can modify significantly the acceleration of a hydromagnetic wind. Modifications arise not only from the direct deceleration of the flow by the drag force, but also from the increased toroidal bending of the magnetic field lines, since the drag can be perceived as increasing the effective inertia of the fluid. The enhanced twisting of the magnetic field when the drag force prevails increases the magnetic energy at the expense of the kinetic energy. This stored energy is eventually released at large distances so as to yield further acceleration of the fluid. A primary effect of drag on the acceleration of the flow is thus to redistribute the locations at which the Poynting flux is converted into kinetic energy. This physical picture is highlighted by the fact shown earlier that the maximum amount of energy converted into kinetic energy is the same for cold flows with drag as it is for drag-free flows.

In § 7.3.1 we consider the situation in which the drag force is small enough to permit an analytical perturbative treatment. In § 7.3.2 we examine the large-drag limit, using a combination of analytic and numerical techniques. § 7.3.3 treats the solution topology of flows in the presence of a drag force with a "realistic" radial dependence. Finally, in § 7.3.4 we estimate the energy dissipated in potentially ultrarelativistic flows (i.e., with large σ) subject to strong drag, and comment on the detectability of the scattered radiation.

7.3.1. Weak-Drag Limit: A Perturbative Approach

In this section we consider a small and localized drag force in an otherwise dissipationless fluid and analyze the linear response of the flow to such a perturbation. The local dissipation due to the drag force $\mathbf{F_D}$ is

$$\Delta\mu = u^2\epsilon_r + vu\epsilon_\phi, \quad \text{and} \quad \Delta\lambda = \gamma u x \epsilon_\phi, \tag{7.32}$$

where the two newly-defined quantities

$$\epsilon_r \equiv -\frac{1}{u^2c^2}\int_{r-\Delta r}^{r} F_{Dr}dr \quad \text{and} \quad \epsilon_\phi \equiv -\frac{1}{u^2c^2}\int_{r-\Delta r}^{r} F_{D\phi}dr \tag{7.33}$$

($\ll 1$) measure the relative amount of work done by the two components of the drag force compared to that done by other forces. For drag of the form given by equation (2.12), acting on a wind, both ϵ_r and ϵ_ϕ are positive. At any given point in the flow, the local specific energy and the angular momentum are related to the local radial velocity by equations (7.2) and (7.3). A straightforward but laborious calculation shows that the response of the radial velocity to localized drag is given by

$$\frac{\Delta u}{u} = \frac{u^3}{\sigma - u^3}\left[(\gamma - \frac{1}{M_A^2})\epsilon_r + \frac{|b_\phi|}{u}\epsilon_\phi\right], \tag{7.34}$$

where M_A ($\equiv x\sqrt{u/\sigma}$) is the Alfvén Mach number (Michel 1969) and b_ϕ is the dimensionless toroidal field strength given by equation (7.6). In equation (7.34) we have taken $\Theta \equiv \mu - \sigma\lambda = 1$, as is the case for a drag-free wind emanating from the origin $x = 0$. In the numerator, the term multiplying ϵ_r switches sign from negative to positive where $M_A^2 = \gamma^{-1}$, i.e., at a point which lies inside the Alfvén radius ($M_A^2 = 1$). The term multiplying ϵ_ϕ is always positive, with a magnitude much smaller than the previous term in

regions $x \ll (1 + u/\sigma)^{-1/2}$ and $x \gg (1 + u/\sigma)^{-1/2}$. Since the fast magnetosonic point in a cold flow (where $u \to \sigma^{1/3}$) lies at $x \to \infty$, the denominator in equation (7.34) is always positive and we conclude that Δu changes sign from a negative value at small radii to a positive value at large radii. The transition occurs inside the radius at which $M_A^2 = \gamma^{-1}$.

Using equation (7.5), we can also determine how localized weak dissipation affects the azimuthal speed v. However, the expression is much more complicated than equation (7.34) and is not so straightforward to interpret.

This model, despite its simplicity, reveals an important property of a dragged hydromagnetic flow: instead of slowing down the flow as one may have anticipated, drag can actually cause the flow to accelerate, in both the radial and the azimuthal directions. In the context of localized dissipation, we can understand the acceleration of the flow across the discontinuity by its analogy with the inverse process of a fluid parcel crossing an MHD shock front in which the magnetic field is strongly inclined to the shock normal (a "quasi-perpendicular" shock). A normally incident parcel of fluid is both slowed down and deflected when passing from the low field side to the high field side of a quasi-perpendicular shock. The parcel's velocity normal to the shock decreases because it must overcome the magnetic pressure gradient, and it is deflected away from the shock normal because of the acceleration by magnetic tension, which arises from the kink of the field line at the shock. In the wind, the drag causes the magnetic field to be wound up to a greater degree inside the drag radius, and an outgoing fluid parcel can be accelerated radially at the discontinuity by the local pressure gradient. The fluid parcel can also be deflected by the tension force resulting from the local bending of the field lines.

Elementary geometrical considerations show why drag leads to deceleration at small radii and acceleration at large radii. At small radii, the magnetic field lines are nearly radial and stiff. The radial drag force acts nearly parallel to the field lines and simply decreases u; the slight bending of the field lines causes the fluid to lag behind rigid rotation, decreasing v. The extra bending of the magnetic field caused by drag does not enhance the magnetic force sufficiently to overcome the direct deceleration of the fluid by drag. The situation is quite different at large radii, where the magnetic field lines are strongly curved (Figure 7.1). At such radii the rotation of the fluid lags substantially behind that of the magnetic field, and the forces exerted by the drag cause the magnetic field to kink as shown in Fig. 7.1. In a frame corotating with the magnetic field in Fig. 7.1, fluid parcels slide along field lines in a clockwise direction; they are accelerated outward by the component of centrifugal force parallel to the field line. As they pass the kink in the field and change direction, the radial speed increases at the expense of the azimuthal speed (as measured in the corotating frame). In the inertial frame, both the radial and azimuthal velocities increase.

One can also interpret the acceleration process in energetic terms. Although the drag removes net energy from the flow, it removes different amounts of kinetic energy and magnetic energy. When the kinetic energy increases in a wind subject to drag, the magnetic energy is actually decreasing by a larger amount. In other words, drag can catalyze the conversion of Poynting flux into kinetic energy.

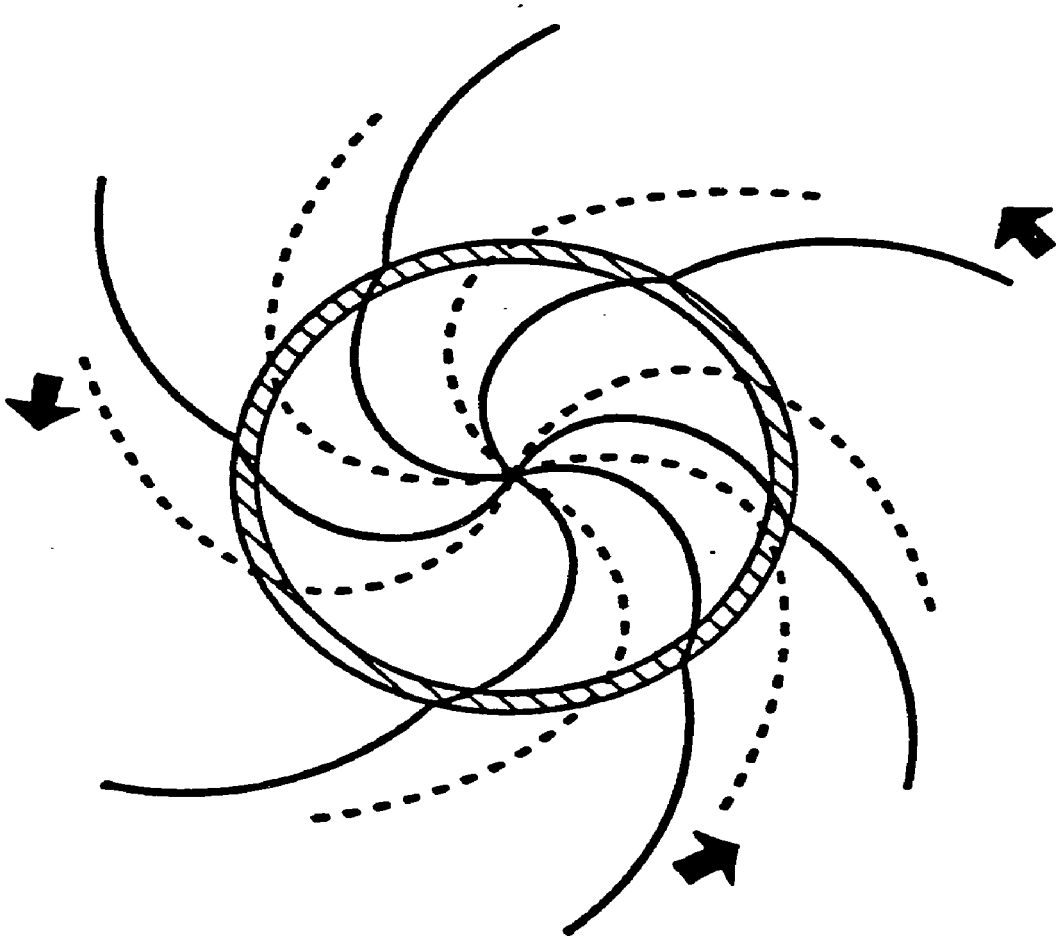


Fig. 7.1–Schematic view of the magnetic field configuration in the presence of a small localized drag force. Dashed lines indicate shape of field lines in the absence of drag. The large back arrows indicate the the sense of the central object's rotation.

7.3.2. Strong-Drag Limit

To learn whether the characteristics of weakly dragged flows persist in the strong-drag limit, it is necessary to integrate equations (7.8) and (7.9) numerically. First, however, we shall present an approximate analysis. The radial component of the momentum equation can be written in the form

$$\frac{du}{dx} = \frac{v^2}{ux} - \frac{d}{dx}\left(\frac{x^2 b_\phi^2}{2\sigma}\right) - \nu_R(x)\gamma^2. \tag{7.35}$$

The first term on the right is the centrifugal force, the second term is a combination of the magnetic pressure and the magnetic tension, and the third is the drag.

At small distances $x \ll (1 + u/\sigma)^{-1/2}$, the magnetic field lines are approximately radial and the fluid motion is dominated by corotation. The radial acceleration is due to the centrifugal force, acting against the drag. Using the assumption of corotation to evaluate v, we can integrate eq. (7.35) once to obtain

$$u = \left[x^2 - 2\int_0^x \nu_R(x) u dx\right]^{1/2} = \left[(1 + \frac{\nu_0^2}{4})^{1/2} - \frac{\nu_0}{2}\right] x + O(x^2) \tag{7.36}$$

where we have taken $\gamma \simeq 1$ and $\nu_R(x) \simeq \nu_0 \equiv \nu_R(0)$ to evaluate the integral. Compared to the ideal case ($u \cong x$), the rise of radial speed is indeed inhibited by the drag near the origin.

On the other hand, at large distances $x \gg (1 + u/\sigma)^{-1/2}$ we have $u \gg v$. The magnetic field lines are tightly wound and it is predominantly the magnetic force that acts against the friction. Since $x \gg (1+u/\sigma)^{-1/2}$, we can immediately write down, from equation (7.6), that $b_\phi \simeq -\mu\sigma/[x(\sigma+u)]$. Substituting $\gamma \simeq \mu u/(\sigma + u)$ to eliminate μ and using the fact that $u \gg v$,

hence $\gamma \simeq (1+u^2)^{1/2}$, we obtain $b_\phi \simeq -\sigma^{1/2}(1+u^2)^{1/2}/ux$. It follows that equation (7.35) becomes

$$(1-\frac{\sigma}{u^3})\frac{du}{dx} = \frac{v^2}{ux} - \nu_R(x)\gamma^2 + O(x^{-3}) \tag{7.37}$$

where the second term on the left side arises from the magnetic force. At large distances, if the drag coefficient $\nu_R(x)$ declines more slowly than x^{-3}, then the centrifugal force (the first term on the right) and the $O(x^{-3})$ correction term are negligible compared to the drag force. The zero of the coefficient of du/dx corresponds to the fast critical point discussed in § 7.2. Since this coefficient is negative over the entire flow, except at ∞, we observe that $du/dx > 0$ and the flow is accelerating. We also note that it is really the magnetic force that overcomes the drag to accelerate the flow. Equation (7.37) can be integrated once to yield

$$Z[u(x)] - Z[u(x_2)] \simeq -\int_{x_2}^{x} \nu_R(x)dx + O(x^{-2}) \tag{7.38}$$

in the case that the centrifugal force is negligible, where

$$Z(u) \equiv \frac{\sigma}{2}\left[\frac{1}{u^2} - \ln(1+\frac{1}{u^2})\right] + \tan^{-1}u, \tag{7.39}$$

and x_2 is the outer radius of the dragged region.

In the following numerical calculations aimed at illustrating the physics of strongly dragged flows, we have avoided the complications arising from the spatial variation of the drag; we find it sufficient to consider a "top hat" drag profile, i.e., $\nu_R(x) = const.$ in some region $x_1 \leq x \leq x_2$ and zero elsewhere. The preceding linear analysis shows that the Alfvén point is approximately the place at which the response of the flow to drag

changes character, therefore we let $x_1 \leq x_A \leq x_2$ in an attempt to capture the features of the transition.

We numerically integrate equations (7.8) and (7.9) as a set of two coupled nonlinear differential equations. The right-hand sides of both equations are expressed in terms of μ, λ, u and x by using equations (7.4)-(7.6), and with the help of the energy equation (7.7), we can solve for u in terms of μ, λ and x at each integration step. As in the ideal problem, the Alfvén critical point is not a singular point of this set of differential equations; in fact, there is no singular point in the differential equations and we are always able to obtain smooth solutions. The boundary conditions are $\mu = 1 + \lambda$ at $x = x_1$ and $\mu = \mu_\infty$ at $x = x_2$. A fourth-order Runge-Kutta scheme has been adopted for the integration, and the results are as follows.

Figures 7.2*a* and 7.2*b* are plots of azimuthal and radial 4-velocity, v and u, as functions of dimensionless radius x. The solid line is for the wind subject to drag. Accompanying the solid line are two reference curves, which correspond to ideal flows with energies equal to the initial energy (dot-dashed line) and the final energy (dashed line) of the dragged flow. Compared with the dot-dashed line, the dragged flow indeed has more terminal kinetic energy than the ideal flow of the same initial energy. We also note that most of the acceleration occurs near the outer edge of the drag profile. Plotted in Figure 7.3 are the different forces in equation (7.35), illustrating how the magnetic force quickly overcomes the drag and accelerates the flow.

For Figures 7.2 and 7.3 we have chosen $\sigma = 10$, a drag coefficient $\nu_R = 5$, and a radial range of the drag profile given by $x_1 = 0.481$, $x_2 = 2.89$. The Alfvén point is located at $x_A = 0.968$. Fig. 7.2*a* shows that the azimuthal speed increases as soon as the drag force appears. The radial velocity

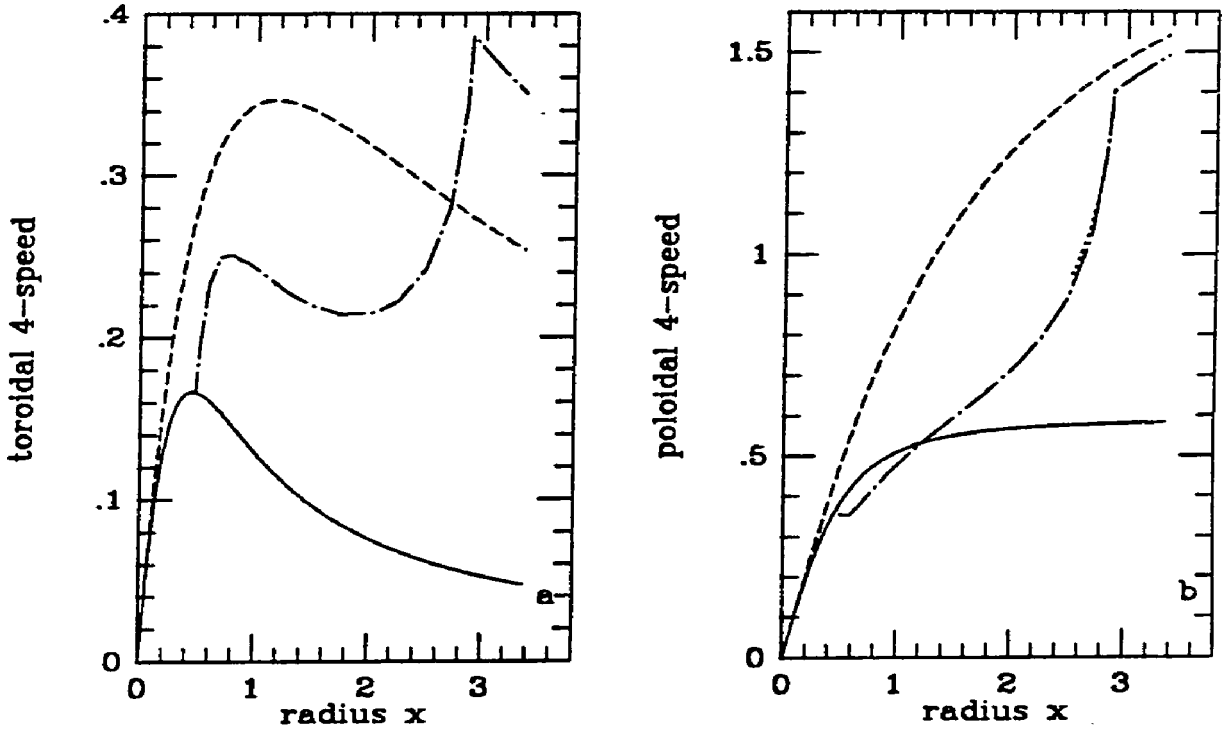


Fig. 7.2–(a) Azimuthal and (b) radial components of the dimensionless 4-speed are plotted against the dimensionless radius x. The dragged solution (solid) has an initial energy $\mu_0 = 17.89$, and a final energy $\mu_\infty = 13.40$. The two drag-free solutions (dot-dashed and dashed) have energy of μ_0 and μ_∞, respectively. The dotted line in (b) is the analytic approximation based on eq. (7.38).

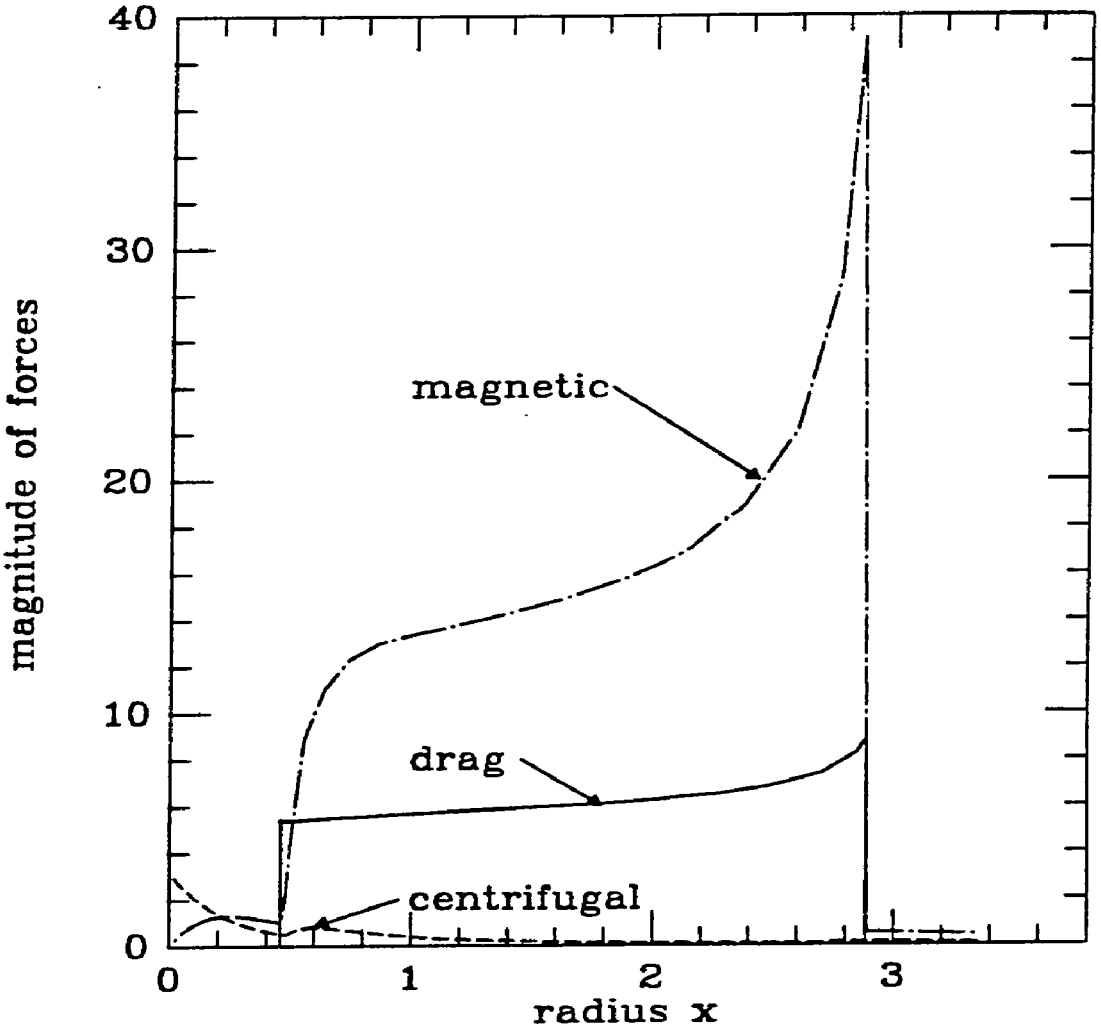


Fig. 7.3–Relative magnitudes of the different forces that are present in a flow with a drag coefficient of step functional form, as calculated from the right-hand side of eq. (7.35). It is the magnetic force that overcomes the drag force and yields flow acceleration.

(Fig. 7.2*b*) has a brief phase of deceleration near x_1, but begins to accelerate before reaching the Alfvén point. After the flow exits into the drag-free zone, the azimuthal velocity always exceeds that of the ideal flow with the same specific energy (dashed line), which is opposite to the situation for the radial velocity. The excess specific angular momentum carried by the magnetic field in the dragged flow must account for this difference. Asymptotically, the radial component of the velocity will approach the dashed line since it gradually becomes independent of the the specific angular momentum and dependent only on σ and μ, as can be shown easily by solving equation (7.7) in the limit $x \rightarrow \infty$ (and as Michel [1969] found). The azimuthal velocity will not approach the corresponding dashed curve, because the specific angular momentum at ∞ is always larger than that of the drag-free solution with the same energy.

The more distant the outer edge (x_2) of the drag profile, the more abrupt the acceleration is just *inside* x_2. Also plotted in Fig. 2*b* is the analytic solution given by equation (7.38) (line of filled circles), to compare with the numerical result. Why is it that the acceleration of the flow is so sudden? It is understandable that under a drag force the flow cannot be greatly accelerated, but why does the flow not pick up the speed in the dissipationless region, rather than doing so almost entirely in the dragged region?

The reason that large acceleration must take place under the influence of the drag force has something to do with the stationarity of the flow and the force balance. Without the drag force, it is difficult to sustain a large magnetic field gradient locally and at the same time to be in a stationary state. As shown in equation (7.37), it is the latter requirement that leads

the magnetic force to be almost exactly proportional to the flow acceleration, which hence becomes large in the dragged region because the field gradient is large there. The suddenness of the acceleration can be explained by the fact that this configuration tends to minimize the dissipation, as the region over which the flow has large velocity is narrow and thus the cumulative loss is small. We recall that it is a minimum-torque solution that maximizes the output kinetic energy for a given σ. (The latter point will be illustrated by Fig. 7.4 in § 7.3.3.)

7.3.3. Solution Topology

In this section and the next, we examine the global structure of winds subject to a radially variable drag coefficient. For an optically thin flow in an isotropic radiation field, the dimensionless radiation drag coefficient ν_R is given by

$$\nu_R = \frac{4}{3}\frac{\sigma_T R_L U_{rad}}{mc^2}, \tag{7.40}$$

where U_{rad} is the ambient radiation energy density, m is the mean mass per electron in the wind, σ_T is the Thomson cross section, and R_L is the radius of the light cylinder radius. Near an accreting compact object, isotropy of the radiation field is likely to be a good approximation only at radii $x < x_{ch} = r_{ch}\sin\theta_0/R_L$ ($\sim [10-20] \times GM_*/R_Lc^2$ for an accreting black hole), within which most of the radiation is generated. As the radiation field becomes increasingly anisotropic at larger radii, the radiation force acting on particles in the wind becomes more complicated (Phinney 1982). However, for winds moving with relativistic speed, a simple drag approximation can suffice even where the radiation field is highly anisotropic. Even if *all* of the radiation were emitted within a radius r_{ch}, the drag force would dominate

over the acceleration force at radii where $r < \gamma r_{ch}$, because of the relativistic aberration of light. However, in this case one can show (Appendix B) that the drag coefficient does not vary with the mean radiation density (which decreases $\propto r^{-2}$) but rather declines much more rapidly ($\propto r^{-6}$). If the flow involves an accretion disk, there will be an approximately isotropic component of the radiation field at all radii, declining $\propto r^{-3}$ at $r \gg r_{ch}$. Equation (7.40) for the drag coefficient is then valid provided that U_{rad} is taken to be the isotropic part of the radiation density. Drag then dominates over radiative acceleration out to radii $\sim \gamma^4 r_{ch}$, i.e., quite far from the central object if the flow is moderately relativistic ($\gamma \gtrsim 3$). If radiation from the central region is intercepted and reprocessed by the disk, or is scattered by diffuse matter above the disk, then the rate of decline of the isotropic component may be as shallow as $\propto r^{-2}$, and the relative importance of drag correspondingly greater.

In this section we wish to illustrate the effects of a slowly declining drag coefficient on the structure of the solution curves, and we therefore adopt the spatial dependence

$$\nu_R(x) = \frac{\nu_0}{1 + (x/x_{ch})^2}. \tag{7.41}$$

In Figure 7.4*a* we present the numerical solution for the case of isotropic drag given by equation (7.41). We repeated our calculations for several anisotropic drag prescriptions and found that no major change occurs in the flow characteristics. For the purpose of comparison we also plot the solution of an ideal flow (Fig. 7.4*b*). In order to observe the asymptotic behavior of solutions more closely, we replace x by $t = x/(1 + x)$, which ranges from 0 to 1 as x ranges from 0 to ∞. Figures 7.4*a* and 7.4*b* are both velocity-radius

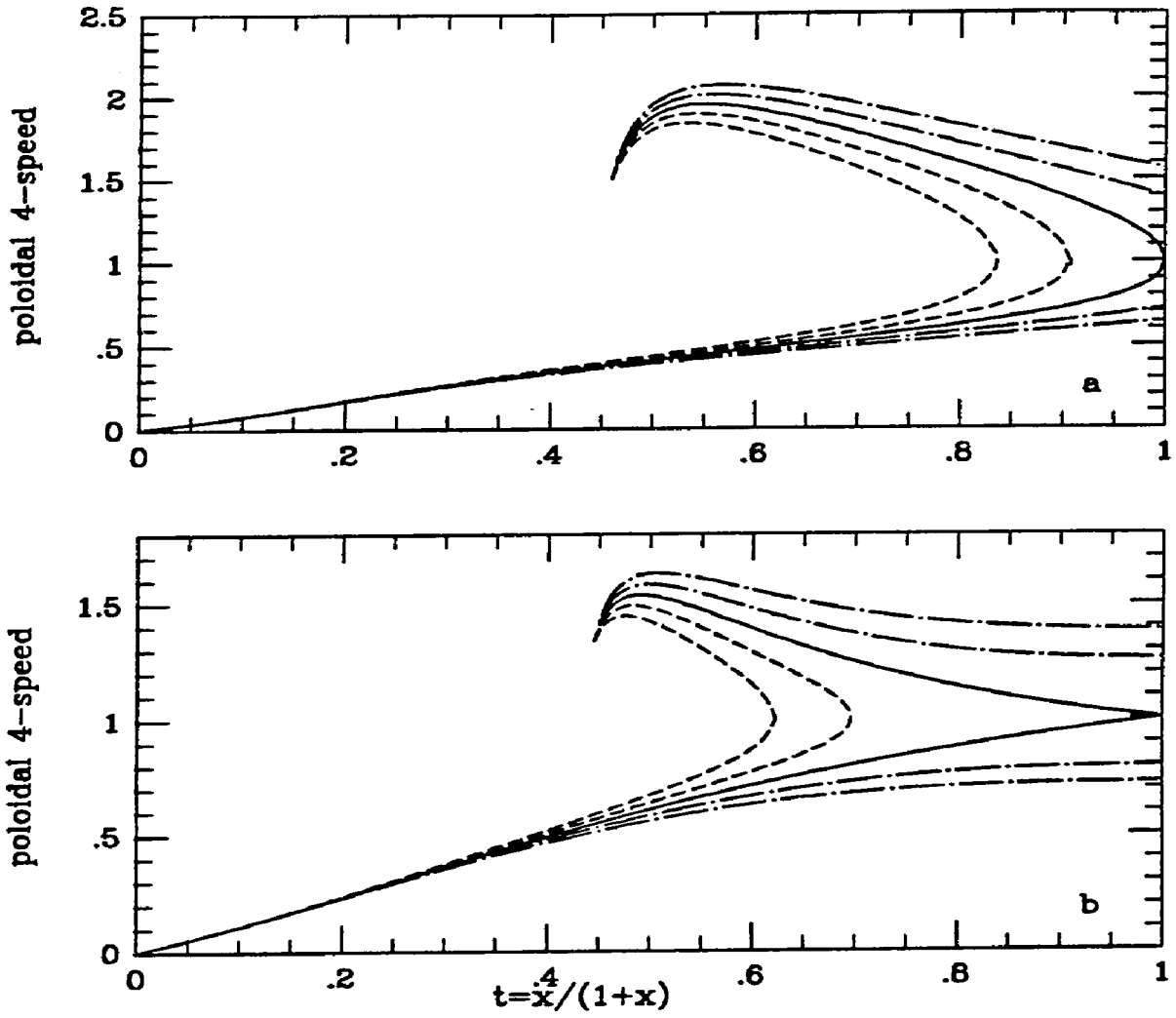


Fig. 7.4–Solution topologies of (a) dragged and (b) ideal flows. Plotted are the curves of radial 4-speed for different initial energy μ_0. Tha radial dependence of the drag coefficient in (a) is given by eq. (7.41). The lower branch of the solid line in each panel is the critical solution with $\mu_0 = \mu_c$ that connects the origin to the infinity. Solutions with $\mu_0 < \mu_c$ (dashed) become singular at finite radius. Those with $\mu_0 > \mu_c$ (chain-dotted) can reach infinity, but they have lower terminal speeds than the critical solution. We use $t = x/(1+x)$ as the abscissa, which varies between 0 and 1 as x varies between 0 and ∞.

plots and σ is chosen to be 1.0 for all solutions. Curves in the plots represent flows emerging from the origin with zero velocity; they have different initial Poynting fluxes and hence different initial energies. In both plots the curves turn back when the initial energy is not sufficiently large, yielding unphysical solutions since $u(x)$ becomes double-valued. As the initial energy increases and approaches a critical value, both plots show that the distance at which the solution turns back moves rapidly to infinity. This critical energy for an ideal flow is precisely the energy of the minimum-torque solution of Michel (1969). The *terminal* energy and velocity of the critical solution for the dragged flow are identical to those for the ideal solution, since these quantities depend only on the magnetization parameter σ. Of course, the *initial* energy of the critical dragged flow is larger than that of the critical ideal flow, the difference equalling the dissipated energy. For flows of greater initial energy than the critical value, only the lower branch of the solutions, which connects to the origin, is the physical one. For these solutions the terminal velocities (and kinetic energies) have smaller values in both the ideal and the frictional cases, indicative of more energy being stored in the form of Poynting flux. As usual, the minimum-torque solution has the maximal kinetic energy output for a given σ.

There are, however, differences between the dragged solutions and the ideal ones. In particular, the manner in which the solutions approach ∞ are distinctly different; this can be understood from equation (7.37). Far from the origin, the drag force in our model decreases $\propto x^{-2}$ whereas the centrifugal force declines $\propto x^{-3}$. The dominance of the former explains the discrepancy between the asymptotes of the dragged and ideal solutions. Note that this difference will go away if the drag coefficient decreases with x more

rapidly than x^{-3}. Otherwise, we observe from Figs. 7.4*a* and 7.4*b* that the drag force does not cause much change in the solution topology: the fast critical point cannot be reached in the dragged flows, any more than it can be reached in cold ideal flows.

In § 7.2.2, we speculated on the possibility that the flow could be driven through a fast critical point by the presence of a large drag. We claimed that this possibility was elusive. This is because a new critical point occurs only if Δ_f is sufficiently positive that it yields a negative value of Y (see eqs. [7.28], [7.29], and the expressions for Y). However, this condition cannot be satisfied in reality because there exists an upper bound for Δ_f, even in the limit of large ν. The critical-point analysis presented in § 7.2 does not reveal this fact because it is a local analysis: this conclusion can be obtained only from a global analysis of the flow, as we now show.

From eqs. (2.13*a*) and (2.13*b*), we can construct a differential equation for $\Delta \equiv \Delta\mu - \Delta\lambda$:

$$\frac{d}{dx}\Delta \equiv \frac{d}{dx}\Delta\mu - \frac{d}{dx}\Delta\lambda = \nu_R\gamma[u + b_\phi v x^2/\sigma]. \tag{7.42}$$

Eliminating b_ϕ and v in favor of γ, μ and λ and using the result that $\Delta = 1 + \lambda - \mu$, we obtain

$$\frac{d\Delta}{dx} = \nu[f(x) - g(x)\Delta], \tag{7.43}$$

where

$$f(x) \equiv \frac{\gamma}{\sigma^2}[\sigma u - (\mu - \gamma)(\gamma - 1)] \tag{7.44a}$$

$$g(x) \equiv \frac{\gamma}{\sigma}(\mu - \gamma). \tag{7.44b}$$

It is obvious that $g(x)$ is positive definite since $\mu > \gamma$. Having realized this we can make use of the Green's function method to solve for Δ, with $f(x)$

being the source. The details of the analysis are given in Appendix C. In the limit of strong drag, the solution is given to first order in $1/\nu_R$ by

$$\Delta = \frac{f(x)}{g(x)}. \tag{7.45}$$

We shall now use equation (4.45) to show that Y is a positive quantity. For large x in the ultrarelativistic limit, we have $\gamma \sim u \lesssim \sigma^{1/3}$, $\mu \sim \sigma$, and find that $\Delta_f \sim \sigma^{-1}$, which is $\ll \sigma^{-2/3}$ and is therefore too small to change the sign of Y. In the subrelativistic limit we find that $\Delta_f \to u_f^2/2\sigma < \sigma^{-1/3}/2$, implying that Y cannot become negative. Our numerical calculations also indicate that Y is always positive and therefore that no new critical points should arise.

7.3.4. Total Dissipated Energy and Scattered Radiation

The amount of dissipation and the intensity of the scattered radiation are strong functions of the Lorentz factor of the flow. Particularly for an optically thin flow, we expect the scattered radiation to stand out from the background only if the flow is relativistic. In this section, therefore, we shall consider only flows that are *potentially* highly relativistic, i.e., with $\sigma \gg 1$. We emphasize the word *potentially* because the flow can remain subrelativistic if the drag is too strong, even if σ is large. Since relativistic flow is possible only beyond the corotation region, we will assume that $x > 1$ (outside the light cylinder) and adopt a drag coefficient of the form

$$\nu_R(x) = \frac{\nu_0}{(1 + x/x_{ch})^{\alpha+1}}\ , \tag{7.46}$$

with $0 < \alpha < 2$. The velocity field is then given by equations (7.38) and (7.39). In the ultrarelativistic limit ($u \gg 1$), we find

$$u \approx \sigma^{1/3} \left[1 + \frac{4\eta\sigma^{1/3}}{\alpha(1+y)^{\alpha}}\right]^{-1/4} , \tag{7.47}$$

where $\eta \equiv \nu_0 x_{ch}$ and $y \equiv x/x_{ch}$. Clearly this limiting case is self-consistent only if $\eta(1+y)^{-\alpha} \ll \sigma$. If this condition is violated, then the flow will be subrelativistic, with

$$u \approx \left(\frac{\alpha\sigma}{2\eta}\right)^{1/2} (1+y)^{\alpha/2} . \tag{7.48}$$

The condition for "strong drag" is that the velocity in equation (7.47) be significantly smaller than $\sigma^{1/3}$, which requires $\eta(1+y)^{-\alpha} \gg \sigma^{-1/3}$.

Now we can estimate the total dissipated specific energy. In the ultrarelativistic limit, the energy dissipated between y_1 and y_2 is

$$\Delta\mu(y_2) - \Delta\mu(y_1) \approx \eta \int_{y_1}^{y_2} \frac{u^2(y)}{(1+y)^{\alpha+1}} dy. \tag{7.49}$$

Given equation (7.47) for u, the integral in equation (7.49) can be done analytically. For $y_2 = \infty$, we obtain the total energy dissipated at $x > x_1$,

$$\Delta\mu(> x_1) \approx \frac{\sigma^{1/3}}{2} \left\{ \left[1 + \frac{4\eta\sigma^{1/3}}{\alpha(1+y_1)^{\alpha}}\right]^{1/2} - 1 \right\} . \tag{7.50}$$

Clearly, the amount of dissipation increases with increasing drag, but the total dissipation *in the relativistic regime* may not exceed $\Delta\mu \sim (\eta\sigma)^{1/2}$ (provided that $\alpha > 1$, which is expected to apply in an accretion flow). If we combine this limit with the self-consistency condition for relativistic flow, we find that $\Delta\mu \lesssim \sigma$, i.e., the total energy dissipated may approach but not greatly exceed the final energy in the flow. Recall that for a relativistic flow with radial geometry, the specific kinetic energy is only a fraction $\sim$

$\sigma^{-2/3} \ll 1$ of the total specific energy, which resides mainly in Poynting flux (Michel 1969). Since only motion of the gas can lead to dissipation of energy through radiation drag, a large value of $\Delta\mu$ implies that a large amount of Poynting flux is being converted to kinetic energy through the unwinding of the magnetic field, and is then being channeled into energy of scattered photons. This result supports our earlier conclusion that radiation drag can catalyze the conversion of Poynting flux into kinetic energy.

The limits quoted above assume that the flow is relativistic down to radii $\lesssim x_{ch}$. This will not be true if $x_{ch} < 1$, or equivalently, if $r_{ch} \sin\theta_0$ is smaller than the light cylinder radius, $R_L \equiv c/\Omega$. Nevertheless, limits analogous to those given above hold if η is rescaled to $\eta/(1+1/x_{ch})^{\alpha}$. If the drag is so strong that the flow is subrelativistic at $x > x_{ch}$, then the total dissipation will be given by $\Delta\mu \sim \eta^{1/2}\sigma^{1/2}$. Since η can exceed σ in the subrelativistic limit, we see that the dissipated energy may actually exceed the total terminal energy of the flow by a large factor.

Since all of the dissipated energy goes into the Comptonized photons, the total luminosity of scattered radiation is given by

$$L_{scatt} = (f\Delta\Omega)c^2\Delta\mu, \tag{7.51}$$

where $f = \rho V_r r^2$ is the mass flux per steradian and $\Delta\Omega$ is the solid angle of the wind. The quantity $f\Delta\Omega$ equals the total mass flux in the wind, $\dot{M}_w$. To compare L_{scatt} with the "ambient" luminosity responsible for the drag, we assume that the mean radiation density at $r \lesssim r_{ch}$ is given by $L_{amb}/4\pi r_{ch}^2 c$. We then have

$$\frac{L_{scatt}}{L_{amb}} \sim \frac{m_p}{m}\frac{r_g}{r_{ch}}\frac{\dot{M}_w}{\dot{M}_E}\left(\frac{\sigma}{\eta}\right)^{1/2}, \tag{7.52}$$

where $\dot{M}_E = L_E/c^2 = 4\pi r_g c m_p/\sigma_T$ is the Eddington mass flux for a central object of gravitational radius $r_g = GM/c^2$, and we have taken $\Delta\mu \approx (\eta\sigma)^{1/2}$. Since σ depends on both $\dot{M}_w$ and the magnetic flux, the numerical value of L_{scatt}/L_{amb} is highly uncertain. To obtain a value $\gtrsim 1$ in the context of standard black hole accretion models, it appears that the mean magnetic energy density at $r \lesssim r_{ch}$ must exceed the ambient radiation energy density. Note that this ratio is also sensitive to the composition of the wind through the factor m_p/m, which ranges from ~ 1 for ordinary electron–ion plasma to ~ 2000 for a flow consisting entirely of electron–positron pairs.

If the flow is collimated to an angle better than $\gamma^{-1} \sim (\eta/\sigma)^{1/2}$, then relativistic beaming may enhance the total scattered *flux* relative to the background by an additional factor γ^2, for an observer situated in the beam. The enhancement in the spectral flux density depends on the spectral slope of the ambient radiation. Techniques for estimating this ratio are discussed in Begelman & Sikora (1987) and Sikora, Zbyszewska & Begelman (1989).

7.4. Conclusions and Discussion

We have demonstrated how a radial MHD wind can overcome the drag force due to an ambient radiation field, to deliver a sizable energy and angular momentum output. Even if the drag coefficient, $\nu_R(x)$, declines with radius as slowly as x^{-2}, the fast magnetosonic point for a cold wind will occur at infinity, just as in Michel's minimum-torque solution. For this reason, the asymptotic total energy output, as well as the final energy partition between the Poynting flux and the kinetic energy flux, is exactly the same as in an ideal flow. The essential effects of the drag force are:

1). to enhance the rate of electromagnetic energy extraction from the central rotating object, to compensate for the dissipation due to the drag. Depending on the intensity of the ambient radiation field, the dissipated energy could be comparable to or even larger than the final total energy that is carried to infinity by the flow and the electromagnetic field. Since all of the dissipated energy must emerge as Compton-boosted photon energy, this effect could be observable, provided that the flow is relativistic and the ambient photon density is large enough (Begelman & Sikora 1987; Melia & Königl 1989). However, for a given mass flux there is an upper limit on the photon energy density above which the flow is no longer optically thin, hence our treatment must be modified in these regions. As an illustration, in Appendix D we derive this limit for a conical jet originating from the region around a black hole or neutron star.

2). to increase the angular momentum flux extracted from the central rotating object as well as that carried out to infinity.

3). to redistribute the locations of flow acceleration. Compared to the ideal case, the velocity rise near the central object is suppressed, but enhanced acceleration occurs where the magnetic force overcomes the drag force.

We emphasize that the fast magnetosonic point is at infinity only for a perfectly *radial*, cold MHD wind. In general, winds from a rotating central object are expected to become collimated about the rotation axis (Heyvaerts & Norman 1989; Chiueh, Li & Begelman 1991; chapter III). If the flux surfaces of a relativistic wind pass through a region in which they

diverge even slightly faster than radially, then the fast point can lie at a few light cylinder radii (Camenzind 1989; chapter IV). In this case there can be significant acceleration outside the fast point, and the kinetic energy flux in the flow could dominate over the Poynting flux at infinity (in contrast to the perfectly radial case, in which the Poynting flux greatly exceeds the kinetic energy flux). If there is significant radiation drag outside the fast point, then the simple picture we have presented for a radial wind may have to be modified. For example, under these circumstances it may be possible for the flow to have more than one fast critical point. Since it is generally impossible to satisfy more conditions than the number of adjustable parameters available, the flow may become nonstationary or may form a shock, as we demonstrate in chapter VIII.

Although the drag force does not enter the cross-field force balance equation directly, it will, nevertheless, affect the structure of the field geometry by changing the flow speed along the flux surfaces. Unfortunately, this effect cannot be addressed in the context of a radial wind.

It has been suggested that the presence of radiation drag can greatly reduce the Lorentz factor of an outflow, or even quench it. From our analysis of radial MHD winds, we infer that this is not necessarily true for winds which are propelled by hydromagnetic forces. The flow properties of such winds are controlled by the conditions at the critical points, particularly at the fast magnetosonic point if the wind is cold. If the region of dynamically significant drag is located well inside the fast point, then the properties of the flow at the fast point and beyond should not be much affected. A highly relativistic wind is then possible, especially because most of the acceleration could occur outside the fast point. On the other hand, if the drag force at

the fast point and beyond is not negligible, then the dissipation outside the fast point cannot feed back to the base of the flow to increase the energy and angular momentum extraction from the central object, to compensate for the losses. As a result, the wind may not reach as large a terminal speed or may even be decelerated. The wind may also become nonstationary or be slowed down by shocks. In these situations, the flow speed may indeed be limited by drag, and the flow might even be quenched. Without a full two-dimensional treatment it is impossible to determine the relationship between the fast point and the region of dissipation; future work including the force balance between flux surfaces will be necessary to resolve this important issue.

CHAPTER VIII

EXTERNALLY-DRAGGED RELATIVISTIC MHD OUTFLOWS OF ARBITRARY GEOMETRY

8.1. Introduction

A longstanding puzzle in QSO research follows from the observation that only a small fraction ($\sim 10\%$) of all QSOs are radio-loud, while the majority of them are radio-quiet. One possible explanation is that the radio-loud QSOs have powerful, highly relativistic jets, whereas such jets are absent for some reason in the radio-quiet QSOs. If all QSOs are capable of producing relativistic jets in the first place, then one has to seek external agents for the suppression of relativistic jets (and thus the radio emission) in the radio-quiet QSOs. It is conceivable that potentially relativistic jets are somehow "dragged" to subrelativistic speeds in radio-quiet QSOs while largely unaffected in radio-loud QSOs. The drag force can be provided by either the Compton scattering of the ambient radiation field or entrainment of ambient matter, both of which could reduce the radio emission by lowering the jet speeds so that the relativistic enhancement is reduced and, possibly, forcing the synchrotron-emitting electrons to stay in a smaller volume (i.e., in slowly-moving bubbles instead of relativistic jets). One can view the importance of the radiation drag on relativistic jets in a slightly different way: given that the Lorentz factor of the jets can be as high as 10 (Begelman, Blandford, & Rees 1984), one might wonder how can they survive the interaction with the intense radiation field expected near the central engines at all.

The investigation of mass-entrainment in relativistic MHD outflows might have important implications for the study of broad absorption line (BAL) QSOs (see Weymann, Turnshek, Christiansen 1985 for a review). If mass loading from the BAL clouds could indeed prevent the relativistic MHD jets from being powerful radio sources, it would naturally explain why all of BAL QSOs are radio-quiet (Stocke et al. 1992). If the magnetized winds are well-collimated on the scale of the BAL clouds ($\sim pc$), then the inferred small covering factor of the BAL clouds (only about 10%-20%) may be explained.

We shall try to address the problem of "dragged" relativistic outflows in the context of the MHD model that we have developed in the previous chapters. The external drag force is expected to dissipate the directed kinetic energy of the flow and thus reduce the flow speed. In doing so, it will also affect the global structure of the flow through modifying the force balance in the cross-field direction and therefore change the rate with which the magnetic energy is converted into kinetic energy (which in turn is partially dissipated by the drag). Apparently, the interaction between the drag and the MHD flow is highly nonlinear and self-consistent solutions are hard to obtain.

In the last chapter, we have confined ourselves to the study of radiation drag on Michel-type (1969) cold, relativistic radial MHD winds for simplicity. Our analysis yielded a somewhat unexpected result: we found that, given a reasonable distribution of the ambient photon energy density, the flow will not be stopped no matter how strong the radiation drag might be. In fact, the final speed in the dragged flow is the same as in the drag-free flow. The reason is that the drag force does not change the location the fast magnetosonic point (which is located at ∞ in an ideal cold radial

wind) and its main effects are: 1) to increase the extraction of energy and angular momentum from the source; and 2) to redistribute the locations of flow acceleration. Radial winds are characterized by a constant flux tube opening function $a = b_p x^2$ along flux tubes. Here, we extend this study to flows with more general geometries. Completely new results arise because of the appearance of critical points at finite distances and their modifications by the presence of external drag force.

We shall generalize the dragged radial wind case with constant a (see the last chapter) to include those with a as an arbitrary function of x. It turns out that, depending on the distribution of $a(x)$ and the radiation field, the drag can "choke" a supermagnetosonic flow by forcing it to encounter a second fast magnetosonic point, or it can cause the flow to remain sub-magnetosonic throughout. In either case, the only solutions that can go smoothly from the base of the flow to infinity are those sub-magnetosonic solutions with much lower (in some cases zero) flow speeds compared to the speed of the supermagnetosonic drag-free flow. It is conceivable that such sub-magnetosonic solutions represent quasi-static bubbles that only slowly expand outwards. This conclusion is reached by assuming that the particular flux tube under consideration is not influenced by the neighboring flux tubes. In reality, however, it is expected that different flux tubes will be affected to different degrees by the applied drag force, and the flux tubes would certainly expand or contract laterally in order to come into a new equilibrium with their neighbors. This extra degree of freedom may enable a realistic dragged flow to avoid the appearance of a second critical point in its flux tubes. But this issue cannot be settled until the cross-field force

balance equation is solved self-consistently in the presence of drag. Such self-consistent treatment of flows under general conditions is beyond the scope of the present study, and we shall concentrate on the very restrictive case of dragged self-similar flows. For this reason, the conclusions drawn in this chapter are tentative. Given the importance of the subject, we feel that our primary exploration here is justified. We hope to highlight the possible difficulties in obtaining unique critical solutions for dragged winds, and to provide guidance for the more complete treatment of the problem in the future.

The plan for the chapter is as follows. We first examine the critical conditions at the critical point in § 8.2.1. The appearance of a second fast point is demonstrated explicitly in § 8.2.2 when the shape of a flux tube, $a(x)$, is prescribed. To demonstrate that the external drag force can indeed modify the global structure of the flow, we treat the idealized case of dragged self-similar flows in § 8.2.3, where the cross-field force balance is taken into consideration self-consistently. We formulate the problem of mass entrainment by relativistic MHD flows in § 8.3.1, and then present illustrative solutions in a simplified case in § 8.3.2. Finally, we summarize our results and discuss their astrophysical implications in § 8.4.

8.2. Radiation-Dragged Relativistic MHD Flows

8.2.1. Analysis of Critical Points

The basic equations for dragged MHD flows with geometry more general than radial are the change of the energy and angular momentum with

radius

$$\frac{d\mu}{dx} = -\gamma\frac{u^2+v^2}{u}\nu_R(x), \tag{8.1}$$

$$\frac{d\lambda}{dx} = -\gamma^2\frac{vx}{u}\nu_R(x), \tag{8.2}$$

and we have defined $\nu_R(x) \equiv \nu(S)/\sin\theta$ (where θ is the angle between the slope of the field line at radius x and the cylindrical axis). These two equations are identical to equations (7.8) and (7.9), except that θ can vary with x, instead of being a constant (as is the case for a radial surface). The energy equation, which relates the local flow speed u at x to μ and λ, is the same as the drag-free case, as long as it is remembered that neither μ nor λ is constant along the flux surface. In the case of a "cold" wind with no gravity, it is given by the dimensionless equation (4.2). The critical points for the energy equation (where critical conditions have to be imposed in order to ensure smooth solutions) are also the same as in the drag-free case: when $a \equiv b_p x^2$ as a function of x is prescribed, the critical point is the fast magnetosonic point (chapter VI).

To obtain the critical conditions at the fast point, we differentiate the energy equation to yield an equation of the form $d\tau/dx = N/D$. The numerator is

$$D \equiv \tau x^4(b_\phi^2 + b_p^2\omega), \tag{8.3}$$

where $\omega \equiv 1-(1+\tau)x^2$ is used for convenience, and the numerator is

$$N = \frac{v}{x}[u_\phi\omega + 2\tau x\lambda\eta] - u_p^2\omega\frac{a'}{a} + [\gamma\omega + \tau x^2\lambda\eta]\mu' - [(\gamma\eta + \frac{u_\phi^2}{\lambda})\omega + \tau x^2\mu\eta]\lambda'. \tag{8.4}$$

The prime denotes a derivative with respect to x. At the fast magnetosonic point, where

$$M_f^2 \equiv \frac{4\pi\rho u_p^2\gamma^2 c^2}{B_\phi^2 + B_p^2(1 - r^2/r_L^2)} = \frac{b_p^2 x^2 \tau}{b_\phi^2 + b_p^2(1 - x^2)} = 1, \tag{8.5}$$

one has $D = 0$. The numerator N has to vanish simultaneously in order for the fast point to be passed smoothly. This is the critical condition that determines the critical energy μ_c (and thus the critical solution) when $a(x)$ is prescribed.

Compared with equation (4.8) for the drag-free case, the numerator N have two additional terms associated with μ' and λ'. Although not evident directly from the expression for N, the combined effects of these two dissipation terms are to push the fast point further away from the light cylinder. We show this in the limit that $x_f^2 \gg 1$, where

$$\eta = \frac{\mu}{\lambda(1+\tau)} + O(x^{-2}), \quad \gamma = \frac{\mu\tau}{1+\tau} + O(x^{-2}). \tag{8.6}$$

from equations (4.3 c) and (2.7). To the leading order of x^{-2}, the coefficient for μ' in the numerator becomes

$$-\frac{\mu\tau^2 x^2}{1+\tau} < 0, \tag{8.7}$$

whereas the coefficient for λ' is approximately zero. Since both μ' and λ' are negative, their contribution to the numerator N is positive, and thus their combined effect is to push the fast point away from the base of the flow. ***This is why the drag force cannot bring the fast point in a radial wind from infinity to a finite radius.*** The same is true even if the fast point is not much beyond the light cylinder, as we shall demonstrate numerically below.

8.2.2. Radiation Drag on Flux Tubes with Prescribed $a(x)$

If a flux tube has its fast magnetosonic point located at a finite radius, and the drag outside the fast point is strong enough, it is possible that the flow speed can be reduced significantly so that a second fast point appears. Since the critical conditions at the first fast point fix the flow completely in the case where $a(x)$ is prescribed, there will be no free parameter left to adjust, i.e., it is impossible to force the numerator N to vanish simultaneously with the denominator D at the second fast point. Therefore, the applied drag would make the flow singular at the second fast point, and thus cause the supermagnetosonic flow either to become unsteady or to undergo a shock transition.

In order for a quantitative study, we adopt a drag coefficient of the following modified power-law form

$$\nu_R(R) = \frac{\nu_0}{(1 + R/R_{ch})^\alpha} = \frac{\nu_0}{(1 + x/x_{ch})^\alpha} \tag{8.8}$$

to imitate the near constancy of the ambient photon energy density inside some characteristic radius R_{ch} (and its dimensionless counterpart x_{ch}) and its rapid decline away from the "central engine" near which most of the ambient photons are produced. In a wind, the quantities ν_0, R_{ch} and α can be different for different flux tubes due to the differences in rotation rate, shape, etc. It has been argued in chapter VII that for a typical radiation field above an accretion disk surrounding a black hole, the exponent α, which controls how fast the photon density declines at large distances ($x \gg x_{ch}$), should be between 2 and 3 for radial flux tubes. We shall take a value of $\alpha = 2.5$ for mathematical definitiveness in the numerical presentation below.

Drag-free flows along given flux tubes are determined completely by the flux tube opening function $a(x)$ according to equation (4.2), and for the fast point to occur at a finite radius the flux tube has to diverge faster than radially, i.e., a has to decrease with x. From this consideration, we shall adopt a simple form

$$a(x) = \frac{a_0 x_a + a_\infty x}{x_a + x}, \tag{8.9}$$

such that the quantity a decreases from its initial value of a_0 to its final value of a_∞ ($< a_0$) quickly near a length scale x_a. The quantities a_0 and a_∞ basically control the total specific energy in the flow and the fraction that will end up in kinetic form, whereas x_a characterizes where most of the flow acceleration occurs. For a terminal Lorentz factor in the range $2 - 20$, as inferred from superluminal motions in extragalactic radio jets, we feel that a value of $a_0 = 40$ should be representive, provided that the radius of the fast point x_f is not too large compared with x_a since $\mu_c \approx a_f = a(x_f)$.

For a given drag distribution, one can integrate equations (8.1) and (8.2) (coupled to the energy equation [4.2]) numerically by the usual Runge-Kutta method. At the fast point, we first guess a value for μ_f, and evaluate x_f and λ_f from the critical conditions. We then apply the L'Hospital's law near the fast point and integrate towards the base of the flow x_0. The quantity μ_f is fixed by the requirement that $u = u_0$ at $x = x_0$. After experimenting with parameters we find that, depending on the position of the fast point relative to the region where the bulk the drag occurs, there are two distinctively different solution behaviors. In the case that x_{ch} is much larger than x_f, there are three types of flow topologies in general. They are illustrated in Figure 8.2*a-f* by plotting the dimensionless poloidal 4-speed

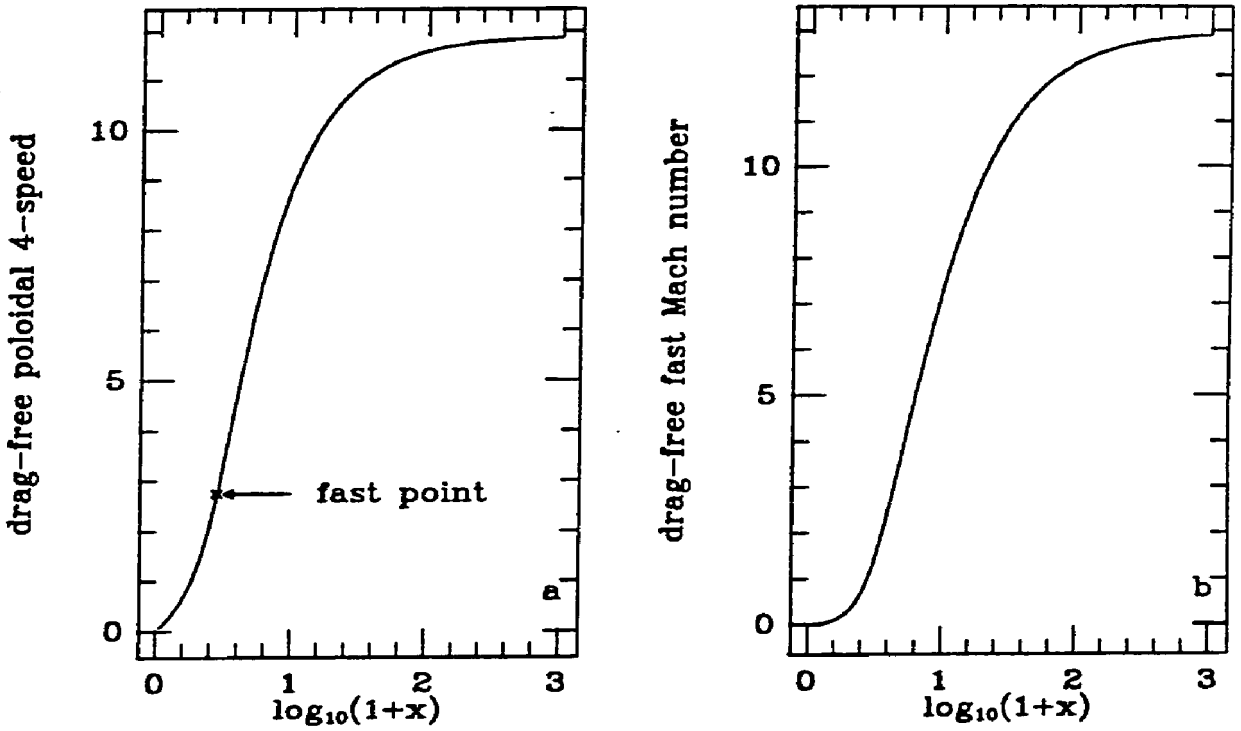


Fig. 8.1–For a set of parameters $a_0 = 40$, $a_\infty = 10$, $x_0 = 0.1$, $u_0 = 0$, $x_a = 1$ and $x_{ch} = 15$, we plot the poloidal 4-speed (a) and the fast magnetosonic number (b) in the drag-free case. These quantities are to be compared with those in Fig. 8.2 where the flow is dragged to various degrees.

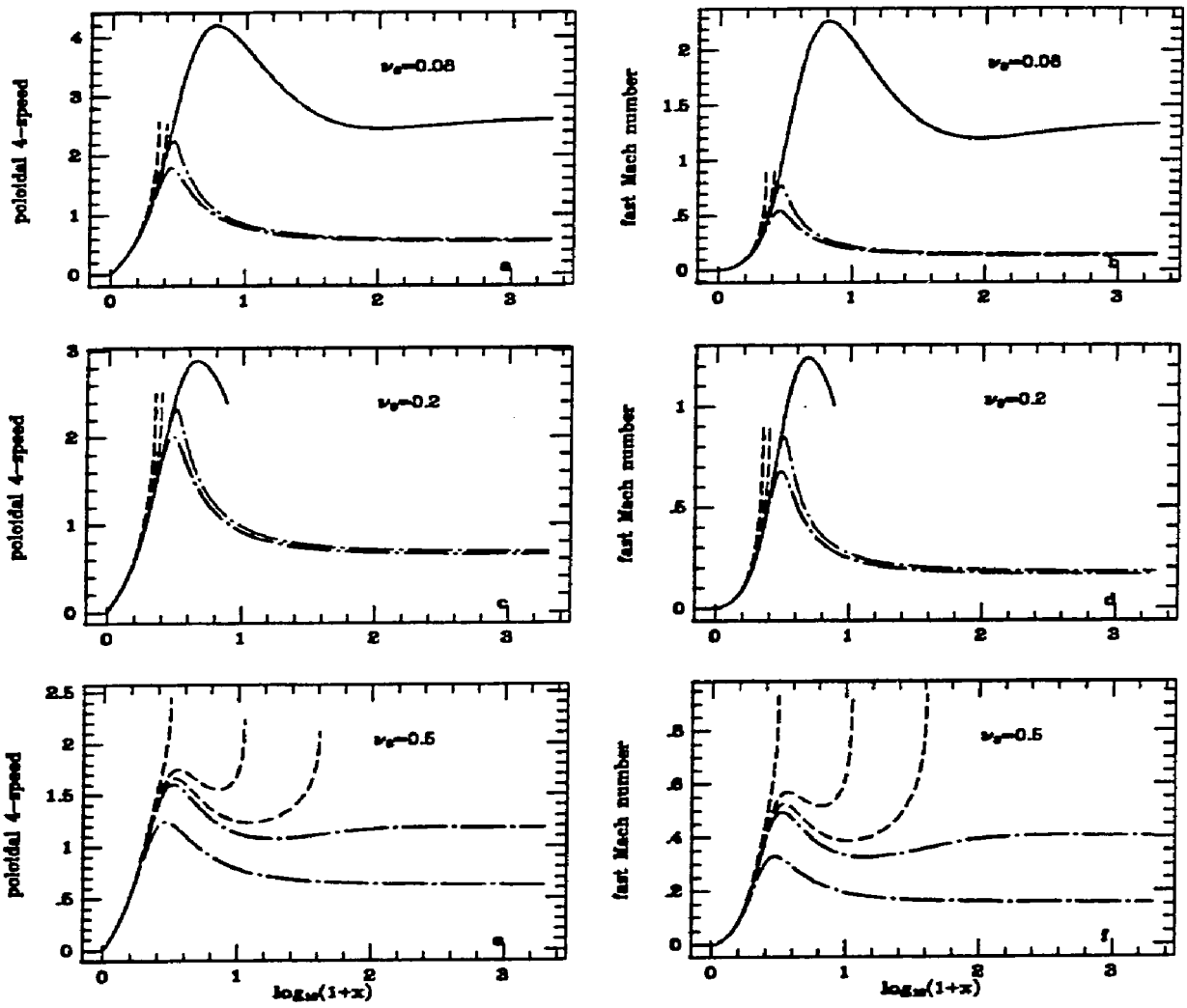


Fig. 8.2–Solution topologies for dragged flows with the same set of parameters as in Fig. 8.1. The poloidal 4-speed and fast Mach number are plotted for each of the three representive cases with $\nu_0 = 0.08$, 0.2, and 0.5. See the text for detailed discussions.

and the fast magnetosonic Mach number M_f against $\log_{10}(1+x)$. In all the panels, we have adopted $a_0 = 40$, $a_\infty = 10$, $x_0 = 0.1$, $u_0 = 0$, $x_a = 1$ and $x_{ch} = 15$. The different curves in each panel correspond to solutions with different initial total energy μ_0, with the critical one with μ_c in solid. Solutions with smaller μ_0 than μ_c (dashed lines) are terminated at the fast point, whereas those with $\mu_0 > \mu_c$ (dash-dotted remain submagnetosonic throughout. For comparison, we have plotted the 4-speed and fast Mach number in the drag-free limit for the same set of parameters in Figure 8.1*a* and *b*. The fast point is located at $x_f = 1.82$ in this case. It is evident that when the drag coefficient is small enough (Fig. 8.2*a*, *b*, $\nu_0 = 0.08$), there exists a unique transmagnetosonic critical solution going smoothly from origin to infinity, as in the drag-free case. Compared to the drag-free case, however, the fast point is at a larger radius ($x_f = 1.97$) and the terminal flow speed is much lower. This reduction in the terminal speed of the flow may facilitate the quenching of the flow by other agents, such as the ambient medium of the flow. When the drag strength is increased, it is possible that (as depicted in Fig. 8.2*c*, *d*, where $\nu_0 = 0.2$) the flow outside the first fast point is retarded by the drag force so much that a second fast magnetosonic point is encountered. In this case, the only solutions that go smoothly from the base of the flow to infinity are the submagnetosonic solutions. We therefore conclude that the smooth critical solution that exists in the drag-free case is quenched. Could the solution that passes the first fast point undergo a fast shock transition before reaching the second fast point and then be reaccelerated to pass smoothly a fast point and become supermagnetosonic again (see discussion by Holzer's [1977] in solar wind context)? We find that the answer is no in general. Solutions that become submagnetosonic behind the

shock always remain submagnetosonic all the way to infinity. This can be demonstrated explicitly with the solutions in Fig. 8.2*c,d*, where one forces the solution to jump from a supermagnetosonic branch to a submagnetosonic branch at some point between the two fast points and continue the integration outward. We find that the solution always stays on the submagnetosonic branch all the way to infinity. It is, however, possible that the flow becomes unsteady instead of being completely stopped. More work is needed to settle this issue. As the drag continues to increase even further, the two fast points move closer and closer toward each other, and eventually degenerate into one single point. After this, they disappear altogether. This latter stage is represented in Fig. 8.2*e, f* (with $\nu_0 = 0.5$), where a transmagnetosonic critical solution does not exist at all and, as in Fig. 8.2*c,d*, the only solutions that go smoothly to infinity are submagnetosonic. In the case that $a_\infty = 0$, submagnetosonic solutions would have zero final speed, and a flow can be regarded as completely "choked" if the critical supermagnetosonic solution is prohibited by drag.

When x_f is located near or outside x_{ch}, quenching of the critical solution is unlikely. We illustrate this case by the choice $x_{ch} = 1.0$ with other parameters the same as in Figure 8.2. The poloidal 4-speed and the fast magnetosonic Mach number M_f are plotted in Figures 8.3*a* and 3*b* respectively, for $\nu_0 = 1$, 5 and 10. The critical solution survives the drag in each case despite the terminal 4-speed being greatly reduced. We see that the stronger the drag is, the larger the radius x_f will be. This reconfirms our earlier statement about the effects of drag force on the location of the fast point. The lower terminal speed in the larger drag case results not only from the larger energy dissipation but also from the lower total critical energy at

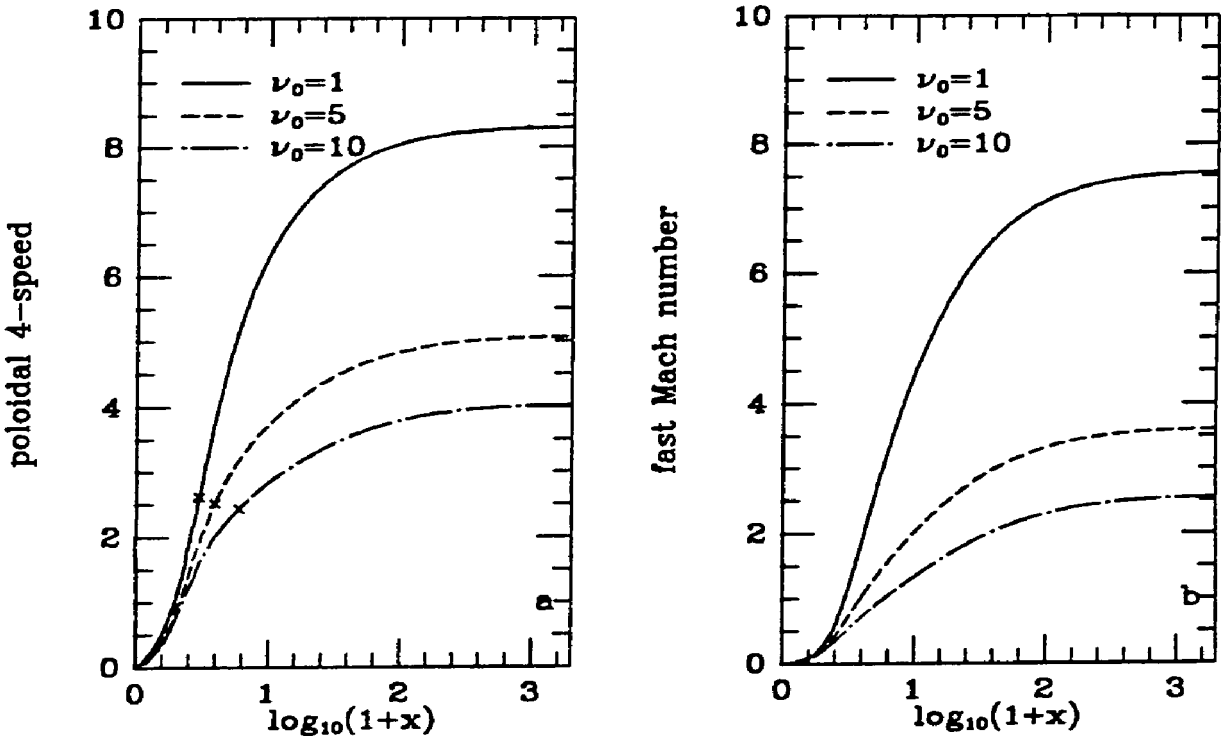


Fig. 8.3–Representive plots of the poloidal 4-speed (a) and the fast Mach number (b) for the case in which the size of the region where most of the photons are generated is comparable to the light cylinder radius. We take $x_{ch} = 1.0$ and all other parameters the same as in Fig. 8.1. The cross on each curve in (a) mark the fast point. It is clear that the second fast point does not appear here.

the fast point (which is pushed by drag further away from the light cylinder, thus giving a lower specific total energy at the fast point, $\mu_f \approx a_f$).

8.2.3. Radiation-Dragged Self-Similar Solutions

To demonstrate the effects of radiation drag on the global structure of the flow, we now consider a dragged self-similar flow, where the cross-field equation can be treated self-consistently. Drag-free self-similar flows have been studied in detail in chapter V. In the presence of a drag force (assumed to be given by equation [2.12]), the two basic self-similar equations, the energy equation and the cross-field force balance equation, have the same appearance as their counterparts in drag-free case, namely, equations (4.2) and (5.8), although the specific energy μ and angular momentum λ are no longer constant along a given flux surface. These two variable quantities are determined from

$$\frac{d\mu}{dx} = -\gamma \frac{u^2+v^2}{u} \nu(x,y)\sqrt{1+\dot{y}^2} \tag{8.10}$$

$$\frac{d\lambda}{dx} = -\gamma^2 \frac{vx}{u} \nu(x,y)\sqrt{1+\dot{y}^2}, \tag{8.11}$$

where y and $\dot{y}$ are the dimensionless height and slope of the reference flux surface defined in chapter V. For optically thin flow in an isotropic radiation field, the dimensionless drag coefficient $\nu(x,y)$ is given by equation (7.40), which is

$$\nu(x,y) = \frac{4}{3}\frac{\sigma_T R_L U_{rad}}{mc^2} \propto R_L U_{rad}. \tag{8.12}$$

Since it is required that any dimensionless quantity has to be the same on all flux surfaces in order for the self-similarity assumption to be self-consistent, we demand that the radiation energy density scale according to $U_{rad} \propto r^{-1}$,

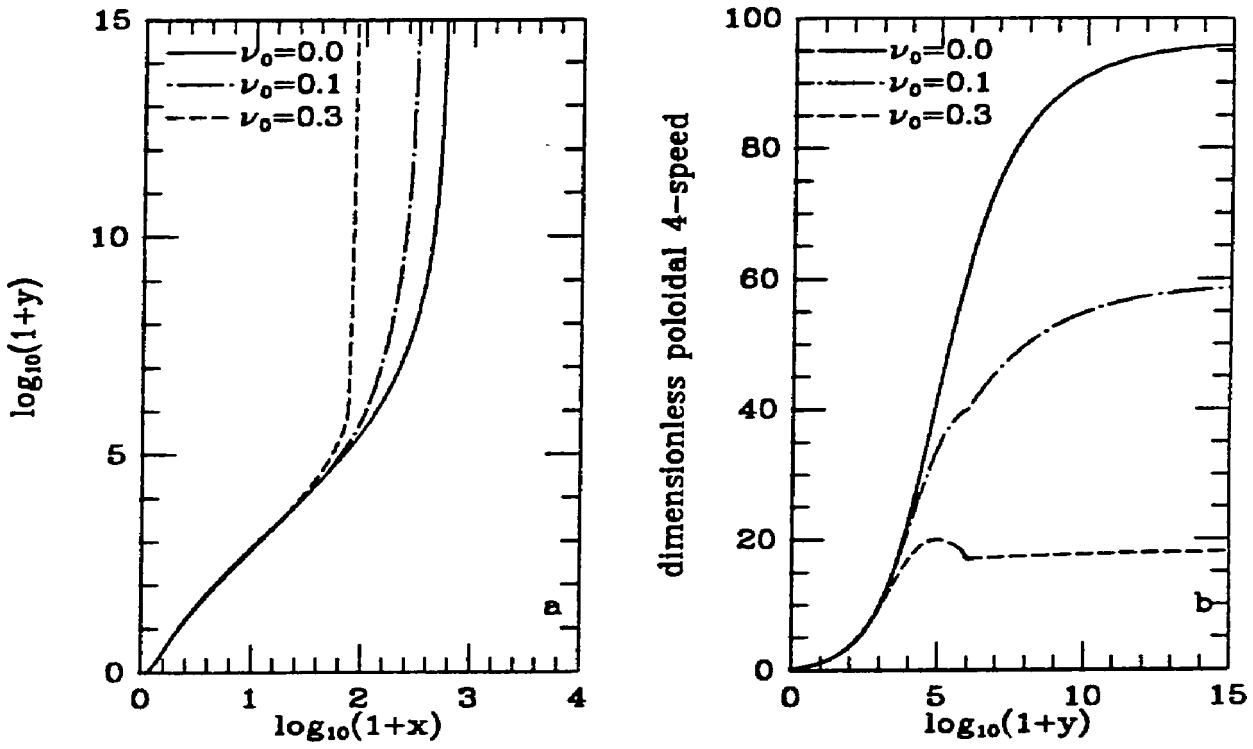


Fig. 8.4–The shapes of the poloidal field line (a) and their associated 4-speeds (b) are plotted for drag-free ($\nu_0 = 0$) and dragged ($\nu_0 = 0.1, 0.3$) self-similar flows. The drag not only lowers the flow speed but also makes the flow better collimated.

that is $\nu(x,y) = \nu_0/\sqrt{x^2+y^2}$. We admit that such an ambient photon field is entirely *ad hoc*, and in order to avoid infinite photon energy we have to truncate U_{rad} at some finite distance. However, we hope that some of the essential features of dragged flows when the cross-field equation is included can be illustrated with such a restrictive radiation energy density distribution.

The numerical procedures for finding critical self-similar solutions for dragged and drag-free flows are nearly the same, except that in the dragged case 1) μ and λ have to be updated at each point by equations (8.10) and (8.11); and 2) when writing down the critical conditions at the Alfvén critical point to initiate the integration of the cross-field equation (5.8), the drag terms have to be included. In Figures 8.4*a* and *b*, we plot the dimensionless poloidal 4-speeds and the shapes of the poloidal reference field line in the case $\sigma = 100$ (defined in eq. [5.2]), $\beta = 0.95$ (defined in eq. [5.3]), $x_0 = 0.1$ and $u_0 = 0$ for three drag coefficient $\nu_0 = 0$, 0.1 and 0.3. The drag coefficient is truncated arbitrarily at a height of $y_e = 10^6$. The reason of choosing such a large σ and y_e is to enhance the drag effects, since in a self-similar flow the plasma is accelerated only gradually and approaches its terminal speed at very large height. It is evident that the drag force not only lowers the flow speed considerably (Fig. 8.4*a*) but also confines the flow to a smaller cylindrical radius (Fig. 8.4*b*).

8.3. Dragging of Relativistic MHD Flows by Mass-Entrainment

8.3.1. Formulation of the Problem

In this section, we use the poloidal flux function Ψ to label field lines and S to measure the length of each poloidal field line from its base. We

consider only the simplest case where stationary ambient material is brought to the local speed of the flow instantaneously from rest at a rate $\epsilon(\Psi, S)$ (with units g cm^{-3} s^{-1}). In this case, the momentum equation, in the limits that thermal pressure and gravity are negligible, is

$$\rho(\mathbf{V}\cdot\nabla)\mathbf{V} = \rho_E\mathbf{E} + \frac{(\nabla\times\mathbf{B})\times\mathbf{B}}{4\pi} - \epsilon(\Psi,S)\mathbf{V}, \tag{8.13}$$

where all of the symbols have their usual meanings (see chapter II). The last term on the right hand side is the drag force due to mass entrainment. One can eliminate the proper density from the continuity equation along a given flux tube between Ψ and $\Psi + \Delta\Psi$. Denoting the area of the cross section at length S by ΔA, one can obtain the mass flux $\Delta\dot{m}$ through ΔA from

$$\rho V_p \Delta A = \Delta\dot{m} = \int_0^S \epsilon(\Psi,S)(\Delta A)dS = (\Delta\Psi)\int^S \epsilon(\Psi,S)/B_p dS, \tag{8.14}$$

where the relation $\Delta\Psi = B_p\Delta A$ has been used. In terms of the mass flux per unit magnetic flux

$$k(\Psi,S) \equiv \int_0^S \epsilon(\Psi,S)/B_p dS + k_0(\Psi), \tag{8.15}$$

we recover the previous relation, $\rho = k(\Psi,S)B_p/V_p$, although it has to be kept in mind that $k(\Psi,S)$ is no longer a function of Ψ only, as is the case for the ideal or radiation-dragged flows. For this reason, we can no longer formulate the problem in terms of the specific energy and angular momentum, as we did before. Instead, the momentum equation (8.13) has to be considered directly. In a conventional cylindrical coordinate system (R,ϕ,Z), the poloidal component of equation (8.13) is

$$\rho\frac{d}{dS}\frac{V_p^2}{2} - \frac{V_R}{V_p}\frac{\rho V_\phi^2}{R} = -\frac{1}{4\pi R^2}\frac{d}{dS}\frac{(B_\phi R)^2}{2} - \epsilon(\Psi,S)V_p, \tag{8.16}$$

and the toroidal component is

$$\rho\frac{V_p}{R}\frac{d(V_\phi R)}{dS}+\epsilon(\Psi,S)V_\phi=\frac{B_p}{R}\frac{d(B_\phi R)}{dS}. \tag{8.17}$$

As in § 8.2, instead of solving the cross-field component of the momentum equation self-consistently, we parameterize the distribution of the poloidal magnetic field strength B_p (or the quantity a). In this case, equations (8.16) and (8.17) can be regarded as two coupled first order ordinary differential equations for V_p and V_ϕ, with the toroidal magnetic field strength B_ϕ given from the flux-freezing condition (2.1), i.e., $B_\phi = B_p(V_\phi - \gamma R\Omega)/V_p$.

8.3.2. Numerical Examples

We shall skip the analysis of the effects of the drag on the critical points, which can be done in a way very similar to § 7.2.2. We shall concentrate on how highly relativistic flows in the absence of drag are brought to subrelativistic speeds by mass entrainment at large radii $R \gg R_L$ from the base of the flow. In this case, since $V_p \gg V_\phi$, one can solve equations (8.16) for the distribution of V_p approximately independently of equation (8.17). If we denote quantities at the length $S = S_0$ by a subscript "0", then one can define the following nondimensional quantities along a given flux surface Ψ: $s \equiv S/S_0$, $b(s) \equiv k(S)/k(S_0)$, $a(s) \equiv B_p R^2\Omega^2/[4\pi k(S_0)c^3]$, and $v = V_p/c$. Equation (8.16) then becomes

$$\left[1-\frac{a(s)}{b(s)v^3}\right]\frac{dv}{ds}=-\left(1+\frac{1}{v^2}\right)\frac{a'(s)}{b(s)}-\frac{vb'(s)}{b(s)}, \tag{8.18}$$

where the derivative with respect to s is denoted by a prime. Given the functions $a(s)$ and $b(s)$, the flow 4-speed as a function of s can be obtained straightforwardly by integrating equation (8.18) using the Runge-Kutta method. An illustrative example is presented in Figure 8.5, where we

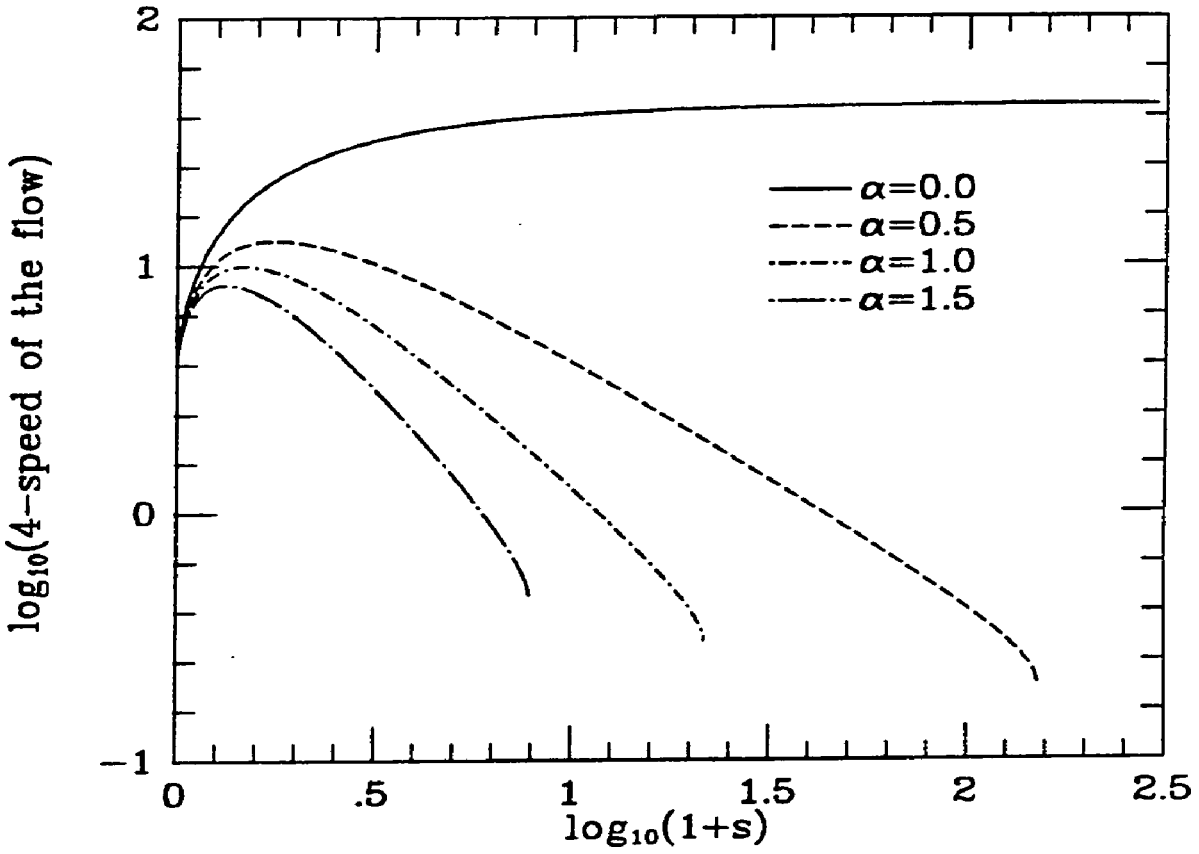


Fig. 8.5–Curves of the 4-speed for the drag-free (solid) and dragged flows. The flux tube is characterized by $a(s) = [a_i s_0 + a_e(s-1)]/[s_0 + (s-1)]$ with $a_0 = 40$, $a_e = 1$, $s_0 = 10$. The drag due to mass entrainment is parametrized by $b(s) = s^\alpha$, which equals 1 at the starting point $s = 1$. The initial 4-speed is taken to be $u_0 = 4$, so that the flow is supermagnetosonic to begin with. All dragged solutions encounter a second fast point where they become singular (i.e., double-valued). The faster the mass is added to the flow (that is, larger α), the sooner the second fast point appears.

plot the velocity curves for different distributions of the mass loading rate $b(s)$ with fixed $a(s)$ and initial speed v_0. Clearly, all but the drag-free case (where $b[s] = 1$) become singular at a second fast point where the coefficient on the left hand side of equation (8.18) is zero. The faster the external mass is entrained (with larger α in Fig. 5.4), the more quickly the singular fast point is encountered. In any case, potentially highly relativistic flows are dragged to subrelativistic speeds as expected, since the momentum of the flow is shared by an ever increasing mass. The only twist is the appearance of the second fast point when the flow is slowed down to a certain degree. The multiple critical point behavior is also found in the usual dissipative, thermal (Holzer 1977) or radiatively-driven winds (Ferrari et al. 1985).

8.4. Conclusions and Astrophysical Implications

8.4.1. Radiation Drag

From the study of dragged flows along flux tubes of given shape we find that it is possible to quench critical solutions that are supermagnetosonic at large distance. For this to occur, there must exist intense radiation beyond the location of the fast magnetosonic point. In the presence of this radiation drag, the unique supermagnetosonic critical solution in the drag-free limit is either forced to encounter a second fast point at which it becomes singular or prevented from passing through any fast point at all. All solutions that can go from the base of the flow to infinity are submagnetosonic and are strongly dominated by Poynting flux. In fact, their final speeds will be less than $a_\infty^{1/3}$. These are the so-called "breeze" solutions (Kennel, Fujimura, & Okamoto 1983) whose nonunique initial specific energies μ_0 and angular momenta

λ_0 are to be determinated by the pressure balance of the flows with their ambient media. If the magnetic pressure inside the flow is required to decline to zero at infinity (to balance a decreasing external pressure, for example), all submagnetosonic solutions have zero speed at infinity since $a_\infty \rightarrow 0$ (see also discussions in § 4.2). In this case, no physical flow solution exists. This may explain the lack of extended radio lobes in radio-quiet AGNs. It is not clear whether the above features for dragged 1-D flow along a fixed flux tube persist in dragged 2-D flows. It is possible that, when the cross-field force balance is considered self-consistently, the second fast point may not appear at all due to the extra degree of freedom in the cross-field direction in a 2-D flow. On the other hand, the drag force may alter the structure of critical surfaces so much that a unique critical, steady-state solution is not obtainable. The dragged self-similar solutions we studied are not good for resolving this important issue since the conventional fast point is modified by the self-similarity assumption so that it is no longer a critical point. These self-consistent self-similar solutions do raise an interesting point: the more strongly dragged solutions tend to be better collimated, as demonstrated in Figure 8.4*a*. Physically, better collimation is achieved when the dragged flow is slowed down and the inertia of the flow plasma becomes smaller, which makes it is easier for the flow to be collimated by the magnetic pinching force.

If most of the angular momentum released in accretion disks around supermassive black holes (Rees 1984) is removed by relativistic MHD winds, but most of energy released in the disks is carried away by radiation field, then the wind-ambient photon field interaction could be self-regulatory. They are related by the accretion rate, $\dot{M}$, of the disk. In particular, a strong

radiation drag on the MHD wind would increase the angular momentum extraction from the disk, and thus enhance the accretion rate $\dot{M}$. This increase in $\dot{M}$ in turn makes the photon field denser (and thus the radiation drag stronger), which will force the wind to extract even more angular momentum from the disk. The radiation field will keep amplifying until it completely shut off the wind. When this happens, the accretion rate is decreased (but maybe not shut off entirely if other processes of angular momentum removal mechanisms are operating too) and a lower photon energy density results which would allow the relativistic MHD wind to escape again. The same process would then repeat itself. Therefore, when the radiation drag is considered self-consistently in the entire wind-disk system, the flow may be intrinsically unsteady. The variation of flow speed maybe responsible for the formation of knots in the VLBI jets. Further work is clearly needed to clarify this point.

Finally, we wish to examine whether drag due to scattered radiation field in the broad line region far away from the base of the flow ($R \gg R_L$) can limit the flow Lorentz factor γ for relativistic flows in AGNs to ~ 10 even if γ is much larger than 10 initially. In this case, we have $\gamma \approx u \approx \mu - a \gg 1 \gg u_\phi$ from the energy equation (4.2). From equation (8.1), one then has the simplified expression

$$\frac{d\gamma}{ds} \approx -\frac{da}{ds} - \gamma^2 \nu(s), \tag{8.19}$$

which basically says that the Lorentz factor of the flow γ is increased by the decline in magnetic energy (the first term on the right-hand side) and decreased by the drag force (the second term on the right-hand side). The

quantity $s \equiv S/R_L$ is the dimensionless length of the poloidal field line measured from the base of the flow. Without solving for the structure of the flow self-consistently, it is impossible to determine $a(s)$. We shall concentrate on the special case where $da/ds \approx 0$, which corresponds to radial (or cylindrical) flows or kinetic energy-dominated flows. In this case one can integrate equation (8.19) to obtain

$$\gamma^{-1}(s) \approx \gamma_0^{-1} + \gamma_d^{-1}(s) \tag{8.20}$$

where the drag (characterized by a Lorentz factor $\gamma_d \equiv 1/[\int_{s_0}^{s} \nu(s)ds)$ reduces the flow from its initial Lorentz factor γ_0 at s_0 to a value γ at s. Therefore, a general condition for radiation drag to be effective in flow speed reduction when the magnetic force is negligible would be $\gamma_d \leq \gamma_0$. Let's examine whether this condition can be satisfied in the broad line region (BLR). For a narrow jet, one can replace s with distance along the axis z, and the drag coefficient due to scattered isotropic radiation (with luminosity of L_{iso}) has the form

$$\nu(z) \approx \frac{\nu_0}{z^2} = \frac{\sigma_T L_{iso}}{3\pi m c^3 R_L} \frac{1}{z^2}, \tag{8.21}$$

where $m = m_e$ for a electron-positron pair plasma and $m = m_p$ for a ordinary electron-proton plasma. Therefore, we have

$$\gamma_d \approx \frac{1}{\nu_0(1/z_0 - 1/z)}, \tag{8.16}$$

which, in the case $z \gg z_0$, yields

$$\gamma_d \approx \frac{z_0}{\nu_0} = \frac{3}{4} \frac{Z_0}{r_g} \frac{m}{m_p} \frac{L_E}{L_{iso}}, \tag{8.17}$$

where Z_0 is the height at which the magnetic acceleration becomes small compared with the drag force, and r_g and L_E are the gravitational radius and

the Eddington luminosity of the central object, respectively. The condition for γ_d in a pair plasma jet to be less than about 10 is that

$$\frac{Z_0}{250 r_g} \frac{0.01 L_E}{L_{iso}} < 1. \tag{8.18}$$

Therefore, if the reprocessed isotropic luminosity in the BLR is about 1% of L_E, then γ will be limited to less than about 10 as long as Z_0 is less than about $250 r_g$, no matter how large γ_0 is at Z_0 initially. Clearly, the speeds for those flows whose acceleration by magnetic processes is limited to a smaller region are more likely affected by the radiation drag.

8.4.2. Mass Entrainment

We have formulated the problem of mass entrainment by relativistic MHD outflows and showed that the flows are dragged to subrelativistic speeds and encounter a second fast magnetosonic point along prescribed flux tubes. More work is needed to treat the cross-field force balance of flux tubes self-consistently. The mass entrainment rate, which is parametrized in our simple treatment, must be obtained from physical considerations. In the case that the entrained mass comes from small cloudlets (for example BAL clouds) in the flows one has to compute self-consistently the mass transfer rate (from evaporation or stripping) from the clouds to the flows and the momentum transfer rate from the flows to the clouds due to both ram-pressure and magnetic interaction (since the clouds lag behind the flows), etc. The evaporation rate should be controlled by the confining pressure of the clouds: presumably, the lower the pressure, the faster the mass is transferred to the underlying flows from the clouds. In the case that confining pressure is supplied mainly by the magnetic field in the underlying flows, as is likely for

BAL clouds (Arav & Li 1993), one has the following self-regulatory behavior for the cloud-flow system: the faster the flow is moving, the lower the magnetic (and thus the cloud-confining) pressure becomes, and the more mass is entrained by the flow, which slows down the flow and builds up the magnetic (confining) pressure. Alternatively, mass can be added to the flows from mixing at the flow boundary due to Kelvin-Helmholtz instabilities, which has been proposed to explain the slowing down of some AGN jets from relativistic speeds on VLBI scale to subrelativistic speeds on larger scales. In addition, mass (from pair production, Phinney 1983) can be loaded continuously on to an electromagnetically-dominated outflow originating from near a rapidly rotating black hole (Blandford & Znajek 1977). It would be interesting to solve self-consistently for the dynamics of the ultrarelativistic jets and the pair plasma production rate.

CHAPTER IX

CONCLUSIONS AND FUTURE WORKS

We have studied the acceleration and collimation properties of steady-state, cold, relativistic MHD outflows associated with AGNs and neutron stars in the ideal MHD limit, assuming axisymmetry. We find that all relativistic MHD flows are collimated asymptotically towards the rotation axis, either to current-carrying cylinders when an ambient medium of finite pressure is present or to asymptotically current-free paraboloidal surfaces when the ambient pressure is negligible. At large distances from the base of the flow, the flow collimation will be extremely slow (in fact, only logarithmic), if the energy density of the electromagnetic field in the flow is small compared to the plasma kinetic energy density. This type of flow structure may exist in free pulsar winds such as that from the Crab pulsar, where the flux of the plasma kinetic energy is inferred to dominate over the Poynting flux in the wind.

To obtain complete flow solutions, one has to solve the modified Grad-Shafranov equation, which describes force balance across the flux surfaces, simultaneously with the energy equation describing flow along each flux surface. This is done first semi-analytically, by assuming that the flux surfaces in the relativistic flow are self-similar, and then numerically for a case in which the field lines are assumed to emerge vertically from an equatorial disk and are collimated parallel to the rotation axis far from the disk. We find that the electromagnetic energy is converted into plasma kinetic energy much more efficiently in a collimated flow than in a purely radial flow (which is the geometry generally assumed in earlier pulsar wind studies), and indeed

more than half of the total energy can end up in the form of plasma kinetic energy. Although the pressure in the flow could be much higher near the axis than close to the outermost edge due to pinching effect of toroidal magnetic fields, the importance of the ambient medium for confining the flow at its outmost edge cannot be ignored. We think that it is mainly the difference in the ambient pressure distribution that differentiates the relatively well-collimated jets in AGNs from the seemingly poorly-collimated Crab-like free pulsar winds.

The magnetic pressure, as well as the magnetic energy flux, in a relativistic MHD jet/wind far beyond the light cylinder radius is determined by the toroidal magnetic field strength, B_ϕ. Since B_ϕ at any point is proportional to the total electric current enclosed by the circle which is centered on the axis and passes through that point (Ampere's law), the return path of the current carried in the flow thus dictates the acceleration and confinement properties of the flow. In other words, the current closure problem, which was recognized in earlier attempts at modelling pulsar winds (Michel 1982), is also a central issue here. We hold the view that the jets in AGNs are always confined by the ambient media at their outermost peripheries. We believe that there should be no toroidal magnetic field induced in the ambient medium that helps confine the flow. This requires that all of the electric current leaving the flow base in one part of the flow has to return through either another part of the flow or a thin current sheet on the outermost surface. When the confining pressure in the external medium drops, the internal magnetic pressure (cancelled to some extent by the electrostatic force) has to decrease as well. This is accomplished by forcing the electric current to flow across the flux surfaces and thus yield flow acceleration at the same time. If

the external pressure asymptotes to some finite value, then the flow will become a current-carrying cylinder with a finite ratio of electromagnetic energy flux to kinetic energy flux, i.e., finite σ. If, on the other hand, the external pressure becomes vanishingly small or there is no external confinement at all, then the radius of the flow will become infinite, and $\sigma \to 0$. The fact that the inferred lower limit on the internal pressure (from equipartition argument) of jets from objects such as M87 seems to be much higher than the ambient pressure is not in conflict with external confinement, as long as the observed features are formed in the inner, high pressure parts of the jets. Our point of view is different from that of Appl & Camenzind (1993), who believed that relativistic MHD jets can be self-collimated, although their analysis is carried out in the unrealistic force-free limit. More works are needed in order to address this important issue adequately.

We contrast briefly relativistic and nonrelativistic MHD outflows. Their fundamental differences include the following. 1) At the fast magnetosonic point, rough equipartition is reached between kinetic and magnetic energy fluxes for nonrelativistic flows, while the kinetic energy flux is only a small fraction of the magnetic energy flux for relativistic flows. Most of the flow acceleration in relativistic flows is achieved beyond the fast point through the so-called "magnetic nozzle" effect. 2) For the same toroidal magnetic field strength at their outer boundaries, the external pressure needed for confinement is much smaller for relativistic flows than for nonrelativistic flows. This is because the induced electric field, which is negligible in nonrelativistic flows, can counter-balance the magnetic pressure to a large degree. We have considered only cold winds, which avoided the complications associated with the slow magnetosonic point and the cusp point (whose properties

remain largely unknown). The price we have paid is that the mass loading rate onto any given magnetic field structure remains unconstrained. How the plasma is loaded onto the rotating magnetic field lines will be a subject of future research.

In order to gain some insights into the fundamental question of why some QSOs are radio-loud, while the majority of them are radio-quiet, we studied the effects of radiation drag on relativistic MHD outflows. In a radial flow geometry, owing to the location of the fast magnetosonic point at infinity, the drag will not change the terminal speed of the flow, but will merely enhance the extraction of energy and angular momentum from the central rotating object and redistribute the locations of flow acceleration. If a flux tube diverges more rapidly than radial in a region where radiation drag is significant, then it is possible for the drag to affect the flow in the supermagnetosonic region and to reduce the terminal flow speed greatly. In some cases, the flow could be stopped completely. It is conceivable that this reduction in flow speed may facilitate the quenching of jets in some QSOs. However, the radiation drag is not likely to affect the slowly-accelerated flows very much. In such cases, one has to rely on other forms of drag, such as that due to entrainment of external material, to slow down the jets, if the central engines of radio-loud and radio-quiet QSOs are the same.

REFERENCES

Abbott, D. C. 1988, in Proceedings of the Sixth Solar Wind Conference, Vol. 1, ed. Pizzo, V. J., Holzer, T. E., & D. E., Sime, 149

Achterberg, A., Blandford, R. D., and Goldreich, P. 1983, *Nature*, 30, 607

Anile, A. M. 1989, Relativistic Fluids and Magneto-Fluids (Cambrigde: Cambridge Univ. Press)

Appl, S., & Camenzind, M. 1993, A&A, 270, 71

Arav, N., & Li, Z.-Y. 1993, ApJ, submitted

Ardavan, H. 1979, *M.N.R.A.S.*, 189, 397.

Arons, J. 1991, in IAU Colloq. 128, Magnetospheric Structure and Emission Mechanisms of Radio Pulsars, ed. J. A. Gil, T. H. Hankins, & J. M. Rankin, in press

Aschenbach, B., & Brinkmann, W. 1975, A&A, 41, 147

Begelman, M. C. 1992, in The Nature of Compact Objects in Active Galactic Nuclei, ed. A. Robison, & R. J. Terlevich (Cambridge: Cambridge Univ. Press), in press

—— 1993, in Astrophysical Jets, STScI Symp. Series Vol. 6, ed. D. Burgarella, M. Livio, & C. O'Dea (Cambridge: Cambridge Univ. Press), in press

Begelman, M. C., Blandford, R. D., & Rees, M. J. 1984, Rev. Mod. Phys., 56, 265

Begelman, M. C., & Li, Z.-Y. 1992, ApJ, 397, 187

Begelman, M. C., & Li, Z.-Y. 1993, ApJ, submitted

Begelman, M. C., Mckee, C. F. and Shields, G. A. 1983, ApJ, 271, 70

Begelman, M. C., and Sikora, M. 1987, ApJ, 322, 650

Bekenstein, J., & Eichler, D. 1985, ApJ, 298, 493

Benford, G. 1984, ApJ, 282, 154

Blandford, R. D. 1976, MNRAS, 176, 465

—— 1989, in Theory of Accretion Disks, ed. F. Meyer, W. J., Duschl, J. Frank, & E. Meyer-Hofmeister (Dordrecht: Kluwer), 35

—— 1990, in Active Galactic Nuclei, ed. Blandford, R. D., Netzer, H., & L. Woltjer (Berlin: Springer-Verlag)

Blandford, R. D., & Payne, D. G. 1982, MNRAS, 199, 88

Blandford, R. D., & Rees, M. J. 1974, MNRAS, 169, 395

Blandford, R. D., and Znajek, R. L. 1977, *M.N.R.A.S.*, 179, 433.

Bridle, A. H., & Perley, R. A. 1984, ARAA, 22, 319

Curtis, H. D. 1918, Lick Obs. Publication, 13, 11

Camenzind, M. 1986, A&A, 162, 32

—— 1987, A&A, 184, 341

—— 1989, in Accretion Disks and Magnetic Fields in Astrophysics, ed. G. Belvedere (Dordrecht: Kluwer), 129

—— 1990, in Reviews in Modern Astronomy 3, ed. G. Klare (Berlin: Springer-Verlag), 234

Chan, K. L., & Henriksen, R. N. 1980, ApJ, 241, 534

Cheng, K. S., Ho, C., & Ruderman, M. 1986, ApJ, 300, 500

Chiueh, T., Li, Z.-Y., & Begelman, M. C. 1990, ApJ, 377, 462 (CLB)

Coroniti, F. V. 1990, ApJ, 349, 538

De Young, D. S. 1993, ApJL, in press

Emmering, R. T., Blandford, R. D., & Shlosman, I. 1992, ApJ, 385, 460

Eichler, D. 1993, preprint

Goldreich, P., & Julian, W. H. 1969, ApJ, 157,869

—— 1970, ApJ, 160, 971

Fanaroff, B., & Riley, J. M. 1974, MNRAS, 167, 31p

Farrari, A., Trussoni, E., Rosner, R., & Tsinganos, K. 1985, ApJ, 294, 397

Heinemann, M., & Olbert, S. 1978, J. Geophys. Res., 83, 2457

Heyvaerts, J., & Norman, C. A. 1989, ApJ, 347, 1055

Holzer, T. E. 1977, J. Geophys. Res., 82, 23

Hughes, P. J. 1991, Beams and Jets in Astrophysics (Cambridge: Cambridge Univ. Press)

Jackson, J. D. 1975, Classical Electrodynamics (New York: John Wiley & Sons)

Kafatos, M. C., & Henry, R. B. C. 1985, The Crab Nebula and Related Supernova Remnants (Cambridge: Cambridge Press)

Kennel, C. F., & Coroniti, F. V. 1984, ApJ, 283, 694

Kennel, C. F., Fujimura, F. S. & Okamoto, I. 1983, J. Ap. Geophys. Fluid Dyn., 26, 147

Königl, A. 1989, ApJ, 342, 208

Koupelis, T., & Van Horn, H. M. 1989, ApJ, 342, 146

Kudritzki, R. P., et al. 1992, A&A, 257, 655

Kulkarni, S. R., & Hester, J. J. 1988, Nature, 335, 801

Lada, C. J. 1985, ARA&A, 23, 267

Landau, L. D. & Lifshitz, E. M. 1959, Fluid Mechanics (London: Pergamon)

Li, Z.-Y. 1993, ApJ, in press

Li, Z.-Y., & Begelman, M. C. 1993, in preparation

Li, Z.-Y., Begelman, M. C., & Chiueh, T. 1992, ApJ, 384, 567

Li, Z.-Y., Chiueh, T., & Begelman, M. C. 1992, ApJ, 394, 459 (LCB)

Lovelace, R. V. E. 1976, Nature, 262, 649

Lovelace, R. V. E., Berk, H. L., & Contopoulos, J. 1991, ApJ, 379, 696

Lovelace, R. V. E., Contopoulos, J., & Romanova, M. 1993, ApJ, in press

Lovelace, R. V. E., Mobarry, C. M. and Contopoulos, J. 1989, in Accretion Disks and Magnetic Fields in Astrophysics, ed. G. Belvedere (Dordrecht: Kluwer), 71

Lovelace, R. V. E., Mechanian, C., Mobarry, C., & Sulkanen, M. E. 1986, ApJS, 62, 1

Lovelace, R., Wang, J. C. L., & Sulkanen, M. E. 1987, ApJ, 315, 504

Lynden-Bell, D. 1978, Phys. Scr., 17, 185

Melia, F., and Königl, A. 1989, ApJ, 340, 162

Mestel, L. 1968, MNRAS, 138, 359

Michel, F. C. 1969, ApJ, 158, 727

—— 1982, Rev. Mod. Phys., 54, 1

Okamoto, I. 1978, MNRAS, 185, 69

Padman, R., Lasenby, A. N., & D. A. Green 1991, in Beams and Jets in Astrophysics, ed. P. A. Hughes (Cambridge: Cambridge Univ. Press), 484

Papaloizou, J. C. B., & Pringle, J. E. 1984, MNRAS, 209, 721

Parker, E. N. 1963, Interplanetary Dynamical Processes (New York: Interscience Publications)

Pelletier, G., & Pudritz, R. E. 1992, ApJ, 394, 117

Phinney, E. S. 1982, MNRAS, 198, 1109

—— 1983, Ph.D. Thesis, University of Cambridge

—— 1987, in Superluminal Radio Sources, ed. J. A. Zensus & T. J. Pearson (Cambridge: Cambridge University Press), 301

Phinney, E. S., Evans, C. R., Blandford, R. D., & Kulkarni, S. R. 1988, Nature, 333, 832

Pizzo, V. J. 1990, ApJ, 365, 764

Press, W. H., Flannery, B. P., Teukolsky, S. A., & Vetterling, W. T. 1989, Numerical Recipes (Cambridge: Cambridge Univ. Press)

Pudritz, R. E., & Norman, C. A. 1985, ApJ, 301, 571

Ray, T., & R. Mundt 1993, in Astrophysical Jets, STScI Symp. Series Vol. 6, ed. D. Burgarella, M. Livio, & C. O'Dea (Cambridge: Cambridge Univ. Press), in press

Rees, M. J. 1984, ARAA, vol. 22, 471

Rees, M. J. & Gunn, J. E. 1974, MNRAS, 167, 1

Safier, P. N. 1993, ApJ, 408, 115

Sakurai, T. 1985, A&A, 152, 121

——, 1987, Publ. Astron. Soc. Japan, 39, 821

Scargle, J. D. 1969, ApJ, 156, 401

Shu, F. H., Lizano, S., Ruden, S. P., & Najita, J. 1988, ApJ, 328, L19

Suess, S. T., & Nerney, S. F. 1973, ApJ, 184, 17

Sulkanen, M. E., & Lovelace, R. V. E. 1990, ApJ, 350, 732

Sikora, M., & Wilson, D. B. 1981, MNRAS, 197, 529

Sikora, M., Zbyszewska, M., and Begelman, M. C. 1989, in BL Lac Objects, ed. L. Maraschi, T. Maccacaro and M.-H. Ulrich (Heidelberg: Springer), 381

Stahler, S. 1993, in Astrophysical Jets, STScI Symp. Series Vol. 6, ed. D. Burgarella, M. Livio, & C. O'Dea (Cambridge: Cambridge Univ. Press), in press

Stocke, J. T., Morris, S. L., Weymann, R. J., & Foltz, C. B. 1992, ApJ, 396, 487

Trimble, V. 1968, ApJ, 73, 535

Wang, Q. D., Li, Z.-Y., & Begelman, M. C. 1993, Nature, in press

Wardle, M., & Königl, A. 1993, ApJ, 410, 218

Weber, E. J. & Davis, L. Jr. 1967, ApJ, 148, 217

Wiita, P. J. 1991, in Beams and Jets in Astrophysics, ed. P. A. Hughes (Cambridge: Cambridge Univ. Press), 379

Weymann, A. J., Turnshek, D. A., & Christiansen, W. A. 1985, in Astrophysics of Active Galaxies and Quasi-Stellar Objects, ed. J. Miller (Oxford: Oxiford Univ. Press), 333

APPENDIX A

EXPANSION NEAR THE LIMITING CYLINDRICAL RADIUS

In this appendix, we shall find approximate solutions to the flow equations (5.22) and (5.23) near the limiting radius x_∞, which has been computed together with τ_∞ for given total specific energy μ and initial conditions in § 5.3. At some radius x close to x_∞, it turns out that we can expand equation (5.23) into

$$\frac{d\ln x}{d\ln y} = \xi(x_\infty - x) = \xi\delta x, \qquad (A1)$$

to the leading order, where $\delta x = x_\infty - x \ll 1$ and ξ is some finite quantity to be determined. This expansion is equivalent to assuming the following power-law form, $y = C(x_\infty - x)^{-\iota}$, for the field line shape in the neighborhood of x_∞ (where the exponent $\iota = 1/(x_\infty\xi)$ and C is an integration constant). We can also write the equation (5.22) in similar form

$$\frac{d\tau}{d\ln y} = \zeta(x_\infty - x) = \zeta\delta x, \qquad (A2)$$

where the finite quantity ζ is obtained by expanding the right hand side of equation (5.22) to the order of δx. Since both $(y\dot{y})^{-1}$ and $\dot{y}^{-2}$ decrease faster than δx as $x \to x_\infty$, terms proportional to them are dropped. Therefore, we have

$$\zeta = -\frac{\tau_\infty^3}{1-\delta/x_\infty^2}\left(\frac{\partial f}{\partial x} + \frac{\partial f}{\partial \tau}\frac{d\tau}{dx} + \frac{\partial f}{\partial a}\frac{da}{dx} - \frac{\xi\delta}{\tau_\infty^2 x_\infty^2}\right), \qquad (A3)$$

where we have defined a function

$$f \equiv \frac{1}{x^2}\left[\frac{\lambda^2(1-\eta)^2}{\tau} + a^2\right] - \frac{(1-\beta)(1-\delta/x^2)a}{\sigma\tau^2},$$

and the derivatives and partial derivatives in equation (A3) are to be evaluated at $x = x_\infty$. We shall drop the subscript "∞" for clarity below. From equation (5.7b), we have

$$\xi = -\frac{\beta}{\sigma}\frac{da}{dx}, \tag{A4}$$

and (da/dx) is related to $(d\tau/dx)$ linearly by

$$\frac{da}{dx} = g_1\frac{d\tau}{dx} + g_2, \tag{A5}$$

where the coefficients g_1 and g_2 are given by equation (4.8) so that

$$g_1 = -\left\{\frac{(\lambda\eta x)^2}{a\tau[1-(1+\tau)x^2]} + \frac{a}{\tau}\right\},$$
$$g_2 = \frac{1}{a\tau}\left\{\frac{\lambda^2(1-\eta)^2}{\tau x^3} - \frac{2\lambda\eta x[\lambda\eta(1+\tau)-\mu]}{1-(1+\tau)x^2}\right\}.$$

If one divides equation (A2) by equation (A1), then one has $(d\tau/dx) = \zeta/(x\xi)$, which can be reduced to the following quadratic equation

$$\left[\frac{x\beta g_1}{\sigma\tau^3}\left(1-\frac{\delta}{x^2}\right)\right]\left(\frac{d\tau}{dx}\right)^2 + \left\{\left[\frac{x\beta g_2}{\sigma\tau^3}\left(1-\frac{\delta}{x^2}\right)\right] - \left(\frac{\partial f}{\partial\tau} + g_1\frac{\partial f}{\partial a}\right) - \left(\frac{g_1\beta\delta}{\sigma\tau^2x^2}\right)\right\}\frac{d\tau}{dx} - \left[\frac{\partial f}{\partial x} + g_2\frac{\partial f}{\partial a} + \left(\frac{g_2\beta\delta}{\sigma\tau^2x^2}\right)\right] = 0, \tag{A6}$$

for $(d\tau/dx)$, where the partial derivatives are given by

$$\frac{\partial f}{\partial x} = \frac{2x[\mu-(1+\tau)\lambda]^2}{\tau[1-(1+\tau)x^2]^2} + \frac{4x^3(1+\tau)[\mu-(1+\tau)\lambda]^2}{\tau[1-(1+\tau)x^2]^3} - \frac{2a^2}{x^3} - \frac{2(1-\beta)a\delta}{\sigma\tau^2x^3},$$
$$\frac{\partial f}{\partial\tau} = -\frac{2\lambda[\mu-(1+\tau)\lambda]x^2}{\tau[1-(1+\tau)x^2]^2} - \frac{[\mu-(1+\tau)\lambda]^2x^2}{\tau^2[1-(1+\tau)x^2]^2} + \frac{2x^4[\mu-(1+\tau)\lambda]^2}{\tau[1-(1+\tau)x^2]^3} + \frac{2(1-\beta)(1-\delta/x^2)a}{\sigma\tau^3}$$
$$\frac{\partial f}{\partial a} = \frac{2a}{x^2} - \frac{(1-\beta)(1-\delta/x^2)}{\sigma\tau^2}.$$

Once $(d\tau/dx)$ is fixed from equation (A6), then one can compute easily other quantities such as (da/dx), etc. In particular, one can compute the exponent $\iota = 1/(x_\infty\xi)$, and thus determine how fast the field line approaches the limiting radius x_∞.

APPENDIX B

ACCELERATION OF A TEST PARTICLE IN THE RADIATION FIELD FROM A SOURCE OF SMALL SOLID ANGLE

Consider the acceleration of a test particle of 4-velocity $u^\mu = (-\gamma, u, \omega, 0)$ in the radiation field from a source of small solid angle. Assume that the source is a disk of uniform intensity so that $I = 1$ for $\theta < \theta_0$ and 0 otherwise, where θ_0 is the half angle of the disk seen by the particle on its axis. Then the stress-energy tensor of the radiation field can be computed easily (with the superscripts 1, 2 and 3 denoting the $z-$, $x-$ and $y-$ components). To order θ_0^6, we obtain:

$$T^{00} = \int I d\Omega \cong \pi \left(\theta_0^2 - \frac{\theta_0^4}{12} + \frac{\theta_0^6}{360} \right) \qquad (B1)$$

$$T^{01} = T^{10} = \int I \cos\theta d\Omega \cong \pi \left(\theta_0^2 - \frac{\theta_0^4}{3} + \frac{2\theta_0^6}{45} \right) \qquad (B2)$$

$$T^{11} = \int I \cos^2\theta d\Omega \cong \pi \left(\theta_0^2 - \frac{7\theta_0^4}{12} + \frac{61\theta_0^6}{360} \right) \qquad (B3)$$

$$T^{22} = T^{33} = \int I \sin^2\theta d\Omega = \frac{1}{2}(T^{00} - T^{11}) \cong \pi \left(\frac{\theta_0^4}{4} - \frac{\theta_0^6}{12} \right) \qquad (B4)$$

and by symmetry all other terms are zero.

The test particle equation of motion is

$$\frac{dp^\mu}{dt} = \frac{1}{\gamma}\frac{dp^\mu}{d\tau} = -\frac{\sigma}{\gamma}\left(u_\alpha T^{\alpha\mu} + u_\alpha T^{\alpha\beta} u_\beta u^\mu \right) \qquad (B5)$$

which, in the limits $u \gg 1, \omega$ and γ^{-1} of order θ_0 or higher, yields the acceleration in $z-$ direction:

$$\frac{dp^1}{dt} \cong \frac{\pi\sigma\theta_0^2}{4\gamma^2}(1 - \omega^4) - \frac{\pi\sigma\theta_0^4}{2}\omega^2 - \frac{\pi\sigma\theta_0^6}{12}\gamma^2. \qquad (B6)$$

Therefore, in the limit $\omega \rightarrow 0$, the leading terms are:

$$\frac{dp^1}{dt} \cong \frac{\pi\sigma\theta_0^2}{4}\left(\frac{1}{\gamma^2} - \frac{1}{3}\gamma^2\theta_0^4\right). \qquad (B7)$$

Hence the drag term dominates when $\gamma > 3^{1/4}/\theta_0$ and the drag coefficient decreases rapidly away from the source ($\propto \theta_0^6 \propto r^{-6}$).

APPENDIX C

THE GREEN'S FUNCTION METHOD

The solution of equation (7.43) can be written formally as

$$\Delta = \int_0^x \nu_R(x') f(x') e^{\int_x^{x'} \nu_R(x'') g(x'') dx''} dx', \qquad (C1)$$

which incorporates the appropriate boundary condition that $\Delta = 0$ at $x = 0$. In the limit of large ν_R, the solution can be evaluated approximately to leading order in ν_R^{-1}.

The Green's function peaks narrowly near $x' = x$, with a width scaling as ν_R^{-1}. Since $f(x')$ and $g(x'')$ have much longer scale lengths, of order the scale of the flow, the integral can be evaluated by treating $f(x')$ and $g(x'')$ as constants over the dissipation length ($\sim \nu_R^{-1}$). The operation $\int dx$ is approximated by the multiplication of the source term by a factor $[\nu_R g(x)]^{-1}$. Since the source scales with ν_R, it follows that Δ is independent of ν_R in the limit of large ν_R. Actually Δ can be expressed in algebraic form since we know how to evaluate the above integral. The expression for Δ given in equation (7.45) follows immediately.

APPENDIX D

CONSTRAINTS ON AMBIENT RADIATION ENERGY DENSITY FOR OPTICALLY THIN CONICAL JET NEAR A COMPACT OBJECT

In order to treat the ambient radiation field as an effective drag force, the flow has to be optically thin. Given the mass flux in the flow, there exists an upper limit to the photon energy density above which the optically thin assumption is no longer valid. This occurs because a higher photon energy density leads to a smaller radial velocity near the base of the flow, which in turn leads to a higher plasma density and hence larger optical depth. Consider a conical jet originating from the region close to a black hole or neutron star. The Thomson optical depth for photons transversing the jet is approximately

$$\tau \cong \frac{\dot{m}\sigma_T \Delta\theta}{m_p v_r r}, \qquad (D1)$$

where $\dot{m}(\equiv f)$ is the mass flux per steradian, $\Delta\theta$ is the half-opening angle of the jet, and the continuity equation has been used. Since the optical depth increases monotonically towards the base of the flow r_b due to the increasing plasma density, it is sufficient to demand $\tau \lesssim 1$ at r_b for the entire jet to be optically thin. For an order of magnitude estimate, we can take r_b to be the gravitational radius $r_g \equiv GM/c^2$. The radial speed (v_r) at r_b must then satisfy

$$v_g \equiv v_r|_{r=r_g} \gtrsim \frac{\dot{m}\sigma_T \Delta\theta}{m_p r_g} \equiv v_{min}. \qquad (D2)$$

In the context of the approximations we made in the text, we assume that $x_g \equiv r_g \sin \Delta\theta / R_L \rightarrow 0$. In this limit we have, from equation (7.36), that

$$v_g = \frac{cu_g}{\gamma} \cong \left[\sqrt{1 + \left(\frac{\nu_R}{2}\right)^2} - \frac{\nu_R}{2} \right] \frac{cx_g}{\gamma}, \qquad (D3)$$

which, when substituted into equation (D2), yield an upper bound for ν_R as

$$\nu_R \lesssim \frac{x_g c}{v_{min}} - \frac{v_{min}}{x_g c}. \qquad (D4)$$

In the limit of strong drag $\nu_R \gg 1$, we have $u_g \ll x_g$, hence we require

$$\nu_R \lesssim \frac{x_g c}{v_{min}} = \frac{m_p c r_g^2}{\dot{m} \sigma_T R_L} \qquad (D5)$$

in order for the jet to be optically thin. Using the expression (7.40) for the drag coefficient, we obtain

$$\dot{m} U_0 \lesssim \frac{3c}{4} \left(\frac{m_p \Omega r_g}{\sigma_T} \right)^2 \qquad (D6)$$

where U_0 is the radiation energy density evaluated at r_b. If $\dot{m}$ is a fraction α of the Eddington accretion rate per steradian $\dot{m}_E \equiv L_E / 4\pi c^2 = GMm_p/(\sigma_T c)$, and $U_0 \equiv \beta L_E / 4\pi c r_g^2$, then the self-consistency condition for optically thin flow is

$$\alpha\beta \lesssim \frac{3}{4} \left(\frac{r_g \Omega}{c} \right)^2. \qquad (D7)$$